

Preliminary Design Review

Team Fontus (10)

Summer 2026 Semester

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Table of Acronyms

Abbreviation	Definition
AI	Artificial Intelligence
BMS	Battery Management System
CAD	Computer Aided Design
CAN	Controller Area Network
CAP	Corrective Action Plan
CCB	Change Control Board
CDAC	Condition–Departure–Asset–Consequence
CDH	Command and Data Handling
CDR	Critical Design Review
CER	Cost Estimating Relationship
CHS	Chassis Subassembly
CNC	Computer Numerical Control
COTS	Commercial Off-The-Shelf
CR	Change Request
DC	Direct Current
DM2	(DM short for DEM) Digital Elevation Model 2, or Nobile Rim 2
DoD	Depth of Discharge
DTE	Direct-to-Earth
EDAC	Error Detection and Correction
EOL	End of Life

EPS	Electrical Power Subsystem
FCU	Flight Computer Unit
F&A	Facilities & Administrative
FMEA	Failure Mode and Effects Analysis
FRR	Flight Readiness Review
FY	Fiscal Year
FSS	Farside Seismic Suite
GaAs	Gallium Arsenide
GIS	Geographic Information System
GDS	Ground Data Systems
GNC	Guidance, Navigation, and Control
HSMD	Human Spaceflight Mission Directorate
ICD	Interface Control Document
IMU	Inertial Measurement Unit
IMS	Integrated Master Schedule
JMARS	Java Mission-planning and Analysis for Remote Sensing
KDP	Key Decision Point
ISRU	In-Situ Resource Utilization
LCC	Life Cycle Cost
LCROSS	Lunar Crater Observation and Sensing Satellite
LiDAR	Light Detection and Ranging
LISTER	Lunar Instrumentation for Subsurface

	Thermal Exploration with Rapidity
LRD	Launch Readiness Date
LRR	Launch Readiness Review
LV	Launch Vehicle
MCOTS	Modified Commercial Off-The-Shelf
MECA	Microscopy, Electrochemistry, and Conductivity Analyzer
M&IE	Meals and Incidental Expenses
MCA	Mission Concept Academy
MCH	Mechanical Subsystem
MCR	Mission Concept Review
MMC	Maximum Material Condition
MLI	Multi-Layer Insulation
MOB	Mobility Subassembly
MPPT	Maximum Power Point Tracking
MR	Mission Requirements
MSolo	Mass Spectrometer Observing Lunar Operations
MTDC	Modified Total Direct Cost
MTD	Mission Task Document
NAND	Not-AND Flash Memory
NASA	National Aeronautics and Space Administration
NeuPAN	Neural Proximal Alternating-minimization

	Network
PAY	Payload Subsystem
PCB	Printed Circuit Board
PDR	Preliminary Design Review
PDU	Power Distribution Unit
PL Z+ Axis	Payload Coordinate Frame Z+ Axis
POL	Point-of-Load
POW	Power Subsystem
PSR	Permanently Shadowed Region
RFAs	Requests for Action
RTG	Radioisotope Thermoelectric Generator
SMD	Science Mission Directorate
SRR	System Requirements Review
STM	Science Traceability Matrix
SQCOTS	Space-Qualified Commercial Off-The-Shelf
SYS	System Requirement
TENG	Triboelectric Nanogenerator
THR	Thermal Requirement
TIM	Thermal Interface Material
TRL	Technology Readiness Level
WBS	Work Breakdown Structure
VIPER	Volatiles Investigating Polar Exploration Rover

1. Mission Overview

1.1. Mission Statement

The primary goal of this mission is to characterize the near-surface thermal energy profiles of the lunar regolith and assess the presence, abundance, and distribution of water ice and other volatile compounds. This investigation targets Permanently Shadowed Regions (PSRs) near the lunar South Pole to evaluate resource accessibility for future exploration goals.

The mission will gather physical temperature gradients across daylight-to-shadow transition zones alongside chemical composition data to identify volatile species. It will also collect surface structural imaging and vehicle mobility feedback metrics to map terrain hazards and operational performance.

For this mission a mobile surface rover will be deployed and targeted to land in a highly sunlit, low-slope area adjacent to a cold trap boundary. The rover will collect environmental data using a generic suite of payload instruments, including a subsurface drill probe, a volatile mass spectrometer, and an autonomous navigation imaging system.

In placing a surface rover on the lunar South Pole, the grasp of the inner solar system’s historic environment will be enhanced by directly measuring how planetary volatile traps preserve ancient compounds over geological time. Furthermore, investigating the natural thermal properties of the regolith provides the core data needed to evaluate the surface material's capacity to store energy for long-term exploration infrastructure.

1.2. Science Traceability Matrix

This mission’s science objectives are organized around a Science Mission Directorate (SMD) goal under Focus Area 3, Lunar Environment and Exploration Effects, and Human Exploration goals supporting In-Situ Resource Utilization (ISRU) and surface mobility/power demonstrations. Under the SMD goal, the mission seeks to determine the thermal properties and heat flow through the lunar regolith in order to quantify the abundance, isotopic composition, and volatile chemistry of water ice deposits within those regions. The Human Exploration objectives evaluate regolith thermal energy storage capacity, NeuPAN autonomous mobility performance, triboelectric energy harvesting, seismic activity, and surface electrostatic conditions. Together, these objectives characterize both the scientific properties of the lunar south polar environment and the environmental conditions, resources, and hazards that must be understood to support future sustained surface operations.

The science objectives are derived directly from the customer requirements and priorities established in the Missions Task Document, which call for a low cost robotic lunar surface system which is capable of exploring permanently shadowed regions, conducting high value lunar science, and reducing uncertainty associated with future human exploration. These priorities drive the mission toward measurements of volatile resources and the thermal conditions which govern their stability while also motivating technology and environmental investigations relevant to mobility, power generation, and surface hazards. Instrument selection reflects the need to obtain these measurements within mass, power, cost, environmental, and operational constraints, favoring flight heritage instruments for primary science measurements while also integrating technology demonstrations whose failure would not eliminate the mission’s core thermal and volatile science return.

The measurement strategy translates each objective into a physical parameter, observable, instrument capability, and mission requirement within the STM. LISTER measures subsurface temperature profile and heat pulse responses to determine thermal conductivity, directly supporting the thermal property and regolith energy storage objectives represented by MR-02 and MR-05. MSolo measures mass-to-charge peak and relative signal amplitudes for targeted volatile species to determine volatile abundance and chemistry in support of MR-03. The geophone array records lunar ground motion and seismic signals for MR-08, while MECA measures surface electrical potential and charge behavior for MR-09 and TENG records voltage, current, and power generated during regolith interaction for MR-07. NeuPAN performance is evaluated through position deviation, heading error, slip, and obstacle clearance measurements during rover traverses in support of MR-06. This measurement approach provides direct traceability from the high level science and exploration goals to measurement observations and the systems that are used to collect them.

Science Goals		Science Objectives	Science Measurement Requirements	
			Physical Parameters	Observables
Science Exploration Goal (SMD)	Science Goal 1 *Focus area 3: Lunar Environment and Exploration Effects: Quantify the present-day lunar environment and assess interactions between the environment and exploration activities, including dust, radiation, and impact hazards.*	Science Obj. 1 Characterize the thermal properties and heat flow of lunar regolith near permanently shadowed regions to determine their influence on the preservation and stability of volatile deposits.	Phys. Para. 1 Categorize subsurface temperature profile and thermal conductivity to 1m depth	Obs. 1 Temperature readings at 0.5m depth intervals up to 1m, 20-120 Kelvin range, and 0.5K resolution with heat pulse decay time recorded at each depth to achieve endogenic heat flux accuracy within 10%.
			Phys. Para. 2	Obs. 2
		Science Obj. 2 Quantify the abundance, isotopic composition, and associated volatile chemistry of water ice deposits within permanently shadowed regions.	Phys. Para. 1 Categorize the chemical composition of the regolith and how it changes over time	Obs. 1 Measured abundance by mass to charge peaks and relative current amplitudes for H20(18), CO(44), NH3(17), CO(28), CH4(16), H2S(34) sampled over discrete intervals at <100 ppm sensitivity
			Phys. Para. 2	Obs. 2
Human Exploration Goal (HSMD)	Science Goal 2 *In-situ resource utilization (ISRU: AS-03/LI-07): Prospecting, characterization, or early stage extraction methods*	Science Obj. 1 Determine the efficiency of thermal energy storage capacity of lunar regolith near permanently shadowed regions and this will support if its feasible to use a thermal energy reservoir to supply power to lunar base utilities.	Phys. Para. 1 Volumetric heat storage capacity and thermal retention time of lunar regolith as a function of compaction density	Obs. 1 Temperature of regolith sample at known density recorded over 24-72 hour heating/cooling cycles
			Phys. Para. 2	Obs. 2
		Science Obj. 2	Phys. Para. 3	Obs. 3
			Phys. Para. 5	Obs. 5
Human Exploration Goal (HSMD)	*Surface Mobility Systems(LI-06) : Demonstration of rover or mobility platform capabilities*	Science Obj. 1 Demonstrate the performance of a NeuPAN-based autonomous motion system	Phys. Para 1 Determine rover navigation accuracy, heading deviation, and wheel slip ratio across variable slope terrain (<15°)	Obs. 1 Measure position deviation relative to inertial baseline, slip angle, and obstacle clearance parameters at 1 Hz logging rate over 50m traversals.

		Science Obj. 2		
Human Exploration Goal (HSMD)	*Surface Power Generation and distribution(LI-01): Demonstration of power systems or energy storage approaches*	Science Obj. 1 Demonstrate the feasibility of harvesting electrostatic charge accumulated at PSR boundary terrain as a supplemental power source for low-power lunar operations	Phys. Para 1 Determine harvested voltage, output current, and cumulative electrical power generated from wheel-regolith triboelectric interactions.	Obs. 1 Log continuous electrical potential across load nodes, charging current into storage capacitors, and generating power across -40 V to +4.5 kV input range.
		Science Obj. 2		
Human Exploration Goal (HSMD)	*Surface Enviorment Categorization (AS-01): measurement of dust, micrometerites or space*	Science Obj. 1 Determine the frequency, magnitude, and spectral characteristics of natural seismic activity and artificial rover-induced acoustic signals near PSR boundaries.	Phys. Para 1 Determine ground vibration frequency, seismic wave propagation velocity, and signal duration across shallow regolith strata.	Obs. 1 Record ground acceleration magnitude and seismic wave arrival times continuously at ≥ 200 Hz sampling rate with sensitivity $<10^{-6}$ m/s^2 across 0.1 -- 100 Hz band.
		Science Obj. 2 Characterize surface electrostatic environment, charge accumulation, and micro-scale grain morphology to identify hazards affecting mechanical hardware.	Phys. Para 2 Determine surface charge density, electric field magnitude, and particle diameter/shape distribution across illuminated and shadowed transitions.	Obs. 2 Measure surface electric potential (-400 V to +10 V), electrometer charge transfer, and capture micro-scale grain images.

Figure 1: Science Traceability Matrix Goals, Objectives, and Requirements

Instrument Performance Requirements	Predicted Instrument Performance	Instrument	Mission Requirement ID #
Temperature Resolution: 0.5K resolution across 20 -120 K range	Better than 0.001K resolution across 100-400K	LISTER (Lunar Instrument for Subsurface Thermal Exploration with	MR-02: The mission shall obtain temperature profiles

Measurement Depth Interval: 0.5m intervals down to 1m depth	0.5m intervals to 1m	Rapidity)	and thermal conductivity in permanently shadowed regions
Heat Pulse Decay Time: Derived thermal conductivity recording at every depth within ± 10% accuracy	PRT needle sensor monitors temperature as it rises and falls		
Mass-to-charge peak detection range: H20(18),CO(28),CO2(44),NH3(17,CH4(16),H2S(34)	Unit mass resolution 1-100 m/z, covers all desired targets	MSOLO (Mass Spectrometer for Observing Lunar Operations)	MR-03: The mission shall qualify the abundance, isotropic composition, and volatile chemistry of water deposits within permanently shadowed regions
Contamination Detection: Ability to distinguish native lunar volatiles from rover volatiles	Onboard calibration Gas system with periodic in-situ calibration to separate native signals from lander signals		
Detection Sensitivity: < 100 pm	95% confidence in measuring differences greater than 15 ppm		
Temperature Measurement over 24 hour cycle:Temperature and derived thermal conductivity measurements at 3 points across a 24-74 hour window s	With a single 130g nitrogen tank given 12 minutes of excavation time, LISTER can support its initial descent plus one additional redeployment	LISTER (Lunar Instrument for Subsurface Thermal Exploration with Rapidity)	MR-05:The mission shall determine the efficiency of thermal energy storage capacity of lunar regolith located near permanently shadowed regions
Position Accuracy: Real-time relative position error <0.5 m over 50m transit	Real-time stereo camera visual odometry combined with chassis IMU feedback; maintains path error within ± 0.3 m across slope boundaries.	Rover GNC System / NeuPAN	MR-06: The mission shall measure the metrics of its NeuPAN-based autonomous motion systems.
Slope Navigation: Maintain stability and control on slopes up to 15 degrees			
Obstacle Clearance: Identify and bypass surface hazards up to 20 cm height			
Detectable electrostatic voltage and current outputs: Measurable voltage within regolith charge accumulation (within the range of 40V/4.5kV)	17.4V open circuit voltage	TENG tribo electrometer	MR-07:The mission shall measure and quantify voltage, current, and power output measurements from charged regolith
Detectable output current: Measurable current within charge accumulation	0.96 short circuit current at 10Hz		
Operating temperature range: Functional across PSR boundary temperature range	Performance degrades above 260K		
Natural frequency of 7.5 HZ for geophones given the Active seismic experiment heritage geophone sensor	Geophone reads from 28 HZ (GS-14-L3), the frequency can be tuned by the company if given a quote with the spring/coil specifications	3 geophones (coil magnet seismometers deployed as a linear array)	MR-08: The mission shall determine the frequency and intensity of seismic activity on the moon

Sensitivity : less than <10^-6 m/s	11.42 V (m/s)-Geopace specs sheet		
Operating temperature range: -256 F to 200 F	Operating temperature of GS-14-L3 geophone is -40 C to 80 C		
Sample rate: 1060 bits/second (about 0.6 seconds per data frame) , sensitivity: 2×10^{-10} m/s² per √Hz at 1 mHz	Operating temperature: -40 C to 60 C, Digitization: 24 bit delta sigma, sample interval: 0.25, 0.5, 1, 2, 4 ms	GSB Data Recorder (Digitizer is included and also found from the Geospace Technologies equipment sheet), a quote from geospace technologies can be done to refine the gain frequency and sampling rate	MR-08: The mission shall determine the frequency and intensity of seismic activity on the moon
Surface Electric Potential: -400V to +10V	1.8 kV/V, 7.4kV full-scale, 3.5V resolution	MECCA/MSI	MR-09: The mission shall characterise surface electrostatic environment and charge accumulation
Charged particle/Ion Detection:Resolve charge differences down to at least 10^6 elementary charges across PSR terrain	1800 plus or minus 0.9 pC full scale; resolution of 5.5x10^6 elementary charge		
Ion Current Sensing: Detect ion currents in pA range at PSR boundary	120 pA full-scale, 60 fA resolution, exceeds requirement by 3 orders of magnitude		

Figure 2: Science Traceability Matrix Instruments and Performance

1.3. Summary of Mission Location

The mission targets the Moon, specifically the lunar south polar region where permanently shadowed regions occur within craters and other topographic depressions. The Moon has an almost negligible atmosphere and an impact generated regolith surface, while polar PSRs can reach temperatures near 40K and act as cold traps capable of preserving water ice and other volatile compounds. These characteristics make the south pole valuable for investigating both the present day lunar environment and conditions relevant to future surface exploration.

This site is located in the Lunar south pole and is referred to as Nobile Rim 2 in the DM2 region located at the coordinates 84.20156 degrees South, 60.699889 degrees East. This coordinate corresponds to the highest ranked candidate identified within Nobile Rim 2 by the multi-criteria landing site analysis of Pena-Asensio, Neira-Acosta, and Sanchez-Lozano (2025), and was used as the reference point for further mission specific site refinement done in JMARS.

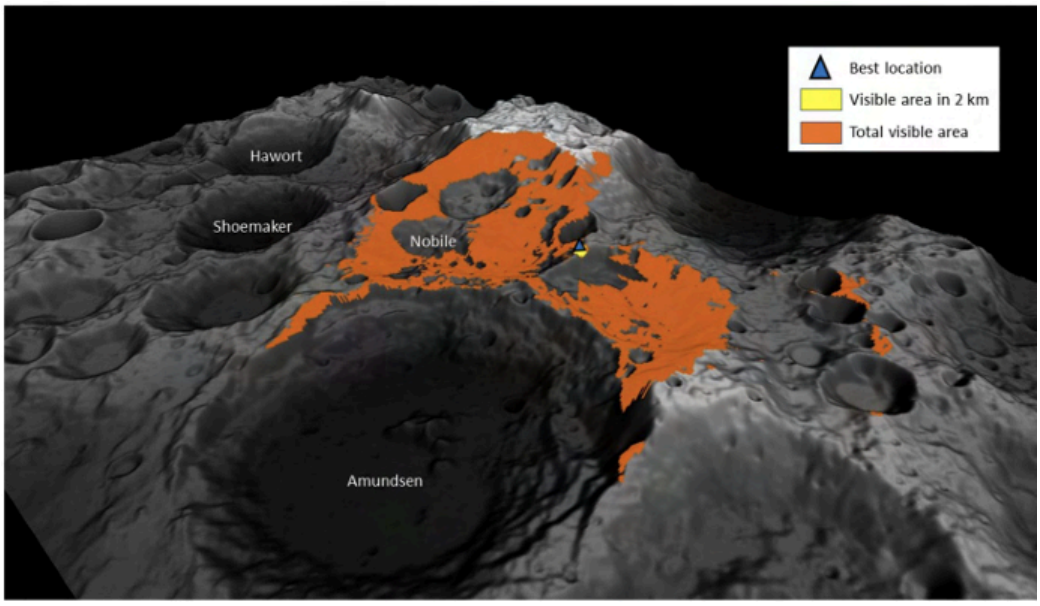


Figure 3: Nobile Rim 2 Localized Context

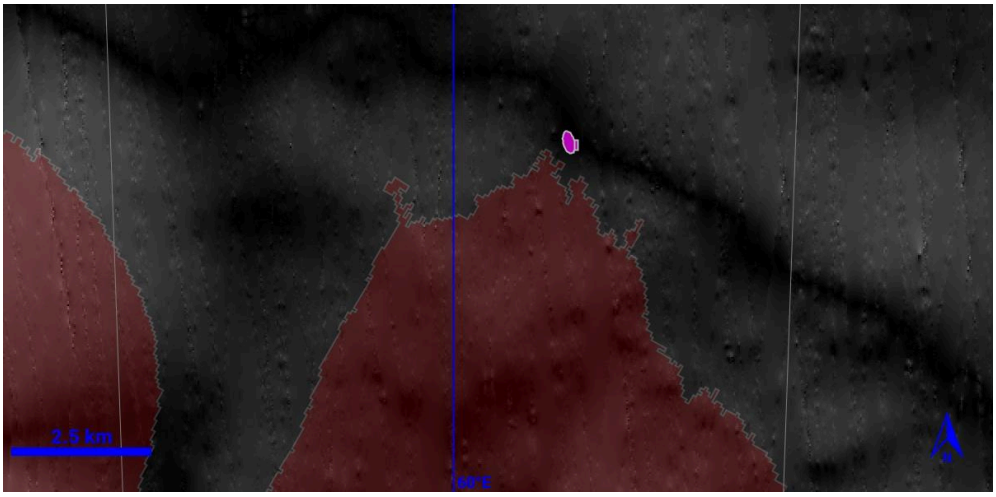


Figure 4: Regional JMARS Context

The pink ellipse in the above Figure X is the landing zone, and the red areas represent PSRs.

Nobile Rim 2 was selected because its proximity to the PSR terrain supports several mission objectives within the same investigation region. The PSR provides the low-temperature environment needed to relate LISTER measurements of subsurface thermal properties to volatile stability and provides MSolo access to terrain where water and other volatiles may be preserved. The transition between illuminated and PSR terrain also provides relevant conditions for electrostatic and seismic characterization, TENG testing, and NeuPAN mobility measurements, allowing the mission to characterize both the scientific value and operational hazards of the lunar south pole environment.

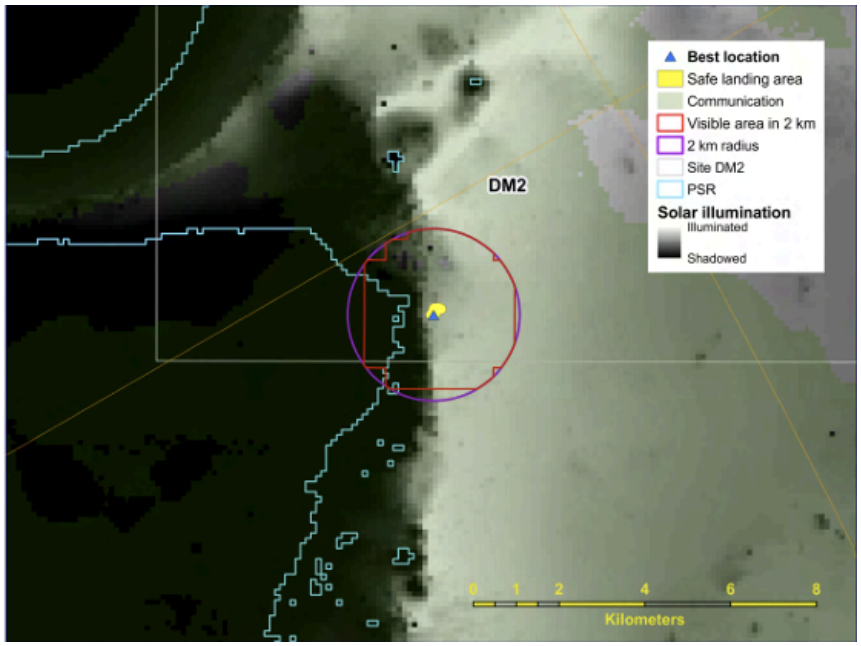


Figure 5: Science Context

The published candidate region was evaluated further in JMARS to determine whether its local terrain could accommodate the mission landing geometry. Initial slope analysis found several nearby areas with more than 5 degree slopes, preventing a 200m radius circular area from remaining entirely below the customer and engineering slope threshold. A 400m major axis landing ellipse was therefore fitted within the surrounding low slope terrain while maintaining access to the PSR within 250m. The refined location also provides favorable solar illumination, while the combination of accessible illuminated terrain and nearby permanent shadow supports repeated rover excursions, recharging, thermal recovery, and science collection.

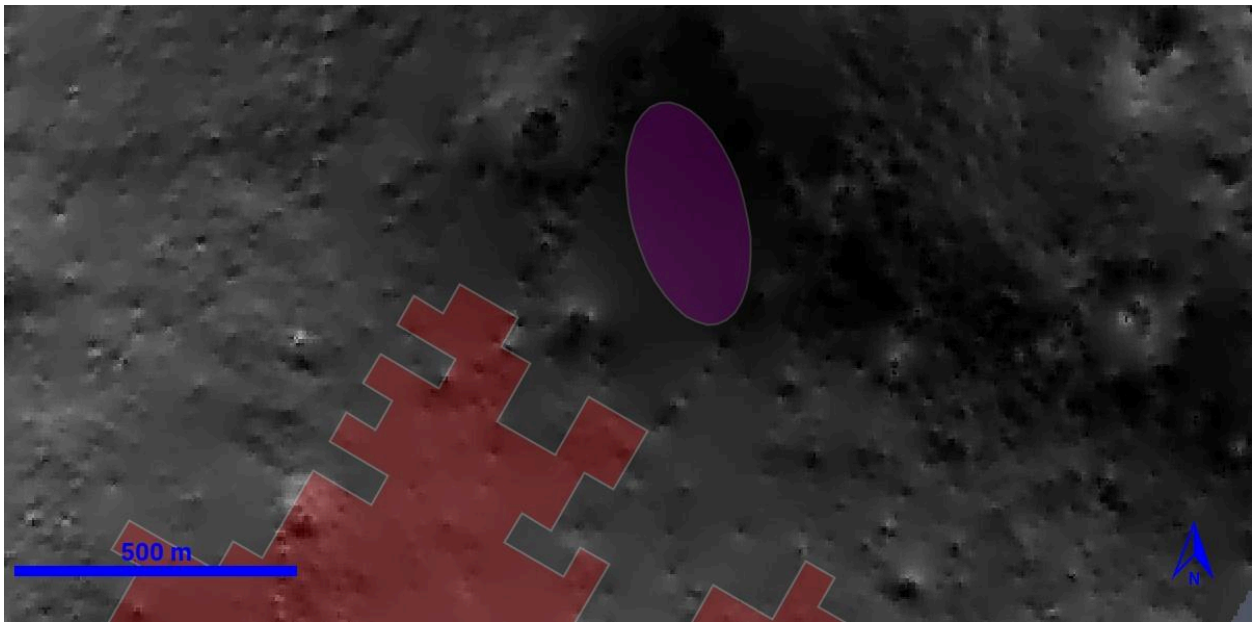


Figure 6: JMARS Slope and Landing Geometry Verification

The above figure shows the selected landing ellipse, its proximity to the PSR, and the geometric challenges that came with slope and engineering considerations. Again, the pink ellipse is the landing zone while the red area is the PSR.

The customer requires a 0.2km radial precision landing ellipse located on terrain with slopes below 5 degrees, solar illumination, and Direct-to-Earth communication during surface operations, with the landing site positions within 250m of the intended Exploration Zone. The refined Nobile Rim 2 ellipse satisfies the current slope, illumination, and EZ-access criteria established through JMARS analysis, while published visibility analysis of the candidate region supports favorable DTE access that can be integrated into the surface operations timeline.

1.4. Mission Requirements

The customer constrains the overall mission to develop a low-cost lunar robotic surface payload system that conducts post-landing science operations in and near permanently shadowed regions of the Moon. The design is directly constrained by a 175 kg total mass limit, a 1.5 m × 1.5 m × 1.0 m stored configuration envelope, a $250 million cost cap, no RTG-derived hardware, no more than 5 g of radioactive material, a December 1, 2028 system integration readiness date, and a February 1, 2029 launch readiness date. In addition, the system must operate within the harsh lunar environment, including the vacuum of space, extreme temperature conditions, and permanently shadowed regions, all of which impose significant constraints on the vehicle's design and operation.

These mission requirements establish the engineering baseline for the development of the lunar robotic surface payload system and provide a framework for evaluating subsystem designs throughout the project lifecycle. Each requirement was developed to ensure that the proposed mission architecture satisfies the customer's scientific objectives while remaining within the established technical, cost, schedule, and operational constraints. Collectively, these requirements support system-level decision making, subsystem integration, and verification activities by providing measurable criteria that guide design development and ensure compliance with the overall mission objectives.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
MR-01	The mission shall characterize environmental and exploration hazards in lunar south pole permanently shadowed regions to support future human lunar surface operations.	Establishes the top-level mission function and objective as described by the Mission Task document and the Science Traceability Matrix, supporting Focus Area 3 as chosen by the team.	Customer constraint, Science Traceability Matrix	SYS-01, 04, 07, 08, 10, 11	Demonstration	Partially Met
MR-02	The mission shall obtain temperature profiles and thermal conductivity in	Understanding the thermal environment and the regolith's	Science Traceability Matrix,	SYS-06	Demonstration	Met

	permanently shadowed regions to a depth of 1 meter.	thermal properties of the PSRs, of at most 1 meter in depth.	Science Exploration Goal			
MR-03	The mission shall quantify the abundance, isotropic composition, and volatile chemistry of water ice deposits within permanently shadowed regions.	Science Objective of categorizing the chemical composition and properties of water ice deposits within PSRs for future research	Science Traceability Matrix, Science Exploration Goal (Focus Area 3)	SYS-11	Test	Met
MR-04	The mission shall quantify the depth and lateral continuity of the subsurface dielectric boundaries within the regolith.	Science Objective to determine if the water ice is possible for in-situ resource utilization	Science Traceability Matrix, Human Exploration Goal (ISRU: AS-01/LI-07)	SYS-11	Test	Partially Met
MR-05	The mission shall determine the efficiency of thermal energy storage capacity of lunar regolith located near permanently shadowed regions.	Science Objective to determine if the regolith is a candidate for in-situ resource utilization to use as a thermal energy reservoir.	Science Traceability Matrix, Human Exploration Goal (ISRU: AS-01/LI-07)	SYS-11	Test	No

MR-06	The mission shall measure the metrics of its NeuPAN-based autonomous motion systems.	HSMD Objective to demonstrate rover or mobility platform capabilities	Science Traceability Matrix, Human Exploration Goal (LI-06)	SYS-05	Demonstration	Met
MR-07	The mission shall measure and quantify voltage, current, and power output measurements from charged regolith.	HSMD Objective to demonstrate the feasibility of harvesting electrostatic charge from regolith at PSR boundary as a supplemental power source.	Science Traceability Matrix, Human Exploration Goals (LI-01)	SYS-11	Test	Not Met - 0.1 Hz low end lost its instrument when the very broad and seismometer was cut; geophones can have a cutoff frequency of 4.5 Hz which

						can be used to test lower frequencies to higher frequency events like shallow moon quakes.
MR-08	The mission shall determine the frequency and intensity of seismic activity on the moon.	HSMD Objective to measure the physical and internal characteristics of the moon.	Science Traceability Matrix, Human Exploration Goals (AS-01)	SYS-11	Test	Met
MR-09	The mission shall characterise surface electrostatic environment and charge accumulation.	HSMD Objective to identify hazards that could affect equipment.	Science Traceability Matrix, Human Exploration Goals (AS-01)	SYS-11	Test	No

MR-10	The mission shall have a total life cycle cost allocation (LCC) of $250 million for all costs post PDR (Phases C through F), excluding an allocated system budget.	Customer derived cost requirement	Customer constraint	SYS-12	Inspection	Met
MR-11	The Lunar Surface Vehicle System shall be ready for system integration by December 1, 2028.	Customer schedule constraint	Customer constraint	SYS-13	Inspection	Met
MR-12	The Lunar Surface Vehicle shall be ready for launch by February 1, 2029.	Customer schedule constraint	Customer constraint	SYS-14	Inspection	Met
MR-13	The Lunar Surface Vehicle System shall remain operational for a minimum of two weeks after landing, with a planning bound of maximum one month, to complete planned science operations.	The mission timeline post-landing was derived by the Science subteam, setting a minimum time requirement such that the system shall survive in order to sufficiently obtain enough science data.	Science and Engineering Derived	SYS-09	Test	Partially Met - Ops concept and recharge strategy defined; site illumination of 11-12 consecutive sunlit

						days requires verification against the 14-day requirement
MR-14	The Lunar Surface Vehicle System shall be compatible with the launch vehicle mechanical mounting interface defined in the MCA LV ICD.	The customer ICD requires that physical payload-to-launch-vehicle mounting interface, complying to this requirement allows for the vehicle to integrate onto and to fly on the LV, avoiding late design changes.	Customer constraint, Launch Vehicle ICD	SYS-15	Analysis (fit check), Inspection	Partially Met
MR-15	The Lunar Surface Vehicle may not exceed the operating temperature of -10°C to 30°C while in the PSRs.	If the Lunar Surface vehicle exceeds the operating temperature then we will need to abort the mission with incomplete data.	Science and Engineering Derived	SYS-06	Test	Partially Met - Design and control strategy defined to the

						required band; compliance pending thermal model validation.

Figure 7: Mission Requirements Table

1.5. Concept of Operations (ConOps)

1. Mission Overview

The overall mission concept aims to establish the area around Nobile Rim 2, as a potential site for future development. The mission’s operational objective is to characterize subsurface thermal properties, volatile chemistries, and seismic activity while using other low cost instruments to interact with and measure the environment. This is done using a rover deployed from a lander positioned at the selected landing site and the use of these instruments will continue until the end of life through passive monitoring and active measurement.

2. Mission Timeline

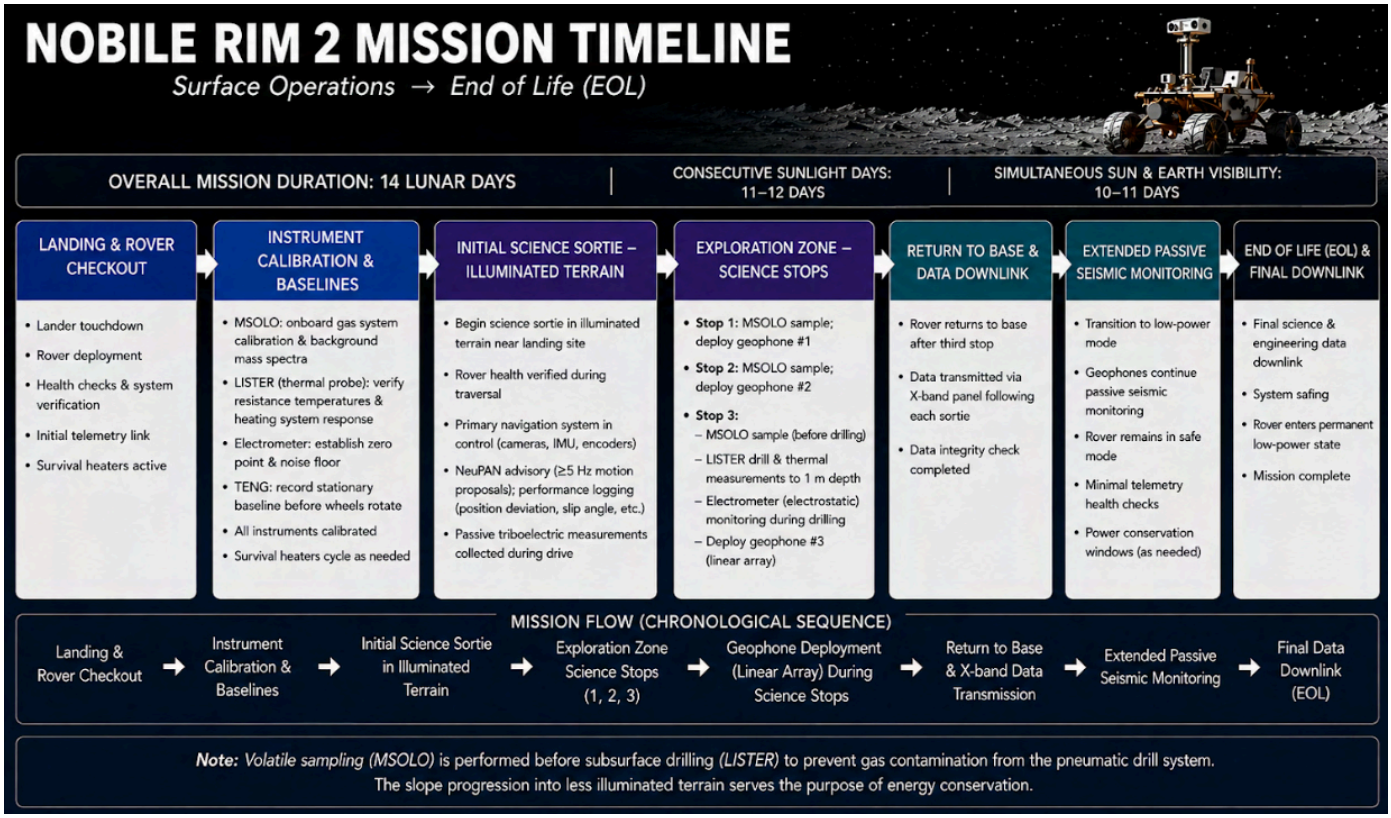


Figure 8: Nobile Rim 2 mission timeline from surface operations through End of Life (EOL).

3. Mission Operations

After the lander touches down the rover completes standard health checks and instrument calibration sequences. The MSOLO completes an onboard calibration of its gas system and records the background mass spectra of the environment to determine lander based materials from materials unique to the lunar surface. The subsurface thermal probe verifies resistance temperatures and confirms that its heating system responds against the appropriate thermal baseline. As well, the electrometer establishes zero point and the noise floor while the rover remains stationary. Finally the triboelectric generator records a stationary baseline before the wheels begin to rotate.

The science sortie begins in illuminated terrain near the landing site; this is where the rover's health is verified. During the traversal, the navigation cameras, hazard cameras, inertia measurement unit and wheel encoders all remain the primary system and only system to have command over the lunar vehicle. However NeuPan operates as a secondary advisory only demonstration running in parallel. It generates motion proposals at a rate of minimum 5Hz. NeuPan's performance is assessed by comparison in that the position deviation, slip angle, and other errors are logged as relative to the primary navigation system. The resulting score percentage deviation serves as a performance metric. Aside

from this, triboelectric measurements are collected passively. Passive collections allow for data to be taken without making extra stops that would deplete energy. The overall aim of this concept is to use every part of the rover that touches the moon to do science. Upon reaching the exploration zone, background measurements are taken. The MSOLO samples at the first two stops and at the third stop the LISTER samples the subsurface temperature and then proceeds to drill cycles to 1m in depth. The electrometer part of the LISTER acts as a constant characterization tool for the electrostatic environment to determine any charging associated with regolith disturbance. The Mass spectrometer is sampled before the LISTER because its drill has a pneumatic evacuation system that draws from the nitrogen supply. If it was done otherwise this would contaminate the volatile measurement. Seismic geophones are deposited at each spot in the linear array configuration. After the rover finishes data collection at the third spot the rover returns back to base. The collected data is transmitted via the X-band panel following each sortie.

Once the science traversal phase is completed the mission transitions to an extended low power passive seismic monitoring phase for the remainder of the mission with a final data downlink at EoL.

4. Operational Rationale

The sequencing supports the mission objective and was made to service the operational constraints. Placing volatile sampling ahead of subsurface drilling protects the MSOLO measurements from gas contamination. The slope progression into less illuminated terrain as well serves the purpose of energy conservation.

1.6. Vehicle Design Summary

1. Vehicle Overview (At least 1 paragraph)

A mobile surface rover will be deployed and targeted to land in a highly sunlit, low-slope area adjacent to a cold trap boundary. The rover will collect environmental data using a suite of payload instruments, including a subsurface drill probe, a volatile mass spectrometer, and an autonomous navigation imaging system. The surface rover supports the mission objectives by measuring volatile traps that can provide information about the geological history of the lunar surface. Other measurements that will be taken include thermal, volatile, subsurface, seismic, electrostatic, and mobility data.

The major design features of the rover include passive or minimally actuated systems, metallic structures, dust-tolerant moving joints, and standard spacecraft/rover integration approaches. The chassis structure is composed of Aluminum 6061-T6, while the body panels or enclosure use an aluminum honeycomb sandwich panel. The rover design

also includes six rigid aluminum grouser wheels and a ground, navigation, and control assembly that includes Light Detection and Ranging (LiDAR) aided navigation sensors, inertial measurement units (IMUs), and supporting sensor hardware. The rover uses a six-wheel rocker-bogie drive architecture with one brushless DC motor per wheel, LiDAR-aided navigation with IMU and odometry, and lock wires/cables for fastener locking at the launch vehicle interface.

2. *Vehicle Summary Table*

Overall dimensions	Total mass	Total maximum power consumption
1.5 x 1.5 x 1.0 m	158.9	317.6

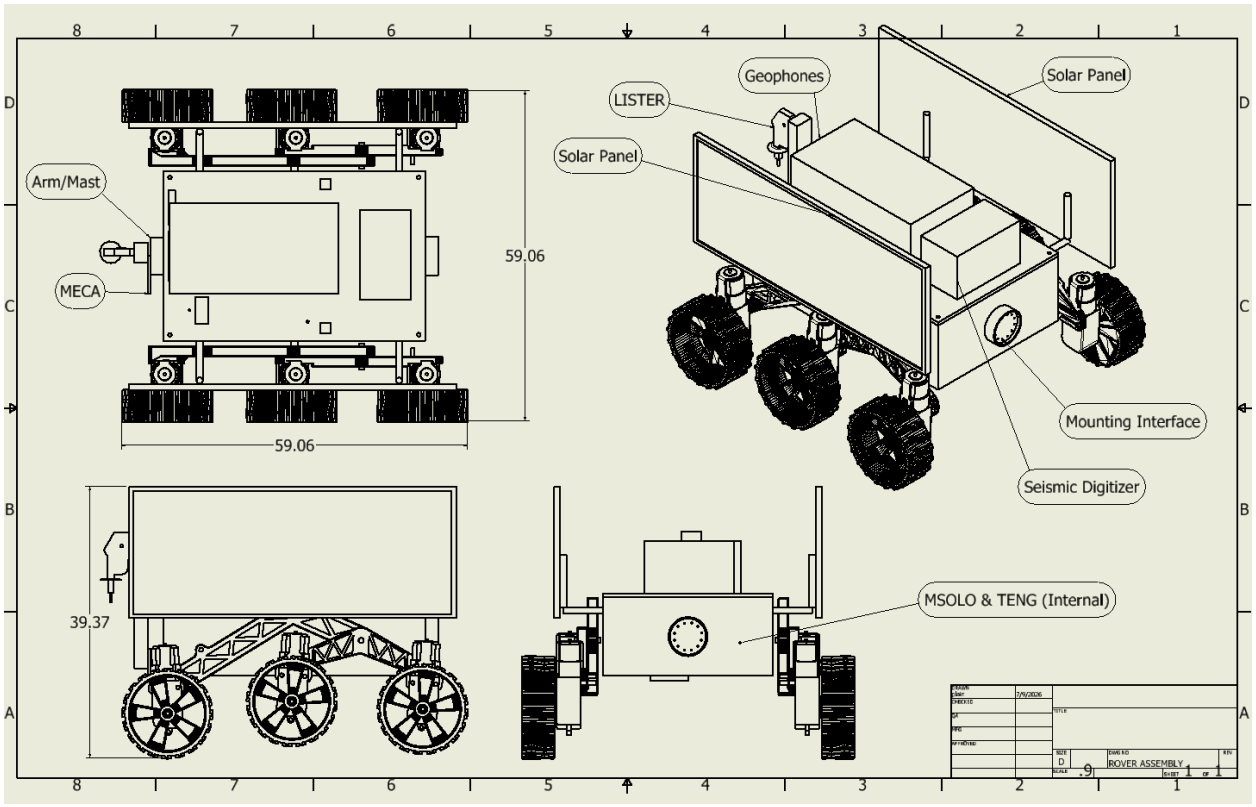
Figure 9: Vehicle Summary Table

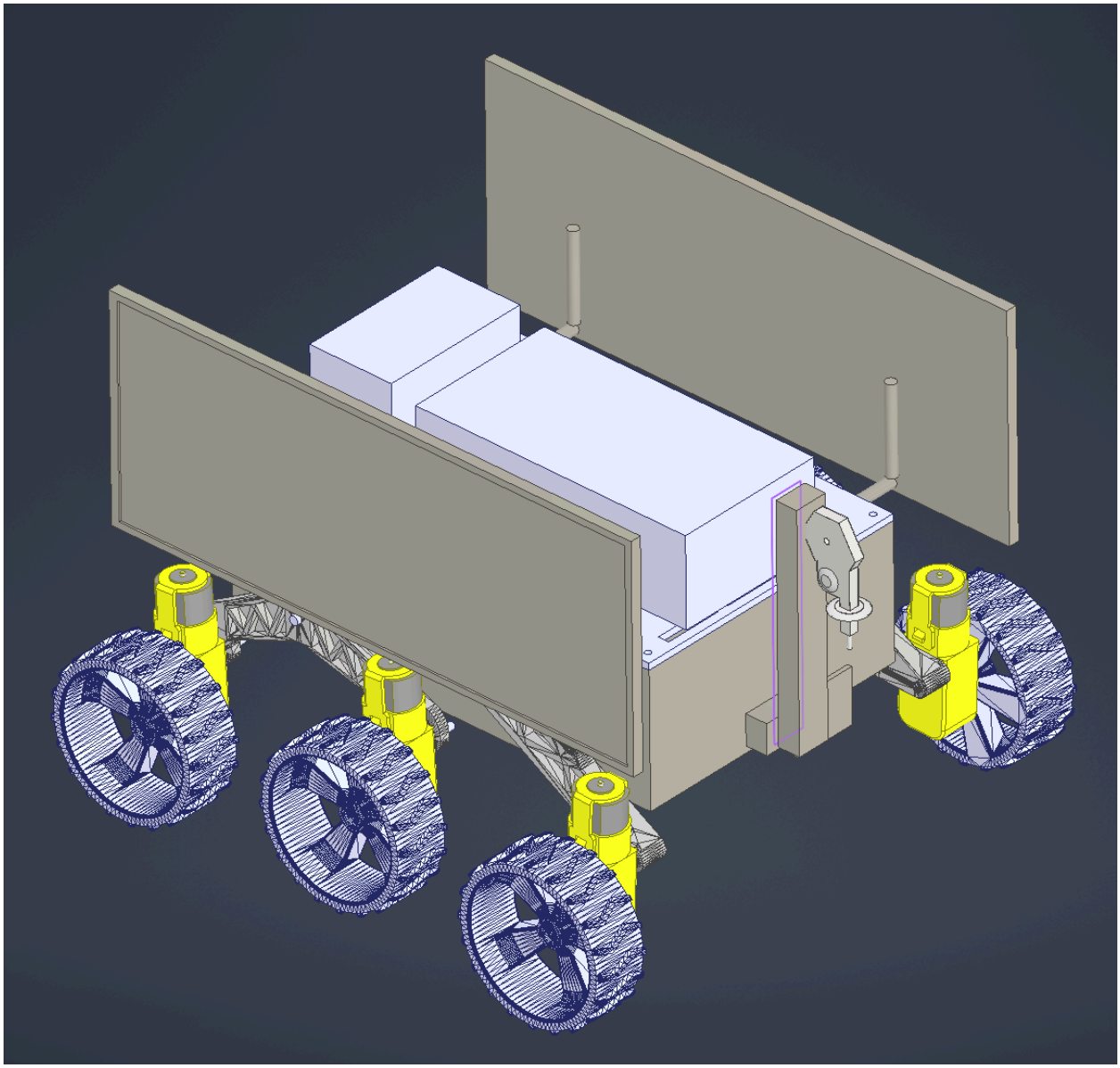
3. *Vehicle Design Narrative*

The final PDR vehicle design incorporates several key features that balance mass, power, operational capability, and risk across the mission. In order to remove a deployable failure mode the team opted to use non-deployable solar panels. This also increased the damage tolerance as the only deployment event in the mission timeline would be the initial stowed configuration. The rocker-bogie suspension and the six rigid aluminium grouser wheels were selected over alternative suspended wheels. This was done to keep the mobility subassembly TRL to 5-6 while also providing the stability needed in order to effectively traverse over difficult regolith terrain. In terms of the technical system the vehicle has a total mass of 158.9 kg which is 16.1 kg under the 175 kg maximum.The payload and mechanical subsystems draw power at respectively 130.6 watts and 65.0 watts. This brings the total maximum power draw to 317.6 W counting additional power drains. The current preliminary configuration is designed to remain within the 1.5 × 1.5 × 1.0 m stowed configuration envelope, with final dimensional verification to be completed against the integrated CAD model.

4. *Labeled CAD Model* *(Required)*

- *Include a labeled image of the complete spacecraft CAD model (or engineering model).*
- *Clearly identify the major subsystems and payload components.*


Arm/Mast
MECA
59.06
59.06
39.37
Geophones
LISTER
Solar Panel
Solar Panel
Mounting Interface
Seismic Digitizer
MSOLO & TENG (Internal)
ROVER ASSEMBLY



1.7. Science Instrumentation Summary

1. Instrumentation Overview

The rover's science payload consists of five instruments: subsurface thermal probe mass spectrometer, geophone, electrometer, and triboelectric nanogenerator. These tools characterize the thermal, chemical, seismic and electrostatic environment of the lunar south pole to determine if it is a potential ground for future science operations. The subsurface thermal probe and mass spectrometer directly address the Science exploration goal by measuring subsurface thermal properties and volatile chemistry. The geophone, electrometer, and triboelectric generator serve the purpose of addressing the Human Exploration goal. The mission will return data and conclusions calibrated from thermal

conductivity profiles specifically from the LISTER, volatile mass spectra specifically from the MSOLO, ground motion waveforms from the geophone sensors; additionally other measurements that the scientific instrumentation will take are the following: surface electric field calculations in the regolith, and triboelectric voltage and current output.

2. Instrument Summary

LISTER (Lunar Instrument for Subsurface Thermal Exploration with Rapidity):

LISTER uses a pneumatic drill to reach a 1m depth across two half meter cycles. It pauses at each interval and measures the thermal gradient and conductivity of the regolith. It does this by using its internal heater and resistance temperature detector sensor. The instrument satisfies MR-02 and MR-05 requirements which respectively concern thermal profiling and thermal energy storage.

MSOLO (Mass Spectrometer Observing Lunar Operation):

The MSOLO is a quadrupole mass spectrometer that provides a mass resolution across 1-100 m/z. It has an onboard calibration gas system that is used to distinguish the lander source gas from lunar volatiles. This instrument helps the mission satisfy the volatile chemistry requirement(MR-03) as it measures the charge to peak ratios of the mass and the amplitudes of signals to classify target species: H20, CO2, NH3,CO,CH4,H2S.

Geophone Array:

Geophones are deployed in a linear array configuration to detect how seismic waves travel within the ground. The functional sensitivity begins at 4.5 Hz. This array satisfied MR-08, the seismic activity requirement and provides P wave and S wave velocity data. This data is usable to constrain the shear modulus of the regolith as well as Young's modulus, which would inform the stability of slopes near the exploration zone. The shear modulus and Young modulus can also help the risk that can be caused over passing by unstable slopes.

MECA (Electrometer):

MECA utilizes charge-sensing needle tips to directly measure electrostatic potential and charge transfer as it interacts with the lunar regolith. It measures surface electric potential across a -400V to +10V range, detects ion currents in the pA range, and resolves charged particle accumulation down to 10^6 elementary charges. The instrument satisfies MR-09 by characterizing the surface electrostatic environment and identifying micro-scale dust hazards to support long-term hardware survivability.

TENG (Triboelectric Nanogenerator):

TENG is a solid-state technology demonstration that leverages wheel-regolith triboelectric interactions to harvest static charge accumulated across PSR boundary terrain. It logs generated electrical outputs within a 40V to 4.5kV input range and records short-circuit charging currents into storage capacitors. This instrument satisfies MR-07 by demonstrating early-stage in-situ resource utilization (ISRU) power harvesting strategies to supplement low-power lunar base utilities.

3. Instrument Summary Table

Instrument name	Primary measurement(s)	Science objective(s) supported	Mass (kg)	Maximum power consumption
LISTER	Thermal Gradient, Thermal conductivity, Heat flow	MR-02, MR-05	4.6	88 W
MSOLO	Volatile Species, Isotopic ratios	MR-03	4.8	35 W
Geophone Array	Ground Motion, Seismic monitoring	MR-08	19	0 W (ground motion makes the coil with the geophone sensor move and this causes an output voltage, the open circuit sensitivity is 11.42 V/m.s found within the Geospace technologies specs sheet.
TENG	Triboelectric voltage/current	MR-07		
MECA	Surface electric potential and ion currents	MR-09	0.95	<250 mW

Figure 10: Instrument Summary Table

The complete instrument suite integrates subsurface, surface, and atmospheric characterization techniques to fulfill both NASA Science Exploration and Human Exploration goals at the lunar south pole. By combining thermal probing (LISTER), volatile mass spectrometry (MSOLO), passive seismic profiling (Geophone Array), electrostatic monitoring (MECA), and triboelectric energy harvesting (TENG), the rover establishes a comprehensive baseline of environmental and resource conditions. These instruments directly complement one another to maximize overall scientific return: for example, thermal gradient data from LISTER combined with volatile composition profiles from MSOLO provides a clearer understanding of volatile preservation mechanics in permanently shadowed regions (PSRs). Simultaneously, physical regolith properties constrained by the Geophone Array inform the structural context of the subsurface, while MECA and TENG jointly assess surface charge mechanics to link electrostatic dust hazards directly with in-situ energy harvesting potential.

1.8. Programmatic Summary

1.8.1. Team Introduction



Simm Vallaban — Temple University, Philadelphia, PA
Role/Discipline: Project Manager
Simm is a senior Electrical Engineering student with previous experience in aerospace mission design through NASA's Virginia Space Coast Scholars, experience in hobby and semi-competitive rocketry through the Team America Rocketry Challenge (TARC), as well as extensive experience in competitive robotics (FRC, FTC, VEX, etc). They currently serve as President of Temple University's newly initiated AIAA Student Branch, where they are leading the development of a Design/Build/Fly competition team and expanding the chapter's hands-on aerospace engineering activities. Their technical experience includes robotics, embedded systems, Python, ROS2, CAD, and systems engineering, with additional experience developing and integrating hardware and software systems.

Yassin Vila — Hunter College, New York City
Role/Discipline: Deputy Project Manager Of Resources
Yassin is a Senior studying Computer Science and Minoring in Math. I specialize in backend software development, AI computation as well as embedded systems and the various interactions between them.



Isabela Chaveztello — Cornell University, Ithaca NY
Role/Discipline: Chief Scientist
Isabela is a Junior studying Physics with a concentration in Philosophy with a minor in electrical engineering. Her academic interests focus on the development of quantum technology in order to address real world problems. Her experience as subteam lead in the quantum computing association, passion for helping others, and experience as an undergraduate researcher focusing on quantum sensing of millicharged particles support the development of the Science mission to determine if the Nobile Rim 2 crater is sustainably habitable for future scientific development.



Mustafa Nazeer — University of Texas at Arlington, Arlington, TX
Role/Discipline: Lead Systems Engineer
Mustafa Nazeer is a senior studying computer science. He recently completed an internship at Red Canyon Software and Engineering as a cFS (core flight system) software intern, where he learned about NASA's cFS software. He is currently working with a professor at UTA to work on undergraduate research in training a graph neural network, and is

interning as a Full-Stack Software Engineering Intern at Lawly, a tech start up that is a law consulting platform for law applicants.



Thomas Meyer — Embry-Riddle Aeronautical University WorldWide, Daytona Beach, FL

Role/Discipline: Bioastronautics

Thomas is a Master of Science student studying Human Factors, as a graduate research assistant in the Small Teams Analog Research Laboratory, he asks a central question: how can human and intelligent systems be designed to sustain performance, resilience, and safety during long-duration, autonomous space missions? Grounded in a background in Industrial-Organizational Psychology, his training emphasizes the social, cognitive, and organizational dynamics that shape how crews adapt, coordinate, and make decisions when operating far from Earth, with limited resources and delayed ground support. He holds technical skills in quantitative analysis and research technologies, including Jamovi, SPSS, Qualtrics XM, RStudio, Python, JavaScript, and AI platforms such as IBM Granite and OpenAI.



Ruby Ledezma — Indiana University South Bend, South Bend, IN

Role/Discipline: Bioastronautics

Ruby has experience in clinical chemistry, hematology, microbiology, blood bank, immunology, and molecular diagnostics. Her background in Medical Laboratory Science,

public health, and environmental biology provides relevant expertise for defining bioastronautics science objectives, assessing environmental hazards like dust and surface charge, and ensuring payload instrument integration aligns with long-term human health and mission survivability goals.



Seamus Daly — University of Illinois at Urbana-Champaign; Urbana, Illinois
Role/Discipline: Planetary Geology
Seamus is an Aerospace Engineering student who is also minoring in philosophy and brings experience in computational analysis, CAD, advanced writing, flight mechanics, and both physics and fluid dynamics. He also works as a mathematics tutor for his previous college, which has strengthened his teamwork, communication, active listening, and ability to explain technical concepts clearly. As a planetary geologist for Team Fontus, he supports lunar south pole landing site characterization, science traceability, and mission planning, including JMARS analysis of the Nobile Rim 2 landing and exploration region.



Viridis Miranda — University of Puerto Rico Mayagüez Campus, Mayagüez Puerto Rico
Role/Discipline: Planetary Geology
Viridis has a background in Geology and experience with lunar seismology as part of her Master’s research she works in adapting receiver functions for the Moon. She also works in the Puerto Rico Seismic Network as a geophysical analyst monitoring the seismic activity of Puerto Rico. These skills contribute to the mission’s objective of having analog sensors (geophones) placed to observe passively seismic activity and also helped in determining location of emplacement of geophone sensors for Nobile Rim2 . She also has experience in

geophysical field work and planetary analog work as part of Project AV's Astrobiology team member. Project AV’s is an extracurricular activity in which she shares her knowledge of Geology and stratigraphy to engineering students to be able to be better equipped in the competition of the university rover challenge 2026.



Abdul Shahid — [University of Illinois Chicago, Chicago, IL]
Role/Discipline: CDH Engineer
Abdul brings a strong computer science and software development background to the team. His experience with C++, Python, data structures and algorithms, backend systems, and data processing supports the missions’s Command and Data Handling subsystem, including on board data management, command processing, CAD design, system communication, and reliable software operations.



Aleena Ajayakumarbini — [University of Maryland, College Park, MD]
Role/Discipline: CDH Engineer
Aleena is a senior Computer Engineering student with experience in hardware and software integration, embedded systems, and scientific research. Her previous experience includes conducting asteroid research that will be published in The Minor Planet Bulletin, cybersecurity research at NIST, and working as a chemistry teaching assistant at the University of Maryland. These roles have strengthened her ability to communicate technical concepts, collaborate with others, and work in multidisciplinary environments. Through her coursework and previous experience, she has developed skills in Python, Java, C, MATLAB, and circuit design. On Team Fontus, she applies these skills to the Power Subsystem by developing the power budget, sizing the batteries and solar panels, creating battery CAD

models, and developing subsystem requirements and verification plans to support rover operations throughout the mission.

Hoang Nguyen — [University, Location]
Role/Discipline: [Role/Discipline]
Team member biography is currently pending and will be updated with relevant academic, professional, and technical experience prior to CDR.



Areg Hovumyan — Cal Poly Pomona, Los Angeles CA
Role/Discipline: CDH Engineer
Areg is a junior computer engineering student at Cal Poly Pomona and brings experience in Electrical, Software and Command and Handling. Areg has developed skills through real life projects working on rovers, drones and robots in University projects sponsored by Lockheed Martin and Northrop Grumman. Areg also gained experience from his research at Autonomous Vehicle Lab working on building an autonomous car working on the main brain of the car using ROS2 in C++. During the research he also implemented cameras and other sensors adding object detection and image recognition on them. This experience is supported.



Angel Tejada — Vaughn College of Aeronautics and Technology, Queens, NY
Role/Discipline: Mechanical Engineer
Angel is a rising senior studying mechatronics Engineering having previous experience doing a one year astrophysic project using unsupervised machine learning for classification

of hot subdwarf candidates. Although Angel has a broad set of technical skills such as multiple programming languages, CNC machining and safety operation, CAD, Electrical components, research methods, data analysis he mostly helped in all the payload sections of all the deliverables. Angel is also part of VCAT Rover nasa club for his school.

Kevin Tieu — Virginia Polytechnic Institute and State University, Blacksburg, VA
Role/Discipline: Payload Mechanical Engineer
Team member biography is currently pending and will be updated with relevant academic, professional, and technical experience prior to CDR.

Alejandro Marote Konecna — [University, Location]
Role/Discipline: [Role/Discipline]
Team member biography is currently pending and will be updated with relevant academic, professional, and technical experience prior to CDR.



Juan Rodriguez Lozada — Universidad Ana G. Mendez, Gurabo Puerto Rico
Role/Discipline: Thermal Engineer
Juan is a Mechanical Engineering graduate with experience in thermal and mechanical analysis, engineering design, CAD, and simulation. His academic background includes thermodynamics, heat transfer, fluid mechanics, mechanical design, and control systems, providing a strong foundation for spacecraft thermal engineering. As a Thermal Engineer for Team Fontus, he contributes to the development of the lunar surface vehicle's thermal control subsystem, including thermal requirements, component selection, heater and insulation concepts, temperature monitoring, thermal interfaces, and verification planning. His work supports the rover's ability to protect critical electronics, batteries, and science instruments while operating across the extreme thermal environments encountered in illuminated and permanently shadowed lunar regions. He also holds Lean Six Sigma White, Yellow, and Green Belt certifications, providing additional experience in systematic problem solving, process improvement, and engineering decision-making.



Parker Roberts — San Jose State University, San Jose, CA
Role/Discipline: Thermal Engineer
Parker has experience in CAD and CFD and knowledge of heat transfer. This experience supports the mission’s thermal control subsystem.



Quinterrel Battle — The University of New Haven, West Haven, CT
Role/Discipline: Mission Assurance & Risk Management Specialist
Quinterrel brings experience as a Doctor of Health Science student, seasoned clinical nurse, and NASA NCAS Project Manager alumnus, with skills in high-reliability systems safety, failure mode analysis, protocol standardization, and operational risk mitigation. His background contributes to the Mission Assurance subsystem by translating clinical safety standards, human factors engineering, and doctoral-level systems analysis into robust risk assessment matrices, compliance tracking, and critical failure prevention for the mission.



Simran Kaur — Cal Poly Pomona, Pomona, CA
Role/Discipline: Mission Assurance & Risk Management Specialist
Simran has a background in biotechnology with experience in glycan cancer research and biomedical robotics research. Their skills in research, data analysis, risk assessment, and interdisciplinary problem-solving support mission assurance by helping identify potential

risks, evaluate technical challenges, and ensure mission objectives are met safely and effectively.



Hope Best — Cornell University, Ithaca, NY
Role/Discipline: Program Analyst
Hope brings experience in computer information science and skills in quantitative risk analysis, lifecycle milestone planning planning, and cross-functional technical communication. Their background supports mission milestone delivery, requirements traceability, and overall schedule and cost analysis.

Zachary Witchez — [Unlisted]
Role/Discipline: Program Analyst
Zachary has experience as a Computer Science student through College course work as well as Robotics research paper. Their skills contribute to mission objectives by being able to analyze, gather, and organize information into a schedule that makes sense for meeting the mission objectives.

1.8.2. Team Management Overview

1. Organizational Chart Narrative

Fontus is structured under a hierarchical matrix framework led by Project Manager Simm Vallaban, supported by Chief Scientist Isabela Chaveztello, Lead Systems Engineer Mustafa Nazeer, and Deputy Project Manager of Resources Yassin Vila. This organization establishes clear reporting lines across science, systems engineering, and programmatic/mission assurance divisions, ensuring balanced resource allocation and technical rigor across all payload and programmatic disciplines.

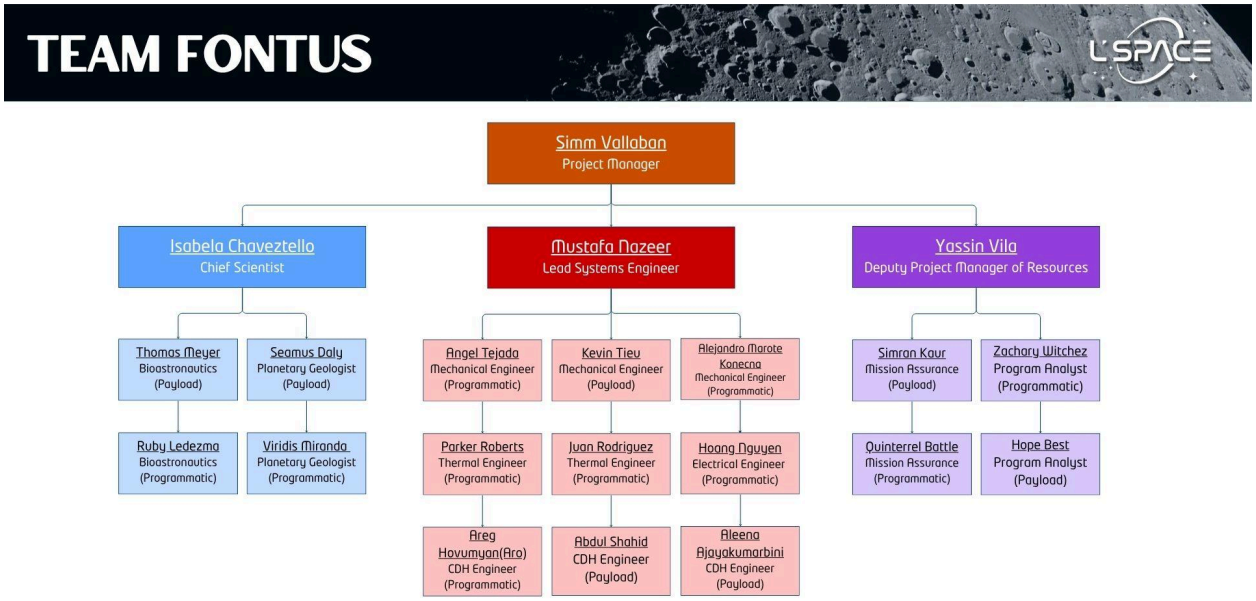


Figure 11: Team Organization Chart

2. Roles, Disciplines, and Availability

Team Fontus consists of 21 multidisciplinary members spanning planetary science, bioastronautics, mechanical, thermal, electrical, and computer engineering, mission assurance, and program analysis. Every member commits approximately 10 to 15 hours weekly to ensure deliverable milestones and technical baseline maturities are achieved on schedule:

Team Fontus Roster & Resource Allocation

Name	Mission Role & Focus Area	Subteam / Division	Committed Availability
Simm Vallaban	Project Manager	Project Leadership & Governance	15 hrs/wk
Isabela Chaveztello	Chief Scientist	Science Division Lead	15 hrs/wk
Mustafa Nazeer	Lead Systems Engineer	Systems Engineering Division Lead	15 hrs/wk
Yassin Vila	Deputy Project Manager of Resources	Programmatics & Resources Division Lead	15 hrs/wk
Thomas Meyer	Bioastronautics (Payload)	Science Subteam	10–12 hrs/wk
Ruby Ledezma	Bioastronautics (Programmatic)	Science & Outreach Subteam	10–12 hrs/wk
Seamus Daly	Planetary Geologist (Payload)	Science Subteam	12 hrs/wk
Viridis Miranda	Planetary Geologist (Programmatic)	Science Subteam	10-12 hrs/wk
Angel Tejada	Mechanical Engineer (Programmatic)	Systems Engineering Subteam	10–12 hrs/wk
Kevin Tieu	Mechanical Engineer (Payload)	Systems Engineering Subteam	10–12 hrs/wk
Alejandro Marote Konecna	Mechanical Engineer (Programmatic)	Systems Engineering Subteam	10–12 hrs/wk
Parker Roberts	Thermal Engineer (Programmatic)	Systems Engineering Subteam	10–12 hrs/wk
Juan Rodriguez	Thermal Engineer (Programmatic)	Systems Engineering Subteam	10–12 hrs/wk
Hoang Nguyen	Electrical Engineer (Programmatic)	Systems Engineering Subteam	10–12 hrs/wk
Areg Hovumyan (Aro)	CDH Engineer (Programmatic)	Systems Engineering Subteam	10–12 hrs/wk
Abdul Shahid	CDH Engineer (Payload)	Systems Engineering Subteam	10–12 hrs/wk
Aleena Ajayakumarbini	CDH Engineer (Payload)	Systems Engineering Subteam	10–12 hrs/wk
Simran Kaur	Mission Assurance (Payload)	Resources & Mission Assurance	12 hrs/wk
Quinterrel Battle	Mission Assurance (Programmatic)	Resources & Mission Assurance	12 hrs/wk
Zachary Witchez	Program Analyst (Programmatic)	Resources Subteam	10–12 hrs/wk
Hope Best	Program Analyst (Payload)	Resources Subteam	10–12 hrs/wk

Figure 12: Roles Table & Reporting Chain

3. Work Delegation

Work assignments flow hierarchically from the Project Manager and division leads down to specialized payload and programmatic functional pairs. Subsystem engineering and science teams hold weekly technical sprints to track action items, coordinate

cross-subsystem interfaces (e.g., thermal-to-power, payload-to-CDH), and ensure deliverables align with customer schedule constraints.

4. Decision-Making Approach

Routine technical and design trades are decided at the subteam level through collaborative consensus and engineering analysis. Time-sensitive decisions impacting system interfaces, critical path milestones, or mass/power margins are escalated directly to Lead Systems Engineer Mustafa Nazeer and Project Manager Simm Vallaban. In instances of divided technical opinions, the Change Control Board evaluates objective trade studies against the Science Traceability Matrix (STM) to issue a final, binding disposition.

5. Team Capability

Team Fontus possesses a balanced distribution of expertise across space systems engineering, software architecture, planetary geology, bioastronautics, quality assurance, and project controls. Any emerging technical or software modeling gaps are actively addressed by leveraging NASA Systems Engineering handbooks, peer-reviewed literature, and mentor guidance.

6. Conflict Resolution

Technical disagreements are evaluated through objective, requirement-driven trade studies to identify solutions that best preserve science return and system margins. Persistent operational or task conflicts are mediated directly by the Project Manager and Deputy Project Manager of Resources to achieve productive, mission-first resolutions.

1.8.3. **Schedule Overview**

The mission schedule that has been developed provides a high-level roadmap for the development, fabrication, integration, testing, launch preparation, mission operation, and closeout of the lunar south pole rover mission. The detailed mission schedule begins with the completion of the Preliminary Design Review (PDR) and focuses on NASA Phases C through F. The schedule establishes major technical reviews, manufacturing activities, integration milestones, verification testing, launch preparation, mission operations, and mission closeout necessary to achieve the planned February 1, 2029 launch.

Pre-Phase A	1	Mission Concept Development	
	1.1	Mission Objectives	
	1.2	Science Traceability Matrix	
	1.3	Mission Requirements	

	1.4	Trade Studies	
	1.5	Schedule Margin	
	1.6	◆ Mission Concept Review	8/31/26
Phase A	**2**	**Requirements**	
	2.1	System requirements	
	2.2	Functional decomposition	
	2.3	Interface definitions	
	2.4	Initial subsystem allocation	
	2.5	Mission Definition Refinement	
	2.6	◆ System Requirements Review	12/31/26
	2..7	Schedule Margin	
	2.8	◆ Mission Definition Review	12/31/26
Phase B	**3**	**Preliminary Design**	
	3.1	Rover mechanical layout	
	3.2	Payload selection	
	3.3	Power architecture	
	3.4	Thermal architecture	
	3.5	Communications architecture	
	3.6	Software architecture	
	3.7	Schedule Margin	
	3.8	◆ Preliminary Design Review	6/30/27
Phase C	**4**	**Detailed Design**	
	4.1	CAD completion	
	4.2	Electrical design	
	4.3	Flight software development	
	4.4	Navigation software	
	4.5	Structural analysis	
	4.6	Documentation	
	4.7	Long-Lead Procurement Planning	
	4.8	Schedule Margin	
	4.9	◆ Critical Design Review	2/29/28
Phase C	**5**	**Fabrication & Procurment**	

	5.1	Order hardware	
	5.2	Manufacture parts	
	5.3	Assemble electronics	
	5.4	Payload fabrication	
	5.5	Schedule Margin	
	5.6	◆ Hardware Complete	6/28/28
Phase D	**6**	**Integration & Verification**	
	6.1	Mechanical integration	
	6.2	Electrical integration	
	6.3	Software integration	
	6.4	Functional testing	
	6.5	Thermal testing	
	6.6	Mobility testing	
	6.7	Environmental testing	
	6.8	Test Discrepancy Resolution & Retesting	
	6.9	Schedule Margin	
	6.10	◆ System Integration Review	12/1/28
Phase D	**7**	**Launch Preparation**	
	7.1	Flight software freeze	
	7.2	Launch vehicle integration	
	7.3	Final verification	
	7.4	Launch Ready	
	7.5	◆ Launch Readiness Review	1/23/29
	7.6	Schedule Margin	
	7.7	◆ Launch	2/1/29
Phase E	**8**	**Mission Operations**	
	8.1	Rover Activation and initial checkout	
	8.2	Surface commissioning	
	8.3	Initial traverse and instrument checkout	
	8.4	Primary science operations	

	8.5	Science Data downlink & preliminary analysis	
	8.6	Schedule Margin	
	8.7	◆ Mission operations Complete	9/28/29
Phase F	**9**	**Mission Closeout**	
	9.1	Mission completion confirmation	
	9.2	Science and engineering data validation	
	9.3	Final data archiving	
	9.4	Post-mission analysis and final reporting	
	9.5	Lessons Learned report	
	9.6	Schedule Margin	
	9.7	◆ Mission Closeout	3/29/30

Figure 13: Mission Schedule Overview

The figure shows the high-level Work Breakdown Structure used to organise the preliminary mission schedule. Any activities shown prior to the Preliminary Design Review are completed works, while the post-PDR schedule will focus on NASA Phase C through F and the activities leading to launch readiness and mission closeout.

Narrative

The schedule will progress through the NASA life-cycle Pre-Phase A through F while using sequenced relationships between concept development, requirements development, preliminary and detailed design, fabrication, integration and verification, launch preparation, mission operations, and mission closeout. With Pre-Phase A focusing on the initial mission concept and concludes with the Mission Concept Review (MCR), following with the requirements development and mission definition activities during Phase A, which include the System Requirements Review (SRR) and Mission Definition Review (MDR). Phase B will focus on the preliminary subsystem design and conclude with the Preliminary Design Review (PDR). Phase C will then focus on detailed design, the Critical Design Review (CDR), and then fabrication and procurement activities necessary to produce the flight hardware. Phase D will progress through system integration verification and environmental testing, resolution and retesting of identified test discrepancies, and launch preparation leading to the planned February 1, 2029 launch. Following launch, Phase E consists of mission operations through September 28, 2029, while Phase F concludes the project through data

validation, archiving, post-mission analysis, lessons learned, and final mission closeout on March 29, 2030.

The schedule was developed using the Work Breakdown Structure (WBS) and the NASA schedule management guidance to identify critical milestones and activity dependencies, and predecessor-successor relationships. With the schedule margin being maintained throughout the mission life cycle to accommodate technical uncertainties, design maturation, hardware and procurement delays, integration issues, testing anomalies, and verification activities. With long-lead procurement planning, fabrication, system integration, environmental testing, test discrepancy resolution and retesting, launch preparation, and the fixed launch date represent the major schedule drivers because any delays to these activities could directly affect downstream milestones and overall mission readiness.

Cost Overview

The total estimated lifecycle cost for the lunar surface vehicle system across NASA Phases C through F is $211,012,673.00, encompassing all labor, travel, outreach, direct hardware procurement, testing facilities, and indirect expenses. The estimate is fully compliant with requirement SYS-12 (MR-10), which mandates a post-PDR total lifecycle cost cap of $250,000,000.00, operating with an unallocated headroom of $38,987,327.00 (15.59% below the cost ceiling). The primary financial contributors driving overall project expenditure are Science Instrumentation ($51.00M), Assembly, Integration, and Test environmental test facilities ($35.70M), Spacecraft Cost Reserves ($17.00M), and the Electrical Power Subsystem ($17.00M).

Mission Budget Summary

Budget Category	Cost (USD)	% of Total Estimated Cost	% of Budget Cap
Mission Budget Cap (Post-PDR SYS-12 / MR-10)	$250,000,000.00	—	100.00%
Final Estimated Mission Cost	$211,012,673.00	100.00%	84.41%
Personnel Costs (Salaries + ERE)	$17,918,705.00	8.49%	7.17%
Travel Costs	$1,884,240.00	0.89%	0.75%

Outreach Costs	$1,872,540.00	0.89%	0.75%
Direct Costs (Subsystems & Facilities)	$178,493,380.00	84.59%	71.40%
• *Spacecraft Direct Subsystems*	$125,438,080.00	59.44%	50.18%
• *Manufacturing & Environmental Test Facilities*	$53,055,300.00	25.14%	21.22%
Facilities & Administrative (10% MTDC Indirect)	$10,843,808.00	5.14%	4.34%
Available Cost Margin (Unallocated Capacity Under Cap)	$38,987,327.00	—	15.59%
Spacecraft Technical Cost Reserve (Funded within Direct Costs)	*$17,000,000.00*	*8.06%*	*6.80%*

Figure 14: Mission Life Cycle Budget Summary across NASA Phases C through F.

Spacecraft direct subsystem development represents the single largest financial commitment at 59.44% ($125,438,080.00) of total mission expenditure. Specialized environmental qualification and manufacturing facility rentals constitute the second largest cost center at 25.14% ($53,055,300.00), driven by mandatory thermal-vacuum (TVAC), vibration, and abrasive dust chamber testing campaigns during Phase D. Personnel costs account for 8.49% ($17,918,705.00) across all engineering, science, management, and technician labor, while travel, public outreach, and indirect costs comprise the remaining 6.93%. Major cost drivers are concentrated in the procurement and qualification of flight-heritage scientific instruments specifically the LISTER thermal probe ($12.00M), seismic geophone array ($8.50M), and MSolo mass spectrometer ($4.50M) alongside the engineering of space-qualified GaAs solar arrays and battery systems to survive cryogenic permanently shadowed regions. Total mission risk is controlled by a $17.00M spacecraft technical reserve embedded within direct costs, supplemented by the $38.99M in unallocated budgetary margin.

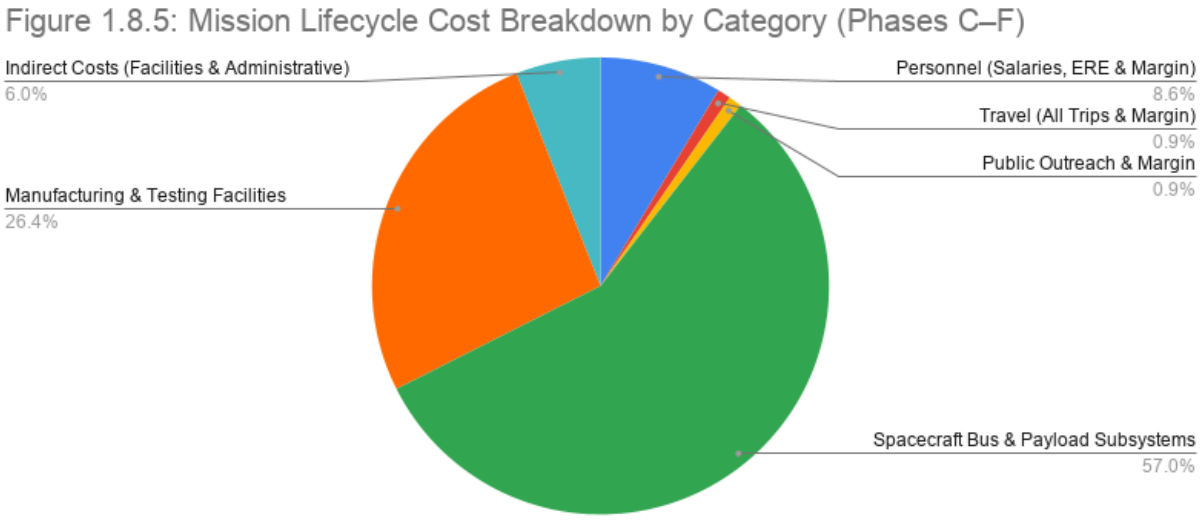


Figure 15: Mission Cost Breakdown By Category (phase C-F)

2. Overall Vehicle and System Design

2.1. Spacecraft Overview

1. *Top-Level Subsystem Requirements*

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
MR-01	The mission shall characterize environmental and exploration hazards in lunar south pole permanently shadowed regions to support future human lunar surface operations.	Establishes the top-level mission function and objective as described by the Mission Task document and the Science Traceability Matrix, supporting Focus Area 3 as chosen by the team.	Customer constraint, Science Traceability Matrix	SYS-01, 04, 07, 08, 10, 11	Demonstration	Partially Met
MR-02	The mission shall obtain temperature profiles and thermal conductivity in	Understanding the thermal environment and the regolith's	Science Traceability Matrix,	SYS-06	Demonstration	Met

	permanently shadowed regions to a depth of 1 meter.	thermal properties of the PSRs, of at most 1 meter in depth.	Science Exploration Goal			
MR-03	The mission shall quantify the abundance, isotropic composition, and volatile chemistry of water ice deposits within permanently shadowed regions.	Science Objective of categorizing the chemical composition and properties of water ice deposits within PSRs for future research	Science Traceability Matrix, Science Exploration Goal (Focus Area 3)	SYS-11	Test	Met
MR-04	The mission shall quantify the depth and lateral continuity of the subsurface dielectric boundaries within the regolith.	Science Objective to determine if the water ice is possible for in-situ resource utilization	Science Traceability Matrix, Human Exploration Goal (ISRU: AS-01/LI-07)	SYS-11	Test	Partially Met
MR-05	The mission shall determine the efficiency of thermal energy storage capacity of lunar regolith located near permanently shadowed regions.	Science Objective to determine if the regolith is a candidate for in-situ resource utilization to use as a thermal energy reservoir.	Science Traceability Matrix, Human Exploration Goal (ISRU: AS-01/LI-07)	SYS-11	Test	No

MR-06	The mission shall measure the metrics of its NeuPAN-based autonomous motion systems.	HSMD Objective to demonstrate rover or mobility platform capabilities	Science Traceability Matrix, Human Exploration Goal (LI-06)	SYS-05	Demonstration	Met
MR-07	The mission shall measure and quantify voltage, current, and power output measurements from charged regolith.	HSMD Objective to demonstrate the feasibility of harvesting electrostatic charge from regolith at PSR boundary as a supplemental power source.	Science Traceability Matrix, Human Exploration Goals (LI-01)	SYS-11	Test	Not Met - 0.1 Hz low end lost its instrument when the very broad and seismometer was cut; geophones can have a cutoff frequency of 4.5 Hz which

						can be used to test lower frequencies to higher frequency events like shallow moon quakes.
MR-08	The mission shall determine the frequency and intensity of seismic activity on the moon.	HSMD Objective to measure the physical and internal characteristics of the moon.	Science Traceability Matrix, Human Exploration Goals (AS-01)	SYS-11	Test	Met
MR-09	The mission shall characterise surface electrostatic environment and charge accumulation.	HSMD Objective to identify hazards that could affect equipment.	Science Traceability Matrix, Human Exploration Goals (AS-01)	SYS-11	Test	No

MR-10	The mission shall have a total life cycle cost allocation (LCC) of $250 million for all costs post PDR (Phases C through F), excluding an allocated system budget.	Customer derived cost requirement	Customer constraint	SYS-12	Inspection	Met
MR-11	The Lunar Surface Vehicle System shall be ready for system integration by December 1, 2028.	Customer schedule constraint	Customer constraint	SYS-13	Inspection	Met
MR-12	The Lunar Surface Vehicle shall be ready for launch by February 1, 2029.	Customer schedule constraint	Customer constraint	SYS-14	Inspection	Met
MR-13	The Lunar Surface Vehicle System shall remain operational for a minimum of two weeks after landing, with a planning bound of maximum one month, to complete planned science operations.	The mission timeline post-landing was derived by the Science subteam, setting a minimum time requirement such that the system shall survive in order to sufficiently obtain enough science data.	Science and Engineering Derived	SYS-09	Test	Partially Met - Ops concept and recharge strategy defined; site illumination of 11-12 consecutive sunlit

						days requires verification against the 14-day requirement
MR-14	The Lunar Surface Vehicle System shall be compatible with the launch vehicle mechanical mounting interface defined in the MCA LV ICD.	The customer ICD requires that physical payload-to-launch-vehicle mounting interface, complying to this requirement allows for the vehicle to integrate onto and to fly on the LV, avoiding late design changes.	Customer constraint, Launch Vehicle ICD	SYS-15	Analysis (fit check), Inspection	Partially Met
MR-15	The Lunar Surface Vehicle may not exceed the operating temperature of -10°C to 30°C while in the PSRs.	If the Lunar Surface vehicle exceeds the operating temperature then we will need to abort the mission with incomplete data.	Science and Engineering Derived	SYS-06	Test	Partially Met - Design and control strategy defined to the

						required band; compliance pending thermal model validation.

Figure 16: Mission Requirements

2. Spacecraft Overview

The mission will use a mobile surface rover designed to operate near a cold trap boundary at the lunar south pole. The rover will collect environmental and science data using LISTER, the MSolo mass spectrometer, TENG, a three-geophone seismic array, and associated seismic data-processing hardware. These instruments provide thermal, volatile, seismic, and electrostatic measurements that support characterization of the lunar regolith and permanently shadowed-region environment.

The rover is designed to support science operations while traveling across the lunar surface and entering permanently shadowed regions. The mechanical design uses mostly passive or minimally actuated systems, metallic structures, and dust-tolerant moving components. The chassis is made of Aluminum 6061-T6 with an aluminum honeycomb body and enclosure. A six-wheel rocker-bogie mobility system with rigid aluminum grouser wheels provides stability and traction across the lunar terrain.

The rover uses a brushless DC motor at each wheel for mobility and a LiDAR-based navigation system supported by IMUs and odometry for autonomous navigation. The rover's Power Subsystem provides electrical power through solar generation and lithium-ion battery storage, while the Thermal Subsystem maintains components within their required operating temperatures. The CDH Subsystem manages commands, telemetry, and rover data. Together, the mechanical, power, thermal, CDH, and payload systems support rover mobility, science operations, and survival throughout the mission.

3. Subsystem Summaries

Mechanical Subsystem

The Mechanical Subsystem consists of four major subassemblies: the chassis/structure subassembly, mobility subassembly, GNC hardware subassembly, and mechanical interface subassembly. These subassemblies collectively provide the physical architecture needed to support rover and science operations, science payload integration,

autonomous mobility, and subsystem mounting. The current design provides the mechanical support and mobility needed for the rover to operate and perform science activities on the lunar surface.

Power Subsystem

The Power Subsystem generates, stores, regulates and distributes electrical power to the rover subsystems. The subsystem consists of GaAs solar panels, a lithium-ion battery system and power regulations and control electronics. The solar panels provide power and recharge the lithium-ion batteries while the rover is in the illuminated regions. The batteries provide the primary power during operations inside the PSR. All of the electrical power is being distributed in the rover through a regulated 28 V DC bus.

Command & Data Handling (CDH)

The Command and Data Handling subsystem is the rover's central computing and data management system. Equipped with two redundant flight computers which receive and validate ground commands, control operations, monitor rover health, and provide fault detection and recovery. CDH also collects, time tags, stores other engineering and science data in the nonvolatile storage until downlink is confirmed. CDH also interfaces with the rover's other subsystems through redundant CAN buses and supports DTE communication through the redundant Iris X-band radio system.

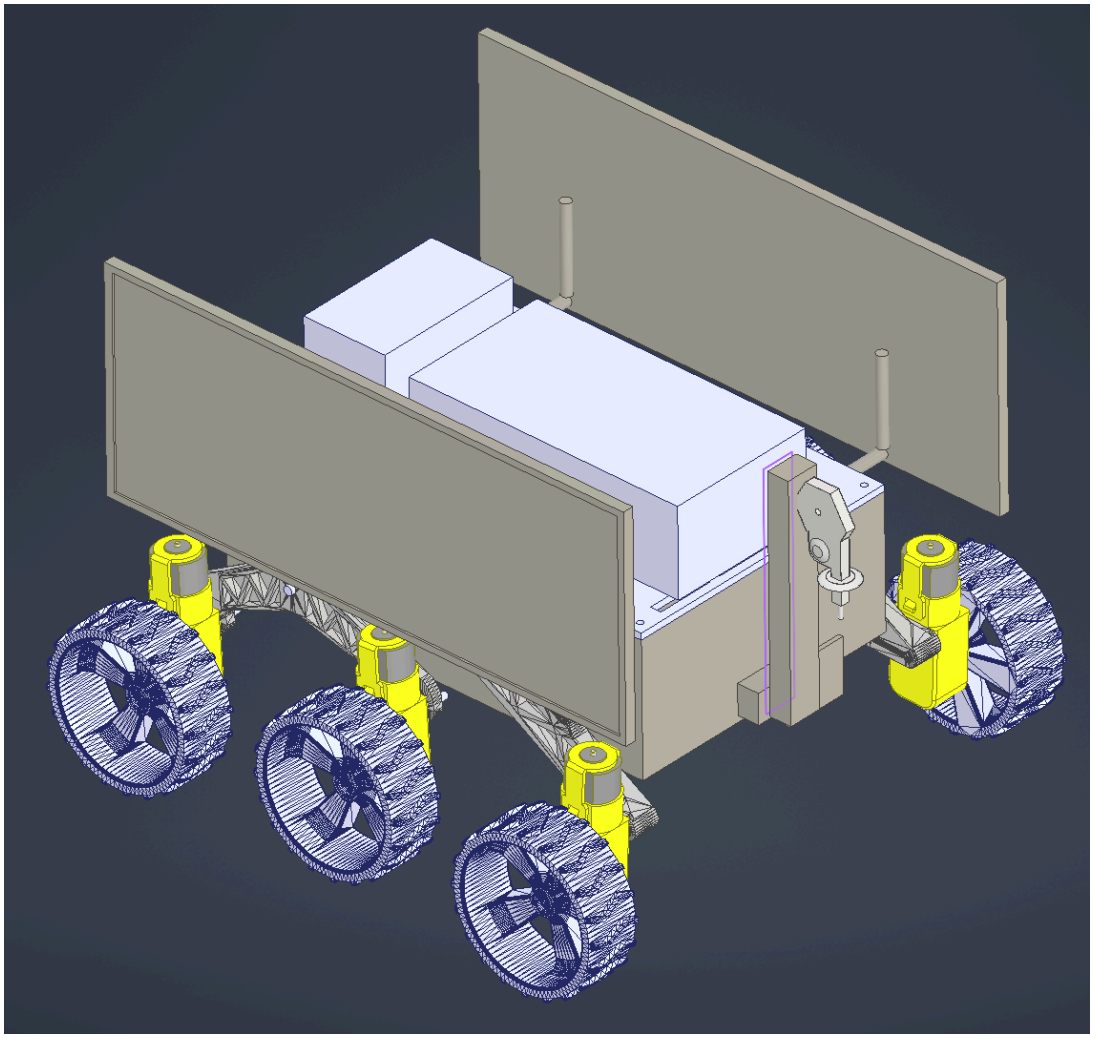
Thermal Subsystem

The Thermal Subsystem keeps the rover components within their required operating temperatures throughout the mission. The rover will experience extreme temperature changes while traveling between the illuminated and permanently shadowed regions. Muilti-layer insulation and surface coatings are used to reduce heat loss and control heat transfer. Heaters also provide additional heat when needed inside the PSR. Furthermore, the subsystem is designed to remove excess heat from the rover while operating in the illuminated regions. Together, these components help maintain safe operating temperatures for the rover and its internal systems throughout the mission.

Science Instrumentation / Payload

The Payload Subsystem provides the scientific instruments needed to support the mission's science objectives. The payload consists of the MSolo mass spectrometer, LISTER thermal probe, TENG, three geophones, and a seismic digitizer. These instruments are used to measure lunar volatiles, thermal properties, heat flow, and seismic activity throughout the rover's traverse inside the PSR. The data collected by the Payload Subsystem will help characterize the lunar regolith and subsurface environment while supporting the overall science objectives.

Preliminary Spacecraft Design Drawings (Figures)



Subsystem	Mass (kg)	Dimensions (m)	Maximum Power Draw (W)
Mechanical	56	1.5x1.5x1	65
Power	44	-	14
CDH	10.05	.500 x .400 x .300	55
Thermal	18	-	53
Payload	30.85	-	130.6
Total Spacecraft	158.9	1.5x1.5x1	317.6

Figure 17: Rover Subsystem Mass, Dimensions, and Power Summary

The current preliminary mass estimate of 158.9 kg provides 16.1 kg of margin against the 175 kg system requirement. The aggregate 317.6 W value represents the sum of the listed subsystem maximum demands and is distinct from the Power subsystem's 255 W continuous-output and approximately 490 W peak-load design capacities.

2.1.1. Mechanical Subsystem Overview

The mechanical subsystem provides the primary physical structure, mobility hardware, and mounting and mechanical interfaces required for the Lunar Surface Vehicle System to conduct post-landing surface operations. The subsystem supports the rover body, science payloads, wheel subassemblies, navigation hardware, and subsystem interfaces while maintaining compatibility with the mission mass, volume, and deployment mechanisms. The mechanical subsystem is responsible for ensuring that the Lunar Surface Vehicle System can transition from its stored configuration to its operational configuration, traverse lunar terrain, maintain stable payload and sensor mounting, and support repeated science operations near the selected Exploration Zone.

The baseline mechanical design consists of a rectangular-prism rover body constructed from Aluminum 6061-T6 with a rocker-bogie suspension system and six rigid aluminum grouser wheels. The rectangular rover body functions as the primary structural enclosure and equipment bay. It provides mounting surfaces for internal subsystem hardware, science instruments, and external sensor hardware. The rocker-bogie suspension enables the rover to traverse uneven lunar terrain while maintaining stability and continuous wheel contact, allowing it to travel from the illuminated landing site toward the Exploration Zone and return as required by the Concept of Operations. The GNC (Guidance, Navigation and Control) hardware features LiDAR (Light Detection and Ranging)-based navigation systems, inertial measurement units (IMUs), and supporting sensor hardware.

The mechanical system design favors high heritage and low complexity architectures. In terms of fulfilling the mission objective, the design philosophy of going for high heritage and low complexity helps in ensuring that the system achieves TRL 6 at least for its baseline mechanical systems while minimizing any unnecessary deployment, actuation, and verification risks. At present, the mechanical system has no use of any hydraulic or active suspensions systems, robotic arms, or rover deployment of support stations. Rather, the design includes passive or minimally actuated systems, metallic structures, dust tolerant moving joints, and normal spacecraft/rover integration approaches.

Subassembly	Mass (kg)	Dimensions (m)	Maximum Power Draw (W)	TRL
Chassis / Body	~12.5	0.65x0.85x0.45	0	6
Vehicle Interface	~6	0.6x0.1x0.8	0	6
Mobility	~35	1.2x0.2x0.5	65	5
GNC Hardware	~2.5	0.2x0.1x0.2	0	6
Total Subsystem	56	< 1.5x1.5x1	65	5

Figure 18: Mechanical Subsystem Summary Table

1. ***Mechanical Subsystem Figure***

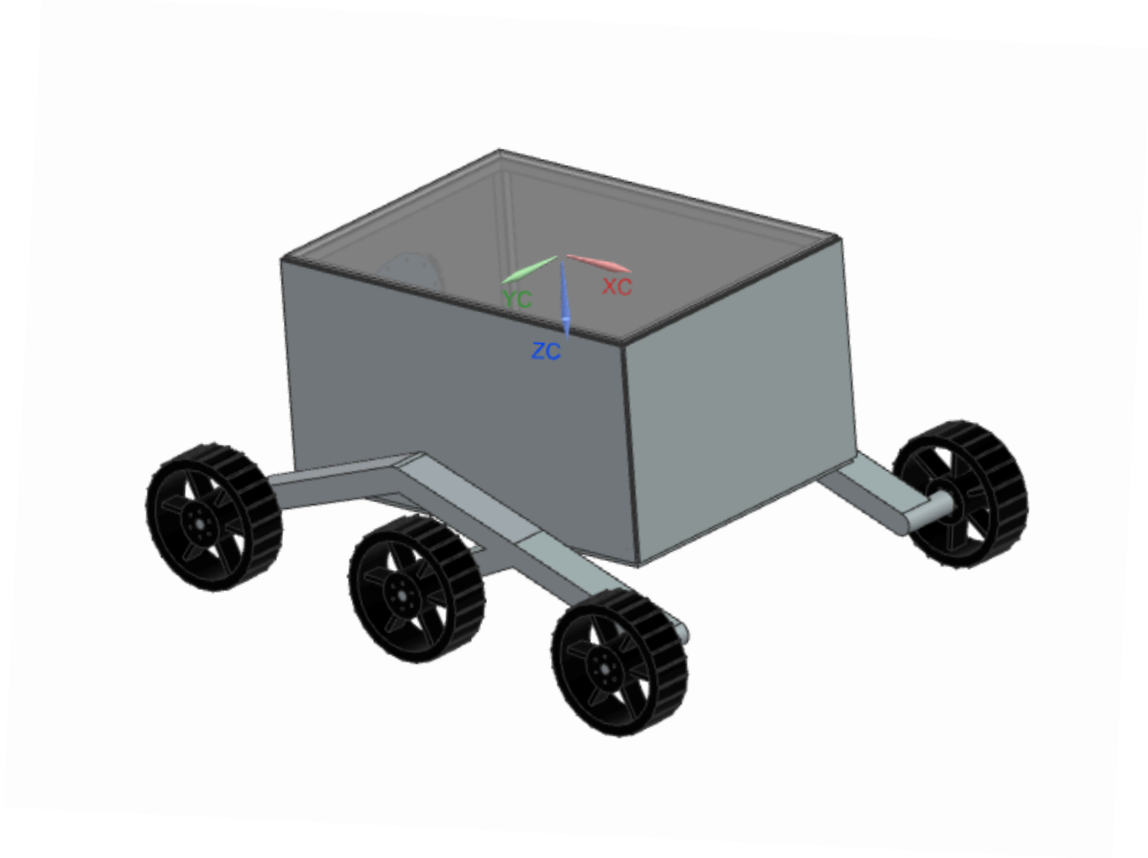


Figure 19: Mechanical Subsystem CAD

2.1.1.1. Mechanical Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
MCH-01	The mechanical subsystem shall have a total mass less than or equal to 56 kg.	Allocates the system mass constraint to the chassis, mobility, deployment, and mounting hardware.	SYS-02	MOB-01	Inspection	No
MCH-02	The mechanical subsystem and all subsequent components shall fit within a stored configuration envelope of 1.5 m × 1.5 m × 1.0 m.	Constraint made by the Mission Task document, confines the overall mechanical subsystem (structure/chassis) and subsequent components.	SYS-03	CHS-01	Inspection	No

MCH-03	The mechanical subsystem shall maintain operational functionality after exposure to PSR environments for at least 2 weeks.	Defines the operational requirement of lasting the minimum mission length requirement.	SYS-09	MOB-01	Demonstr...	No
MCH-04	The mechanical subsystem shall enable the rover to traverse at least 300 meters on lunar terrain.	Provides a 20% margin on the constraint defined by the Landing Zone and Exploration Zone distance.	SYS-05	MOB-01	Test	No
MCH-05	The mechanical subsystem shall enable the Lunar Surface Vehicle System to traverse lunar terrain slopes up to 15° without loss of static stability or mobility function.	Provides a 100% margin on static stability, adding a factor of safety to ensure the rover does not tip even in conditions far worse than expected by the Mission Task document.	SYS-10	MOB-06	Test	No
MCH-06	The mechanical subsystem shall enable the Lunar Surface Vehicle System to deploy to the operational configuration within 30 minutes of landing on the Moon.	Allows for a swift and efficient deployment to accelerate science operations.	SYS-04	MOB-01	Demonstr...	No
MCH-07	The stowed mechanical subsystem shall maintain structural integrity and functional mechanical interfaces after exposure to launch,	Ensures the Lunar Surface Vehicle System is functional prior to its surface mission.	SYS-04	MOB-01	Test	No

	transit, and landing loads.					
MCH-08	The mechanical subsystem shall support a fully integrated rover mass with a minimum static factor of safety of 2.0 under 1-g ground handling loads on Earth.	Ensures the Lunar Surface Vehicle System is functional prior to launch.	SYS-10	MOB-01	Test	No
MCH-09	The mechanical subsystem shall provide mechanical mounting interfaces for all required subsystems and subassemblies (power, CDH/communications, thermal, payload).	Ensures the integration of all subsystems for surface operations.	SYS-01	MOB-01	Demonstr...	No
MCH-10	The mechanical subsystem shall protect exposed moving interfaces from lunar regolith during nominal surface operations.	Reduces the risk of deployment (to operational configuration) and mobility function caused by abrasive regolith.	SYS-01	MOB-01	Test	No

Figure 20: Mechanical Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
MOB-01	The mobility subassembly shall have a total mass less than or equal to 35 kg.	Ensures subassembly mass does not go over allocated mass budget.	MCH-01	N/A	Inspection	No

MOB-02	The mobility subassembly shall fit in the system stored configuration within the 1.5 x 1.5 x 1.0 m envelope.	Confines the mobility subassembly to ensure subassembly volume remains within overall system and subsystem volume constraint.	MCH-02	N/A	Inspection	No
MOB-03	The mobility subassembly shall use a six wheel passive suspension architecture to enable the minimum rover traverse distance of 300 meters.	Defines the number of wheels the rover shall have in order to move and conduct science operations.	MCH-04	N/A	Test	No
MOB-04	The mobility subassembly shall use wheels with an outer diameter of 270 m ± 0.01 m.	Provides obstacle-clearance capability while preserving packaging within the rover envelope.	MCH-06	N/A	Inspection	No
MOB-05	The mobility subassembly shall provide a minimum underbody ground clearance of 0.125 m in deployed configuration.	Reduces risk of chassis grounding on rocks and allocates a minimum clearance for science instrumentation.	MCH-06, MCH-10	N/A	Inspection	No
MOB-06	The mobility subassembly shall maintain controlled motion at commanded low-speed traverse rates between .01 m/s and .05 m/s.	Defines the operational velocity of the rover inside PSR terrain.	MCH-03	N/A	Test	No
MOB-07	The mobility subassembly shall remain mechanically operational within the temperature range of -133°C to +121°C.	Ensures that the mobility subassembly will continue to operate within a range of temperatures expected to encounter in PSR	MCH-06, MCH-10	N/A	Test	No

		terrain.				
CHS-01	The chassis subassembly shall have a total mass less than or equal to 15 kg.	Ensures subassembly mass does not go over allocated mass budget.	MCH-01	N/A	Inspection	No
CHS-02	The chassis subassembly shall fit within an allocated 1.3 x 0.75 x 0.9 m such that the integrated system in the stored configuration is within the 1.5 m x 1.5 m x 1.0 m envelope.	Confines the chassis subassembly to ensure subassembly volume remains within overall system and subsystem volume constraint.	MCH-02	N/A	Inspection	No
CHS-03	The chassis subassembly shall maintain operational functionality after exposure to PSR environments for at least 2 weeks.	Defines the operational requirement of lasting the minimum mission length requirement.	MCH-03	N/A	Demonst...	No
CHS-04	The chassis subassembly shall support subsystem placement such that the integrated vehicle CG remains within 5 cm of the geometric centerline in the lateral direction, and no higher than 50 cm above the lunar surface in its deployed configuration.	Enforces the placement of the vehicle CG for stability, and to inform chassis design.	MCH-05	N/A	Test	No
CHS-05	The chassis subassembly shall provide the mounting interfaces for the mobility, GNC, sample collection, thermal,	Subsystem/subassembly interfaces should transfer loads to the chassis for the structural integrity of the vehicle.	MCH-09	N/A	Demonst...	No

	CDH, and power subsystems.					
CHS-06	The chassis subassembly shall provide at least 0.5 m × 0.5 m × 0.75 m of internal or external usable volume for the scientific instruments and supporting hardware of the payload subsystem.	Enforces the minimum volume for scientific instruments and hardware to meet the STM science objective requirements.	SYS-01	N/A	Inspection	No
CHS-07	The chassis subassembly shall withstand launch, landing, deployment, and surface operation loads of at least 10 g in the axial direction, and 5 g in the lateral direction, with a factor of safety of at least 2.0.	Defines the minimum loads the chassis must bear in the stowed configuration.	MCH-03	N/A	Test	No
CHS-08	The chassis subassembly shall transfer the mobility system's loads up to 2,000 N in the vertical direction, 1,000 N in the lateral direction, and a 500 Nm moment at each chassis-mobility subassembly interface.	Defines the minimum loads the chassis must bear in the deployed configuration.	MCH-03	N/A	Test	No
CHS-09	The chassis subassembly in the vehicle stowed configuration shall maintain structural integrity and mounted subassembly alignment	Defines the minimum loads the chassis interfaces must bear in the stowed configuration.	MCH-07, MCH-08	N/A	Analysis	No

	within 0.5 deg after exposure to vibrations from 20 Hz to 20,000 Hz.					
CHS-010	The chassis subassembly and its mounting interfaces shall remain mechanically operational within the temperature range of -133°C to +121°C.	Defines the thermal range the chassis interfaces must survive in the deployed configuration.	MCH-03	N/A	Test	No
CHS-11	The chassis subassembly and its mounting interfaces shall remain mechanically operational when subjected to the thermal gradients of traversing in/out of the PSR.	Defines the thermal gradient the chassis interfaces must survive in the deployed configuration.	MCH-03	N/A	Test	No
CHS-12	The chassis subassembly shall allow for installation, inspection, and application/removal of mounted subsystems/subassemblies within 4 hours.	Defines assembly, verification, and disassembly time constraints in order to stay aligned with the mission schedule.	SYS-13, SYS-14	N/A	Demonst...	No
GNC-01	The GNC hardware subassembly shall have a total mass less than or equal to 5 kg.	Ensures subassembly mass does not go over allocated mass budget.	MCH-01	N/A	Inspection	No
GNC-02	The GNC hardware subassembly shall fit within an allocated 0.25 m × 0.25 m × 0.1 m such that the integrated system in the stored	Confines the GNC hardware subassembly to ensure subassembly volume remains within overall system and subsystem volume	MCH-02	N/A	Inspection	No

	configuration is within the 1.5 m x 1.5 m x 1.0 m envelope.	constraint.				
GNC-03	The GNC hardware subassembly shall provide the mechanical mounting for the localization and navigation sensors defined by GNC/CDH.	LiDAR, cameras, and navigation sensor selection must be integrated into the mechanical subsystem by the GNC hardware subassembly.	MCH-09	N/A	Demonst...	No
GNC-04	The GNC hardware assembly shall provide a datum (alignment reference point) that allows sensor orientations to be verified relative to the rover body coordinate frame within 0.25 deg.	Ensures that sensor alignment can be verifyable, which is especially important for navigation and localization sensors.	MCH-09	N/A	Inspection	No
GNC-05	The GNC hardware subassembly shall ensure that the sensor mounts maintain an alignment within 0.25 deg after undergoing launch, transit, landing, and deployment loads.	Ensures that loads do not shift GNC sensor alignment prior to the rover's deployment.	MCH-07	N/A	Demonst...	No
GNC-06	The GNC hardware subassembly shall ensure that the sensor mounts maintain an alignment within 0.25 deg after 2 weeks of exposure to lunar PSR environments.	Ensures that thermal cycling during the traverse in and out of PSRs does not cause mount loosening or misalignment.	MCH-03, MCH-07	N/A	Demonst...	No
GNC-07	The GNC hardware subassembly shall provide an	Sensors such as LiDAR or cameras must have unobstructed views to	MCH-03, MCH-07	N/A	Demonst...	No

	unobstructed field of view for terrain sensing/mapping sensors during nominal surface operations.	obtain accurate terrain sensing/mapping.				
GNC-08	The GNC hardware subassembly shall incorporate lunar regolith and dust mitigation for optical or sensing hardware.	Dust mitigation for sensing hardware will decrease risk of failure due to regolith contamination.	MCH-10	N/A	Test	No
INT-01	The spacecraft structure shall incorporate 12 through-holes on an 8.000 in [203.20 mm] bolt circle, spaced 30 degrees, first hole 15 degrees off the PL Z+ axis, per MCA Interface Control Document (ICD) Figure 2.	Matches the standard Launch Vehicle (LV) mounting hole pattern, allowing the vehicle to physically mate with the interface.	MCH-07	N/A	Inspection	No
INT-02	Each mounting through-hole shall have a diameter of 0.273 to 0.278 in [6.93 to 7.06 mm], per MCA ICD Figure 2.	This size allows for the clearance holes to accept the standard interface fasteners.	MCH-07	N/A	Inspection	No
INT-03	Each mounting through-hole shall hold a true position tolerance of 0.020 in at MMC to datums A and B, per MCA ICD Figure 2.	Guarantees that the hole pattern can locate just enough to align with the LV interface holes.	MCH-07	N/A	Inspection	No
INT-04	The mating face (datum A) shall be flat to within ±0.004 in [0.10 mm], per MCA ICD Figure 2 Section A-A.	Ensures a flat mating plane, allowing the interface to seat evenly and load to transfer uniformly across the	MCH-07	N/A	Inspection	No

		bolt pattern.				
INT-05	Each fastener shall conform to the standard interface, sized to the Figure 2 hole pattern, per MCA ICD Table 1.	ICD defers to the standard interface for the fasteners size, so the fasteners are sized to the same pattern, maintaining compatibility and qualification.	MCH-07	N/A	Inspection	No
INT-06	Each fastener shall incorporate at least one locking feature that does not rely on preload, using, in order of preference: lock wire or lock cable; a staked head with a hardness check; or a thread locker with a hardness check; per MCA ICD Table 1.	Prevents fastener back out under launch vibration, independent of clamp load.	MCH-07	N/A	Inspection	No
INT-07	Each fastener shall provide a minimum of two thread pitches protrusion past the nut, per MCA ICD Table 1.	Ensures complete and full thread engagement, allowing the joint to carry the load.	MCH-07	N/A	Inspection	No

Figure 21: Mechanical Subsassembly Requirements

2.1.1.2. Mechanical Sub-Assembly Overview

The mechanical subsystem is divided into four major subassemblies: the chassis/structure subassembly, mobility subassembly, GNC hardware subassembly, and mechanical interface subassembly. These subassemblies collectively provide the physical architecture needed to support rover and science operations, science payload integration, autonomous mobility, and subsystem mounting. The design remains preliminary; therefore, the descriptions below define the current baseline functions and mechanical responsibilities without finalizing all detailed geometry or component selections.

Chassis/Structure Subassembly

The chassis/structure subassembly acts as the load bearing structure of the rover. The main purpose of this subassembly is to act as the mounting for the internal hardware, payload, wheel attach points, sensors, and mechanical interface systems of the rover while ensuring that the mass and volume requirements of the mission are met. The chassis acts as the main load path for ground handling, launch and landing configuration, deployment, traversing the surface, and stationary and dynamic science operations.

The chassis forms the underlying structural frame that carries loads and supports major mechanical interfaces. Mounted on top of this chassis is the rover body, which is currently designed as a compact rectangular-prism enclosure. This body geometry was selected because it provides a simple structural enclosure, efficient packaging volume, and flat mounting surfaces for subsystem integration. The rectangular body also simplifies manufacturing, payload deck layout, and structural analysis. The chassis structure is currently baselined to use Aluminum 6061-T6, whereas the body panels or enclosure material has been trade studied to use an Aluminum Honeycomb Sandwich panel, which is currently being used on NASA's VIPER mission.

Mobility Subassembly

The mobility subassembly provides the rover's mechanical surface traversal capability. Its purpose is to allow the rover to move from the illuminated landing site toward the Exploration Zone, traverse lunar regolith terrain, maintain traction, and support stable low-speed autonomous operations. For the current design, the Mobility Subassembly is focused primarily on the wheel assemblies, drive motors, and their structural integration with the rover body. The baseline mobility concept uses six rigid aluminum grouser wheels arranged along the left and right sides of the rover body. Each wheel is driven by an electric motor integrated into the mobility system, providing the torque required for rover locomotion and allowing controlled low-speed traversal across the lunar surface. Power is supplied to the drive motors by the rover's electrical power subsystem, while motor commands and speed control are coordinated through the rover's control system. The use of six independently driven wheels improves load distribution, terrain contact, and mobility capability compared with a simpler four-wheel or rigid-wheel configuration. The wheels are intended to operate in vacuum and abrasive lunar regolith without relying on pneumatic tires or rubber-dominated components. Grousers or equivalent traction features are included to improve traction and reduce excessive slip on loose regolith.

Guidance, Navigation, and Control (GNC) Subassembly

The GNC hardware subassembly provides the mechanical mounting, alignment, and protection features required for rover navigation and mobility-state hardware. This

subassembly does not include the GNC algorithms or software. Its role is to ensure that selected GNC hardware can be mounted in mechanically useful locations with adequate stiffness, field of view, access, and protection.

The current design includes Light Detection and Ranging (LiDAR) aided navigation sensors, inertial measurement units (IMUs), and supporting sensor hardware. The exact sensor package remains under development, so the mechanical design must preserve flexibility. The accommodation approach should provide elevated mounting options for terrain-observing sensors, protected internal mounting surfaces for inertial sensors, and lower body or wheel-adjacent provisions for mobility-related sensors as needed.

Mechanical Interface Subassembly

The mechanical interface subassembly includes the physical mounting features that allow the system structure to mechanically integrate with the Launch Vehicle interface and other major system elements. Its primary purpose is to provide a reliable, repeatable, and verifiable structural connection between the vehicle and the standardized Launch Vehicle mounting interface defined in the MCA Interface Control Document. Because this interface controls how the vehicle physically mates to the launch vehicle, it is a critical mechanical design feature for integration, launch loading, and mission compatibility.

The baseline mechanical interface is defined by the MCA ICD 8-inch diameter mechanical interface. The spacecraft structure shall incorporate twelve through-holes located on an 8.000 in [203.20 mm] bolt circle, with holes spaced 30 degrees apart and the first hole positioned 15 degrees off the PL Z+ axis. Each mounting through-hole shall have a diameter between 0.273 and 0.278 in [6.93 to 7.06 mm] and shall meet the required true position tolerance of 0.020 in at MMC relative to datums A and B. The mating face, identified as datum A, shall be flat within ±0.004 in [0.10 mm] to ensure that the interface seats evenly and transfers load uniformly across the bolt pattern.

Technology Readiness Level (TRL)

The Mobility Subassembly is rated at TRL 6 because its rigid metallic grouser wheels and rocker-bogie-based mobility architecture are derived from demonstrated NASA Mars rover technology. Rocker-bogie suspension has operational heritage extending from Mars Pathfinder's Sojourner rover through Spirit and Opportunity, Curiosity, and Perseverance. Curiosity and Perseverance also use rigid aluminum wheels with raised grousers, with Perseverance's wheels machined from flight-grade aluminum and incorporating 48 curved grousers for traction. These technologies have therefore been demonstrated in an operational planetary environment, corresponding to a high level of maturity for the underlying mobility concept. However, the mission-specific wheel geometry, suspension configuration, deployment characteristics, and chassis integration have not yet been

demonstrated in a representative lunar environment, supporting a system-level rating of TRL 5–6 rather than a higher TRL.

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The GNC Hardware Subassembly is rated at approximately TRL 6 because mechanical integration techniques for LIDAR, IMUs, cameras, and other navigation sensors including rigid mounting, alignment features, and harness-routing provisions are established spacecraft and planetary rover practices. The individual sensing technologies and their mounting approaches have been demonstrated in relevant environments, but the specific sensor arrangement and mechanical integration proposed for this rover have not yet been demonstrated as a complete subsystem in representative lunar conditions. Therefore, the subassembly is considered consistent with TRL 6 only where representative subsystem-level integration and testing can be demonstrated.

The Mechanical Interface Subassembly is rated at TRL 6 because the launch vehicle mounting concept has progressed beyond a conceptual design and is defined within the MCA Interface Control Document through a 12-hole bolt pattern, specified hole sizing, and flatness and true position tolerances. The approach uses conventional spacecraft mechanical attachment and launch interface practices rather than a novel technology. However, because the mission-specific interface has not yet undergone representative integrated structural or launch-load testing, it does not yet fully satisfy the TRL 6 criterion of a system or subsystem model or prototype demonstrated in a relevant environment.

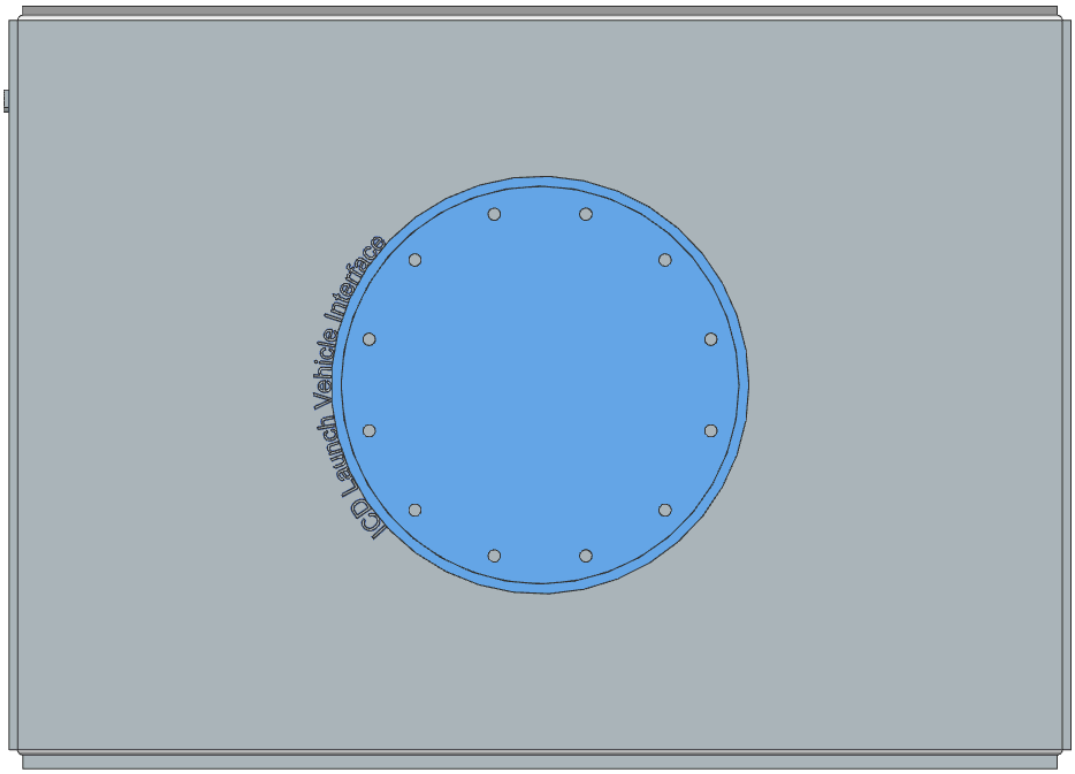


Figure 22: Launch Vehicle Interface Subassembly CAD

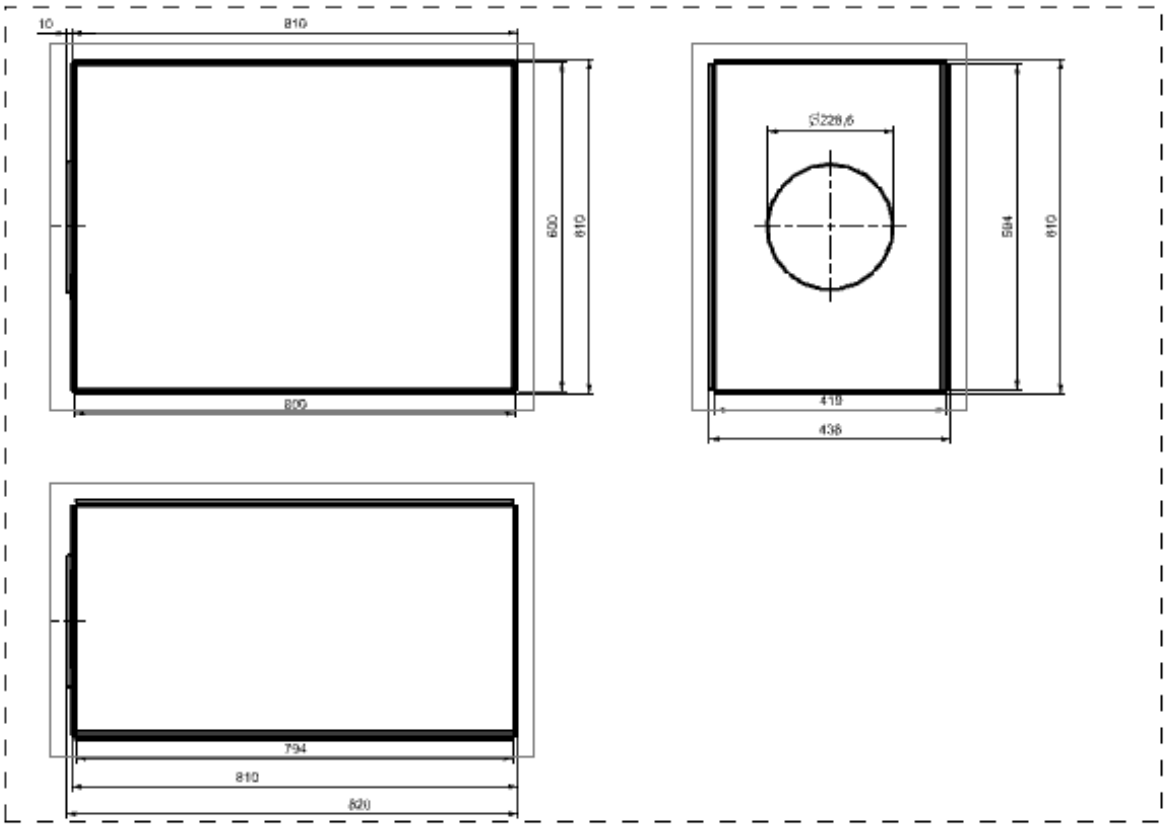


Figure 23: Body Panel Subassembly Engineering Drawing

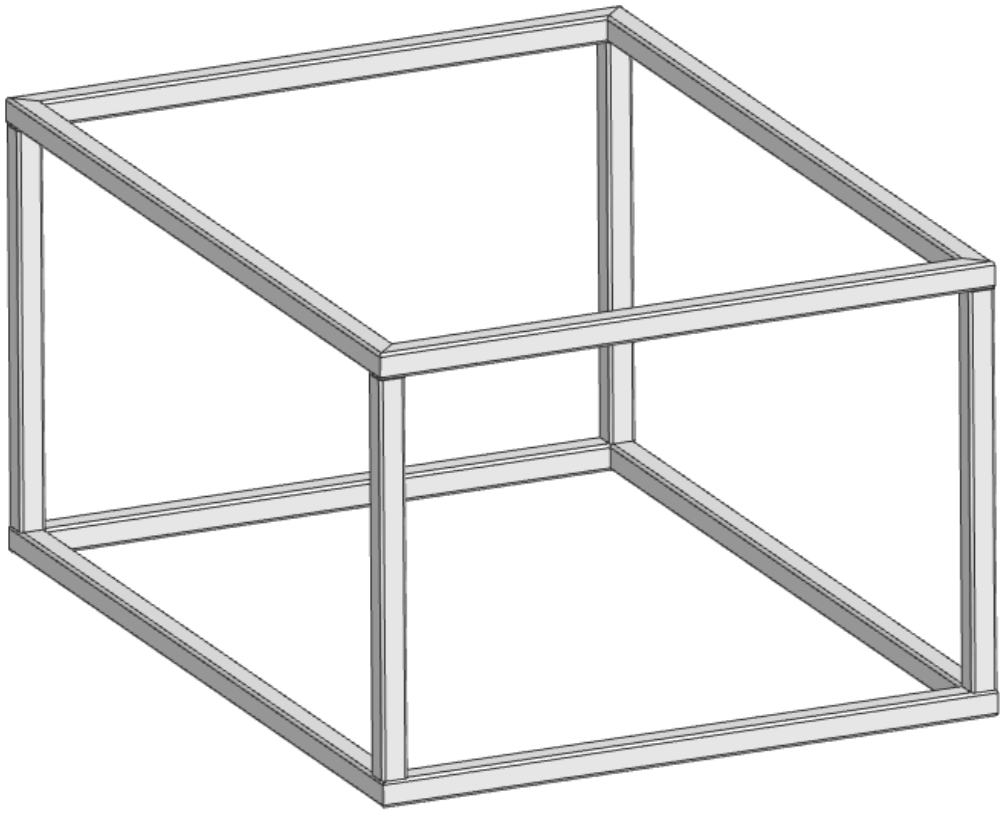
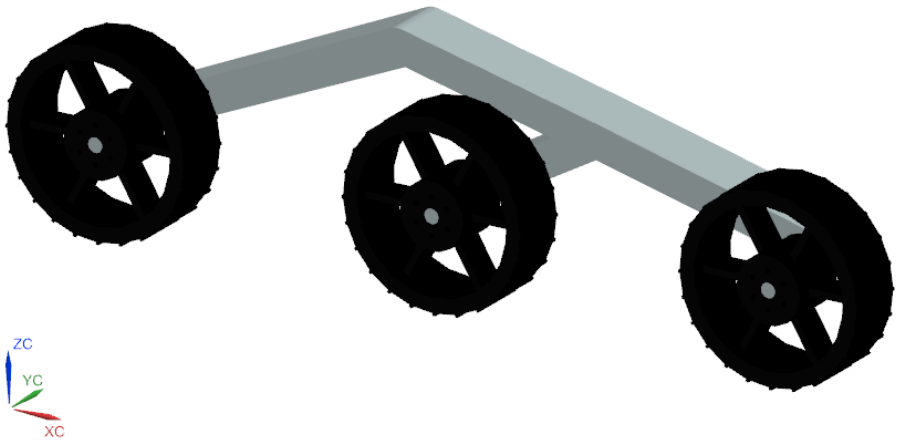


Figure 24: Chassis Subassembly CAD



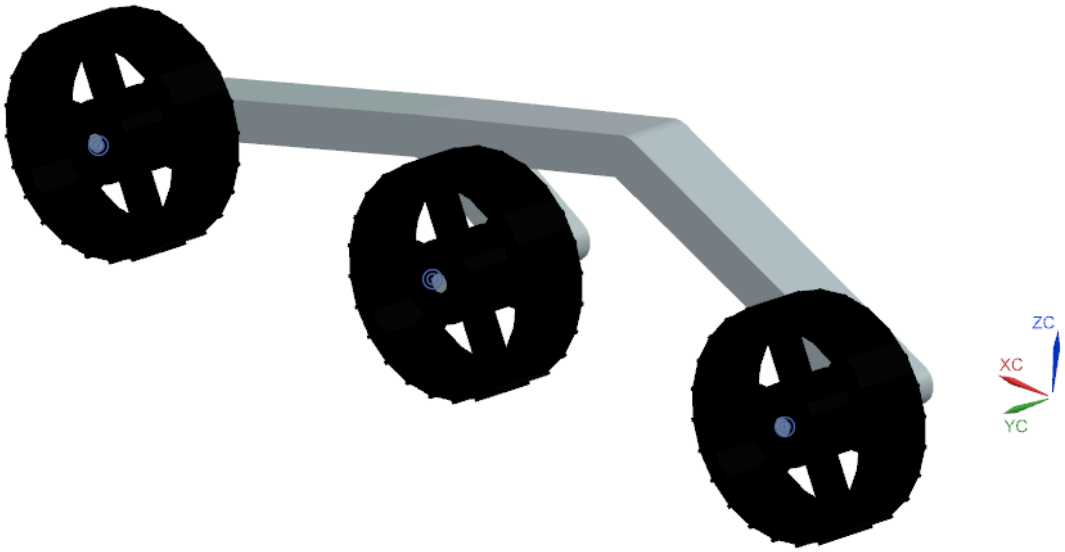


Figure 25: Mobility Subassembly CAD

2.1.1.3. Mechanical Subsystem Recovery and Redundancy

Redundancy

The mechanical subsystem uses a risk-reduction approach rather than full duplication of major mechanical components. The chassis, rover body, launch vehicle interface, and wheel assemblies are not duplicated because adding backup structures or duplicate mobility hardware would increase mass, volume, and mechanical complexity. Instead, mechanical risk is reduced through verified load paths, simple passive mechanisms, and inspection of critical interfaces. The six-wheel rocker bogie suspension layout provides some inherent tolerance because the rover load is distributed across six wheels; however, the design does not assume that a failed wheel is fully redundant. The

launch vehicle interface is also not redundant, but risk is controlled through compliance with the MCA ICD hole pattern, fastener-locking features, thread engagement, flatness, and installation requirements.

Recovery

Failure of mechanisms will be indicated by changes in rover performance or subsystem telemetry, including wheel slip, unusual rover attitude, wheel loading and current beyond acceptable levels, deployment not achieved, or motion not following the command. Recovery of problems will be done operationally, not through reconfiguration of hardware. If the rover's mobility is impaired, it will move at a slower pace, avoid steeper inclinations, make gentler turns, choose better terrain, and perform science on targets which are accessible. If a deployment or interface issue occurs, the rover should remain in a safe configuration until the issue is understood rather than attempting nominal traversal.

2.1.1.4. Mechanical Subsystem Manufacturing and Procurement Plans

Procurement Overview:

The manufacturing procurement methods for the mechanical subsystem's primary components are build-to-print, build-to-spec, and COTS/SQCOTS. Most of the components in this subsystem require specific geometry, mass, load paths, and interfaces under the team's discretion. For this reason, build-to-print procurement will be utilized for the chassis, body panels, mounting rails and brackets, wheels, suspension system (which includes links, pivots, shafts, bearing housings, retainers, bearings, and supporting hardware), and launch vehicle to rover interface plate. Due to the exhaustive nature of fully designing brushless motors and a reduction gearbox, a build-to-spec procurement will be used with mass, volume, torque, and power draw constraints. This approach leverages vendor expertise so that time, and resources such as personnel can be diverted to the mobility system and subsystem integration. The lock cable, fasteners, and supporting hardware components for mating the launch vehicle interface to the rover will be procured using SQCOTS (COTS/MCOTS as backup), thus standardizing fasteners and locking mechanisms with proven flight heritage. This will reduce complexity and costs, and improve predictability, assembly times, and inspectability.

The primary and secondary suppliers were chosen for specific components or assemblies based on weighing the following factors from highest to lowest importance: field expertise and manufacturing ability, heritage and certifications (AS9100/ISO), availability, and lead times. Land Aero Group LLC was chosen as the primary supplier for the chassis, body panels, mounting rails and brackets, and launch vehicle interface plate

due to their specialization in precision manufacturing of aerospace grade aluminum, and space heritage (based on hardware manufactured for the ISS). TOMI Engineering is the backup supplier of Land Aero LLC cannot fulfill the order, as it is a similar company with the main difference being less space heritage. Mistequay Group LTD is identified as the primary supplier for the motors and reduction gearbox due to their field expertise in electromechanical systems (such as motors and planetary gears). MIC Group LLC is chosen as the primary supplier of the rocker-bogey suspension system due to their expertise in complex mechanical systems, and previous contracts with SpaceX (although exact parts/systems are not publicly known). LAN Research specializes in military hardware, thus it was chosen as the primary supplier for the fasteners, lock cables, and supporting hardware.

Subsystem / Component	Procurement Method	Primary Supplier	Backup Supplier	Estimated Lead Time	Justification
Primary Chassis Frame (6061-T6)	Build-to-Print	Land Aero Group, LLC	TOMI Engineering	12-18 weeks	AS9100D/ISO9001 /HUBZone certified. Build-to-print and precision manufacturing of aluminum expertise in the space sector.
Aluminum Honeycomb Body Panels	Build-to-Spec	Land Aero Group, LLC	TOMI Engineering	14-20 weeks	AS9100D/ISO9001 /HUBZone certified. Build-to-print and precision manufacturing of aluminum expertise in the space sector.
Mounting Rails and Brackets	Build-to-Print	Land Aero Group, LLC	TOMI Engineering	6-12 weeks	AS9100D/ISO9001 /HUBZone certified. Build-to-print and precision manufacturing of

					aluminum expertise in the space sector.
Brushless DC Motors with Reduction Gearbox	Build-to-Spec	Mistequay Group LTD	Aero Space Controls Corporation	24-30 weeks	Expertise in electro-mechanical systems and gears.
Rigid Aluminum Grouser Wheels	Build-to-Print	Land Aero Group, LLC	TOMI Engineering	16-24 weeks	AS9100D/ISO9001 /HUBZone certified. Build-to-print and precision manufacturing of aluminum expertise in the space sector.
Rocker-bogie Suspension Assembly	Build-to-Print	MIC Group, LLC	Mistequay Group LTD	24-30 weeks	Expertise in complex mechanical systems, ISO 9001:2015/ AS9100D certified. SpaceX heritage.
Launch Vehicle Interface Plate	Build-to-Print	Land Aero Group, LLC	TOMI Engineering	8-12 weeks	AS9100D/ISO9001 /HUBZone certified. Build-to-print and precision manufacturing of aluminum expertise in the space sector.
Lock Cable & Supporting Hardware	COTS / SQCOTS	LAN Research	Thermal Management Technologies	3-6 weeks	Field expertise in fasteners, fast lead times.

Figure 26: ___

Lead Time Summary:

The mechanical subsystem is expected to be fully procured within 30 weeks. The "weakest links" that pose the highest risk to the mission schedule are the procurement of the brushless DC motors with reduction gears, and rocker-bogie suspension system. This is due to the complex nature of these components, and tight tolerance requirements which prolong lead times. There would be medium level scheduling risks in the case of delays from the primary and secondary vendors for these critical components, but this risk cannot be mitigated through alternative manufacturing such as MCOTS because of the technical risk that this would pose. Therefore, the scheduling risks primarily associated with the lead times of the motors and suspension system should be monitored, and possibly mitigated by finding second backup space qualified suppliers. The scheduling risks for all other components remain low, due to the ease of manufacturing, and substantially lower lead times than the higher lead time risk components.

2.1.1.5. Mechanical Subsystem Verification Plans

Verification Overview

The Mechanical Subsystem verification approach uses a combination of inspection, test, and demonstration to confirm that all subsystem requirements have been satisfied. Inspection will be used for requirements that can be verified through CAD models, dimensional checks, mass tracking, interface drawings, and physical measurements, such as stored envelope compliance, mass limits, and mounting-interface compatibility. Tests will be used for requirements that require measured performance under representative conditions, including structural loading, slope traversal, regolith exposure, launch/landing load survival, and long distance mobility. Demonstration will be used for functional requirements where the subsystem must show that it can perform an operation, such as deploying to operational configuration, maintaining mechanical functionality after environmental exposure, or integrating required subsystem hardware. Together, these methods verify that the Mechanical Subsystem can fit within the required envelope, survive expected loads and environments, support rover mobility, protect key interfaces, and provide the required mechanical structure and mounting features for mission operations.

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
MCH-01	The mechanical subsystem shall	Inspection	Verify by reviewing the mechanical subsystem mass budget, CAD	Maintain a mechanical subsystem mass budget and compare updated CAD mass

	have a total mass less than or equal to 80 kg.		mass properties, and measured or vendor-provided component masses to confirm the total mechanical mass is ≤80 kg.	properties, vendor/component masses, and measured masses against the 80 kg allocation throughout design maturation. Final verification will confirm the total mechanical subsystem mass is ≤80 kg before integration.
MCH-02	The mechanical subsystem and all subsequent components shall fit within a stored configuration envelope of 1.5 m × 1.5 m × 1.0 m.	Inspection	Inspection is appropriate because the stored envelope can be verified directly using CAD dimensions, packaging layouts, and physical measurements. This confirms that the mechanical subsystem and all attached components remain within the required 1.5 m × 1.5 m × 1.0 m stored configuration before integration.	Perform CAD envelope checks with all mechanical components in stowed configuration. Confirm final dimensions by physical measurement before integration.
MCH-03	The mechanical subsystem shall maintain operational functionality after exposure to PSR environments for at least 2 weeks.	Demonstrat...	Demonstration is appropriate because functionality must be confirmed after representative PSR exposure. This verifies the hardware can still operate after the required environment duration.	Expose mechanical hardware to a 2-week PSR-relevant thermal/vacuum test. Demonstrate wheel rotation, suspension articulation, and interface functionality afterward.
MCH-04	The mechanical subsystem shall enable the rover to traverse at least 300 meters on lunar terrain.	Test	Test is appropriate because traverse capability must be verified through actual mobility performance over representative terrain. This confirms the rover can	Conduct a 300 m mobility test over lunar regolith simulant or representative terrain. Verify wheel function, traction, stability, and completion of the required traverse distance.

			complete the required distance under expected surface conditions.	
MCH-05	The mechanical subsystem shall enable the Lunar Surface Vehicle System to traverse lunar terrain slopes up to 15° without loss of static stability or mobility function.	Test	Test is appropriate because slope stability and mobility must be verified under representative terrain conditions. This confirms the rover can traverse a 15° slope without tipping, losing traction, or losing mobility function.	Conduct mobility testing on a representative 15° inclined surface using lunar-regolith simulant or equivalent terrain. Verify rover attitude, wheel traction, slip, and continued mobility throughout the test.
MCH-06	The mechanical subsystem shall enable the Lunar Surface Vehicle System to deploy to the operational configuration within 30 minutes of landing on the Moon.	Demonstrat...	Demonstration is appropriate because deployment must be shown as a timed functional sequence. This confirms the system can transition from stored to operational configuration within 30 minutes.	Perform a timed deployment demonstration using the representative mechanical configuration. Verify all required wheels, suspension, interfaces, and deployable hardware reach operational configuration within the required time.
MCH-07	The stowed mechanical subsystem shall maintain structural integrity and functional mechanical interfaces after exposure to launch, transit, and landing loads.	Test	Test is appropriate because structural integrity and interface function must be verified under representative launch, transit, and landing loads. This confirms the stowed system can survive loading without damage or interface failure.	Perform representative vibration, shock, and load testing on the stowed mechanical configuration or critical structural interfaces. Inspect for damage and verify required mechanical interfaces remain functional after testing.
MCH-08	The mechanical subsystem shall support a fully integrated rover	Test	Test is appropriate because the structure must physically demonstrate it can	Apply representative 1-g ground handling loads to the integrated rover structure or critical load paths. Verify no structural

	mass with a minimum static factor of safety of 2.0 under 1-g ground handling loads on Earth.		support the integrated rover under ground handling loads. This verifies the mechanical subsystem meets the required 2.0 static factor of safety.	failure, permanent deformation, or interface damage occurs.
MCH-09	The mechanical subsystem shall provide mechanical mounting interfaces for all required subsystems and subassemblies (power, CDH/communications, thermal, payload).	Demonstrat...	Demonstration is appropriate because the mounting interfaces must be shown to physically accommodate the required subsystems. This verifies that power, CDH/communications, thermal, and payload hardware can be mechanically integrated.	Demonstrate subsystem fit using CAD integration checks or representative hardware mockups. Verify mounting locations, clearances, fastener access, and interface compatibility for each required subsystem.
MCH-10	The mechanical subsystem shall protect exposed moving interfaces from lunar regolith during nominal surface operations.	Test	The test is appropriate because regolith protection must be verified through representative dust exposure and operation. This confirms exposed moving interfaces can continue functioning during nominal surface use.	Expose representative joints, wheel hubs, bearings, or deployment interfaces to lunar-regolith simulant. Verify required motion and interface function after exposure.

Figure 27: Insert figure caption here

2.1.2. Power Subsystem Overview

1. *Power Subsystem Overview*

The Power Subsystem provides the electrical power required for the Lunar Surface Vehicle System to conduct post-landing surface operations. The subsystem generates, stores, regulates, and distributes electrical power to the mechanical, CDH, thermal, and payload subsystems through a regulated 28 V DC bus. During operations in the illuminated regions of the lunar south pole, the Gallium Arsenide (GaAs) solar panels will generate electrical power for the rover while also recharging the onboard lithium-ion battery system. The power regulation and control electronics will manage the power generated by the solar panels, control battery charging and discharging, and distribute electrical power to the rover subsystems. When the rover enters the PSR, the lithium-ion battery system will become the primary electrical power source for the rover. Overall, the Power Subsystem will provide the power needed to support rover operations in both illuminated and permanently shadowed regions.

2. *Power Capabilities*

The Power Subsystem is designed to provide a minimum continuous output power of 255 W during nominal rover operations and support peak loads of up to approximately 490 W. Based on the current subsystem estimates, the rover has a total maximum power draw of approximately 318 W. The lithium-ion battery system provides approximately 1,296 Wh of usable energy to support the planned 6.07-hour PSR sortie, which requires approximately 1,064 Wh. Power regulation and distribution electronics maintain the nominal 28 V DC bus and monitor battery voltage, current, temperature, state of charge, and overall system health. The battery system is limited to a maximum Depth of Discharge (DoD) of 60% to maintain an energy reserve during PSR operations. High-power activities are scheduled sequentially whenever possible to reduce peak power demands and stress on the power electronics. The subsystem is also designed to maintain safe and reliable power operation within the harsh lunar south pole environment.

3. *Power Subsystem Summary Table*

Subassembly	Mass (kg)	Dimensions (m)	Maximum Power Draw (W)	TRL
Power Generation	16	1.0 x 0.9 x 0.03	0	6
Power Storage	22	0.22 x 0.13 x 0.17	4 W (Delivers >490 W Capacity)	5

Power Regulation and Control Electronics	6	0.2 x 0.2 x 0.1	10 W (Operational)	6
Total Subsystem	44	-	~14 W internal (Supports ~490 W peak load)	

Figure 28: Power Subassembly

4. *Power Subsystem Figure*

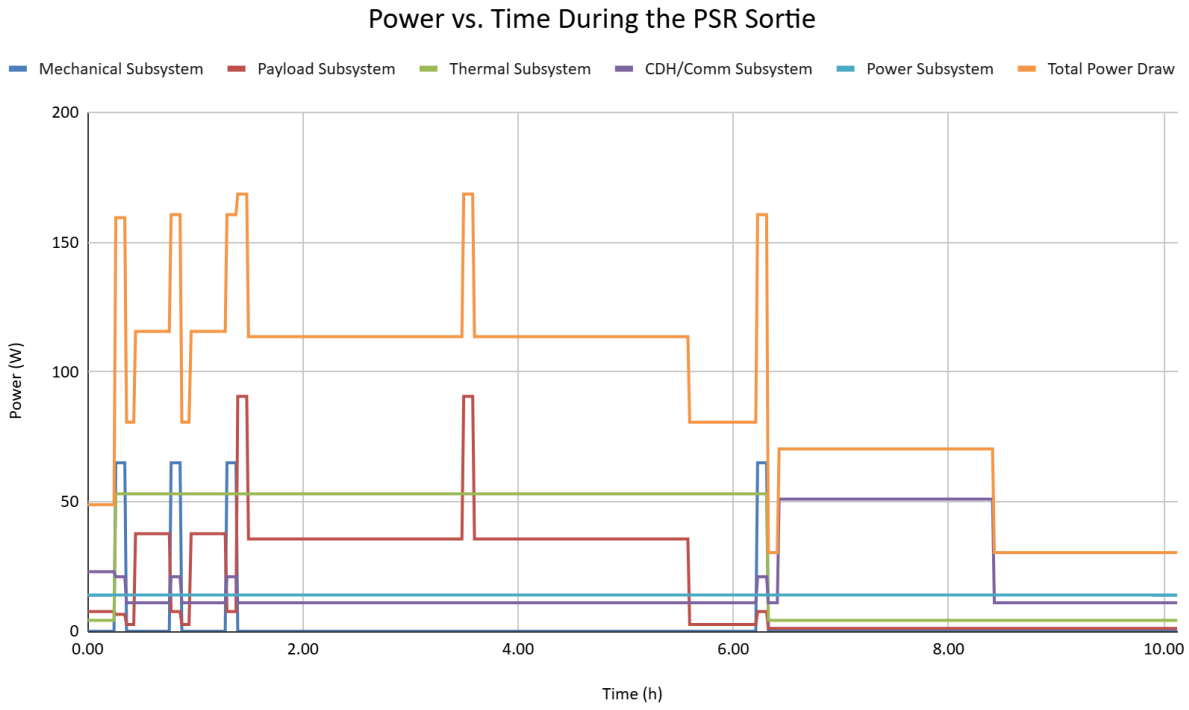


Figure 29: Rover Subsystem Power vs Time Chart

2.1.2.1. Power Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
EPS-01	The Power Subsystem mass shall not exceed 44 kg.	Allocates the system mass constraints to ensure the overall 175 kg rover mass limit is maintained.	SYS-02		Inspecti...	Y...
EPS-02	The energy storage subassembly shall provide at least 1,064 Wh of usable energy for one 6.07 hour sortie.	Necessary to ensure there is sufficient stored energy to complete the planned PSR sortie.	SYS-08		Analysis	Y...

EPS-03	The battery subassembly shall operate safely without exceeding the rover's maximum thermal constraints during surface operations.	Necessary to ensure the battery remains within the rover's required operating temperature and does not overheat the thermal constraints of the rover.	SYS-06		Analysis	Y...
EPS-04	The energy storage subassembly shall not exceed an allocated cost of $17.0 M.	Allocates the system cost constraints to ensure the total system cost remains within $250 million.	SYS-12		Analysis	Y...
EPS-05	The battery subassembly shall provide a minimum continuous output power of 255 W during nominal rover operations and support peak system loads of up to 490 W.	Necessary to ensure the battery provides sufficient electrical power to support rover operations and short-duration peak loads without exceeding bus limits.	SYS-08		Test	No
EPS-06	The battery enclosure shall withstand the mechanical shock loads experienced during landing and surface operations without damage or loss of functionality.	Ensures battery safety and prevents electrical/mechanical failure during landing and surface mobility.	SYS-01		Test	No
EPS-07	The energy storage subassembly shall store and provide a minimum of 100 W to support rover subsystems during emergency and safe-mode operations.	Ensures there is sufficient power available to maintain critical rover operations if a power system failure were to occur.	SYS-08		Test	No
EPS-08	The battery subassembly shall support a minimum of 2,000 charge/discharge cycles without degrading below 80% of its nominal	Since the rover will rely on multiple recharge cycles (via solar panels), the battery must survive the full mission duration without fatal capacity loss.	SYS-08		Test	No

	capacity.					
EPS-09	The battery subassembly shall provide the charging state, voltage, and temperature telemetry to the CDH subsystem.	Crucial for the power monitoring and fault protection systems to manage the rover's power budget and detect failures before they become critical.	SYS-08		Demon...	No
EPS-10	The energy storage and regulation subassembly shall supply a nominal regulated DC bus voltage of 28 V.	Required to ensure electrical compatibility with the rover power-distribution architecture and point-of-load converters.	SYS-08		Test	No
EPS-11	The power regulation electronics shall prevent the battery from exceeding a maximum Depth of Discharge of 60%.	Deeply discharging lithium-based batteries can cause permanent chemical damage so this protects the asset and ensures recovery.	SYS-08		Test	No
EPS-12	The Power Subsystem shall protect the battery subassembly and exposed electrical connections from lunar regolith during surface operations.	Crucial for preventing damage to electrical components from abrasive lunar regolith and maintaining mission operations.	SYS-01		Inspecti...	No

Figure 30: Power Subsystem Requirements

2.1.2.2. Power Sub-Assembly Overview

1. ***Power Generation***

The power generation subassembly provides the primary source of electrical power for the rover subsystems. Its main function is to convert solar radiation into usable electrical energy and recharge the power storage subassembly while the rover is in the illuminated lunar regions. The power generation subassembly design consists of two fixed solar panels containing GaAs solar cells mounted on the upper surface of the rover chassis, sized to provide at least 500 W of peak power. While operating in the illuminated regions, the rover will primarily be powered by the solar panels. Electrical power generated by the solar panels is first transferred to the power regulation and control subassembly. There it is regulated by Distributed Maximum Power Point Tracking (MPPT) controllers before being used to power active rover loads and recharge the power storage subassembly.

The power generation subassembly also includes the structural mounting hardware, electrical wiring, and connectors required to integrate the solar panels with the rover. The subassembly interfaces with the mechanical subsystem through the solar panel mounting structure and with the power regulation and control electronics through its electrical connections. Solar panel voltage and current telemetry are also provided to the CDH subsystem to monitor power generation and overall system health. GaAs solar cells have demonstrated high conversion efficiency, radiation tolerance, and thermal performance through their flight heritage, making them suitable for the harsh lunar environment. The solar panels will be able to withstand extreme thermal conditions, lunar regolith exposure, and mechanical loads expected during launch, landing, and lunar surface operations.

2. ***Power Storage***

The power storage subassembly stores the electrical energy generated by the GaAs solar panels and provides the primary source of electrical power during operations inside the PSR. The power storage design consists of two rechargeable lithium-ion battery modules located inside the rover body. The battery system uses a total of 16 lithium-ion cells, with eight cells contained within each battery module. Each cell has a nominal voltage of 3.6 V and a nominal capacity of 43 Ah. The battery system is designed to provide approximately 1,296 Wh of usable energy to support the planned 6.07-hour PSR sortie, which requires approximately 1,064 Wh. The additional available energy provides margin for changes in rover power consumption and for recovery operations. Lithium-ion batteries have demonstrated high energy density, reliable performance, and thermal survivability in previous space missions, making them suitable for powering the rover.

The battery system includes an insulated enclosure, structural supports, battery management and protection electronics and electrical wiring and connectors. The Battery Management System (BMS) monitors battery voltage, current, temperature, state of charge, and overall battery health while managing battery charging and discharging. The power storage subassembly interfaces with the mechanical subsystem through its structural hardware and battery enclosure and with the thermal subsystem to maintain safe operating temperatures. It also interfaces with the power regulation and control subassembly for battery charging, discharging, and power delivery and with the CDH subsystem to provide battery health information throughout the mission. The battery system will be designed to withstand the extreme thermal conditions, mechanical loads, and lunar regolith exposure expected throughout the mission.

3. ***Power Regulation & Control***

The power regulation and control electronics subassembly utilizes a Distributed Maximum Power Point Tracking (MPPT) and Point-of-Load (POL) architecture to efficiently manage power. It regulates the raw electrical energy harvested from the GaAs solar arrays to maintain a stable regulated 28V DC bus. This configuration ensures that power is optimally tracked and converted, significantly reducing electrical transmission losses across the rover.

To accommodate the rover's estimated peak power demands, driven by mechanical, thermal, and payload operations, the power regulation electronics distribute power through the regulated 28 V DC bus. The BMS manages the charging and discharging of the power storage subassembly while monitoring battery conditions. By sizing the battery system to provide approximately 1,296 Wh of usable energy, the regulation electronics can support the planned 6.07-hour PSR sortie without discharging the lithium-ion cells beyond the maximum 60% DoD limit. POL converters located at each subsystem interface step down the main 28 V DC bus to the specific operating voltages required by individual components.

4. Technology Readiness Level (TRL)

The overall Power Subsystem is rated at TRL 5. Although the GaAs solar cells, lithium-ion batteries, and MPPT/POL controllers have flight heritage, the specific power system developed for the rover has not yet been fully integrated and tested. The power generation and power regulation and control subassemblies are rated at TRL 6 because their main technologies have already been demonstrated in space environments. However, the integration of the solar panels and battery system with the MPPT controllers still requires additional testing under expected lunar and PSR conditions.

The power storage subassembly is rated at TRL 5 because the 16-cell, two-module battery configuration is specific to this rover. The battery modules, BMS, enclosure, electrical connections, and thermal integration still need to be tested together under the expected PSR conditions. The complete Power Subsystem will also need to be tested using the expected mission power profile. This includes mobility, science operations, peak loads, and battery discharge and recharging to verify its performance throughout the mission. Since the current power system design is mainly based on data from previous missions, the overall Power Subsystem has a TRL of 5.

5. ***Supporting Figures***

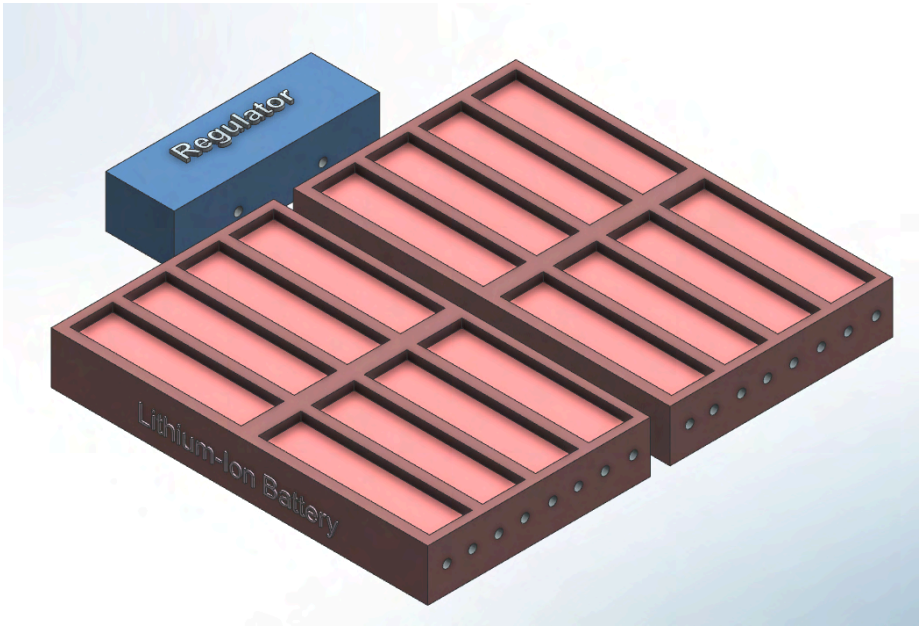


Figure 31: Battery Modules with Regulator

2.1.2.3. Power Subsystem Recovery and Redundancy Plans

1. ***Redundancy***

The Power Subsystem uses partial redundancy to reduce the risk of a single component failure affecting the mission. The rover uses two GaAs solar panels, allowing some power generation to remain available if one panel fails. The power storage subassembly also consists of two lithium-ion battery modules, allowing the rover to maintain limited power capability if one battery module experiences a failure. The central power regulation and distribution electronics, including the MPPT controllers, POL converters, and the regulated 28 V DC bus, have limited redundancy. This is because adding additional hardware would increase the mass and complexity of the subsystem. Instead, space-qualified components and continuous system monitoring are used to reduce the risk of these single-point failures. This partial redundancy provides additional reliability for power generation and storage while remaining within the rover's mass and integration constraints.

2. ***Recovery***

Failure of power components will be detected by changes in system telemetry, including bus voltage drops, irregular current draw, reduced charging efficiency, or battery

cell temperatures exceeding operational limits. Power issues will be managed by automatically reducing nonessential loads and adjusting rover operations as needed. If a system error occurs or the battery charge falls below its operational threshold, the rover will enter safe mode. During safe mode, non-essential systems, such as the scientific payload, will be powered down while essential systems, including CDH, thermal, and mechanical, remain operational. This will help reduce energy consumption and support recovery. If a permanent hardware failure occurs, the mission profile will be adjusted. For example, if one battery module fails, the rover will reduce its operating time inside the PSR to stay within the 60% DoD limit. If the battery charge becomes critically low or the rover remains inside a PSR for an extended period, it will enter survival-power mode. During this mode, only the systems required to maintain rover health and safely exit the PSR will remain powered. Once the rover returns to an illuminated region and the batteries are sufficiently recharged, it will exit survival-power mode and resume nominal operations.

2.1.2.4. Power Subsystem Manufacturing and Procurement Plans

1. ***Procurement Overview***

The power subsystem is mainly supplied through space-qualified commercial off-the-shelf (SQCOTS) components and prime contractor development. The distributed Maximum Power Point Tracking (MPPT) controllers were selected as SQCOTS components because they are commercially available and already qualified for space environments, reducing development time and risk. The lithium-ion battery system will use space-qualified lithium-ion cells integrated through prime contractor development to meet the rover's energy storage, voltage, thermal, and mechanical requirements. The GaAs solar panels will also be produced through prime contractor development because they require a custom design to meet the rover's power requirements. This combination of existing space-qualified components and custom development helps reduce technical risk while meeting the specific requirements of the rover.

Boeing was selected as the primary supplier for the GaAs solar panels because of its extensive flight heritage and high-efficiency GaAs solar cell technology. EaglePicher Technologies was selected for the lithium-ion battery system because of its experience supplying space-qualified lithium-ion cells and battery systems for aerospace applications. AAC Clyde Space was selected for the distributed MPPT controllers because of its experience providing flight-proven spacecraft power electronics. Suppliers were selected based on their ability to meet the mission requirements, technical capability, expected schedule and cost, and previous spaceflight experience. Backup suppliers were also selected based on their flight heritage and product reliability.

2. ***Lead Time Summary***

The estimated lead time for the power subsystem ranges from approximately 6 to 12 months. The lithium-ion battery system has an estimated lead time of 6 to 9 months due to battery assembly, integration, and testing, while the distributed MPPT controllers require approximately 6 to 8 months. The GaAs solar panel assembly has the longest estimated lead time of 6 to 12 months due to the custom design, manufacturing, integration, and testing required for the rover. These lead times will need to be considered early in the project schedule to ensure all power subsystem components are available for system integration and testing. If a backup supplier becomes necessary, additional procurement delays may occur depending on supplier availability, manufacturing schedules, and any additional integration or testing.

3. ***Procurement Summary Table***

Subsystem / Component	Procurement Method	Primary Supplier	Backup Supplier	Estimated Lead Time	Justification
Lithium-Ion Batteries	Prime Contr...	EaglePicher Technologies	EnerSys	6-9 mo	Space qualified batteries with high reliability.
GaAs Solar Panels	Prime Contr...	Boeing Company	Redwire Space, Inc	6-12 mo	Flight heritage and high-efficiency GaAs technology.
Distributed MPPT Controllers	SQCOTS (Sp...	AAC Clyde Space	GomSpace	6-8 mo	Flight-proven modules providing high-efficiency tracking, optimized for the variable illumination and critical power recharge cycles at the lunar south pole.

Figure 32: Power Procurement Summary Table

2.1.2.5. Power Subsystem Verification Plans

1. ***Verification Overview***

The team's overall verification approach for the Power Subsystem uses a combination of analysis, testing, inspection, and demonstration to verify that the subsystem requirements are met. Analysis will mainly be used to verify requirements

related to energy capacity, thermal performance, and cost. Testing will be used to verify battery performance, power output, mechanical shock, cycle life, the regulated 28 V DC bus, and battery protection. Inspection will be used to verify the subsystem mass and protection from lunar regolith. Finally demonstration will be used to verify that the required battery telemetry is provided to the CDH subsystem. If these verification methods are successful, then they will show that the Power Subsystem requirements have been satisfied.

2. *Power Verification Matrix*

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
EPS-01	The power subsystem mass shall not exceed 44 kg.	Inspection	The mass of the power subsystem components can be measured to verify that it remains within the allocated mass limit.	Weigh the fully assembled power subsystem components using a scale. The requirement is met if the total mass does not exceed 44 kg.
EPS-02	The energy storage subassembly shall provide at least 1,064 Wh of usable energy for one 6.07 hour sortie.	Analysis	The battery and solar panel specifications, the system's usable capacity, voltage, efficiency, and load requirements can be analyzed to verify that it provides enough usable energy.	Calculate the usable battery energy using the battery capacity, voltage, using component data and the power budget. The requirement is met if at least 1,064 Wh is available for the 6.07-hour sortie.
EPS-03	The battery subassembly shall operate safely without exceeding the rover's maximum thermal constraints during surface operations.	Analysis	Using the power dissipation and known thermal environment, the battery's thermal performance can be analyzed to verify that it remains within the rover's thermal constraints.	Perform a thermal analysis of the battery under the expected lunar surface and PSR conditions. The requirement is met if the battery remains within its specified operating temperature limits.

EPS-04	The energy storage subassembly shall not exceed an allocated cost of $17.0 M.	Analysis	Using known component quantities and costs, the total cost can be calculated to verify that the energy storage subsystem remains within the allocated cost.	Calculate the total expected cost by adding together the estimated procurement costs for each power subsystem component. The requirement is met if the total does not exceed $17.0M.
EPS-05	The battery subassembly shall provide a minimum continuous output power of 255 W during nominal rover operations and support peak system loads of up to 490 W.	Test	The power output is a quantitative performance requirement that must be measured under controlled electrical loads.	Connect the battery system to a controlled electrical load and measure its output power. The requirement is met if the battery provides at least 255 W continuously and supports peak loads up to approximately 490 W.
EPS-06	The battery enclosure shall withstand the mechanical shock loads experienced during landing and surface operations without damage or loss of functionality.	Test	Mechanical shock must be tested in a controlled environment to verify that the battery enclosure can withstand the required shock loads.	Apply mechanical shock to the battery enclosure and perform checks before and after testing. The requirement is met if no structural damage or loss of battery functionality occurs.
EPS-07	The energy storage subassembly shall store and provide a minimum of 100 W to support rover subsystems during emergency and	Test	Testing allows the available emergency power to be directly measured while the rover is operating in its safe-mode configuration.	Operate the battery system using the expected safe-mode loads and measure the available output power. The requirement is met if the system can provide at least 100 W for essential rover functions.

	safe-mode operations.			
EPS-08	The battery subassembly shall support a minimum of 2,000 charge/discharge cycles without degrading below 80% of its nominal capacity.	Test	Battery capacity degradation and cycle life are measurable performance characteristics that must be tested in a controlled environment.	Charge and discharge the battery for at least 2,000 cycles and measure its capacity. The requirement is met if the battery maintains at least 80% of its original capacity.
EPS-09	The battery subassembly shall provide the charging state, voltage, and temperature telemetry to the CDH subsystem.	Demonstra...	This requirement is about functionality and can be demonstrated to verify the required telemetry is provided to the CDH subsystem.	Operate the battery system and demonstrate that charging state, voltage, and temperature telemetry are provided to CDH.
EPS-10	The energy storage and regulation subassembly shall supply a nominal regulated DC bus voltage of 28 V.	Test	Testing allows the regulated DC bus voltage to be directly measured during operation.	Operate the power system under expected rover loads and measure the DC bus voltage. The requirement is met if the system maintains the required nominal 28 V DC bus.
EPS-11	The power regulation electronics shall prevent the battery from exceeding a maximum Depth of Discharge of 60%.	Test	This must be tested under controlled electrical conditions to verify that prohibited current flow does not occur.	Operate the battery through a controlled discharge and monitor the battery state of charge and DoD. The requirement is met if the regulation and protection electronics prevent discharge beyond the 60% DoD limit.
EPS-12	The power subsystem shall protect the battery	Inspection	Protection for exposed battery components and electrical	Inspect the battery enclosure, covers, seals, and exposed electrical connections to verify

	subassembly and exposed electrical connections from lunar regolith during surface operations.		connections can be verified by inspecting the enclosure and design.	that they are protected against lunar regolith.

Figure 33: Power Verification Matrix

3. Verification Method Rationale

The verification methods selected for the power subsystem are inspection, analysis, test, and demonstration. Inspection is used for requirements that can be verified by directly examining or measuring the subsystem, such as the total mass and the protection of exposed electrical components. Analysis is used when design information and mathematical calculations can determine whether a requirement is met. For example, calculating usable battery energy, thermal performance, and total cost. Test is used when requirements need to be measured by operating the system under controlled conditions. This includes power output, mechanical shock, cycle life, regulated bus voltage, and reverse-current protection. Demonstration is used for the telemetry requirement because it is based on functionality. The requirement can be verified by operating the system and showing that the required power telemetry is provided to the CDH subsystem.

4. Preliminary Verification Plan

Verification of the Power Subsystem requirements will occur during the development and testing of the subsystem. EPS-01 will be verified by weighing the fully assembled power subsystem components using a calibrated scale. For this verification, the batteries, solar panels, power regulation electronics, wiring, connectors, mounting hardware, and other power subsystem components will be needed. The requirement is met if the total mass does not exceed 44 kg. This verification assumes that all final subsystem components are included in the mass measurement. EPS-02 will be verified by calculating the usable battery energy using the battery capacity, voltage and expected rover loads. For this verification, the battery cell specifications and the rover power budget will be needed. The requirement is met if the battery provides at least 1,026 Wh of usable energy for the 6.07-hour sortie. This analysis assumes that the battery specifications and expected rover power loads accurately represent the final system.

EPS-03 will be verified using thermal analysis of the battery under the expected lunar surface and PSR conditions. For this verification, the expected lunar thermal conditions, battery operating temperature limits, battery power dissipation, and thermal properties of the battery enclosure will be needed. The requirement is met if the analysis

shows that the battery remains within its required operating temperature range during surface operations. EPS-04 will be verified by calculating the total expected cost by adding the estimated procurement costs for each power subsystem component. For this verification, the supplier costs will be needed. The requirement is met if the total expected cost does not exceed $17.0M. This verification assumes that the estimated supplier costs are final procurement costs.

Testing will be performed once the power subsystem hardware is available. EPS-05 will be verified using a controlled electrical load and voltage/current measurement equipment. The requirement is met if the battery provides at least 255 W continuously and supports peak loads of approximately 490 W. This assumes the test loads represent expected rover power loads. EPS-06 will be verified by testing the battery enclosure under the expected mechanical shock loads from landing and surface operations. For this verification, the battery enclosure and mechanical shock testing equipment will be needed. The requirement is met if there is no structural damage or loss of functionality after testing. This assumes the shock loads represent expected mission conditions.

EPS-07 will be verified by operating the subsystem under the expected safe-mode loads and measuring the available power. For this verification, the battery system, controlled electrical loads, and power measurement equipment will be needed. The requirement is met if the battery system can provide at least 100 W for essential rover functions. This verification assumes that the simulated safe-mode loads are similar to the expected loads during actual safe-mode operations. EPS-08 will be verified by charging and discharging the battery for at least 2,000 cycles and measuring its capacity. For this verification, the battery system and battery cycling equipment will be needed. The requirement is met if the battery maintains at least 80% of its nominal capacity after 2,000 cycles. A limitation of this verification is the amount of time required to complete the full 2,000-cycle test.

EPS-09 will be verified during subsystem integration by demonstrating that state of charge, voltage, and temperature telemetry are provided to the CDH subsystem. The battery system, BMS, CDH interface, and monitoring equipment will be needed. The requirement is met if all required telemetry is correctly provided. This assumes the test interface represents the final CDH interface. EPS-10 will be verified by operating the power system under expected rover loads and measuring the DC bus voltage. For this verification, the power regulation electronics, controlled electrical loads, and voltage measurement equipment will be needed. The requirement is met if the system maintains the required nominal 28 V DC bus. This assumes the test loads represent expected rover loads.

EPS-11 will be verified by charging and discharging the battery while monitoring the Depth of Discharge (DoD). For this verification, the battery system, power regulation and

protection electronics, battery cycling equipment, and monitoring equipment will be needed. The requirement is met if the power regulation and protection electronics prevent the battery from exceeding the 60% DoD limit. This verification assumes that the BMS accurately measures the DoD. EPS-12 will be verified by inspecting the battery enclosure, covers, seals, and exposed electrical connections. For this verification, the final battery enclosure and electrical connections will be needed. The requirement is met if the required protection from lunar regolith is included in the final design. This assumes the inspected hardware represents the final rover configuration.

2.1.3. CDH Subsystem Overview

1. CDH Subsystem Overview

The PDR changed the earlier SRR/MDR architecture by replacing the single flight computer with two identical redundant GR712RC FCUs because one computer was a mission critical single point failure, and its two processor cores do not provide independent redundancy. The second FCU fits within the 12.50 kg CDH allocation, and only one radio string is powered at a time. The earlier 128 Gb storage assumption is also replaced by 48 GB raw nonvolatile storage because each 128 Gb module was found to provide only 16 GB per device. So three devices were chosen which provide 48 GB raw and at least 32 GB user capacity after error correction and recovery which provides around 1.65x margin over the calculated 19.44 GB maximum mission data volume.

The Command and Data handling subsystem is the rover's main control and information backbone. It converts ground commands into bounded subsystem actions, collects and protects science and engineering data, manages Direct to Earth communication, provides telemetry, maintains operating state, and provides recovery when onboard conditions become hostile. Two identical GR712RC based flight computer units host the mission critical flight software, with an EPS supervisor powering exactly one paired FCU string at a time. Either FCU can independently command the rover, control safe mode, store data, and communicate with Earth. The Aitech S-A1760 NeuPan processor is separately powered and isolated from mission critical paths, and is only approved during demonstration windows. In those windows it produces mobility proposals that the active FCU validates before forming any actuator command. The science and vehicle components interface through two independent CAN buses, while a radiation tolerant recorder preserves mission data. Two Iris X-band communication chains, one fixed high rate panel antenna, and two opposed low gain antennas provide nominal contingency communication capability.

2. CDH Capabilities

CDH receives telecommands through either low gain antenna and the selected Iris radio authenticates them using Space Data Link Security, and rejects corrupt / unauthorized commands. Payload and engineering data are time tagged, packetized, prioritized, and stored in a 48 GB raw recorder storage, including a 2 GB protected volume for fault, health, command history, minimum science records and security. Error detection and correction, and redundant metadata writes protect stored information. The telecommunications architecture supports a nominal 1 Mbps X-band downlink through the fixed panel, 8kbps command reception through either low gain antenna, and at least 1 kbps contingency health return following loss of the fixed panel. CDH generates telemetry and uses proper delivery, data remaining deletion is inhibited until the downlink is confirmed by the ground. Fault detection, isolation, and recovery monitors the active computer and radio, recorder, memory, power state, time, and other critical flags. Detected failures invoke bounded application restart, hardware rest, component isolation or a Safe entry. Autonomous mobility remains contained: NeuPAN can submit proposals but has no electrical or logical path to either CAN bus or any actuator, and the only active GR712RC FCU can issue mobility commands

1. CDH Subsystem Summary Table

Subassembly	Mass (kg)	Dimensions (m)	Maximum Power Draw (W)	TRL
Flight computer / software	2.40	Two units - each: .180 x .130 x .060	<= 6	4
NeuPAN demonstration processor	1.00	.129 x .129 x .052	<= 10	4
Nonvolatile data storage	.50	.160 x .100 x .040	<= 3	4
Internal networks / harness	1.45	Distributed	<= 2	4
X-band telecommunications	3.20	Distributed antennas	12 receive, <= 40 transmit	4
Shielding, mounts, straps	1.50	Distributed	Passive	5
Total Subsystem	10.05	.500 x .400 x .300	55 max	4

Figure 34: CDH Subsystem Summary Table

2. *Software Architecture Flowchart*

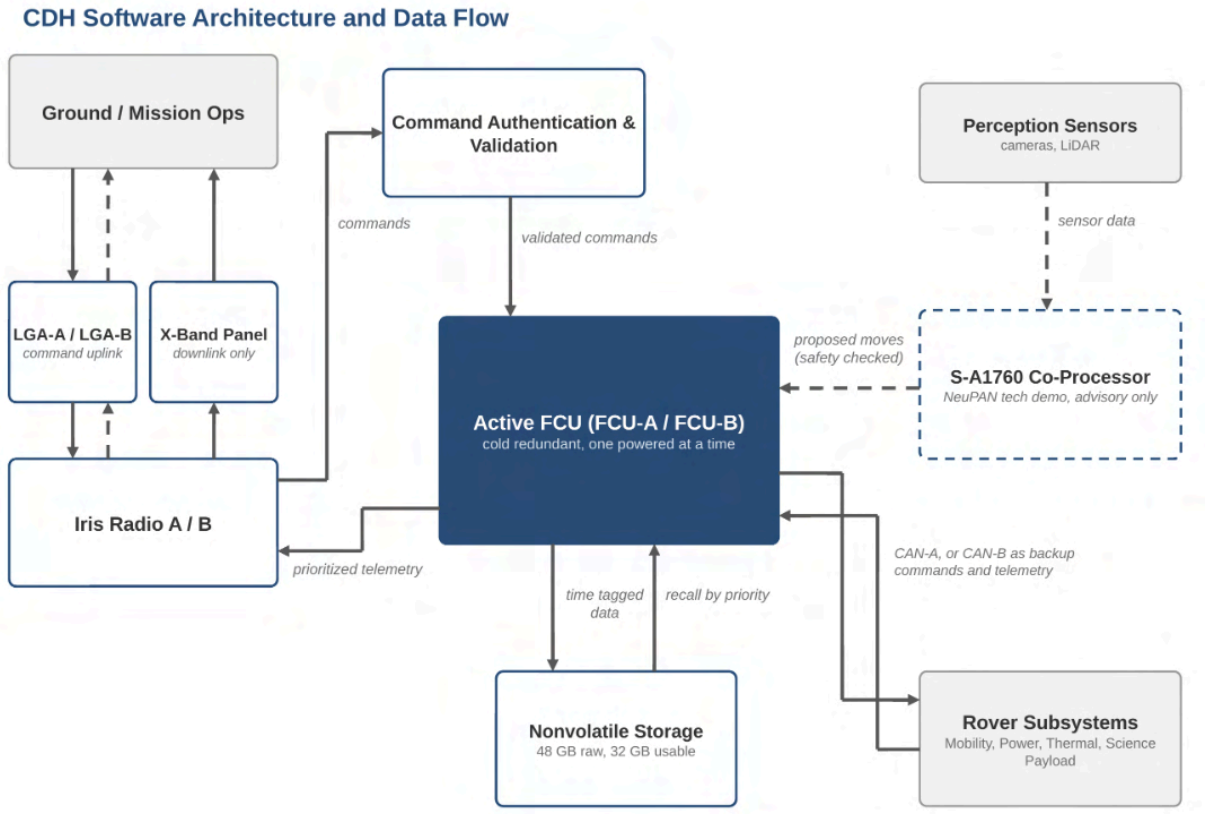


Figure 35:

3. *Data Flow Narrative*

This figure shows the PDR baseline for the command and data flow, centered on the active flight computer unit, and also of a redundant GR712RC pair with a single FCU and radio string powered at a time. Ground commands arrive at DTE through a low gain antenna and the selected Iris radio, pass authentication, protection, and validation, and are then executed by the active FCU. Then subsystem commands are routed over CAN-A nominally or CAN-B as needed. Science and engineering data return over the active CAN bus to the FCU, where they are all time tagged, packetized, and written to non volatile storage until delivery is confirmed; during communication windows the FCU recalls stored data in priority order, generates AOS telemetry frames, and manages CFDP delivery thought the selected Iris radio and the fixed, downlink only X-band panel. The S-A1760 co-processor receives perception data over an isolated 1GbE link and Reuters advisory motion proposals over RS-422 to the FCU.

2.1.3.1. CDH Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
CDH-1	CDH flight hardware shall have mass <= 10.05 kg, mass with 20% growth <= 12.06 kg.	Remains in the allocated mass budget.	SYS - 02		Inspecti...	Y...
CDH-2	CDH shall draw <= 16 W in stationary mode, <= 26 W during NeuPAN traverse, <= 28 W while receiving, and <= 55 W during DTE.	Maintains bounded operation loads.	SYS-06		Analysis	S...
CDH-3	The CDH hardware shall fit within .500 x .400 x .300 m, with the antennas and other external equipment being kept in the 1.5 x 1.5 x 1.0 m space.	Maintains space within the allocated vehicle configuration.	SYS-03		Inspecti...	No
CDH-4	CDH shall validate all authorized commands, and reject invalid commands; and at least store 72 hours of all received command sequence for windows without DTE.	Provides secure ground navigation and mission integrity.	SYS-09		Test	No
CDH-5	CDH shall time tag all engineering and science data within 1-3 ms.	Keeps all engineering, science and telemetry data preserved.	SYS-11		Test	No
CDH-6	The CDH subsystem shall packetize >= 32 kbps of payload and engineering data	Keeps the storage and downlink demand bound.	SYS-11		Test	No

	without packed loss.					
CDH-7	The storage sub assembly must provide >= 48 GB raw and >= 32 GB user nonvolatile storage.	This provides ~1.5x margin over the calculated 19.44 maximum data allocation.	SYS-09		Inspecti...	No
CDH-8	The CDH subsystem shall retain data until DTE is confirmed, and all data should be prioritized in order of health data before science data.	Prevents the loss of important data during unreliable contact windows.	SYS-09		Demon...	No
CDH-9	When a critical fault is detected, CDH shall inhibit mobility and enter SAFE mode within 1 minute, and complete recovery transfers within the same window as required.	Provides mechanism for mission recovery.	SYS-06		Test	No
CDH-10	The Iris radio and X-band panel shall provide >= 1 Mbps downlink for at least 2 hours per 24 hours. 256 kbps shall always remain available as backup.	Allows for the most daily data packaging with appropriate capacity and margins.	SYS-11		Analysis	No
CDH-11	The Iris radio shall support 8kbps command uplink and >= 1 kbps health downlink.	Preserves health data and crucial commands margin.	SYS-09		Analysis	No
CDH-12	CAN-A shall be the 1 Mbps control bus and CAN-B will be the independent	Redundant communication while preventing NeuPAN from	SYS-05		Inspecti...	No

	backup bus with minimal command latency. NeuPAN shall remain isolated from CAN and provide advisory proposals only.	obtaining direct control.				
CDH-13	Each CDH equipment must have qualified radiation tolerance.	Protects all important CDH functions from radiation.	SYS-07		Analysis	No
CDH-14	NeuPAN shall be a separate demonstration, and shall not receive mission survival credit and shall remain power isolated.	Contains the low rated technology demonstration isolated.	SYS-07		Demon...	No

Figure 36 : Insert figure caption here

2.1.3.2. CDH Sub-Assembly Overview

Flight Computer / Processing:

Two identical GR712RC based flight computer units provide the rovers mission critical command and control authority, removing the single computer failure point in the earlier design. Only one unit is powered at a time, while the second remains available for recovery. The active computer validates commands, executes stored sequences, processes telemetry, supervises NeuPAN, manages data storage, and controls safe state entry. Each unit interfaces dual CAN buses with radio, recorder, watchdog, and NeuPAN RS 422 link. There is also a timekeeper which will preserve accurate mission time through resets and computer transfers. Thermal conduction, radiation shielding, and recovery support operation in the lunar environment.

NeuPAN Technology:

The Aitech S-A1760 co-processor runs the NeuPAN navigation demonstration during the approved power managed traverse periods at a proposal rate of at least 5 Hz. Flash LiDAR and hazard camera data enter though an isolated 1 GbE connection, and NeuPAN returns advisory motion proposals to the active flight computer. The S-A1760 has no direct

CAN connection, and the GR712RC will independently check the proposals information and integrity. Because NeuPAN has lower radiation tolerance and technology readiness, it is separately powered and isolated so its failure affects only the demonstration, not the commanding or minimum mission success.

Data Storage System:

Science data, engineering telemetry, command history, even logs, time state, fault information, recovery files are all stored in the recorder until downlink is confirmed. Three 128 Gb radiation tolerant SCL NAND devices provide 48 GB raw capacity and at least 32 GB usable capacity. This exceeds the calculated 19.44 GB maximum mission data volume by a factor of 1.65. A radiation tolerant FPGA controller provides error correction, wear management and isolated access by the FCU. At least 2 GB of critical data is triplicated, and it cannot be overwritten by lower priority items.

Communications:

The comms network carries science data, health information, commands, engineering telemetry, thermal limits, payload health. There are two independent 1 Mbps CAN buses which provide the mission important path. CAN A operates nominally and CAN B only becomes active after a failure is detected. To account for duplicate commands being sent, commands are only sent on the selected bus while either bus can preserve safe state and ground communication ability. Separate low priority links connect the perception sensors to the S-A1760 through an independent 1 GbE. SpaceWire is no longer used because the selected hardware does not require it.

Flight Software:

The flight software is built on RTEMS, with hardware support, key services, and controlled image recovery. Key functions include command processing, sequence storing, time tagging, storage management, data priority, telemetry monitoring, event logging, and Safe rover mode. The software will communicate through the rover subsystems through the chosen CAN bus and the ground through the active Iris radio. NeuPAN operates separately on the S-A1760 and supplies only advisory proposals to the GR712RC for safety check. The protected architecture allows for recovery after resets and power interruptions.

Fault Detection / Recovery:

The fault management system monitors watchdog status, memory, storage health, CAN communication, radio process, power, thermal status, mobility status, and payload health. When a critical fault is detected, CDH inhibits mobility and less valued payload

activity instantly, and enters a SAFE mode right after that while preserving the important data and ground command capability. If the FCU string cannot recover, an independent hardware check removes its power and transfer control to the backup FCU. SAFE mode is still enabled until all the faults are fully diagnosed and a way is achieved to recover.

Telecommunications:

This subassembly uses X-band DTE communication with redundancy, telemetry, and data communication. There are two Iris V2.1 radio chains which are paired with the two FCU’s. With only one string active at a time. Right hand circular polarization communication X-band panel is being used to supports 1 Mbps downlink and 256 kbps degraded mode, and two opposed RHCP low gain antennas provide an 8 kbps command uplink and at least 1 kbps contingency health downlink. Stored data is prioritized and retained until delivery is confirmed.

Technology Readiness Level:

The CDH subsystem is rated a TRL 4, which is the lowest readiness rating the subassembly established. The GR712RC processor has reliable flight heritage, but the new dual FCU configuration is TRL 4 due to it not being built and tested yet. The custom internal network, timekeeping hardware, software are also TRL 4 because not enough environment testing has been done on them. The NeuPAN demonstration and its isolated S-A1760 integration are also TRL 4, mainly due to the NeuPAN software not being extensively tested in this environment, though the S-A1760 hardware does have better reliability. The Iris radio and X-band are rated TRL 6, and the installation harness is at TRL 4. Overall, the CDH subsystem hardware and software remain at TRL 4 at the PDR. To get to TRL 6, more testing is needed to show storage integrity, recovery, communication, and mainly radiation tolerance. NeuPAN integration is treated as a demonstration with respect to its TRL rating and not mission critical.

1. *Supporting Figures*

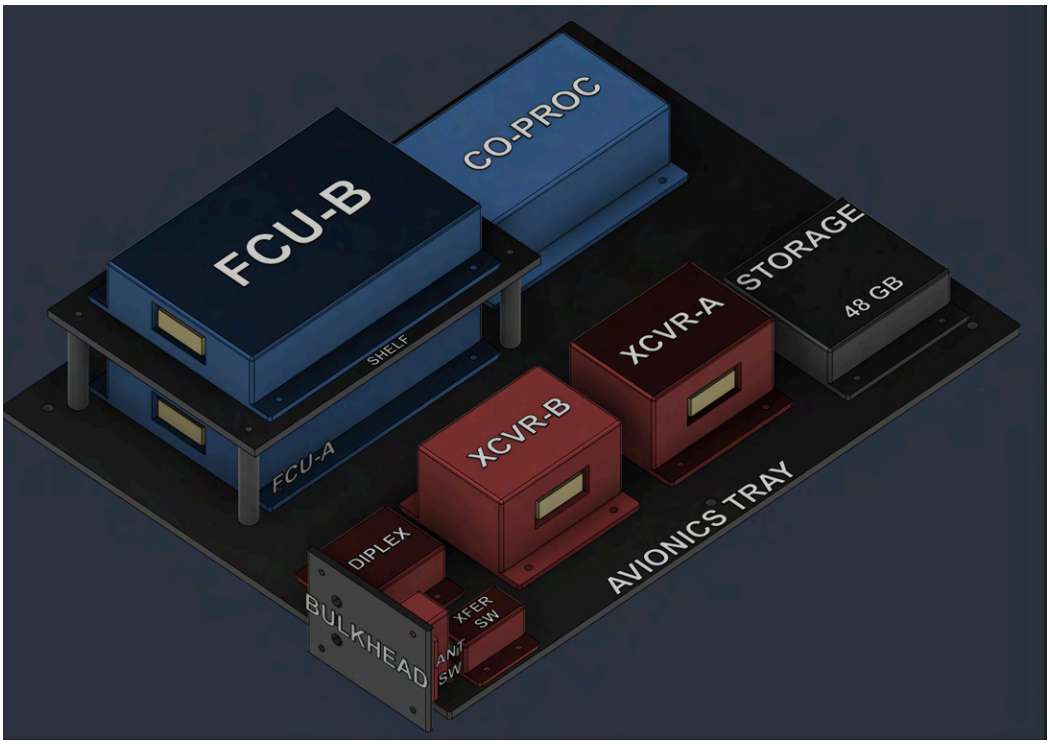


Figure 37: Labeled isometric view of CDH components, showing two flight computers, X-band transceivers, isolated co-processor, data recorder, and routing hardware.

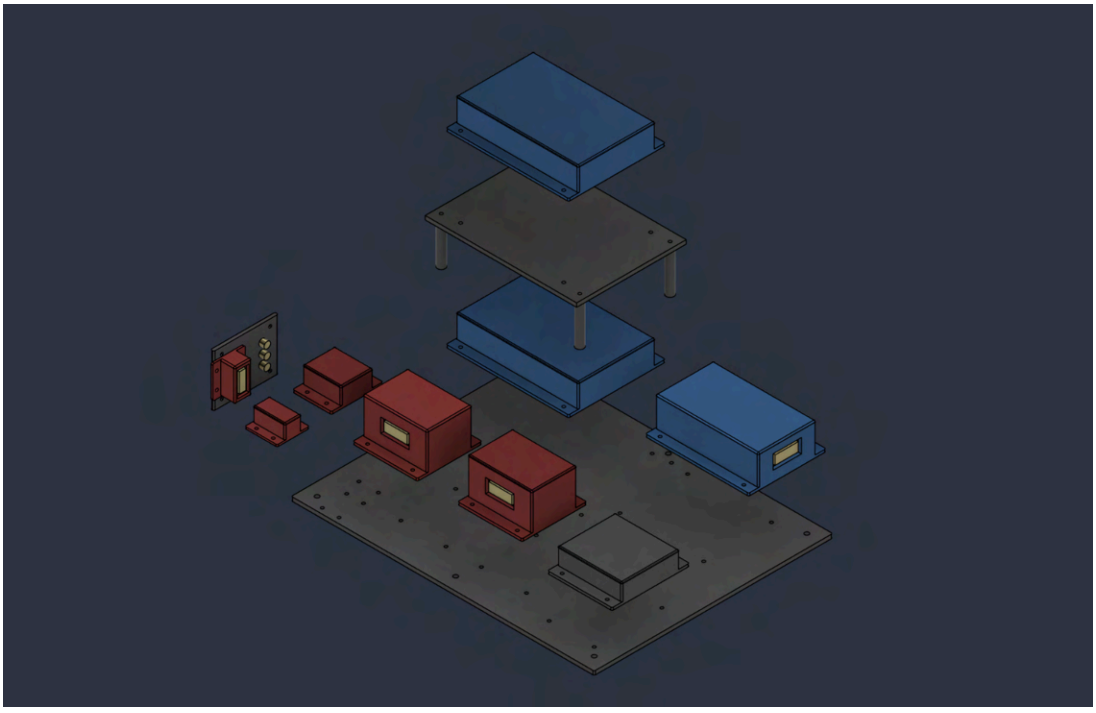


Figure 38: Exploded isometric view of the CDH assembly showing the component arrangement.

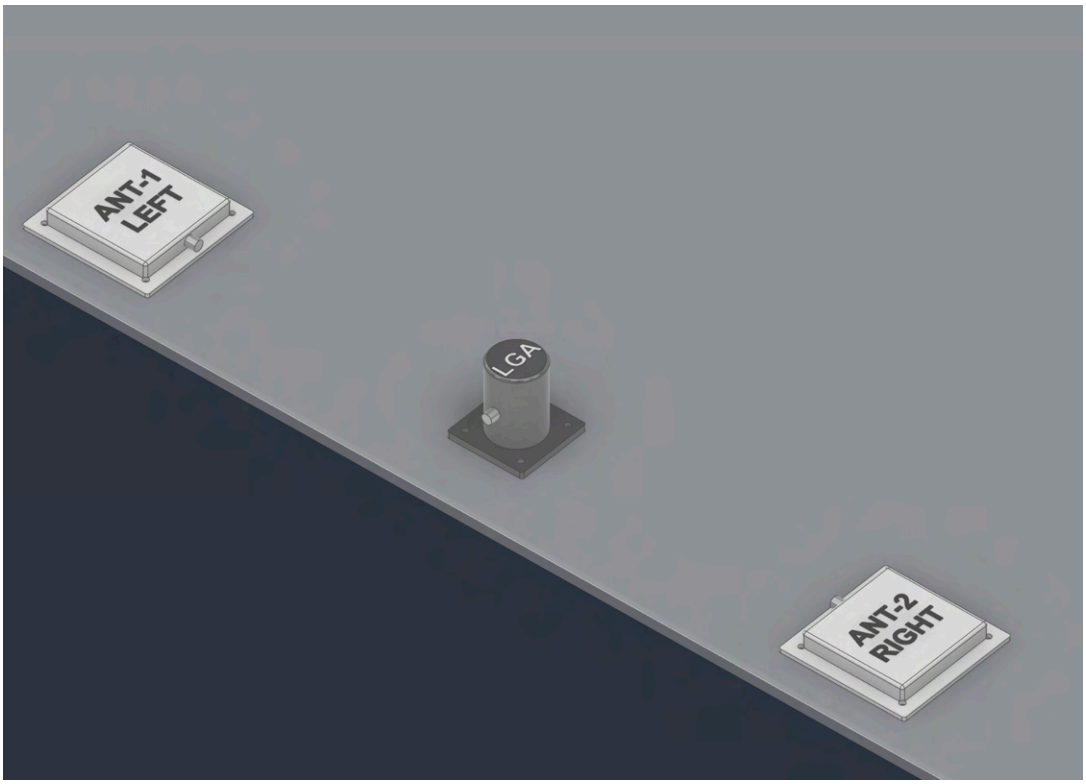


Figure 39: Roof arrangement of the X-band panel and two low gain antennas.

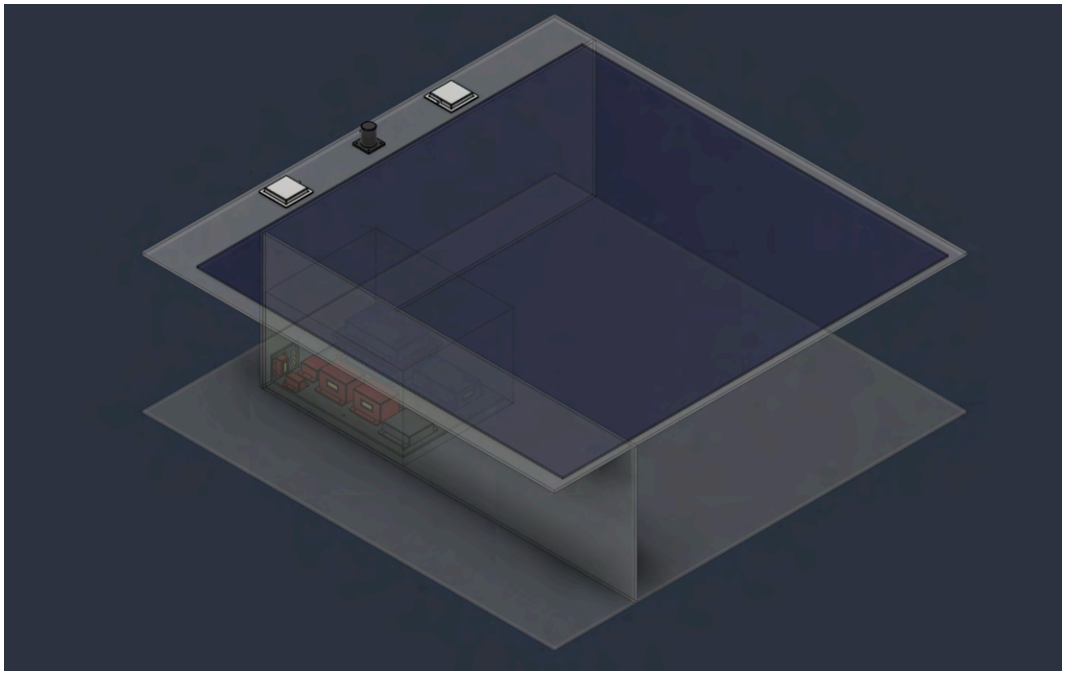


Figure 40: Full arrangement of CDH assembly including the communication hardware.

Together, these figures show the CDH design for component, assembly, antenna placement. FCU-A and FCU-B are the two flight computer units, while XCVR-A and XCVR-B are the corresponding Iris X-band transceivers chains used to receive commands. The co-processor the S-A1760 is isolated and used for NeuPAN demonstration, and the storage until provides 48 GB raw capacity. For the radio frequency hardware, the diplexer is used to separate transmit and receive signals, the transfer switch XFER SW selects the active transceiver, and the antenna switch ANT SW routes that transceiver to the X-band panel or antennas. The bulkhead panel provides the cable connections between the internal bay and the externally mounted antennas, though the wiring is not displayed in these diagrams.

2.1.3.3. CDH Subsystem Recovery and Redundancy Plans

The Command and Data Handling (CDH) subsystem uses radiation-tolerant hardware, error-correcting memory, fault detection, and operational recovery procedures to ensure a safe mission. It also includes a recovery architecture, designed around the radiation-hardened GR712RC flight computer, Aitech S-A1760 co-processor, 128 GB nonvolatile storage system, dual-path Controller Area Network (CAN) command bus, SpaceWire payload data path, and redundant Direct-to-Earth (DTE) communications architecture.

Redundancy

The GR712RC, the primary flight computer, is responsible for command processing, telemetry, communication between subsystems, and spacecraft fault-management functions. A second complete flight computer is present and is included in the baseline design. It is crucial in case of an error. Error Detection and Correction (EDAC) and hardware watchdog recovery. This way, we reduce mass and power consumption while still fully protecting our system against the radiation-induced upsets identified in the mission FMEA.

The Aitech S-A1760 co-processor operates separately from the primary flight computer and runs NeuPAN for full autonomy. Failure of the S-A1760 does not inherently result in loss of the primary flight computer or basic CDH functionality. If the co-processor becomes unavailable, NeuPAN’s autonomous-navigation activities will be stopped or reduced while the GR712RC continues with the mission and communication between the rover and the ground. This separation prevents loss of the system from automatically moving on to a mission-ending CDH failure.

The internal command network uses a dual-path CAN architecture. Failure of one CAN path, transceiver can be prevented by moving traffic into a healthy path. Payload science data use the SpaceWire interface, separating high-volume payload data transfer and preventing payload data traffic from unnecessarily loading the command bus.

The 128 GB nonvolatile storage system is sized to cover more than 100% of the requirements. Rather than relying exclusively on duplicated storage hardware, the architecture reduces data-loss risk using radiation-tolerant NAND storage, ECC, and store-and-forward operations. When the communications opportunities are available, Science and engineering data will be transmitted, so there is as little as possible information kept directly on the rover.

The DTE communications architecture uses redundant communications links so that a single electronic or software failure does not eliminate the rover's communication window.

Failure Detection and Recovery

CDH fault detection will use processor watchdog status and subsystem health telemetry to identify anomalous conditions. The logs flagged as faulty will be selected according to the severity of the detected condition. A flight-computer upset caused by radiation will initiate watchdog-based recovery and will recreate the previous commands. Nonessential loads and activities can be suspended while the system keeps operating. If the onboard storage throws an error, ECC will be used, affected data will be flagged in a case where the data was not recovered, and the unaffected data will be preserved for downlink.

2.1.3.4. CDH Subsystem Manufacturing and Procurement Plans

Procurement Overview

The CDH subsystem will use ready-made, space-qualified hardware rather than creating custom computer hardware from scratch. The main parts include the GR712RC flight computer, S-A1760 co-processor, 128 GB radiation-tolerant storage, CAN transceivers, SpaceWire, and the DTE communication system. Most project work will focus on inspecting, testing and installing the hardware to the whole system. The GR712RC will handle the main spacecraft computing, the S-A1760 will run the NeuPAN navigation software, the 128 GB storage will save mission data, CAN will support command communication, and SpaceWire will carry payload data.

1. Procurement Summary Table

Subsystem / Component	Procurement Method	Primary Supplier	Backup / Alternate	Estimated Lead Time	Justification
GR712RC Flight Computer	COTS / space-qualified	Frontgrade Gaisler	Frontgrade GR740-based board	9–12 months	Radiation-hardened LEON3FT processing and extensive flight heritage. Primary CDH long-lead item. An alternate processor would require software and interface requalification.
Aitech S-A1760 Co-processor	COTS / space-qualified	Aitech Systems	No qualified drop-in alternate identified in the MDR baseline	4–6 months	Space-packaged Jetson TX2i platform selected to host NeuPAN processing. Early procurement and developmen

Subsystem / Component	Procurement Method	Primary Supplier	Backup / Alternate	Estimated Lead Time	Justification
					t-unit testing reduce single-source schedule risk.
128 GB Nonvolatile Storage	COTS / space-qualified	3D Plus	Teledyne e2v	6–9 months	Radiation-tolerant NAND storage sized to approximately 150% of predicted mission data volume and suitable for store-and-forward operations.
CAN Bus Transceivers	COTS	Texas Instruments	Frontgrade	3–6 months	Supports the dual-path CAN command architecture and has multiple procurement sources, reducing supply-chain risk.

Subsystem / Component	Procurement Method	Primary Supplier	Backup / Alternate	Estimated Lead Time	Justification
SpaceWire Interface Hardware / Harnessing	COTS / integration hardware	Native GR712RC interface with procured interconnect hardware	Compatible qualified interconnect source	3–6 months	Supports high-rate payload data transfer while maintaining separation from the primary command bus.
X-band DTE Communications Chain	MCOTS	Space Dynamics Laboratory baseline candidate	Safran baseline candidate	9–12 months	Baseline communications architecture uses redundant chains so that a single chain failure does not eliminate DTE communications. Final contractual source will be configuration-controlled before release.

Figure 42: Procurement Summary Table

Manufacturing and Integration Approach

CDH manufacturing will mainly focus on putting together and connecting the electronics. All purchased flight hardware will be inspected before installation, and connections such as power, CAN, etc will be checked before implementing them on the system. Development units will also be used to test software without putting extra stress on the actual flight hardware. Hardware and software versions will be carefully tracked to make sure we have a proper working system.

Lead-Time and Schedule Management

CDH procurement needs to happen early because some parts take a long time to arrive. The GR712RC flight computer and DTE communication hardware may take about 9–12 months, the 128 GB storage about 6–9 months, and the S-A1760 and CAN hardware about 3–6 months. Since system integration must be ready by December 1, 2028, the longest-lead parts should be ordered during Phase C. Backup suppliers may also add several months, as new hardware would require additional testing and software changes.

2.1.3.5. CDH System Verification Plans

1. ***Verification Overview***

Verification of the Command and Data Handling subsystem will demonstrate that the internal system and the communications interfaces satisfy their assigned tasks. Verification will use a combination of inspection, analysis, test, and demonstration since verifying each of the systems one by one would guarantee a successful mission.

The verification program will begin with bench-level testing and will progress through electrical integration, software integration, thermal/environmental testing, and final integrated verification. We will use Ground Support Equipment (GSE) before final flight-hardware testing.

2. ***CDH Verification Matrix***

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
CDH-01	CDH shall provide primary onboard command processing, telemetry generation, and spacecraft control using the GR712RC flight computer.	Test / Demonstration	Functional performance must be demonstrated with representative commands and subsystem interfaces.	Execute nominal and off-nominal command sequences on the integrated GR712RC configuration and confirm correct command execution, telemetry generation, and subsystem response.
CDH-02	CDH shall provide processing capability for the NeuPAN autonomous-navigation software using the S-A1760 co-processor while maintaining core mission functions independently of that processor.	Test / Demonstration	NeuPAN operation and graceful degradation following loss of the co-processor are functional behaviors.	Execute NeuPAN processing with representative navigation inputs, then remove or disable the co-processor and verify that the GR712RC retains essential command, telemetry, storage, and subsystem-control functions.
CDH-03	CDH shall provide 128 GB of nonvolatile storage sized to at least 150% of	Inspection / Analysis / Test	Physical capacity is inspected, margin is analyzed, and usable	Verify installed capacity and part configuration; compare usable storage against the

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
	predicted mission data volume.		capacity/readback requires testing.	current mission-data-volume estimate; perform write, readback, fill, and file-retrieval testing.
CDH-04	CDH shall protect flight-computer and stored data against correctable memory errors using EDAC/ECC functions.	Test / Demonstration	Error-handling performance requires controlled error injection or equivalent vendor-supported test methods.	Inject or simulate correctable memory/storage errors and verify detection, correction, fault reporting, and preservation of unaffected data.
CDH-05	CDH shall provide a dual-path CAN command network capable of maintaining command and telemetry functions following loss of one available path.	Test / Demonstration	Redundancy is verified by deliberately removing one path under controlled conditions.	Operate representative subsystem traffic on the nominal CAN path, disable or disconnect that path, transition to the alternate path, and confirm continued command and telemetry exchange.
CDH-06	CDH shall provide the SpaceWire payload data path required to acquire and transfer	Test	Interface timing, data integrity, and compatibility require	Connect representative payload hardware or simulators, transfer representative

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
	science data from the payload subsystem.		integrated hardware testing.	science-data products through SpaceWire, store the data, and verify complete error-free retrieval.
CDH-07	CDH/communications power consumption shall remain approximately within the allocated 26 W nominal and 61 W transmission operating cases.	Test / Analysis	Electrical power compliance must be measured under representative operational loads.	Measure voltage, current, and power during nominal CDH processing and the transmission operating case using integrated power-system test equipment; compare results to allocated limits and power margin.
CDH-08	CDH shall detect and recover from recoverable processor or software faults using watchdog recovery and fault-management functions.	Test / Demonstration	Recovery performance can only be established by introducing representative faults.	Inject processor hangs, software task faults, and selected interface faults; verify detection, watchdog/reset response, safe-state behavior, restoration of command/telemetry , and generation of fault records.
CDH-09	CDH shall preserve mission data through	Test / Demonstration	End-to-end data handling must	Generate representative

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
	store-and-forward operation and prioritized downlink when immediate DTE transmission is unavailable.		be demonstrated across storage and communications functions.	engineering and payload data while DTE transmission is disabled, store the data, restore the communications path, and verify priority-based retrieval and downlink without uncorrected data loss.
CDH-10	The redundant DTE communications architecture shall retain communications capability following loss of one communications chain.	Test / Demonstration	The redundancy claim requires a controlled failover demonstration.	Establish DTE communications through the active chain, disable the active electronics path, transition to the redundant chain, and verify restoration or continuation of required command and telemetry capability.

Figure 44: CDH Verification Table

Verification Method Rationale

Inspection will be used to check items such as part numbers and software versions. Analysis will be used for things that require calculations or supplier data, such as storage margin and environmental limits. Testing will be the main method and will check processor operation, command and telemetry handling, CAN and SpaceWire connections, storage performance, and communication links. Demonstration will be used to show that

important functions work correctly, including autonomous navigation, backup bus switching, communication failover, watchdog recovery, safe-mode operation, and store-and-forward data handling.

Preliminary Verification Plan

Verification will happen in steps so problems can be found early. During electrical integration in August–September 2028, the team will check power, wiring, grounding, and other crucial communication systems. During software integration from August to early October 2028, the GR712RC, S-A1760/NeuPAN, telemetry, commands, storage, and fault-management software will be tested together. Functional testing from October 6–20, 2028 will check normal operations, data storage, payload data transfer, backup bus switching, watchdog recovery, communication handling, and power use. CDH will also be tested during thermal and environmental testing to make sure the electronics still work under mission conditions. Any problems will be fixed and retested from November 24–28, 2028, and all CDH verification must be completed before System Integration Readiness on December 1, 2028.

2.1.4. Thermal Control Subsystem Overview

1. Thermal Subsystem Overview

The thermal control subsystem (TCS) is meant to keep all components, regardless of thermal environment, within operating temperature limits of -10°C to -30 °C. This becomes more significant as the rover undergoes, on this mission, extreme hot and cold temperatures due to the illuminated regions and permanently shadowed regions (PSR). Additionally, the TCS has the capability of heat rejection which is needed for transitioning from PSRs to illuminated regions where the rover can charge utilizing solar panels. The heat rejection is performed through the top of the rover due to only inserting on the sides and bottom of the lunar surface vehicle. Moreover, the insulation is durable enough to withstand multiple thermal cycling of illuminated regions to PSRs.

2. Thermal Subsystem Summary Table

Subassembly	Mass (kg)	Dimensions (m)	Maximum Power Draw (W)	TRL
Multi-layer Insulation	1.2	1.4 x 1.4 x 0.015	0W	9

Thermal Interface Material	3.5	0.30 x 0.30 x 0.001	0W	9
Kapton Polyimide Heater	0.4	0.25 x 0.20 x 0.0003	50W	9
Temperature Sensors and Controller	0.2	0.15 x 0.10 x 0.04	3W	9
Total Subsystem	**18**	**-**	**53W**	**9**

Figure 46: Thermal Subsystem Summary

3. Heat Flow Maps

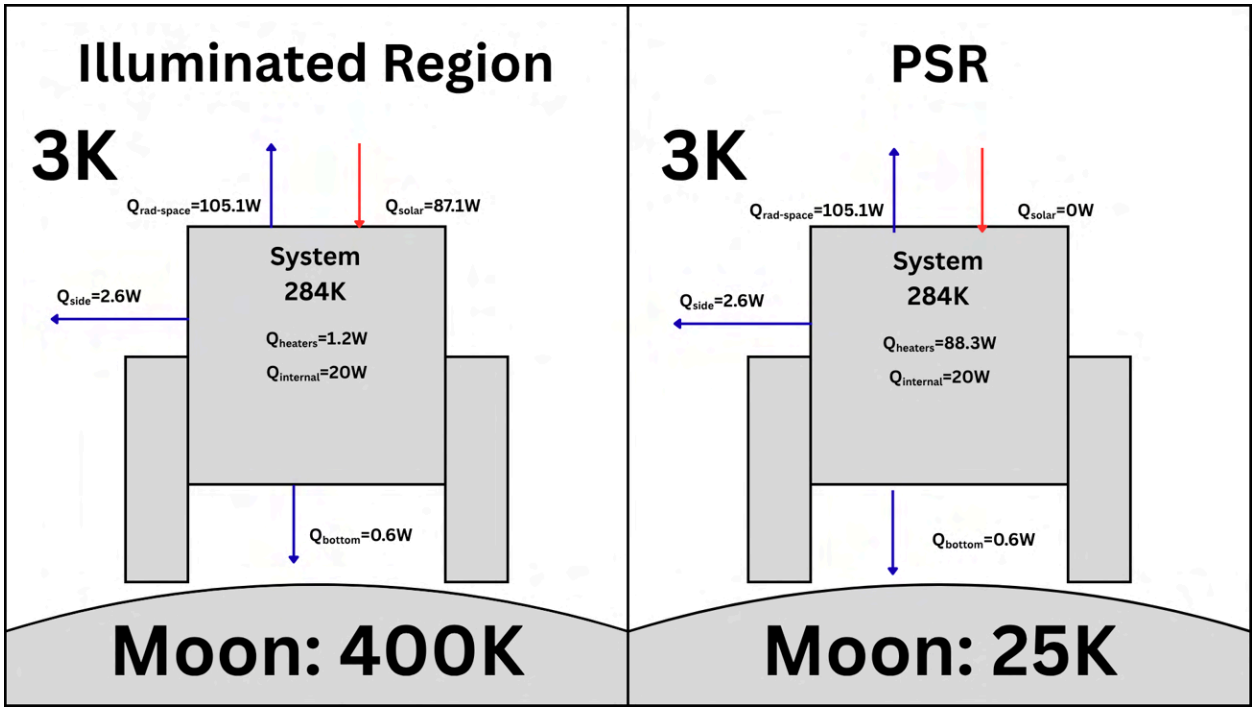


Figure 47: Thermal Heat Transfer Diagram

Heat Transfer:

There will be a MLI Blanket surrounding the bottom and sides of the lunar surface vehicle while having the top of the vehicle have no insulation. While in illuminated regions the vehicle will experience 87.1W, as you can see in Figure X, resulting in using only 1.2W from our kapton polyimide heaters. During the voyages of the PSRs, the heaters will output 88.3W, shown in Figure X, to make up for the heat leaving the lunar surface vehicle. The internal heat is allocated for the electronics, payloads, and other components being used during the mission.

2.1.4.1. Thermal Control Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
TCS-01	The thermal control subsystem shall maintain critical rover electronics within their allowable operating temperature range during mission operations.	Prevents electronics failure during illuminated and PSR operations.	SYS-06		Analysis	Y...
TCS-02	The thermal control subsystem shall provide thermal protection for short-duration excursions into PSRs.	PSRs can reach extremely low temperatures, so the rover must survive cold exposure.	SYS-06		Analysis	Y...
TCS-03	The thermal control subsystem shall minimize heater power demand during PSR operations.	Power is limited while the rover is inside the PSR with no sunlight.	SYS-06		Analysis	Y...
TCS-04	The thermal control subsystem needs to not surpass the mass of 18 kg.	Allocates the system mass constraint to the thermal control subsystem and its components.	SYS-02		Inspecti...	No
TCS-05	The thermal control subsystem shall not surpass the allocated total cost of $25.7M.	Allocates the system cost constraint to guarantee that the total system cost to be within $250 million.	SYS-12		Analysis	Y...
TCS-06	The thermal control subsystem shall use passive thermal control methods where practical before relying on active heater power.	Passive methods such as insulation, coatings, and thermal isolation reduce power demand during PSR excursions where solar power is unavailable.	SYS-06		Analysis	Y...
TCS-07	The thermal control subsystem shall maintain thermal compatibility with the science instruments during data collection	Instrument measurements can be affected by extreme cold or thermal gradients, so the TCS must support reliable	SYS-06		Test	No

	inside or near PSRs.	science data collection during mission operations.				
TCS-08	The thermal control subsystem shall reject excess heat during rover operations in illuminated regions.	Prevents overheating of electronics and batteries while operating and charging in sunlit areas.	SYS-06		Analysis	Y...
TCS-09	The thermal control subsystem shall survive repeated thermal cycles between illuminated terrain and permanently shadowed regions.	The baseline rover concept requires multiple excursions into and out of PSRs throughout the mission.	SYS-06		Analysis	Y...
TCS-10	The thermal control subsystem shall provide temperature monitoring for critical rover components during mission operations.	Temperature sensors allow the rover to verify that electronics, batteries, and science instruments remain within acceptable thermal limits during sunlit operations and PSR excursions.	SYS-06		Analysis	Y...
TCS-11	The thermal control subsystem needs to maintain a minimum operating temperature for the external instruments including Msolo at −10°C to 45°C and LISTER ~30°C.	Prevents our external instruments from failure due to temperature conditions.	SYS-06		Test	No

Figure 49: Thermal Control Subsystem Requirements

2.1.4.2. Thermal Control Sub-Assembly Overview

1. Sub-Assembly Descriptions

Multi-Layer Insulation (MLI):

The Multi-Layer Insulation (MLI) provides passive thermal protection by minimizing radiative heat transfer between the rover and the lunar environment. It reduces heat loss while in permanently shadowed regions (PSRs) and limits heat gain in illuminated regions, decreasing heater power requirements and maintaining stable internal temperatures.

The MLI will be installed around thermally sensitive rover components where passive isolation from the lunar environment is required. Its low effective emissivity reduces radiative coupling between protected hardware and the surrounding environment, helping limit both heat loss during cold PSR operations and unwanted heat absorption when the rover is exposed to illuminated regions. MLI therefore reduces the amount of active heater power required to maintain critical hardware within allowable temperature limits. The insulation interfaces primarily with the rover structure, battery, avionics, science instruments, and other thermally sensitive hardware. Its placement must account for attachment points, electrical connections, mechanical interfaces, and areas where conductive heat paths are required. The MLI design supports the thermal subsystem requirements by reducing environmental thermal loads and helping maintain critical rover components within their required operating and survival temperature ranges.

Thermal Interface Material (Indium Foil):

The Thermal Interface Material (TIM), consisting of indium foil, is placed between heaters and critical components to improve conductive heat transfer. It minimizes thermal contact resistance, allowing heat generated by the heaters to be transferred efficiently to the battery, avionics, and science instruments.

Indium foil provides a controlled conductive interface between the heaters and the hardware requiring thermal control. Because the effectiveness of an active heater depends on transferring the generated heat into the protected component, minimizing thermal contact resistance is an important design consideration. The interface material must remain compatible with the expected lunar thermal environment and must not interfere with mechanical mounting or electrical interfaces. Indium foil will be incorporated where efficient conductive coupling is necessary between the thermal-control hardware and components such as the battery, avionics, and science instruments. This interface supports subsystem performance by improving

Kapton Polyimide Heaters:

Kapton polyimide heaters provide active thermal control for components that require supplemental heating during cold lunar operations. The heaters are bonded directly to critical hardware and are activated whenever component temperatures approach their lower operating or survival limits.

The heater configuration provides active thermal control when passive insulation alone cannot maintain the required component temperatures. The current design incorporates eight Kapton heaters, distributed among thermally vulnerable hardware according to the heating requirements of the rover. Heater operation interfaces directly

with the electrical power subsystem because heater activation creates an additional power demand, particularly during PSR operations when solar energy is unavailable. The heaters also interface with the temperature sensing and control hardware, which determines when supplemental heating is necessary. Environmental considerations include repeated exposure to lunar temperature extremes and thermal cycling as the rover transitions between illuminated and permanently shadowed regions. The heaters support the subsystem requirements by providing controllable heat input when component temperatures approach their established lower operating or survival limits.

Temperature Sensors and Controller:

The temperature sensing and control subsystem continuously monitors temperatures at critical thermal zones, including the battery, avionics, and external science instruments. Sensor feedback is processed by the heater controller, which autonomously activates or deactivates the Kapton heaters to maintain components within their specified operating temperature limits while minimizing power consumption.

The current thermal design incorporates eight temperature sensors to provide feedback from critical thermal zones. Sensor placement will prioritize thermally sensitive hardware and locations where temperature measurements are necessary to determine whether heater activation is required. The controller receives these measurements and uses them to manage heater operation, creating a closed-loop relationship between thermal monitoring and active thermal control. This subassembly therefore interfaces with the heaters, protected rover hardware, electrical power, and the rover's control architecture. The sensors must remain capable of providing reliable measurements throughout the expected lunar temperature range and repeated thermal cycles. Continuous temperature monitoring allows the subsystem to detect approaching thermal limits and respond before critical hardware exceeds its allowable operating or survival range.

Technology Readiness Level (TRL) :

The Technology Readiness Level for all of the thermal control subsystem’s major components is a 9. The TRL rating of 9 is mainly due to each individual subassembly having heritage regarding spacecraft missions. Multi-Layer Insulation has been in every spacecraft since the 1960s and is favored for its performance in vacuum and temperature extremes. Indium foil is a non-outgassing thermal interface material that is manipulable in cryo temperatures with history being used in spacecrafts. Kapton heaters are the standard for spacecraft heaters that bond to temperature sensitive hardware. Temperature sensors have been used in every single type of space mission.

Together, the passive and active thermal-control subassemblies provide a layered approach to thermal management. MLI reduces radiative heat transfer with the lunar environment, indium foil improves conductive heat transfer between heaters and protected components, Kapton heaters provide supplemental thermal energy when required, and the temperature sensors and controller provide the feedback necessary to regulate heater operation. This architecture is particularly important during PSR excursions, where the absence of solar heating can expose rover hardware to extremely low temperatures. Integration of these elements allows the thermal subsystem to protect critical rover electronics, batteries, and science instruments while minimizing unnecessary heater power consumption.

2.1.4.3. Thermal Control Subsystem Recovery and Redundancy Plans

Redundancy:

There will be 8 kapton heaters, 8 temperature sensors, and a temperature controller that will be a part of the lunar surface vehicle. All of these mechanical components will have duplicates in case something malfunctions or be completely unusable.

Thermal control underperformance: This failure mode occurs if the thermal subsystem cannot maintain thermally vulnerable hardware within safe operating or survival limits. The likely causes are heater damage or degraded insulation performance. If this occurs, batteries, avionics, electronics, or payload instruments may degrade, shut down, or fail, reducing the rover's ability to operate near permanently shadowed region conditions.

Thermal cycling damage: This failure mode occurs when the rover repeatedly transitions between illuminated regions and permanently shadowed region boundary conditions. The cause is repeated exposure to hot-to-cold and cold-to-hot thermal changes. These transitions can stress sensors, insulation, joints, mechanisms, and payload interfaces, which could result in uncertain science measurements, increased power demand, reduced mobility performance, or reduced confidence in volatile measurements.

Recovery:

Failure Detection and Management:

Thermal-control failures will be detected primarily through the eight temperature sensors continuously monitoring critical thermal zones throughout the rover. Temperature measurements that approach or exceed established operating or survival limits will

indicate potential heater, sensor, insulation, or controller underperformance. The temperature controller will use sensor feedback to determine whether additional heating is required. Abnormal temperature trends, unexpected heater operation, or inconsistent sensor readings can also be used to identify degraded thermal-control performance. Once a failure is detected, the affected component and associated thermal-control hardware will be evaluated to determine whether redundant hardware can maintain the required thermal conditions.

Operational Workarounds:

If a heater or temperature sensor becomes unusable, the redundant thermal-control hardware will be used where available to continue monitoring and protecting critical components. If full thermal-control capability cannot be restored, rover operations will be adjusted to reduce exposure of thermally vulnerable hardware to extreme conditions. This may include reducing the duration of operations within permanently shadowed regions, limiting activities that increase thermal or electrical loads, or relocating the rover to a more thermally favorable environment when mission conditions permit. Temperature measurements from the remaining operational sensors will continue to be monitored to verify that critical hardware remains within acceptable limits.

Recovery from Thermal Cycling Damage:

Potential degradation caused by repeated hot-to-cold and cold-to-hot transitions will be identified through temperature trends and changes in thermal-control performance. If degraded insulation, sensors, heaters, or interfaces are suspected, the rover will rely on the remaining functional thermal-control hardware while exposure to additional extreme thermal cycles is minimized. Operational planning can then be adjusted to reduce repeated transitions between illuminated and permanently shadowed regions until the subsystem's available thermal margin has been evaluated.

2.1.4.4. Thermal Control Subsystem Manufacturing and Procurement Plans

Procurement Overview:

The majority of the thermal subsystem is made up with components that work right out of the box without any modifications. For the sole exception of the insulation which will be made up of MLI Blankets. The reason why this particular component needs to be

modified is because it needs to not only be tailored to the main body of the lunar surface vehicle but also to the two other external payloads, the LISTER and Msolo. Our main priority when finding the primary and backup suppliers was time and meeting our requirements, especially for the MLI Blanket.

Lead Time Summary:

The estimated procurement lead time for the thermal control subsystem is 10 to 14 weeks due to needing a custom manufactured MLI blanket. The other components take less time to procure, as you can see in Figure 50 (the Components Supplier Table), starting with indium foil taking 4 to 6 weeks, the resistance temperature detectors taking 6 to 8 weeks, the electronics taking 8 to 10 weeks, and the kapton polyimide heaters taking 8 to 12 weeks to obtain.

Backup Supplier and Schedule Impact:

If a primary supplier is unable to meet the required delivery schedule, the identified backup supplier will be used to reduce the risk of major procurement delays. However, switching suppliers may increase lead time because component availability, specifications, manufacturing schedules, and compatibility with the thermal subsystem would need to be confirmed before procurement. The MLI blankets represent the most schedule-critical thermal component because their estimated 10–14 week lead time is the longest within the subsystem and they require customization for the rover and external payloads. Therefore, MLI procurement should begin as early as possible. The remaining thermal components should also be ordered early enough to allow time for integration and verification before final assembly. Procurement progress will be monitored throughout the project so that backup suppliers can be used before a component delay affects thermal subsystem integration or the overall project schedule.

Subsystem / Component	Procurement Method	Primary Supplier	Backup Supplier	Estimated Lead Time	Justification
Kapton Polyimide Heaters	COTS (Com...	Minco Products	Omega Engineering	8-12 weeks	Proven aerospace heater technology with high reliability.
Multilayer Insulation Blankets	MCOTS (M...	Sheidahi	Dunmore Aerospace	10-14 weeks	Flight-qualified insulation materials specifically designed

					for spacecraft thermal protection.
Resistance Temperature Detectors	COTS (Com...	Honeywell	TE Connectivity	6-8 weeks	High accuracy, wide operating range, extensive spaceflight heritage.
Thermal Interface Material (Indium Foil)	MCOTS (M...	Indium Corportation	Goodfellow	4-6 weeks	Excellend thermal conductivity and widely used for spacecraft electronics cooling.
Thermal Control Electronics	COTS (Com...	Texas Instruments	Analog Deivces	8-10 weeks	Reliable controller electronics with established aerospace applications.

Figure 50: Component Suppliers Table

2.1.4.5. Thermal Control Subsystem Verification Plans

1. Verification Overview

The Thermal Control Subsystem will be verified using a combination of analysis, inspection, and testing to demonstrate compliance with all thermal subsystem requirements. Analysis will be the primary verification method for requirements involving predicted temperatures, heater power, heat rejection, cost, and thermal performance because these parameters can be evaluated using the rover thermal model under expected and worst-case mission conditions. Inspection will be used for requirements that can be verified through physical or design characteristics, such as subsystem mass. Testing will be used where actual hardware performance must be demonstrated, including thermal cycling and thermal balance testing under simulated lunar environmental conditions. Together, these methods will verify that the thermal design can maintain critical rover components and science instruments within their required operating and survival limits throughout the mission.

2. Thermal Verification Matrix

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
TCS-01	The thermal control subsystem shall maintain critical	Analysis	Must hold over a range of terrain and thermal cycling. Analysis	Build a thermal model of the rover and run it for the worst-case hot and worst-case cold mission scenarios. Compare the predicted

	rover electronics within their allowable operating temperature range during mission operations.		identifies the worst hot and cold cases.	temperature of each critical electronics box to the operating range of -10°C to 30°C. Pass if every box stays inside its range with positive margin.
TCS-02	The thermal control subsystem shall provide thermal protection for short-duration excursions into PSRs.	Analysis	Analysis provides information for not just one case but covers all potential issues during the PSR excursions.	Run a thermal analysis of a PSR excursion using the coldest PSR environment. Pass if no component drops below -10°C at any point during the excursion or the recovery back into sunlight.
TCS-03	The thermal control subsystem shall minimize heater power demand during PSR operations.	Analysis	Analysis is the only method capable of evaluating configurations that have not been built yet.	Use the thermal model to predict heater power draw during PSR operations for the selected design. Compare the result to the heater allocation in the power budget. Pass if predicted demand is at or below the allocation.
TCS-04	The thermal control subsystem needs to not surpass the mass of 18 kg.	Inspection	The mass properties information that will be given with the materials is satisfactory information.	Inspect CAD mass properties, and sum the mass of every component. Pass if the total mass is below 18 kg.
TCS-05	The thermal control subsystem shall not surpass the allocated total cost of $25.7M.	Analysis	Cost analysis is the only method to properly know if the thermal control subsystem is within the budget.	Roll up the cost estimate from vendor quotes, labor hours, and test costs in the project cost model. Compare the total, including reserves, to the $25.7M allocation. Pass if the estimate is at or below the allocation.
TCS-06	The thermal control subsystem shall use passive thermal control methods where practical before relying on active heater power.	Analysis	Analyzing trade studies is the only way to satisfy this constraint.	Review the thermal design trade studies, which compares passive options against active heating for each thermal zone. Pass if passive control was selected wherever it meets the temperature requirements, and each use of heater power has a documented justification.
TCS-07	The thermal control subsystem shall maintain thermal	Test	A thermal balance test of the lunar surface vehicle with the instruments installed is the only	Perform a thermal balance test of the integrated rover with the science instruments installed in a thermal vacuum chamber that

	compatibility with the science instruments during data collection inside or near PSRs.		method to observe the behavior.	simulates the PSR environment. Operate the instruments and record all interface temperatures. Pass if every instrument stays within -10°C to 30°C and returns valid data.
TCS-08	The thermal control subsystem shall reject excess heat during rover operations in illuminated regions.	Analysis	This requirement is focused on measurements which are better suited for analysis.	Run a steady-state heat transfer analysis at maximum solar load. Confirm the radiator area rejects the full heat load. Pass if all components remain below 30°C.
TCS-09	The thermal control subsystem shall survive repeated thermal cycles between illuminated terrain and permanently shadowed regions.	Test	The requirement addresses durability accumulated over life, and only running cycles on real hardware shows that durability.	Thermal cycle the rover in a thermal vacuum chamber between the hot and cold extremes for the qualification number of cycles. Perform functional checks and visual inspection before and after. Pass if there is no cracking, delamination, or loss of thermal performance.
TCS-10	The thermal control subsystem shall provide temperature monitoring for critical rover components during mission operations.	Analysis	Sensor placement must be justified against predicted thermal gradients. Only analysis can establish that a measured location represents the component temperature of interest to within the required tolerance.	Compare the predicted temperature at each sensor location to the predicted temperature of the component it represents across the hot and cold cases. Pass if every critical component has at least one sensor.
TCS-11	The thermal control subsystem needs to maintain a minimum operating temperature for the external instruments including Msolo at −10°C to 45°C and LISTER ~30°C.	Analysis	Analysis is required to show the many operating cases that determine the operating temperatures are met.	Run the thermal model with Msolo and LISTER included, for all operating cases. Pass if Msolo stays between -10°C and 45°C and LISTER stays at its required ~30°C in every case.

Figure 52: Thermal Verification Matrix

3. Verification Method Rationale

Verification methods were selected according to the type of performance being evaluated. Analysis is used for most thermal requirements because thermal models can evaluate temperatures, heat transfer, heater power demand, and component performance across mission environments that cannot all be practically reproduced during early design phases. Inspection is appropriate for requirements such as the thermal subsystem mass because compliance can be confirmed directly using CAD mass properties and component information. Testing is used for requirements involving thermal cycling and integrated thermal performance because physical testing provides direct evidence that the hardware can withstand repeated temperature changes and operate under simulated lunar thermal conditions. Using these methods provides appropriate verification while reducing unnecessary testing during the preliminary design phase.

4. ***Preliminary Verification Plan***

Verification will occur throughout subsystem development and will become progressively more detailed as the thermal design matures. During preliminary design, thermal models, CAD data, trade studies, supplier information, and subsystem budgets will be used to verify requirements primarily through analysis and inspection. Once representative or flight-like hardware becomes available, thermal balance and thermal cycling tests will be performed using appropriate thermal-vacuum facilities to simulate hot, cold, illuminated, and permanently shadowed region conditions. Temperature sensors and data-acquisition equipment will be used to monitor critical components during testing. Each requirement will be considered successfully verified when its specific acceptance criteria in the Thermal Verification Matrix are satisfied. Current verification results remain dependent on preliminary thermal-model assumptions and will be updated as the rover geometry, material properties, component specifications, and mission operating conditions mature.

2.1.5. Payload Subsystem Overview

The payload subsystem provides the necessary scientific measurement capabilities to meet the investigation's thermal, volatile, sub surface and seismic objectives. The selected payload includes the MSolo mass spectrometer, LISTER thermal measurement, TENG triboelectric nanogenerator, three geophones and a seismic digitizer. The combination of these payload elements enables the rover to perform characterization of volatiles, including their chemistry, measure thermal properties at depth, detect and

analyze seismic events, assess the link between rover motion and TENG energy harvesting and record time tagged data while on the surface. The MSolo, LISTER , three geophones and digitizer combination therefore contribute to the primary science objectives of characterizing the near surface thermal profile of the lunar regolith and determining the water compound inventory (abundance and chemistry) of the PSRs. They also address the secondary objectives of characterizing subsurface and any ice distribution, assessing thermal properties and storage and demonstrating technologies enabling future missions.

The payload elements perform complementary functions to one another while addressing different science investigations. LISTER makes direct thermal measurements at specific locations on the surface, while MSolo analyzes low molecular weight volatiles related to surface and subsurface processes. The geophones allow passive seismic measurements to take place and when combined with GEOthree digitizer, which also provides the analog to digital conversion, timing, storage and telemetry, it becomes possible to measure seismic events. TENG provides a technology demonstration by measuring the amount of energy harvested from the motion of the rover. The payload therefore not only supports the various science measurements but also provides technology demonstrations while remaining within the mass, power, thermal and operational constraints.

The payload subsystem provides chemical, thermal, seismic, subsurface, environmental, and technology-demonstration capabilities. MSolo provides direct, in situ volatile characterization by obtaining mass spectra during 20-min science stops. LISTER provides thermal measurements of the regolith, including thermal conductivity and the near-surface thermal environment. The geophone array provides passive seismic measurements by detecting seismic signals at multiple locations and conditioning them for the seismic digitizer. TENG provides a measurement of the electrical output due to mechanical motion of the rover, thus providing information on the potential for mechanical energy harvesting near the PSR boundary. Together, these instruments provide time- and location-tagged science data that can be correlated with the rover location, vehicle operations, and environmental measurements. The payload is compatible with the current mission concept, wherein science instruments are turned on in a serial fashion when possible, so that the overall payload power is constrained to about 40 W during traversal and about 42 W during science stops, with a peak power (for duty-cycling electronics) of about 161.5 W for electrical system sizing purposes.

1. ***Payload Summary Table***

Instrument	Quantity	Mass (kg)	Dimensions (m)	Maximum Power Draw (W)	TRL
Msolo	1	4.8	0.155 × 0.200 × 0.460	35	8
LISTER	1	4.60	0.350 × 0.250 × 0.200	88	6
TENG	1	0.5	0.20 × 0.15 × 0.10	5	4
MECA	1	0.95	0.125 × 0.10 × 0.085	1.5	7
Geophone	3	19	0.050 × 0.180 × 0.050	0	5
Seismic Digitizer	1	1	0. 169 x 0.088	1.1	6
Total Subsystem	8	30.85		130.6	4

Figure 53: **Payload Subsystem Instruments**

2.1.5.1. Science Instrumentation Requirements

The Payload subsystem requirements are derived from the team's Science Traceability Matrix. The STM defines two main science goals: characterizing the near surface thermal profile of the lunar regolith, and determining the abundances and chemical make up of water compounds in the PSRs. Additionally, the STM defines science goals related to subsurface and ice distributions, thermal storage and the demonstration of rover autonomous functions. As such, the science requirements for the payload are centered on acquiring thermal, chemical, seismic, and technology data while being cognizant of the mass, power, thermal, communications, and operational constraints placed upon the spacecraft.

The payload shall provide measurement capabilities contributing to the mission's thermal and volatile science objectives. The payload shall additionally provide seismic measurements supporting science related to subsurface characterization, and shall provide instrument and engineering data necessary to relate science findings to rover location and operations. Operations of the science payload shall be compatible with the rover's sortie timeline and power budget. The science payload's data storage and communications capabilities shall be sufficient to allow for science data collected during a sortie to be stored for later transmission during a Direct-to-Earth communications opportunity.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
PAY-01	The Payload shall measure near-surface regolith thermal properties within the PSR.	Directly supports the thermal-profile science objective in the STM.	SYS-06		Test	Y...
PAY-02	The Payload shall measure the chemical composition of low-molecular-weight volatiles during PSR operations.	Supports determination of water compounds and other volatiles.	SYS-06		Test	Y...
PAY-03	MSolo shall complete a health and calibration sequence before science acquisition.	Ensures measurements can be distinguished from instrument/background effects.	SYS-06		Test	Y...
PAY-04	MSolo shall produce time- and location-referenced mass spectra during designated science stops.	Provides traceable volatile measurements.	SYS-06		Test	Y...
PAY-05	LISTER shall obtain thermal measurements at designated PSR science locations.	Supports characterization of the near-surface thermal profile.	SYS-06		Test	Y...
PAY-06	LISTER shall support measurement sequences associated with regolith thermal equilibration and response.	Allows thermal properties to be derived from the instrument measurements.	SYS-06		Test	Y...
PAY-07	The Payload shall provide passive seismic measurements using three geophones.	Supports characterization of the PSR/subsurface environment.	SYS-06		Demon...	Y...
PAY-08	The seismic digitizer shall acquire and time-reference data from the geophone array.	Provides the digital data product required from the geophones.	SYS-02		Test	Y...

PAY-09	The seismic digitizer shall support a sampling range sufficient to characterize the geophone measurement band.	Ensures compatibility with the selected geophones rather than the removed broadband FSS.	SYS-12		Test	Y...
PAY-10	The Payload shall record science data during rover operations and retain the data until downlink.	Prevents loss of science data between communication opportunities.	SYS-06		Test	Y...
PAY-11	TENG shall measure electrical output generated during rover mechanical motion.	Supports the technology demonstration of mechanical energy harvesting.	CDH-05		Test	Y...
PAY-12	Payload instruments shall operate within allocated spacecraft power and thermal limits.	Ensures payload operation does not compromise spacecraft survival or sortie completion.	SYS-02		Test	Y...
PAY-13	Payload instruments shall provide health/status information sufficient to identify instrument faults.	Supports fault detection and recovery.	CDH-05		Test	Y...

Figure 54: Payload Subsystem Requirements

2.1.5.2. Payload Instrument Overview

The Lunar Instrument for Subsurface Thermal Exploration with Rapidity (LISTER) provides the mission's primary subsurface thermal measurements by characterizing vertical temperature gradients, thermal conductivity, and heat flow in regolith near the PSR. LISTER operates by pneumatically penetrating the regolith to a mission target depth of 1 m and uses platinum Resistance Temperature Detector sensors to measure temperature at a depth. Controlled heating pulses are applied at measurement locations and the subsequent thermal rise and decay are then used to determine regolith activity. The current STM requires measurements at 0.5 m depth intervals to 1 m, temperature coverage across the 20-120 K science range with 0.5 K resolution, and thermal conductivity measurements derived to approximately +/- 10% accuracy. The instrument interfaces mechanically with the rover through its deployment and drilling assembly, electrically with

the Power subsystem, and with CDH for command, telemetry, and science data storage. Its pneumatic system also requires a finite nitrogen supply, making deployment location and operating sequence important design considerations. MSolo measurements are therefore performed before LISTER excavation to reduce the chance of contaminating volatile measurements. LISTER directly supports MR-02 by characterizing subsurface thermal properties and also supports MR-05 by providing the thermal measurements needed to evaluate regolith energy storage behavior.

MSolo provides in-situ chemical characterization of volatile compounds associated with lunar surface and subsurface material. The instrument operates as a quadrupole mass spectrometer that ionizes sampled gases, separates the resulting ions according to mass-to-charge ratio, and measures their relative signal intensities to identify volatile species and isotopic signatures. The current STM identifies target species including H20, CO2, NH3, CO, CH4, and H2S, while MSolo provides unit mass resolution across approximately 1-100 m/z and uses an onboard calibration gas system to distinguish lunar signals from gases associated with the rover or instrument itself. MSolo interfaces with the Power subsystem for electrical supply, CDH for command, telemetry, data storage, and downlink, and the Thermal subsystem because its operating temperature must be maintained within the instrument's required range. The mission sequence places MSolo sampling before LISTER drilling so that volatile measurements are collected before pneumatic excavation disturbs the local regolith and releases nitrogen. These measurements directly support MR-03 by quantifying volatile abundance, composition, and isotropic characteristics within PSR terrain.

The Geophone Array provides passive high bit rate mode and active seismic measurements used to characterize ground motion, seismic wave behavior, and the mechanical properties of the shallow lunar regolith. The current mission concept uses multiple coil and magnet geophones deployed from the rover to form an array, with each sensor converting ground motion into an analog electrical signal proportional to mechanical motion. The selected geophone concept has useful response beginning near approximately 4.5 Hz, reflecting the current hardware capabilities after the earlier very-broadband seismometer was removed from the payload. The array interfaces mechanically with the deployment system, electrically with the rover power and payload electronics, and with CDH for timing, digitized data acquisition, storage, and eventual downlink. Environmental design considerations include maintaining sensor performance under lunar vacuum, extreme temperature, abrasive regolith, and rover generated mechanical noise that may corrupt data. The resulting seismic measurements support MR-08 by determining the frequency and intensity of lunar seismic activity and by characterizing local ground response near Nobile Rim 2.

The Triboelectric Nanogenerator (TENG) is a technology demonstration instrument intended to determine whether mechanical rover regolith interaction can generate useful electrical output through triboelectric charging. Its operating principle relies on contact and relative motion between materials producing charge separation, which is then measured as electrical voltage and current during rover wheel activity. The current mission concept records a stationary electrical baseline before wheel motion and then compares that baseline with measurements collected during controlled or nominal rover traversal. The STM identifies voltage, current, and generated electrical power as the primary observables, while the present performance reference uses experimental values near 17.4 V open circuit voltage and 0.96 micro amperes short circuit current at 10 Hz as a scale for expected performance rather than as a guaranteed flight output. TENG interfaces mechanically with the rover mobility system because wheel motion provides the excitation, with Power and payload electronics for electrical measurement, and with CDH so TENG output can be time correlated with wheel speed and rover operating state. Dust exposure, temperature, mechanical wear, and changes in regolith contact conditions are important design considerations because they can alter the triboelectric response. TENG directly supports MR-07 by measuring voltage, current, and power generated from charged regolith interactions and evaluating the feasibility of electrostatic energy harvesting.

The MECA electrometer package characterizes the electrostatic environment near the lunar surface by measuring electric potential, charge transfer, and low level currents associated within illuminated, shadowed, and mechanically disturbed regolith. The instrument uses sensitive electrometer channels and exposed sensing elements to detect variations in surface potential and charge accumulation, particularly during rover motion and LISTER drilling when regolith disturbance may generate triboelectric charging. The current STM requires characterization of surface electric potential from approximately -400 V to +10 V and ion current measurements in the picoampere range, with the selected sensing approach providing substantially finer current resolution than the mission requirement. MECA interfaces with the Power subsystem for regulated electrical supply. It interfaces with CDH for continuous data acquisition and time correlation with rover operations, and the Mechanical subsystem through placement of sensing elements where local electrical conditions can be measured without excessive vehicle interference. Environmental and design considerations include electrical noise from rover hardware, sensor contamination by lunar dust, baseline drift, and the need to distinguish naturally occurring electrostatic signals from charging caused by mission operations. The instrument therefore establishes quiescent background measurements before active operations and continues monitoring during regolith disturbance. MECA supports MR-09 by characterizing

surface electrostatic conditions and charge accumulation that may present hazards to future lunar hardware and surface operations.

Technology Readiness Level (TRL)

The MSolo instrument has been assessed at a TRL 8 for this mission as the underlying instrument architecture has progressed beyond laboratory and relevant environmental validation. MSolo has been demonstrated in the lunar environment by way of the Intuitive Machines 2 mission (IM-2). The MSolo had originated as a commercial residual gas analyzer and was subsequently modified by NASA for spaceflight through the addition of the space-rated electronics, environmental qualifications, contamination controls, calibration capability and lunar-specific system integration. NASA documentation reports that MSolo completed verification and validation testing prior to its IM-2 flight and subsequently operated on the lunar surface during IM-2. The instrument was powered on, completed instrument checkouts and detected gases associated with the lander’s propulsion system. NASA subsequently reported that MSolo successfully demonstrated the full range of its hardware in lunar conditions. This established the direct operational environment heritage for the instrument. IM-2 ended prematurely, thus, the TRL of 8 has been selected as the conservative maturity level for MSolo.

In understanding the flight history of LISTER, this instrument has been assessed to sit at the TRL of 6. While the underlying subsurface thermal probe architecture has been demonstrated in the lunar environment, the specific mission configuration and required 2 to 3 m subsurface deployment capability require additional integration, qualification and verification. LISTER was developed at Texas Tech University in collaboration with Honeybee Robotics as a pneumatically drilled subsurface thermal probe designed to measure lunar thermal gradient and thermal conductivity at multiple depths. In 2025, a LISTER flight configuration was successfully deployed aboard a Firefly Aerospace’s Blue Ghost (MIssion 1). During Mission 1, LISTER operated on the lunar surface, reached approximately 1 m depth, and collected eight temperature and thermal conductivity measurements, providing direct lunar-environment heritage for the instrument's drilling and thermal measurement architecture. NASA identifies LISTER as a pneumatically drilled thermal probe capable of targeting depths of approximately 2 to 3 m and describes its measurement approach as determining heat flow from the product of subsurface thermal gradient and thermal conductivity. The selected TRL 6 therefore reflects the maturity of the demonstrated LISTER technology while conservatively accounting for the deeper deployment depth, mission-specific mechanical and electrical integration and qualification of the configuration proposed for this mission. The instrument;s pneumatic excavation system, thermal probe, temperature sensors , deployment mechanisms and electronics must be verified under the environmental, thermal and operational conditions expected for this mission.

TENG's technology readiness level sits lower than the others at a TRL of 4. TENG's underlying triboelectric energy generation principle has been demonstrated through laboratory prototypes, but the proposed lunar surface implementation has not yet been validated in a relevant lunar environment. TENG devices generate electrical energy through contact electrification and electrostatic induction and have been demonstrated using a variety of polymer and electrode materials. Research on the subject has specifically demonstrated triboelectric nanogenerators incorporating lunar regolith simulants, including a prototype that produced about 17.4 V open-circuit voltage and 0.96 μA short-circuit current at 10 Hz. However, such demonstrations do not establish flight readiness because the technology has not been demonstrated under the combined vacuum, thermal, dust, radiation, mechanical and operational conditions expected in the lunar surface. The TENG proposed for this mission will require custom fabrication and mission specific development using materials such as PTFE, Kapton, copper electrodes, and flexible polymer layers, followed by electrical characterization, and impedance. Power measurements include thermal vacuum testing, vibration testing, dust exposure and extended cycling to establish performance and durability under representative conditions. TRL is a conservative and supportable assessment because of its lack of lunar environment demonstration.

MECA's TRL sits between 6 and 7, based on the established maturity and environmental heritage of its electrostatic measurement technology and the instrument owner's documented technology assessment. The electrometer technology was developed through NASA planetary surface research to measure the electric field strength, charged particles, triboelectric interactions and surface temperature - testing performed under simulated conditions. NASA technical documentation describes the electrometer's demonstrated capability to characterize electrostatic and charged-particle environments relevant to planetary surfaces though NASA's Electrostatics and Surface Physics Laboratory identified the technology as applicable to lunar surface electrostatic measurements. The selected TRL range reflects the distinction between the maturity of the established electrometer technology and the specific mission configuration proposed for this lunar mission. A TRL 6 assessment is supported when consideration of the demonstration in technology within a relevant planetary environment. TRL 7 is applicable if the specific integrated configuration is considered sufficiently representative of the intended flight system and has been demonstrated in an operational planetary environment.

The geophone and GSB recorder would sit at technology readiness level between 6 and 7 because field testing for the natural resonant frequency of 7.5 HZ is needed in order to be similar to the geophones used for the sensor heritage of the active seismic experiments used in Apollo mission 16. Also calibration tests, shock wave testing to ensure

it can mechanically couple directly and resistant to the electrostatic regolith, gain frequency tests and measuring the spring to coil ratios is also needed to ensure the prototype is being tailored adequately for the lunar environment. At TRL 6 the prototype would need to be tested first in an analog simulant making sure of its proficiency. Once its tailored with similar parameters as the sensor's heritage similar to the geophones used in previous Apollo missions it can reach a TRL of 7.

Supporting Figures

LISTER:

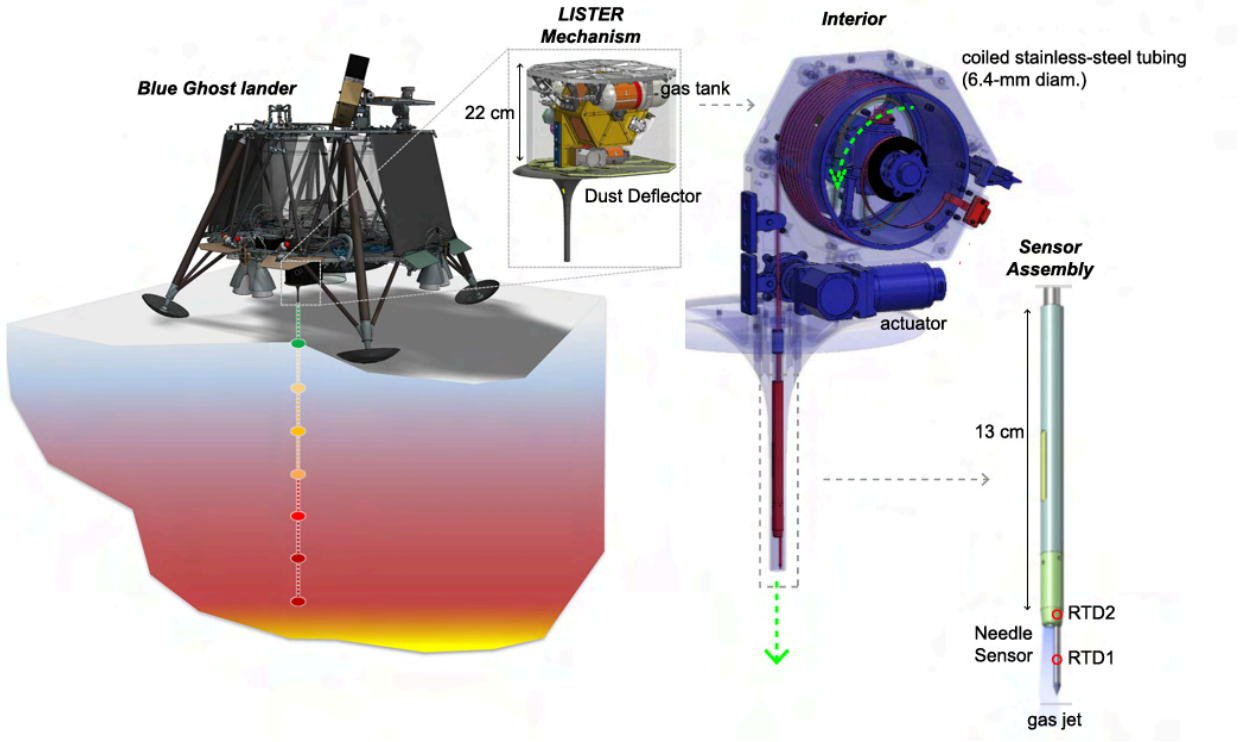


Figure 55: A conceptual drawing of the LISTER deployment mechanism accommodated below the platform of the Blue Ghost lander as it makes thermal measurements at multiple depths into the subsurface (left). Schematic drawings of the interior of LISTER's deployment mechanism (middle) and sensor assembly (right) are also shown.

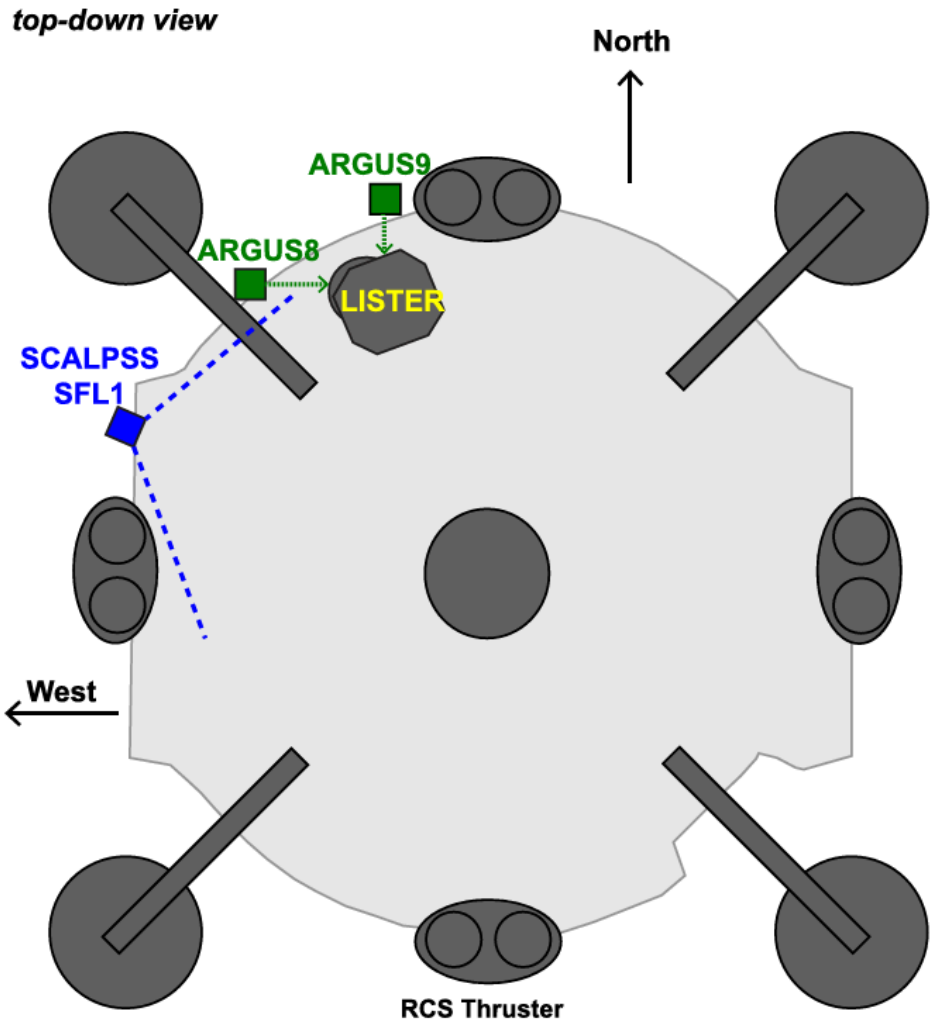


Figure 56: A top-down view drawing of the positions of the below-deck cameras used to monitor LISTER's operations. Argus 8 and 9 are video cameras operated by Firefly Aerospace. SFL1 is a wide-angle, short-focus black-and-white camera of the SCALPSS payload. The blue dashed lines indicate its field of view.

MECA:

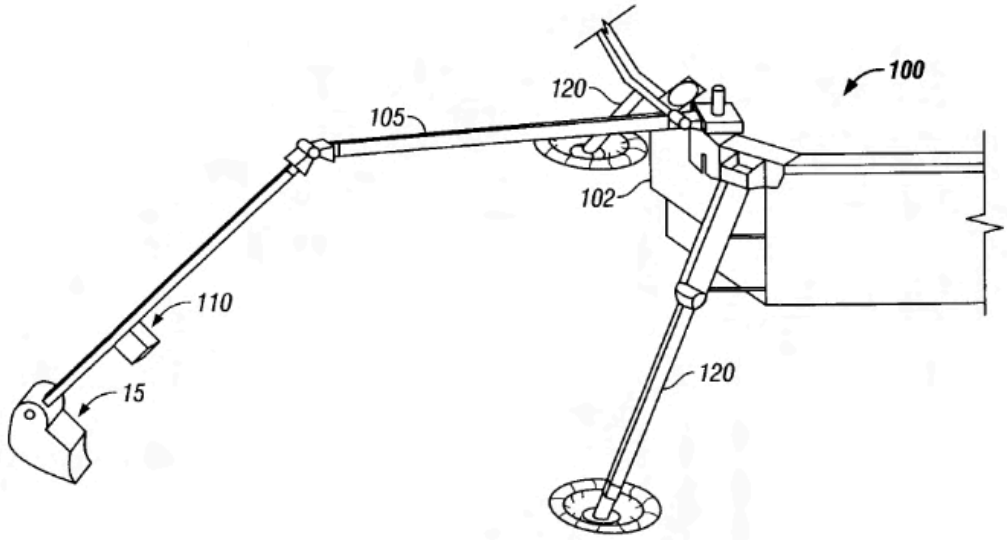


Figure 57: Mechanical schematic diagram of the MECA payload deployment architecture.

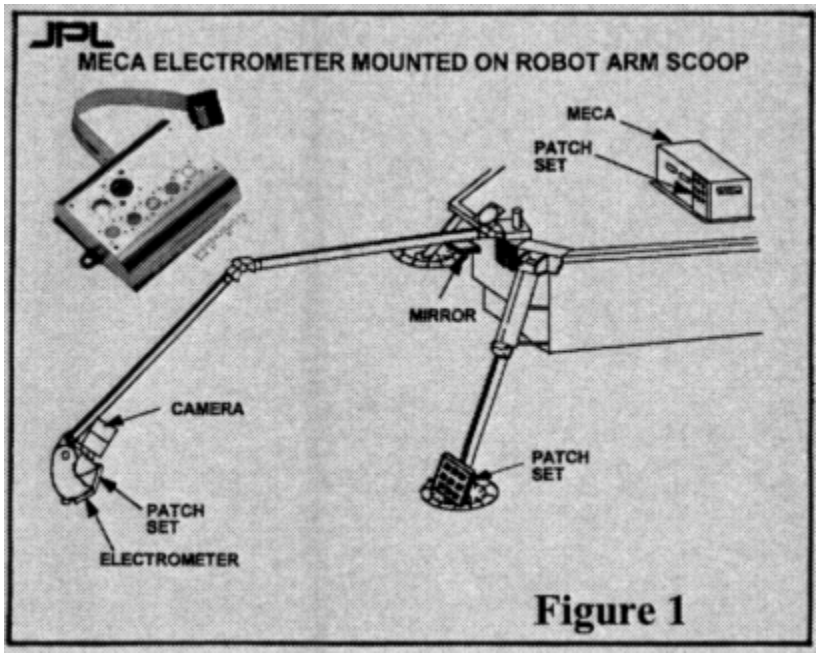


Figure 58: Functional diagram showing the JPL MECA electrometer instrument layout, highlighting sensor placement mounted directly on the robotic arm scoop alongside the associated control box and patch set electronics.



Figure 59: Image of a GS-14-L3 which is the model that is sold by Geospace technologies, quote must be made towards the company with specific tuning instructions to be more tailored prior to deployment of prototype.



Figure 60: GSB recorder that includes a digitizer that can enable the conversion from analog to digital signal. Quotes must be done towards the company to specify calibration, frequency gain to ensure good recordings.

2.1.5.3. Payload Subsystem Recovery and Redundancy Plans

Science Measurement Requirements		**Baseline Performance Requirements**		**Instrument**		
Physical Parameters	Observables			**Inst. 1**	**Inst. 2**	**Inst. 3**
Phys. Para. 1	Obs. 1	Inst. Per. Req. Parameter 1	Instrument Performance Req. Value 1			
		Inst. Per. Req. Parameter 2	Instrument Performance Req. Value 2			
		Inst. Per. Req. Parameter 3	Instrument Performance Req. Value 3			

		Inst. Per. Req. Parameter 4	Instrument Performance Req. Value 4			
Phys. Para. 2	Obs. 2	Inst. Per. Req. Parameter 5	Instrument Performance Req. Value 5			
		Inst. Per. Req. Parameter 6	Instrument Performance Req. Value 6			
		Inst. Per. Req. Parameter 7	Instrument Performance Req. Value 7			
		Inst. Per. Req. Parameter 8	Instrument Performance Req. Value 8			
Phys. Para. 3	Obs. 3	Inst. Per. Req. Parameter 9	Instrument Performance Req. Value 9			
		Inst. Per. Req. Parameter 10	Instrument Performance Req. Value 10			
		Inst. Per. Req. Parameter 11	Instrument Performance Req. Value 11			
		Inst. Per. Req. Parameter 12	Instrument Performance Req. Value 12			

Figure 61: Payload Redundancy Plans Table

Redundancy

The Payload subsystem does not make extensive use of hardware redundancy, since the duplication of its primary instruments would lead to unacceptable increases in the mass, power, volume, and complexity, without commensurate benefits to the science return. MSolo, LISTER, and TENG are single-string instruments, while the seismic system is inherently redundant due to having three spatially separated geophones. The latter provides additional measurement location redundancy, so that the loss of one geophone will not result in an inability to make measurements at the desired location. The seismic digitizer is a single point of failure, since duplication would add significant weight and power to the payload. However, since the telemetry power of the digitizer is only about 1.1 W, and its mass is 1.2 kg, this should be reconsidered if the analysis indicates that it is a

critical element. For MSolo and LISTER, any level of redundancy is inappropriate, since the two instruments make fundamentally different primary science measurements, which cannot be substituted for one another. Finally, the acceptable single point failures for the subsystem are dictated by the limited payload allocations, and by the need to prioritize the instruments which make unique measurements contributing to the science objectives of STM.

Recovery

Payload malfunctions will be identified by health housekeeping telemetry, power-state checking, communication, calibration, and comparison of sensor outputs to operational ranges. The recommended procedure in the case of an MSolo malfunction is to cycle the instrument power and reestablish the health and calibration baseline before further measurements are attempted. If this proves unsuccessful, background measurements and subsequent observations may be used to assess the potential value of any measurements taken. LISTER malfunctions may be bypassed by either restarting the instrument or, if sortie time permits, repeating the thermal measurement in a later stationary period. The loss of one geophone in the seismic system may be bypassed by checking the other channels for consistency while performing passive seismic monitoring with the remaining accelerometers. With the loss of a digitizer, seismic observations are unavailable until the electronics can be reset; the other two payload instruments, however, may continue to operate independently. Loss of a channel in TENG is less serious, as this instrument is primarily a technology demonstrator, and the goals of the mission are unaffected by its loss. In any case, the recovery sequence will attempt to preserve rover safing, science data, and the ability to restart other systems as much as practical.

2.1.5.4. Payload Subsystem Manufacturing and Procurement Plans

1. ***Procurement Overview***

The procurement strategy for the Lunar Surface Vehicle's payload subsystem—which includes MSolo, LISTER, MECA, the Geophone Array, and TENG is multi-faceted, balancing technical flight heritage with commercial capability to minimize cost, development lead times, and overall mission risk. This approach is specifically structured to mitigate technical risk while satisfying the project's stringent $250 million total cost cap, mission technical requirements, and strict launch readiness deadlines. Following NASA sourcing guidelines, commercial off-the-shelf (COTS), space-qualified COTS (SQCOTS), and modified COTS (MCOTS) solutions are prioritized over custom in-house

development whenever viable hardware exists on the market or through established research institutions. Suppliers were selected following market surveys and trade studies evaluating four critical criteria: cost, technical capability, flight heritage, and schedule reliability. Prioritizing mature or commercially available instruments drastically reduces overall development and qualification costs compared to ground-up custom builds. Custom in-house manufacturing is strictly reserved for specialized structural integration, custom mounting brackets, and electrical harnesses necessary to interface the instruments with the rover bus, ensuring internal budget allocations are optimized. Maintaining alternate commercial suppliers further mitigates supply chain risk and provides procurement flexibility without requiring substantial interface modifications or cost overruns.

2. ***Component Suppliers***

MSolo (Mass Spectrometer): Procured via a Modified Commercial Off-the-Shelf (MCOTS) approach, the MSolo mass spectrometer is derived from the commercially available INFICON Transpector MPH quadrupole mass spectrometer, which was adapted by NASA for the VIPER lunar mission. INFICON serves as the primary commercial supplier, with NASA performing custom flight packaging, environmental qualification, software integration, and rover interface development to achieve TRL 7. Extrel CMS is designated as the backup supplier capable of providing an equivalent commercial quadrupole mass spectrometer. Preflight calibration and standard NASA payload qualification testing (TVAC, vibration, EMC) are required prior to rover integration. Utilizing a commercially mature baseline significantly reduces development cost, technical risk, and schedule compared to developing a new flight mass spectrometer.

LISTER (Heat-Flow Probe): Sourced via a Sole Source / Commercial Flight Contractor model, Honeybee Robotics (a Blue Origin company) serves as the primary supplier in collaboration with Texas Tech University. Honeybee uniquely provides the compact pneumatic spool drive and deployment system required to drill 2 to 3 meters into regolith with proven flight heritage from Mars and Moon missions (TRL 6). The backup strategy utilizes In-House Systems Integration using commercial pneumatic spooling hardware paired with outsourced precision machining (e.g., NASA Kennedy Space Center or university space research partners).

MECA (Electrometer Package): Sourced through an Inter-Agency Transfer / Custom Contract Build from NASA Jet Propulsion Laboratory (JPL), which jointly designed the instrument suite with Kennedy Space Center (KSC). JPL is the primary supplier due to extensive flight heritage from the Mars Phoenix Lander (TRL 6–7) and its specialized Microdevices Laboratory (MDL) for custom micro-electronics fabrication. Subcontracted

Academic Labs or specialized space electronics vendors (e.g., Space Lab Technologies) serve as secondary backup suppliers.

Seismic Instruments (Geophones & Digitizer): Sourced via COTS / SQCOTS procurement. The primary setup consists of Geospace Technologies models, specifically the GS-ONE LF geophone paired with a Geospace GSB series recorder digitizer to satisfy TRL 6 requirements (heritage from Apollo passive and active seismic experiments). Backup hardware includes Fiber Optic Seismic Vector Sensors by Paulsson Inc. or traditional gold-coated wired geophones.

TENG (Triboelectric Nanogenerator): Procured as a Custom Contract Build / Research Partnership because no flight-qualified commercial TENG exists for lunar surface operations. The primary supplier is the Georgia Institute of Technology, leveraging university microfabrication cleanrooms to fabricate the unit from commercially available materials such as Kapton and PTFE (TRL 4). NASA-approved commercial MEMS fabrication facilities, including JPL, serve as the backup supplier to mitigate development and facility availability risks.

Instrument Mounting Hardware & Harnesses: Manufactured In-House by Team Fontus using a Build-to-Print strategy. Land Aero Group LLC acts as the primary precision machining supplier for structural components and mounting rails, with TOMI Engineering as the backup supplier. LAN Research is the primary supplier for SQCOTS fasteners, lock cables, and hardware.

3. Lead Time Summary

The procurement timelines across the payload subsystem hardware are governed by a mix of commercial off-the-shelf lead times, custom microfabrication constraints, and rigorous flight qualification testing requirements. Custom-developed hardware represents the primary schedule risk on the mission critical path. Specifically, the custom TENG technology demonstration demands the longest procurement horizon at 15 to 24 months due to required cleanroom microfabrication, iterative prototype refinement, and low-TRL environmental testing. The contractor-built LISTER probe and the NASA-integrated MSolo mass spectrometer follow closely with acquisition, customization, and integration lead times of 12 to 18 months. Standard flight-board fabrication and sensor coating for the MECA Electrometer Package are estimated at 8 to 12 months. While commercial off-the-shelf geophones offer relatively short hardware acquisition times from suppliers such as Geospace Technologies, final delivery dates remain subject to formal vendor quotes. Following hardware delivery, all payload hardware incorporates a mandatory 2 to 3 month schedule buffer for environmental qualification testing—including Thermal Vacuum

(TVAC) chamber runs, vibration testing, electromagnetic compatibility (EMC) verification, and end-to-end sensor calibration—prior to final mounting on the rover chassis.

Transitioning to backup suppliers or secondary manufacturing pathways introduces notable schedule delays and cost risks that must be managed to protect the critical path. For instance, transitioning the LISTER heat-flow probe to an in-house build would add 4 to 6 months to the schedule due to the required re-engineering of the pneumatic spool assembly and full environmental flight re-qualification. Similarly, switching MSolo to an alternate commercial manufacturer like Extrel CMS requires 3 to 6 months of schedule reserve to accommodate custom interface redesign and re-qualification. Transitioning the MECA package to a backup university subcontractor introduces a 3 to 5 month delay for printed circuit board (PCB) redesign, component re-sourcing, and secondary calibration runs. Finally, maintaining a qualified backup MEMS fabrication facility for TENG serves to mitigate potential manufacturing delays associated with university cleanroom scheduling. To ensure these secondary supplier transition timelines do not compromise the launch readiness date, long-lead procurement contracts must be initiated early in Phase C to preserve adequate schedule margin.

4. *Procurement Summary Table*

Subsystem / Component	Procurement Method	Primary Supplier	Backup Supplier	Estimated Lead Time	Justification
MSolo Mass Spectrometer	MCOTS (Mo...	INFICON (NASA Adaptation)	Extrel CMS	12–18 Months	Derived from commercial INFICON unit; VIPER flight heritage; low cost & low technical risk.
LISTER Heat-Flow Probe	Prime Contr...	Honeybee Robotics	In-House Systems Integration	12–18 Months	Unique commercial capability for pneumatic drilling; deep Mars and lunar surface heritage.
MECA Electrometer Package	Technology ...	NASA JPL / KSC	Academic Lab / Space Lab Technologies	8–12 Months	Deep NASA institutional heritage (Mars Phoenix Lander); cost savings via inter-agency build.
Seismic Geophones	COTS (Com...	Geospace Technologies	Paulsson Inc.	6–9 Months	Commercial availability (GS-ONE LF & GSB series); Apollo experiment

					heritage; low procurement cost.
TENG	University-L...	Georgia Institute of Technology	NASA / JPL MEMS Cleanroom	15–24 Months	Custom microfabrication research partnership; low baseline R&D cost for novel technology demonstration.
Mounting Hardware & Harnesses	Build-to-Prin...	Land Aero Group LLC	TOMI Engineering	4–6 Months	Custom internal manufacturing to guarantee precise structural fit and mass/volume optimization.

Figure 62 :Payload Subsystem Procurement Summary Table

2.2. Interface Control

System Architecture Overview

The Lunar Surface Vehicle System consists of five primary subsystems: Mechanical, Power, Command and Data Handling, Thermal Control, and Payload. Mechanical provides the rover structure and mobility platform, Power generates and distributes electrical energy, CDH manages commands and data, Thermal Control protects hardware from the lunar environment, and Payload performs the mission's science measurements. Together, these subsystems support rover traversal, science operations near the Exploration Zone, system health monitoring, and return of science data back to Earth.

System Block Diagram Narrative

The system block diagram's purpose is to show the primary power, data, thermal, and mechanical interfaces between rover subsystems. The Power subsystem supplies electrical power to Mechanical, CDH, Thermal, and Payload hardware. CDH distributes commands through the rover's CAN network and receives subsystem telemetry and science data from Payload for storage and downlink. The Thermal subsystem regulates temperature sensitive hardware throughout the rover and receives electrical power and

control commands. The Mechanical subsystem provides the structural mounting and deployment interfaces required by the other subsystems. CDH communicates with Earth through the X-band Direct-to-Earth link. The most critical interfaces are the Power connections and the CDH data paths, since the loss of either would prevent rover operation and thus scientific measurements. These interfaces allow the five subsystems to operate together during rover traversal, science collection, thermal management, and data return.

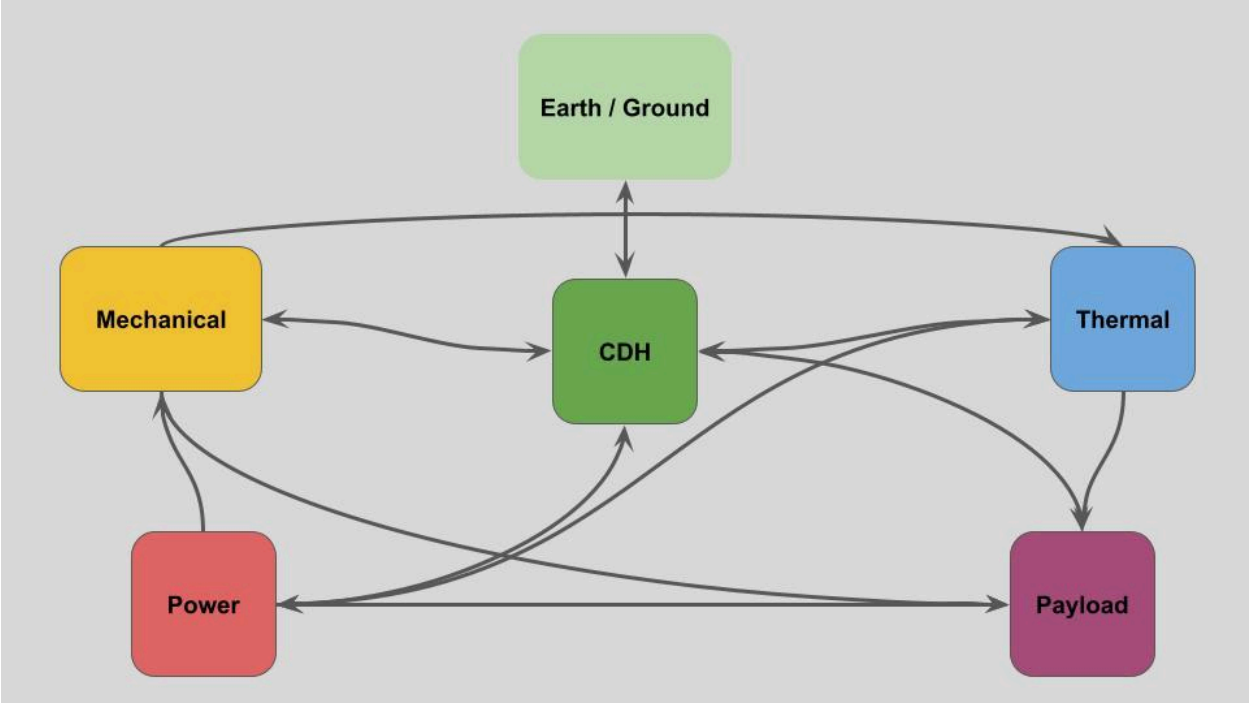


Figure x: System Block Diagram

N^2 Chart Narrative

The N^2 chart identifies the primary interfaces and dependencies among the Mechanical, Power, CDH, Thermal, and Payload subsystems. Rows represent the subsystem providing a resource or information, while columns represent the subsystem receiving it. Major interfaces include electrical power supplied by EPS, commands and synchronization from CDH, thermal management from TCS, mechanical mounting and deployment support from Mechanical, and science data returned by Payload. The Power–Payload interface is critical because payload operations, particularly LISTER drilling and MSolo sampling, create significant and time-dependent electrical loads that must be coordinated with available battery energy. The CDH–Payload interface provides instrument commands,

synchronization, telemetry, and science-data transfer required to execute and interpret measurements. The Mechanical–Payload interface is particularly important for LISTER deployment and penetration, geophone emplacement, and instrument mounting because these interfaces must accommodate operational loads without exceeding the rover's structural or stowed-envelope constraints. The Thermal–Power and Thermal–Payload interfaces must also be coordinated because heater demand increases during PSR operations while instrument temperature limits constrain when measurements can be performed.

Provider ↓ / Receiver →	**MECH**	**EPS**	**CDH**	**TCS**	**PAY**
MECH	MECH	Battery, solar array, & PDU mounting	Avionics/ GNC mounting & alignment	Mounting surfaces & conductive paths	Instrument mounting & deployment
EPS	Motor and actuator power	POWER	Avionics/ comms power & telemetry	Heater & controller power	Instrument power
CDH	Drive/ GNC commands via CAN	Power mode & load commands	CDH	Thermal mode commands	Instrument commands & time syncing
TCS	Structural thermal control	Battery thermal managemen t	Temperature & thermal health telemetry	THERMAL	Instrument thermal control
PAY	Deployment/ penetration mechanical loads	Electrical load demand	Science data	Instrument heat loads	PAYLOAD

Figure 64: Engineering N^2 Chart Subsystem Interfaces

3. Science Mission Plan

3.1. Science Objectives

The mission's science objectives are organized around two complementary goals, which are to characterize the lunar environment and evaluate environmental conditions relevant to sustained human activity at the lunar south pole. Under the Science Exploration Goal, the mission will characterize the thermal properties and heat flow of regolith near PSRs to determine their influence on volatile preservation and will quantify the abundance, isotropic composition, and associated volatile chemistry of water ice deposits within PSRs. These objectives establish the fundamental relationship between the extreme thermal environment of the lunar south pole and the volatile resources preserved there. Under the Human Exploration Goals, the mission will determine the thermal energy storage capability of lunar regolith, demonstrate NeuPAN based autonomous rover performance, evaluate electrostatic charge harvesting as a supplemental power source, characterize seismic activity, and characterize the surface electrostatic environment and charge accumulation near PSR boundaries.

These objectives were selected because they connect the scientific characterization of Nobile Rim 2 with the environmental and operational information needed for future lunar surface activity. Thermal and volatile measurements address the resource potential and stability of PSR materials, while thermal storage, mobility, and electrostatic harvesting objectives evaluate technologies or surface properties that could support future operations. Seismic and electrostatic measurements further characterize environmental conditions that may present hazards to equipment and surface systems. Together, the objectives support the overall mission goal of characterizing the lunar south polar environment while assessing how its resources, terrain, and environmental hazards could affect future robotic and/or human exploration.

Characterizing the thermal and volatile environment near lunar south pole PSRs is scientifically important because the low temperatures of these regions allow volatile compounds to remain stable over long periods of time. Diviner observations have demonstrated the presence of exceptionally cold south polar cold traps, establishing the thermal condition necessary for volatile preservation (Paige et al. 2010). Subsequent orbital observations provided direct evidence of exposed water ice within lunar polar cold traps as well, while also showing that ice is not uniformly distributed among them (Li et al. 2018). Measuring subsurface temperatures, thermal conductivity, and volatile abundance and composition at Nobile Rim 2 can therefore determine how local thermal conditions influence volatile stability and reveal spatial variations in the composition and potential accessibility of polar resources. These measurements directly address the mission's

primary thermal and volatile objectives established in the STM.

The remaining objectives extend this characterization towards conditions important for sustained lunar operations. Measuring regolith thermal energy storage behavior evaluates whether local material properties could support future thermal storage systems; peer reviewed modeling has identified lunar regolith as a potentially useful component of such systems (Liu et al. 2023). Evaluating NeuPAN mobility performance provides data on autonomous traversal in the uneven and obstacle rich terrain expected during lunar exploration, where reliable autonomous path planning is an important capability. The TENG objective evaluates whether mechanical rover activity can be converted into supplemental electrical energy, an approach supported by recent lunar regolith TENG experiments (Yohannan et al. 2026). Finally, seismic measurements can characterize local ground motion hazards associated with active lunar faults, while electrostatic measurements can determine surface potential and charging conditions that influence dust behavior and surface hardware. Both phenomena have been identified in peer reviewed studies as relevant concerns for south polar exploration and lunar surface operations (Mishra and Senthil Kumar 2022, Halekas et al. 2008).

The Science Traceability Matrix links each mission objective to specific measurements and planned observations. The thermal objective is traced to subsurface temperature profiles and thermal conductivity measurements, while volatile characterization is traced to mass spectrometer measurements of water and other volatile species. The Human Exploration objectives are traced to measurements of regolith thermal energy storage, NeuPAN navigation performance, TENG voltage and power output, seismic activity, and surface electrostatic conditions.

For example, the STM traces the Science Exploration Goal of characterizing the present day lunar environment to the objective of quantifying the abundance and chemistry of PSR volatiles. That objective is translated into the physical parameter of regolith chemical composition and then into the observable mass-to-charge peaks and relative signal amplitudes for H2O, CO2, NH3, CO, CH4 and H2S at less than 100 ppm sensitivity. These measurement requirements lead to MSolo, whose 1-100 m/z unit mass resolution range is designed to detect these targeted species, and ultimately to MR-03, which requires the mission to characterize the abundance, isotopic composition, and volatile chemistry of water deposits within PSRs. This progression demonstrates direct traceability from the high level science goal through a measurable observation, instrument capability, and mission requirement.

The payload suite directly supports the mission's primary science objectives by combining chemical, thermal, seismic, and electrical characterizations of the lunar south pole regolith to resolve the origin, distribution, and preservation of polar volatiles while

assessing local operational hazards.

The Mass Spectrometer Observing Lunar Operations, (MSolo), conducts in-situ mass spectrometry to measure volatile chemical composition, isotopic ratios, and mass-to-charge distributions of volatile species such as H_2O, CO_2, and CO, which quantifies volatile abundance and distribution in permanently shadowed regions to establish the origin and stability of volatile deposits. Meanwhile, the Triboelectric Nanogenerator, (TENG), measures mechanical-to-electrical energy conversion efficiency, voltage and current output under dynamic movement, and localized surface electrostatic charge accumulation, directly evaluating electrostatic charge harvesting for supplemental energy generation and assessing triboelectric effects caused by dust interaction during mobility.

To evaluate the subsurface, three passive seismic Geophone Arrays, or GA units, record three-component surface seismic wave arrivals, ground velocity, ambient seismic noise, and surface acoustic vibrations, processing these acoustic arrivals to measure regolith mechanical properties and subsurface layering structures while differentiating local physical terrain hazards. Concurrently, the Lunar In-Situ Thermal and Electrochemical Determination instrument, (LISTER) measures subsurface vertical temperature gradients, thermal conductivity, thermal diffusivity, and heat flux as a function of depth, enabling the team to quantify the heat capacity and thermal energy storage of regolith near cold traps to reveal how thermal environments preserve ancient volatile compounds over geological time.

Finally, the Materials Adhesion and Environmental Components Analyzer, (MECA), completes the suite by assessing regolith physical properties, particle size distributions, abrasive wear metrics, and dust electrostatics through targeted material exposure and microscopy observations, capturing critical environmental interaction metrics that answer key questions regarding local regolith reactivity, environmental degradation, and long-term hardware survivability for polar exploration infrastructure.

3.2. Experimental Logic, Approach, and Method of Investigation

The mission investigates how the environmental conditions of lunar south polar permanently shadowed regions influence volatile stability, regolith thermal behavior, surface hazards, and the feasibility of future robotic or human operations. The investigation will measure subsurface temperature and thermal conductivity, volatile abundance and composition, regolith thermal energy storage behavior, seismic activity, surface electrostatic conditions, electrostatic energy harvesting performance, and NeuPAN based rover navigation performance. These measurements directly address the mission objectives established in the Science Traceability Matrix and characterize both the scientific

value and operational challenges of PSR and PSR-boundary terrain. Nobile Rim 2 supports this investigation by providing access to a nearby permanently shadowed region while retaining comparatively favorable landing slopes and solar illumination. Its close proximity to both illuminated and PSR terrain creates a strong environmental gradient that benefits several investigations simultaneously. The cold PSR environment provides LISTER with a scientifically relevant location for relating thermal properties to volatile stability and provides MSolo access to terrain where water and other volatiles are most likely to remain preserved. Rover and drilling activity allow both MECA and TENG to characterize electrical effects associated with regolith disturbance. The payload combines LISTER for subsurface thermal measurements, MSolo for volatile characterization, a geophone array for seismic measurements, a geophone array can help measure measured P and S wave velocities which can be of advantage in in helping to derive measurements of mechanical properties like Shear modulus and the Young Modulus of the regolith. These two mechanical properties can help measure slope stability of lobate slopes that can pose a risk of landslides due to seismic activity (Watters et al., 2024). MECA for surface electrostatic characterization, and a TENG system for electrostatic power generation testing, while the rover navigation system records NeuPAN mobility performance metrics. Together, the selected site and measurement systems allow conditions to be compared across illuminated, PSR boundary, and permanently shadowed terrain while maintaining access to an illuminated staging region for rover recovery and recharge.

1. ***Describe the mission's science operations in chronological order.***

01. Target acquisition and positioning

Following contractor provided landing at Nobile Rim 2, the rover will establish its position within the selected landing ellipse, complete initial vehicle checkout, and prepare for the traverse to the Exploration Zone. The landing ellipse was refined using JMARS around the highest ranked Nobile Rim 2 candidate region and has a 400 m major axis positioned entirely within terrain below the 5 degree slope requirement. The selected geometry also provides favorable solar illumination and places the PSR-facing portion of the landing ellipse within the 250 m of the intended Exploration Zone. These characteristics allow the rover to begin operations in illuminated terrain before moving toward the more thermally and energetically challenging PSR environment.

The Exploration Zone and primary approach route will be identified before landing using JMARS slope, illumination, topography, and PSR mapping. During surface operations, the rover will use its navigation cameras, hazard cameras, inertial measurement unity, wheel encoders, and NeuPAN based autonomous motion system to refine the preplanned route and avoid local obstacles that cannot be fully resolved from existing orbital data. Rover position and terrain geometry will be confirmed before stationary instrument

measurements begin so that each observation is associated with a known location relative to the PSR boundary. Three geophones mirroring the Apollo station 16 active seismic experiment would be spaced at 45 meters each and 144 meter round trip. It would be arranged in a linear array and 10 meters away from the rover and 101.8 meters off the craters rim of Nobile Rim 2. The coordinates where it could be placed would be 84.202 N, 60.572 E.The geophones would be planted vertically and make sure that they are mechanically coupled correctly.

02. Instrument checkout and calibration

Following landing and initial rover checkout, the payload instruments will complete in-situ health checks and calibration activities before nominal science collection begins. Each instrument will be checked for proper power draw, command response, telemetry, communications with the CDH subsystem, and acceptable operating conditions. MSolo will then use its onboard calibration gas system to verify instrument response and will collect background mass spectra to characterize signals produced by the instrument and rover environment before lunar volatile measurements are interpreted. LISTER will verify the electrical continuity of its Resistance Temperature Detector channels, pneumatic and deployment system status, and command response before penetration into the regolith. An ambient thermal baseline and controlled heater pulse will then be used to confirm that the temperature sensors respond consistently and that the expected thermal rise and decay behavior can be measured. These checks establish reference conditions that can later be compared with science measurements collected in and near the Exploration Zone. The MECA electrometer will establish its electrical baseline by checking sensing channel response, amplifier stability, sensor temperature, and background noise while the rover is in a relatively quiescent state. Internal reference and zero potential measurements will be used to confirm that later electrostatic signals can be distinguished from rover generated electrical noise or sensor drift. The TENG system will similarly establish a stationary electrical baseline with the rover wheels stopped before controlled wheel rotation is used to verify that measured voltage and current are repeatable and correspond to mechanical excitation. Initial health checks for the geophones to ensure it can have a safe deployment are the following: The geophone linear array must be mechanically coupled correctly within the surface that is chosen to be deployed. Previous calibration checks within the spring/coil within the geophone sensor must be checked and tailored for the lunar surface and tested in an analog environment to make sure frequency, gain and shock parameters can withstand impact without damaging the geophone sensor.

Instruments will be considered ready for science operations once they respond properly to commands, maintain stable telemetry, complete their required calibration or reference measurements, and produce repeatable outputs that can be distinguished from

background noise. These post landing procedures are intended to confirm that the instruments remain functional after landing and to establish the local reference measurements needed for reliable interpretation of data collected at Nobile Rim 2.

03. Data collection sequence

Science collection begins during the traverse toward the Exploration Zone, with NeuPAN performance metrics and TENG electrical measurements collected during rover motion. At designated science stops, background measurements are collected first, followed by MSolo volatile sampling before LISTER drilling to avoid contamination from LISTER's pneumatic nitrogen system. LISTER then performs subsurface thermal measurements to 1 m depth while MECA monitors electrostatic conditions associated with regolith disturbance, and the geophone array records seismic activity throughout the science phase.

04. Data handling

Science and engineering data are time tagged, packetized, and stored in onboard nonvolatile memory by the CDH subsystem until successful transmission is confirmed. The current architecture provides at least 48 GB of raw storage and 32 GB of usable nonvolatile storage, with science data transmitted during Direct-to-Earth communication windows using the Iris radio and X-band downlink system.

05. Repeat observations or follow-up measurements

Measurements are repeated at multiple science stops so that changes in thermal, volatile, electrostatic, seismic, and mobility conditions can be compared across the Exploration Zone. Repeated background instrument measurements can also help distinguish actual environmental variation from instrument noise, contamination, or temporary operational effects.

06. End-of-mission or final data pass

Near the end of the mission, the rover transitions to a low power passive monitoring phase, with the geophone system continuing seismic observations at high bit rate while remaining instruments are preserved as power allows. Before End of Life, stored science and engineering data are prioritized for a final downlink, ensuring that the highest value thermal, volatile, environmental and seismic measurements are returned before the rover enters its permanent low power state.

3. Experimental Logic

The operational sequence is designed to maximize science return while limiting the power, thermal, mobility, and data risks associated while operating near permanently shadowed terrain. Beginning each cycle in illuminated terrain allows rover health verification to occur before energy is committed to the PSR. During the traverse, NeuPAN and TENG measurements can be collected without delaying stationary science operations. After reaching the exploration zone, quiet background measurements should be collected before major mechanical activity begins, followed by MSolo volatile measurements before LISTER deployment. This sequencing protects MSolo measurements from contamination caused by LISTER’s pneumatic excavation system, which uses a finite 130 g nitrogen supply, while also conserving that gas until the rover has confirmed a scientifically valuable drilling location. LISTER can then perform its subsurface drilling and thermal measurements, while MECA operates during drillings to characterize electrostatic charging associated with regolith disturbance. Higher power measurements are sequenced rather than conducted simultaneously because the baseline PSR sortie has a 1.1% energy margin.

The payload measurements are complementary instead of independent. Where LISTER characterizes the thermal environment controlling volatile stability, MSolo determines which volatile species are present and their subsequent abundances. Electrostatic measurements characterize conditions that may affect equipment and surface operation, while TENG evaluates whether charge accumulation can be exploited as a supplemental energy source. Seismic measurements characterize local ground motion and environmental activity, and NeuPAN mobility measurements quantify how effectively the rover can traverse the same PSR terrain being characterized. Repeating these measurements across different locations and environmental conditions allows the mission to relate resource potential and physical properties to the operational hazards future lunar missions might experience.

4. Site Context and Visuals

Nobile Rim 2 supports the experimental plan because it combines access to permanently shadowed terrain with nearby illuminated terrain that can serve as a staging, recovery, communicative, and recharge area. The region was selected initially from published lunar south pole landing site analyses and was refined afterwards through mission specific JMARS investigation. Rather than adopting the published highest ranking coordinate without further analysis, local terrain was evaluated against the customer landing constraints.

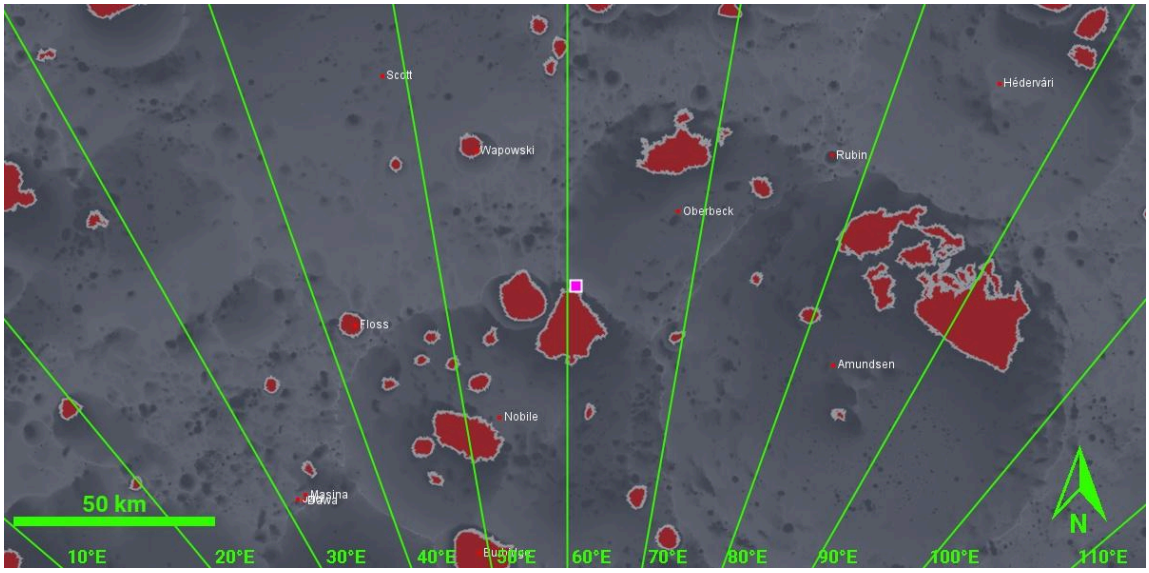


Figure 65: Surrounding Nobile Crater Environment

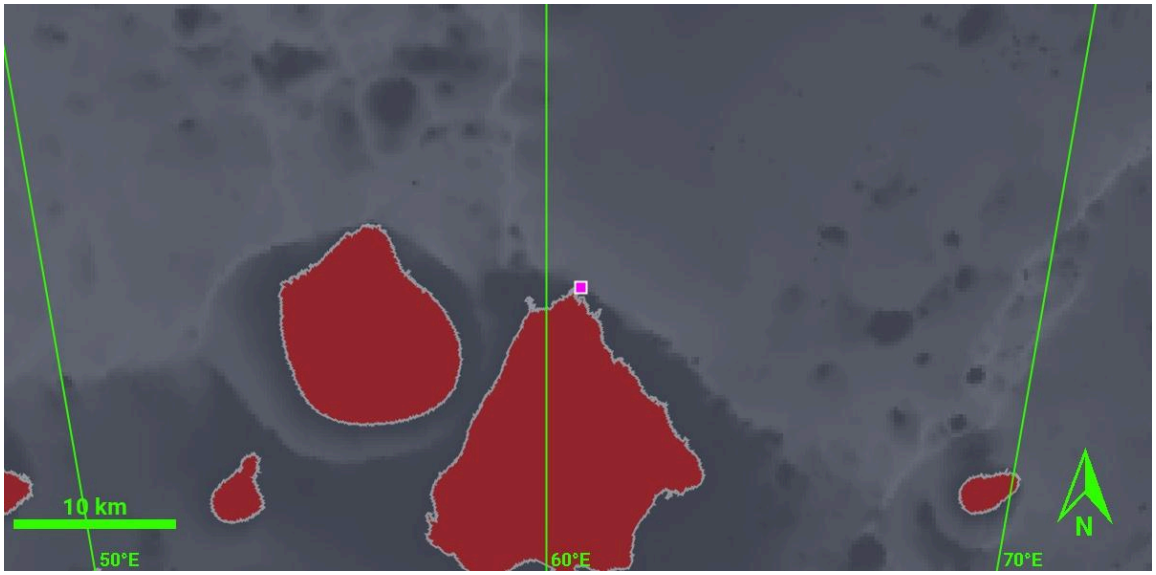


Figure 66: Nobile Rim 2 Local Environment

Figure X shows the surrounding context of the Nobile crater, and the pink dot is a pixelated representation of the landing zone. This helps contextualize the local environment where the mission will take place. The Nobile crater is surrounded by various significant craters, such as Amundsen, Floss, Wapowski, and Oberbeck. Figure X shows the more detailed Nobile Rim 2, where the landing zone is. Again represented by a pink square in Figure X, the landing zone sits right outside the permanently shadowed region and demonstrates the context in which the mission aims to take place. Important features from this figure show the rim of the Nobile crater where the mission will land. Next to this rim is another, smaller crater that belongs to the same Nobile Rim 2 region.

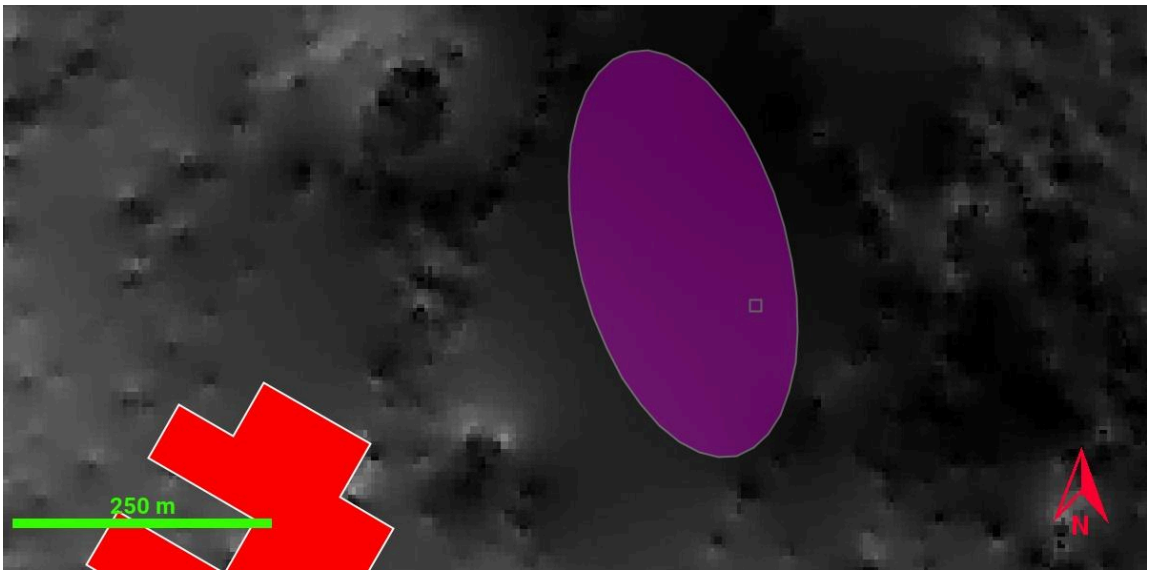


Figure 67: Landing Zone Ellipse

The initial 200 m radial circular interpretation intersected several regions with slopes greater than 5 degrees, and therefore could not satisfy the requirement that the landing area slopes are less than 5 degrees. A 400 m major axis landing ellipse was instead fitted to the nearby, low slope corridor. The resulting ellipse shown in Figure X remains within terrain below the required slope threshold, retains favorable illumination, optimizes direct to Earth communication, and places its PSR facing portion within 250 m of the intended Exploration Zone. Figure X also showcases the adequacy of the landing zone’s slope requirement. Darker regions represent lower slopes, while white regions represent slopes greater than 5 degrees. From the figure, the landing ellipse resides within a darkened slope area.

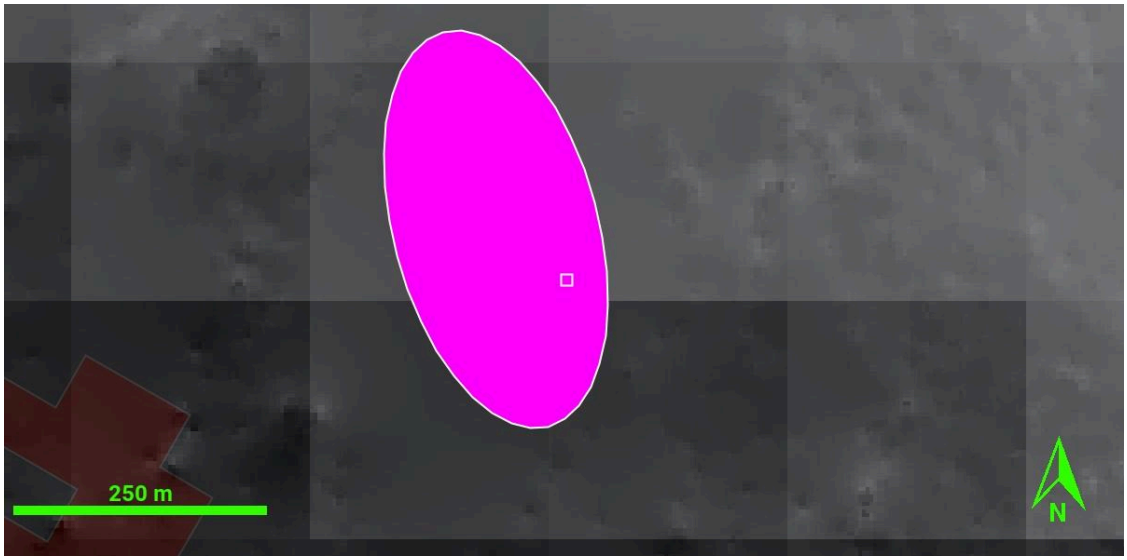


Figure 68: Landing Zone Illumination Showcase

The above figure, Figure X, shows another customer and engineering constraint being met. The shaded squares represent percentage values of illumination where lighter means higher percent illumination and darker means lower percent illumination. Within the landing ellipse, the lowest percentage is around 14% and the highest reaches 35%. These values will allow for surface operation recharging, and during planned mission time-frames, the sun will be easily visible in this region outside of PSR terrain.

These JMARS results directly shaped the experimental plan. The landing geometry allows the rover to begin in illuminated terrain while keeping the science target close enough that repeated excursions can be performed without excessive traverse distance or energy expenditure. The nearby PSR boundary also enables measurements to be compared across illuminated, transitional, and fully permanently shadowed environments. This distinction is valuable for thermal, volatile, electrostatic, and mobility objectives.

3.3. Payload Success Criteria

1. ***Payload Success Overview***

The Payload subsystem success criteria define the core operational, power, thermal, data, and mechanical baselines required to validate the mission's scientific payload on the lunar surface. The successful execution of payload operations directly enables the achievement of primary mission science objectives by ensuring that the suite of instruments including the Mass Spectrometer Observing Lunar Operations, Triboelectric Nanogenerator, Lunar Instrumentation for Subsurface Thermal Exploration with Rapidity, Materials Adhesion and Environmental Components Analyzer, and Geophone linear Array can reliably perform chemical, thermal, seismic, and mechanical characterizations of the

regolith. By executing scheduled surface traverses and stationary science acquisitions within designated permanently shadowed regions and adjacent zones, the payload suite gathers critical empirical data on volatile composition, structural regolith profiles, thermal transport, and electrostatic dynamics. Mission success is bounded by a minimum science return threshold, which requires successful volatile identification and quantitative surface analysis across a minimum operational profile without exceeding hardware performance boundaries. Achieving this minimum return ensures that even under constrained surface operations, core science traceability goals are met, establishing the origin, abundance, and preservation mechanisms of lunar polar ice while proving operational survivability for future exploration.

Detailed Payload Success Criteria

The payload subsystem must operate within nominal continuous power profiles, drawing approximately 40 W while the rover is traversing and 42 W while stationary, with allowable operational fluctuations strictly bounded between 40 W and 45 W. The subsystem power delivery network must accommodate a transient peak draw of 161.5 W, which is duty-cycled and managed as a power electronics sizing requirement rather than a continuous energy draw. During mobility segments, the Navigation Sensor Package remains the primary active payload element to ensure exact pathfinding and payload position logging. All operations must conform to a standardized 6.07-hour sortie structure, consisting of 0.4 hours of driving and 5.67 hours of stationary science acquisition. Total energy consumption per sortie must not exceed 1,064 Wh against a usable surface battery capacity of 1,296 Wh, maintaining a minimum energy margin of 232 Wh, or approximately 22 percent.

During the execution of each 6.07-hour sortie, the Mass Spectrometer Observing Lunar Operations, or MSolo, must successfully complete a minimum of 20 minutes of active mass spectrometry acquisition at each of two planned sampling stops to capture volatile species such as water, carbon dioxide, and carbon monoxide along with their isotopic distributions. For subsurface thermal profiling, the Lunar Instrumentation for Subsurface Thermal Exploration with Rapidity, known as LISTER, must complete two drill cycles during its roughly 5-hour stationary allocation. Because LISTER descends approximately half a meter per cycle, these two cycles achieve the required total depth of one meter into the regolith to measure vertical thermal gradients, thermal conductivity, and heat flux. Simultaneously, the subsystem must successfully emplace three passive seismic geophone units in a linear 45-meter configuration during the sortie to perform acoustic wave arrival and ground velocity tracking throughout the lunar subsurface.

Data acquisition and telecommunications require the payload network to sustain a continuous science acquisition stream of 16 kbps during active operations. A single full

sortie yields approximately 43.7 MB of raw science data, totalizing approximately 55 MB once engineering data and telemetry overhead are included. To ensure that a missed or delayed downlink opportunity does not cost the mission science data, the payload flight computer must maintain at least 110 MB of onboard non-volatile storage, which accommodates two full sorties of data. The telecommunications architecture supports at least one Direct-to-Earth downlink per sortie, operating on a recurring cycle roughly every 6 to 7 hours.

Thermal control limits require MSolo to maintain an operational temperature of at least 281 K, operating within a general range of -10 °C to 45 °C, which represents the highest minimum operating temperature requirement among all payload instruments. LISTER payload elements must maintain a steady-state operational temperature of approximately 30 °C during active drilling and subsurface measurement sequences. Regarding structural mass allocations, the total integrated payload mass must strictly conform to an upper envelope of 27.55 kg. Individual geophone unit mass figures must be resolved to ensure compliance; while a 7.1 kg per unit figure results in a 28.4 kg total that exceeds the entire payload budget on its own, utilizing the 131 g per unit configuration reduces total array mass to roughly half a kilogram, easily fitting within system constraints. Additionally, geophone success criteria are defined against the true physical response of the hardware, establishing functional acoustic sensitivity starting around 4.5 Hz rather than the historical 0.1 Hz target associated with cut broadband seismometer hardware. Finally, mission-level spatial access is governed by system requirement SYS-05, requiring the payload suite to conduct characterization operations within an exploration zone located within 250 m of the landing site, explicitly excluding mechanical capability targets like MCH-04 from science success definitions.

Minimum Science Return for Mission Success

To achieve minimum primary mission success, the payload subsystem must execute at least one complete 6.07-hour sortie within 250 m of the landing site, delivering verified data streams across all core operational modes. Under this baseline, MSolo must complete at least one 20-minute analytical run at a single station to acquire mass spectra confirming the presence or absence of volatile species in the upper regolith. LISTER must complete a minimum of one 0.5-meter penetration cycle to record vertical thermal gradients and regolith thermal conductivity. The Geophone Array must successfully emplace units and record surface acoustic wave arrivals within its native frequency band starting at 4.5 Hz to bound upper regolith mechanical properties. To be able to record seismic activity throughout the regolith the estimated period established of the mission. Finally, the payload data system must capture, store, and transmit at least one primary data payload of approximately 55 MB back to Earth or preserve it in onboard memory without corruption,

ensuring that foundational insights into lunar polar regolith properties and volatile preservation are secured.

2. Instrument Success Criteria

The MSolo will be considered successful if it completes its health and calibration sequence and is able to produce mass spectra that can be distinguished from instrument and spacecraft background. The primary science deliverable will include a time and location referenced dataset of volatile measurements obtained during rover operations. NASA identifies the MSolo's primary function as indicating low-molecular weight volatiles and studying gases associated with lunar surface and subsurface materials. Therefore, a successful MSolo measurement will directly support the objective of characterizing the composition and distribution of volatiles near the lunar south pole. The ideal outcome would be the detection and characterization of multiple volatile species across more than one terrain type. A stretch goal would perhaps be the identification of spatial trends or isotopic signatures that provide additional information about volatile sources and behavior.

The TENG will be considered successful if it demonstrates repeatable conversion of electrostatic charge from lunar regolith into measurable electrical output. Voltage and current measurements must be sufficient to calculate harvested power and cumulative energy.. The minimum science deliverable in this instance would be a demonstration of the relationship between mechanical rover activity and measurable TENG output. The optimum success would be repeatable electrical output across multiple charged-regolith conditions and terrain locations with a quantitative relationship between electrostatic excitation and harvested power.. The recent development of lunar-regolith-simulant-based TENG provides a relevant experimental basis for this objective. Peer-reviewed work demonstrates an all-3D-printed TENG using lunar regolith simulant incorporated into PLA and reported approximately 17.4 (±) 0.4 V open-circuit voltage and 0.96 (±) 0.2 μA short-circuit current at 10 Hz. This provides useful evidence that the lunar-regolith based TENG construction is technically and feasibly possible and may provide a useful reference point for ground testing and mission expectations.

The primary objective of LISTER is to quantify subsurface thermal gradients, regolith thermal conductivity (k), and geothermal heat flow at the Nobile Rim 2 target site. Successful operation requires pneumatic probe deployment into the regolith to a target depth of 1.0 m, full activation of Resistance Temperature Detector (RTD) arrays, and execution of active heating pulse experiments across a minimum operational duration of 14 lunar days. Expected science deliverables include calibrated depth-versus-temperature profiles, calculated thermal conductivity curves, and local heat flux estimates. These deliverables directly support Mission Objective 1 by resolving thermal energy transport within polar regolith. The scientific significance of these results lies in providing the first

high-precision, in-situ polar heat flow measurements, constraining models of lunar thermal evolution and cold-trap volatile stability.

The primary objective of the MECA electrometer is to characterize electrostatic field magnitudes, surface electrical potentials, and triboelectric charging mechanisms induced by regolith movement and plasma interactions. Criteria for successful operation include continuous sensing electrode voltage sampling, noise-floor stabilization, and real-time logging during mechanical drilling activity. Expected science deliverables consist of electric field intensity time-series data, charge accumulation profiles during mechanical contact, and local surface potential curves. These data support Mission Objective 2 by establishing empirical baselines for high-latitude surface-plasma dynamics. The results hold major scientific significance for understanding dust levitation dynamics and providing critical operational risk frameworks for future surface systems and human exploration hardware.

The geophone sensors would be arranged in a linear array and would be successful if the instrument records seismic velocities passively at a continuous sampling rate that is consistent with a sensitivity rate less <10 ^6 m/s^2. Expected science deliverables is to retrieve wave velocities from passive monitoring and to be able to perform telemetry to earth in order for data to be later processed. Also, the difference between acoustic signals from the rover and seismic activity can aid further research on how to enhance and optimize the geophone sensor hardware for future objectives in measuring different seismic activity on the lunar surface. This data supports mission objective 1 in determining natural seismic activity from induced acoustic signals from the rover in order to differentiate. Also this instrument aligns with the human exploration goal.

3. Instrument Failure Modes

MSolo failure modes include loss of power or communication, detector or ion-source failure, sampling-system malfunction, contamination of the measurement path and failure of the calibration gas system. MSolo uses a quadrupole mass spectrometer with Faraday cup and electron multiplier detectors as well as an ion source and an onboard calibration gas system. A failure of the detector or ion source would prevent the instrument from producing usable mass spectra and would directly affect its ability to identify the mission's target volatile species. Failure of the calibration gas system would increase uncertainty in the measurements because the instrument would lose its primary onboard reference for checking and adjusting its response. Another failure mode discussed involves contamination as a potential data-quality risk. Because MSolo is designed to measure gases around the rover, significant contamination could produce false-positive detections or obscure low-abundance lunar volatiles. If the MSolo were to be out-of-commission, it would threaten the mission's ability to carry out Science Objective 2, as MSolo is the primary instrument selected for volatile characterization. Mitigation will include post-landing health

checks, calibration measurements, background measurements, continuous instrument telemetry monitoring and comparison of anomalous measurements with known space operations.

A TENG failure mode could occur due to mechanical wear or damage to the triboelectric surfaces, loss of mechanical coupling to the rover wheel, electrical connection failure, short circuits or leakage. Measurement-electronics failure and degradation caused by dust or environmental conditions could also create a failure mode. These occurrences could either prevent the TENG from generating an electrical response or make the measured output unreliable. Because the proposed TENG is intended to harvest electrostatic energy associated with charged regolith, loss of contact or coupling with the regolith would be particularly important as the generator could remain electrically functional while receiving insufficient electrostatic excitation. A TENG failure therefore, would primarily affect the energy-harvesting technology demonstration over the mission's primary volatile-science objectives. If the TENG produces no measurable electrical response under charged-regolith conditions, the mission would be unable to determine the feasibility of electrostatic energy harvesting from the lunar surface. The remaining instruments could continue operating independently. Mitigation includes pre-launch mechanical and electrical testing, post-landing continuity checks, stationary electrical-baseline measurements, repeated measurements under controlled regolith-contact conditions and simultaneous recording of relevant rover operational parameters. The TENG is also to be evaluated against ground-test measurements so that low output caused by an actual device failure can be distinguished from low output caused by insufficient mechanical excitation.

Potential failure modes for LISTER include pneumatic drill binding or mechanical jamming during regolith insertion, cable deployment entanglement, or RTD sensor array lead open-circuits. A mechanical jam prior to reaching the target 1.0 m depth would prevent full vertical profiling, reducing the accuracy of subsurface heat flow calculations. If insertion fails entirely at the surface, subsurface thermal gradient measurements cannot be acquired, directly impacting the primary baseline of Objective 1. Mitigation strategies include autonomous drill retraction and re-pumping sequences, variable torque and pressure commands, and redundant thermal sensors along the upper probe housing to preserve partial depth measurements at shallower levels (0.3–0.5 m).

Potential failure modes for the MECA electrometer include electrode saturation due to sudden high-voltage arc discharges, physical surface contamination by thick dust layers, or amplifier channel failures. Electrode saturation or dust blinding would degrade voltage sensitivity and produce clipped or noisy field magnitude data, limiting the ability to capture subtle ambient electric field fluctuations. While this would reduce the fidelity of Objective 2,

overall mission operations would remain intact. Mitigation strategies involve high-voltage protection circuitry with auto-resetting shunts, multi-channel electrode redundancy positioned across varied geometric locations on the lander, and digital filtering algorithms to isolate sensor noise from true plasma potential signals.

Potential failures of the geophone sensors can include not performing a correct mechanically coupling within the lunar surface. Also geophone sensors could be damaged upon impact if not tailored with the correct vibration/shock parameter. Another factor to take into consideration is overheating due to the lunar temperature fluctuations in temperature when it reaches sunrise. However, the outer coating and specific hardware can be tailored to survive these temperatures in order to reduce overheating. Also since the regolith has abrasive material it could have an issue with the vibration damping when planted vertically. However a quote to the manufacturer could be done to inquire over material that has an outer coating more resistant to abrasive material by running tests on lunar simulant material and to make adjustments with the spring/coil to better tune the frequency to differentiate ambient noise from seismic activity.

4. Success Criteria Summary Table

Category	Minimum Success	Optimum Success	Stretch Goal
LISTER Payload	Deploy probe to a minimum subsurface depth of 0.5 m; capture passive temperature profile across 1 diurnal cycle.	Full deployment to target depth of 1.0 m; execute active thermal conductivity tests across 14 lunar days.	Deep penetration (> 1.5 m); continuous thermal gradient logging spanning multiple lunar day/night transitions.
MECA Electrometer Payload	Measure ambient surface potential and baseline electric field magnitudes during initial deployment.	Log dynamic triboelectric charge build-up during drilling and monitor continuous electrostatic variations over 14 days.	Capture high-rate, multi-channel electric field transients during lunar terminator and solar storm plasma events.
MSolo Mass Spectrometer Payload	Complete health/calibration and at least one 20-minute mass spectrometry acquisition at a sampling station; produce distinguishable spectra	Finish at least two 20-minute acquisitions at planned sampling stops; identify and characterize volatile species and isotopic distributions across sampled	Detect low-abundance volatile species across multiple terrain types and identify spatial trends or isotopic signatures which indicate volatile sources and

	for volatile species.	locations.	transport.
TENG Payload	Demonstration of repeatable lunar regolith energy conversion during rover movement, produce measurable voltage/current output.	Quantify harvested electrical power and cumulative energy across representative rover speeds and terrain conditions, establishing a repeatable relationship between regolith excitation and TENG output.	Demonstrate enhanced energy harvesting across a wider range of wheel speeds and terrain conditions, establishing performance suitable for meaningful supplemental rover power generation.

Figure 69: Insert figure caption here

5. Mission Science Impact

The integrated payload suite comprising the Mass Spectrometer Observing Lunar Operations, Triboelectric Nanogenerator, Geophone Arrays, Lunar In-Situ Thermal and Electrochemical Determination instrument, and Materials Adhesion and Environmental Components Analyzer directly advances the mission's primary science goals by delivering a comprehensive, multi-disciplinary characterization of the lunar south pole regolith. By combining in-situ chemical mass spectrometry, high-resolution subsurface thermal profiling, seismic velocity profiling, and surface electrostatics analysis, the complete payload addresses longstanding questions regarding volatile origin, migration, and long-term retention within permanently shadowed cold traps. The expected science deliverables, which include isotopic ratios of trapped ice, one-meter vertical thermal conductivity profiles, acoustic velocity distributions, and granular adhesion metrics, significantly deepen our understanding of the Moon's volatile compounds, regolith physical structure, and surface-environment interactions. Achieving all payload success criteria delivers exceptional scientific and operational value, as it not only maps the chemical and physical state of polar ice deposits to model ancient solar system volatile transport, but also establishes fundamental geotechnical and environmental baselines necessary to design durable hardware, harvest local resources, and ensure long-term human survivability for future lunar surface exploration.

3.4. Testing and Calibration Measurements

1. Testing & Calibration Overview

Following landing and rover deployment, the science payload will complete instrument health checks, baseline measurements, and in-situ calibration before nominal science operations begin. These activities verify that each instrument remains functional after launch, transit, and landing and establish reference measurements under the actual lunar environmental conditions at Nobile Rim 2. Calibration measurements are used to identify instrument offsets, background signals, electrical noise, or environmental effects that could otherwise be mistaken for scientific observations. This process differs from pre-launch Verification and Validation, which demonstrates that the instruments satisfy their design and performance requirements before flight. In-situ testing instead confirms that those capabilities have been preserved after arrival and establishes the lunar surface baselines needed to interpret the mission data correctly.

2. Instrument Testing & Calibration

The MSolo will undergo an initial health and functionality check after landing and deployment to verify power, communication, telemetry and overall instrument status prior to science measurement starting. MSolo operates as a quadrupole mass spectrometer over approximately 0-100 m/z and uses both a Faraday cup and an electron multiplier for detection. Following its health check, MSolo will perform its onboard calibration procedure using the calibration gas system. This system has been identified as a means for MSolo to check and adjust its measurements on the lunar surface. After calibration, the background measurements will be collected prior to nominal science operations to establish the baseline contribution from the instrument, rover and lander environment. These background measurements are important because MSolo must distinguish lunar volatiles from gases introduced by spacecraft operations. The primary reference measurements, therefore, will be the known calibration-gas response and repeated background spectra. During nominal operation, MSolo is expected to produce stable and repeatable mass spectra capable of resolving the nominal mass locations of the mission's target volatile species. MSolo is to be considered operating correctly if it responds to commands, maintains stable telemetry, completes the calibration sequence, produces a consistent response to the calibration gas, and produces repeatable background measurements. These checks will establish that the instrument is functioning properly before measurements are used for volatile characterization.

TENG testing begins with an electrical continuity and functionality check of the generator, wiring and measurement electronics. A stationary measurement will then be

collected with the rover wheels stopped to establish the electrical noise and baseline output of the system in the absence of intentional mechanical excitation. In-Situ characterization will then be performed under controlled contact with charged regolith with TENG voltage and current recorded simultaneously with river operating state and relevant environmental conditions. Regolith contact conditions and electrostatic measurement will provide the primary reference for distinguishing generated electrical signals from measurement noise. Repeated measurements under comparable regolith and operational conditions will be used to determine the repeatability of the TENG response. Therefore, TENG will be considered to be operating correctly if it produces repeatable voltage and current signals under charged-regolith conditions where the signals are distinguishable from the stationary noise baseline, and measured output provides a consistent relationship between electrostatic excitation and harvested electrical power.

LISTER testing begins with power-on telemetry verification of internal electronics, motor drive resistance, thermal sensor loop continuity, and pneumatic gas valve actuation. An ambient pre-penetration thermal measurement will be collected at the surface to record initial probe thermal equilibration against ambient surface temperature models before mechanical excavation begins. In-situ calibration occurs during the initial penetration cycle; as the pneumatic gas jet excavates regolith and the spooling motor lowers the probe, thermal resistance detectors (RTDs) embedded in the needle sensor pause at target depths to establish zero-heat-flux baseline readings. The control variables are pneumatic gas pressure, motor insertion force, and active heating pulse power (e.g., 50 mW thermal pulses), while pre-drilling surface temperatures and thermal transport models serve as primary reference standards. Expected performance includes smooth mechanical descent per cycle and predictable thermal pulse dissipation into surrounding regolith. LISTER will be considered operating correctly if deployment mechanism telemetry confirms uninterrupted descent, sensor RTDs record stable thermal equilibration curves during pauses, and active heating sequences yield repeatable thermal conductivity response curves.

MECA testing begins with an initial health check verifying power delivery, electrometer bias voltage stability, optical camera alignment, and data bus communication. A stationary reference measurement will be acquired prior to surface operations by capturing optical images of an integrated high-contrast calibration target and recording open-air electrometer baseline voltages in the ambient lunar vacuum environment. In-situ calibration will be conducted by monitoring regolith particle accumulation on exposure plates and measuring electrostatic charge shifts during initial rover movement. Control variables include camera exposure times, illumination levels, and rover movement states, while the onboard optical calibration target and initial open-air electrometer readings serve

as primary reference baselines. Expected performance includes sharp optical image resolution capable of particle size discrimination down to micro-scale dimensions and clear electrometer response to dust electrostatics. MECA will be considered operating correctly if camera telemetry delivers focused target images, the electrometer records stable voltage signals distinguishable from ambient electronic noise, and optical sensors demonstrate repeatable data capture during surface exposure.

After landing and deployment from the geophone sensor, outer coating material must be checked to make sure the abrasive material from the regolith did not have any damage and the sensor is vertically planted correctly ensuring correct mechanically coupling. Also calibration testing to achieve fine tuning of frequency ranges that can passively monitor the seismic activity depends on testing until finding a frequency range that matches the resonant frequency of the Active Seismic experiment heritage sensor. This can achieve a better gain and frequency range of each sensor. Also coil resistance and shock resistant upon landing will heavily weigh on how resistant the sensor would be upon landing.

3. Data Quality Assessment

To ensure that collected science data are accurate, repeatable, and scientifically valid, the team executes a sequential data quality assessment protocol across the payload suite. First, data from the Lunar Instrumentation for Subsurface Thermal Exploration with Rapidity (LISTER) are validated by comparing passive surface temperature profiles against orbital thermal infrared radiometer data from the LRO Diviner instrument at the Nobile Rim 2 site, while active thermal conductivity calculations are cross-referenced against laboratory conductivities of Apollo regolith simulants tested under lunar vacuum conditions. Next, measurements from the Microstructure and Electrostatic Carrying Array (MECA) electrometer package are validated by comparing open-circuit sensor baselines against theoretical solar wind plasma potential models and verifying that triboelectric charge spikes correlate directly with active mechanical drill motion logged by lander telemetry. Msolo data will be considered scientifically valid when measured mass peaks are consistent with the known response of the instrument to its calibration gas and remain stable across repeated measurements. Background spectra collected under similar conditions will also be compared to identify persistent spacecraft or instrument-related signals, while changes in gas abundance will be evaluated against rover activity and regolith disturbances. A measurement which falls outside expected calibration range will be flagged for investigation using instrument-health and rover telemetry before being accepted as lunar volatile detections. TENG data will be assessed by comparing measured electrical output with ground-test results and with repeated measurements obtained under similar mechanical conditions during flight. Consistent relationships between electrical output and

recorded wheel motion, vibration or other mechanical excitation will provide evidence that the signal is produced by expected response rather than electrical noise / instrument artifact. Anomalies in amplitudes, isolated spikes or poor repeatability will be identified as anomalous and checked against operational telemetry and stationary baseline measurements before being included in the science analysis. Seismic Geophone Array data will be validated by first establishing a stationary background noise baseline and then comparing signals recorded across multiple geophone channels. Signals associated with known rover motion or instrument activity will be compared with operational telemetry to confirm that expected rove induced vibrations are detected consistently. Candidate natural seismic events will be evaluated for consistent timing and waveform characteristics across the deployed sensors, whale isolated spikes or signals that correspond with rover activity will be flagged as possible operational noise. Measurements that cannot be confidently distinguished from background noise will be treated with lower confidence during scientific analysis. To ensure quantitative ground motion variables from the geophone sensors repeated measurements must be taken at consistent meters within the array and compared to ambient noise in order to distinguish each frequency range. To be able to passively listen to seismic activity grain and frequency range must be also tuned for the digitizer in order for telemetry to be possible and obtain a digital signal to ensure data is collected in a known format. The seismic activity within the lunar surface is challenging given the long ringing the seismicity has and the scattering properties the lunar regolith has. This causes seismic signals to attenuate.

4. Testing & Calibration Summary Table

Instrument	Initial functionality test	Calibration method	Calibration method	Success criteria
LISTER	Telemetry read of drive electronics, motor resistance, RTD loop continuity, and gas valves	Pre-drilling thermal equilibration read against ambient environment to establish zero-flux point		Uninterrupted drill descent, Linear thermal pulse sensor response, stable resistance during 0.5 m descent cycles, and thermal conductivity accuracy
MECA Electrometer	Electrometer bias check, power	Test image acquisition of		Sharp focus on calibration target,

	verification, optical camera alignment, & data bus check	onboard optical grid and ambient open-space electrometer zero-reading		electrometer signals distinct from background noise, & repeatable particle/charge data
MSolo	Test power, communications, telemetry, ion source, detector and instrument health	Run onboard calibration gas to establish a known mass-spectrum response		Stable telemetry, successful calibration-gas response, repeatable background spectra
TENG	Verify electrical continuity, wiring, measurement electronics, and baseline functionality	Record stationary voltage output to establish electrical noise and zero-motion baseline		Repeatable voltage and current signals during wheel rotation, distinguishable from stationary noise, output showing consistent relationship to wheel rotational conditions
Geophones	Test tuning of frequency range and compare to the resonant frequency of the heritage design Apollo mission 16, gain and sensitivity rate, test damping ratio, test coil resistance (coil within the geophone)	Record ambient noise and noise generated from a known sensor (could be a mechanical tap from within the own rover design) compare each frequency range and then determine the natural frequency of the sensor. Finding the absolute sensitivity by comparing it with the sensitivity		Natural resonant frequency similar to the heritage sensor and quantitative measurements of ground motion at each sensor is obtained.

		of the heritage sensor.		

Figure 70: Insert figure caption here

3.5. Precision and Accuracy of Instrumentation

1. Precision & Accuracy Overview

Precision and accuracy are considered essential for the PSR mission because the science objectives depend on distinguishing the real variations in the lunar environment from normal measurement variability. The LISTER thermal measurements must be sufficiently precise in order to resolve changes in subsurface temperature and thermal properties. The MSolo measurements must reliably distinguish lunar volatile signals from background and contamination. Geophone linear Array should distinguish genuine seismic events and ground motion from instrument or rover-generated noise. The MECA electrometer must accurately characterize changes in surface electric potential and triboelectric charging while the TENG measurements need to reliably determine whether changes in electrical output are associated with mechanical activity or regolith, thereby demonstrating the feasibility of electrostatic energy harvesting. Across all instruments, measurement uncertainty is likely to affect the confidence with which the team can compare observations between locations, operating conditions and repeated trials. If uncertainty is large relative to the physical changes being investigated, then genuine scientific trends may be obscured or incorrectly identified, reducing the ability of the mission to meet its objectives. Despite this, sufficiently precise and accurate measurements will allow the team to separate meaningful lunar signals from measurement variability and integrate results from the different instruments into a scientifically defensible assessment of the lunar surface environment.

2. Instrument Performance

LISTER (Lunar Instrumentation for Subsurface Thermal Exploration with Rapidity) The primary LISTER measurements are absolute regolith temperature, subsurface vertical thermal gradient, and in situ regolith thermal conductivity collected at multiple depths using a deployment mechanism and a sensor-equipped probe. These measurements will be used to determine the local geothermal heat flux and evaluate subsurface thermal properties near the lunar landing site. Specific sensor accuracies for thermal gradient and thermal conductivity measurements are established through ground-based calibration,

thermal-vacuum testing, and heritage performance models from the Apollo 15 and 17 Heat Flow Experiments, achieving a target cumulative endogenic heat flux accuracy of within 10%. Measurement uncertainty stemming from sensor calibration limits, depth placement tolerances, or localized thermal disturbances introduced during deployment primarily affects the precision of calculated geothermal flux and the ability to isolate internal heat flow from surface diurnal temperature variations. LISTER's demonstrated measurement methodology and thermal sensitivity are sufficient for the mission's objective of characterizing the lunar thermal environment, although accurate endogenic heat flux estimates will depend on sufficient thermal equilibration time and reliable sensor contact within the regolith.

MECA (Microscopy, Electrochemistry, and Conductance Analyzer) The primary MECA measurements include aqueous chemical parameters such as solution pH, oxidation-reduction potential, electrical conductivity, and specific dissolved ion concentrations collected by its Wet Chemistry Laboratory, alongside micro-scale physical imaging provided by its Optical Microscope and Atomic Force Microscope. These measurements will be used to determine the chemical composition, toxicity, and physical particle morphology of regolith samples. MECA's baseline performance characteristics, including electrochemical sensor response ranges and an Optical Microscope spatial resolution of 16 micrometers per pixel, were established through extensive ground characterization, certified calibration solutions, and flight performance documented during NASA's Phoenix Mars Lander mission. Measurement uncertainty in ion concentration or optical resolution will primarily affect the precision of soluble salt abundance calculations and the fine structural classification of grain boundaries. MECA's verified measurement capabilities are sufficient for the mission's objective of evaluating soil geochemistry and particle morphology, though absolute concentration values will depend on calibration stability and sample processing performance.

Main measurements collected by MSolo are mass spectra used to identify and characterize low-molecular-weight volatile species released from the lunar surface and subsurface. NASA describes the MSolo as a quadrupole mass spectrometer operating over approximately m/z 1-100 with unit mass resolution, allowing measurements of volatile species and isotopic signatures such as D/H and O^{18}/O^{16}. Under specified operating conditions, NASA documents minimum reports minimum detectable partial pressures of approximately 1.5×10^{-12} Torr using the Faraday cup and 2×10^{-15} Torr using the electron multiplier. These performance characteristics were established through instrument development and ground-based characterization of the modified commercial instrument. Measurement uncertainty will primarily affect the ability to distinguish weak volatile signals from background contamination and to convert measured signal intensity into reliable

abundance estimates. MSolo's demonstrated mass resolution and sensitivity are sufficient for the mission's objective. That being of identifying and comparing lunar volatile signatures, although absolute abundance estimates will depend on calibration stability, background conditions and detector responses.

The primary TENG measurements are electrical voltage and current recorded as functions of time during exposure to electrostatic charge from lunar regolith. These measurements will be used to calculate electrical power and cumulative harvested energy and determine whether charged regolith produces a repeatable electrical response. These measurements will be used to calculate electrical power and cumulative harvested energy and to determine whether mechanical activity associated with rover operation produces a repeatable electrical response. TENG performance depends strongly on mechanical excitation, this means that there is not a single published accuracy specification that can be directly applied to the proposed flight device. Instead, expected performance will be established from repeated ground measurements and published lunar-regolith-simulant experiments. Recent peer-reviewed studies have demonstrated outputs on the order of 10-17 V and approximately 1-2 μA under controlled mechanical excitation, with reported uncertainties in voltage and current measurements. The research on 17.4 (±) 0.4 V open-circuit voltage and 0.96 (±) 0.2 μA short-circuit current at 10 Hz provide a reference scale to be evaluated (as opposed to a formal flight requirement). TENG precision will therefore be evaluated through repeated measurements under comparable electrostatic and regolith-contact conditions, while measurement uncertainty in voltage and current will be propagated into calculated power and energy.

The primary measurements collected from the geophone sensors would be seismic velocities recorded passively in a high bit rate mode. It would be precise if spoken to the manufacturer to make adjustments in tuning for the internal spring and coil to replicate heritage geophone sensors from the Active seismic experiment from Apollo mission 16. A high shock range of 5000 G would be resistant upon impact and reduce damage to the sensor, however measurements within different cutoff frequencies can ensure to differentiate low frequencies from different seismic events.

3. Measurement Uncertainty

Measurement uncertainty across the science payload is going to arise from instrument calibration and drift, environmental conditions, background signals, electrical and mechanical noise as well as limitations in the ability of each instrument to resolve small changes in the measured quantity. For LISTER, uncertainty in temperature measurements and thermal-property calculations will be evaluated using calibration data and repeated measurements. MSolo background spectra and calibration-gas measurements will be used

to distinguish lunar volatile signals from instrument or spacecraft contamination. The Geophone linear Array, instrument noise and rover-generated vibrations will be considered when identifying genuine seismic signals. MECA electrometer, baseline measurements and instrument-health data will be used to separate electrostatic signals from electrical noise and operational effects. Finally, TENG repeated measurements will be compared against electrostatic conditions, regolith-contact, motion and environment excitation.. These uncertainties will be minimized through the pre and post operation calibration checks. During analysis, measurement uncertainty will be included when comparing observations and determining whether an observed difference is scientifically meaningful so that normal instrument variability is not incorrectly interpreted as change in the lunar environment.

4. Instrument Performance Summary Table (Required)

Instrument	Primary measurement(s)	Accuracy	Precision (if available)	Source of performance values or stated assumption	Associated science objective(s)
LISTER	Absolute regolith temperature (T), thermal conductivity (k), and geothermal heat flux.	Measure heat flux at landing site to 10%		ENGINEERING AND TEST DEVELOPMENT OF HEAT FLOW PROBE AND PNEUMATIC DRILL FOR LUNAR LANDER MISSION TO MARE CRISIUM	Quantify heat flow, thermal conductivity, & volatile preservation depth
MECA Electrometer	Surface potentials, dynamic electric field intensity, and triboelectric charge accumulation.	Optical resolution of 16 µm/pixel		NASA Washington University Analyst's Notebook – Phoenix MECA OM Data Sets	Evaluate regolith grain adhesion, reactivity, & dust hazards
MSolo	Mass spectra of low-molecular-weig	Unit mass resolution;		NASA MSolo instrument	Characterize the

	ht volatile species and isotopic signatures.	min detectable partial pressure of ~ 1.5 x 10^{-12} Torr using the Faraday cup and 2 x 10^{-15} Torr (electron multiplier)		specifications and ground based instrument characterizatio n (VIPER and INFICON)	composition and distribution of lunar volatiles near the south pole
TENG	Harvested voltage, current and electrical power generated from electrostatic charge on regolith	No formal flight accuracy specification identified; performance established through ground testing and comparison to published results		Ground testing and comparison with published results; 17.4 (±) 0.4 V open-circuit voltage and 0.96 (±) 0.2 μA short-circuit current at 10 Hz	Demonstration on the feasibility of harvesting electrostatic charge on a PSR boundary terrain as a supplemental power source for low-power lunar operations.
Geophone	Passive monitoring in high bit rate mode in order to perform telemetry to earth and ensure seismic data to be recorded from analog to digital signal.	Location accuracy approximate to 50 meters, other accuracy can be within high frequency and can be attenuated by the instrument damping. Geophone		Publication from Weber et al 2018 from AGU2018 P53B-05 (Seismic expression of thermal	Differentiate between acoustic signals from rovers and seismic activity. Seismic velocities recorded can be used to derive Shear modulus and Young modulus parameters which can help determine

		prototype designs must be tested out for different resonant frequencies and vibration damping to be consistent with accuracy presented by geophone sensor heritage of active seismic experiment Apollo 16 mission.		degradation on the Moon)	slope stability within the craters rims and neighboring slopes.

Figure 71: Insert figure caption here

3.6. Expected Data & Analysis

1. Expected Data Overview

The primary science payload will gather continuous and discrete environmental datasets, including volatile mass spectra, subsurface regolith thermal profiles, electrostatic charge accumulation levels, triboelectric energy generation rates, and active surface acoustic waveforms. These complementary data products directly support the primary mission objectives by establishing quantitative baselines for polar volatile abundance, geothermal heat flow, regolith physical mechanics, and plasma-driven dust mobility at Nobile Rim 2. Raw telemetry transmitted from the surface will undergo systematic ground processing, including sensor calibration corrections, background noise filtering, and signal deconvolution. To validate instrument accuracy and evaluate local anomalies, analyzed surface datasets will be modeled directly against established orbital thermal maps from the NASA LRO Diviner Radiometer and historical seismic surface baselines compiled during the Apollo Passive Seismic Experiment.

2. Instrument Data & Analysis

The LISTER payload will produce time-series subsurface temperature profiles versus depth (Kelvin vs. meters) down to one meter, alongside active heating pulse dissipation curves (temperature change over time) to calculate bulk thermal conductivity (k). Thermal needle probe operational physics and heat-pulse regression models follow the flight instrument design and testing framework established for NASA's Commercial Lunar Payload Services program. While historical Apollo heat flow probes evaluated sunlit equatorial locations, our mission's dataset will capture extreme cryogenic background temperatures and localized thermal shifts characteristic of high-latitude micro-cold traps at Nobile Rim 2. Analysis entails fitting transient temperature rise curves from internal probe heaters to one-dimensional radial heat conduction equations, enabling precise derivation of regolith thermal conductivity and vertical geothermal flux. The expected scientific outcome is a direct measurement of local geothermal heat flow, establishing critical subsurface thermal boundary conditions required to model ice stability depth profiles.

The MECA electrometer package will generate time-stamped voltage decay curves, dynamic electric field intensity metrics (volts per meter), and cumulative charge transfer levels. Electrometer charge sensitivity, amplifier gain scaling, and electronic noise floor baselines are integrated directly from planetary electrometer qualification benchmarks established by NASA Kennedy Space Center. Unlike atmospheric planetary measurements, our mission's dataset will quantify triboelectric charge accumulation specifically during regolith-drilling contact within a high-latitude shadowed plasma sheath. Data analysis will entail filtering baseline electronic offset noise, converting raw telemetry bits to absolute charge levels using open-circuit calibration sweeps, and correlating electrostatic voltage spikes with simultaneous drill telemetry. The expected scientific conclusion is an empirical characterization of regolith charging thresholds, providing crucial data to refine risk models for spacecraft dust adhesion, electrostatic discharge, and mechanical wear during surface operations.

3. Error & Uncertainty Analysis

Instrument accuracy and precision directly affect the confidence with which the collected data can be interpreted. For LISTER, uncertainty in temperature and thermal conductivity measurements may limit the ability to distinguish small subsurface thermal differences, while MSolo uncertainty may make weak volatile signals more difficult to separate from the noise of other instruments or from possible spacecraft contamination. MECA measurements may be affected by electrical interference, baseline drift, and sensor limitations, and TENG measurements may vary with both electrical measurement uncertainty and changes in rover mechanical excitation. Seismic measurements are very similarly sensitive to instrument noise and rover vibrations, which can complicate

identification of genuine lunar ground motion. These effects mean that apparent differences between measurements cannot automatically be treated as real environmental variation.

Measurement uncertainty will therefore be considered during interpretation by comparing observed changes against the known precision, accuracy, background variability, and repeatability of each instrument. Calibration measurements, background spectra, baseline measurements, repeated observations, and instrument health data will be used to determine whether or not a signal is consistent and distinguishable from expected measurement variability. Results that fall close to an instrument's uncertainty or background level will be treated with much lower confidence and may require repeat observations before being interpreted as meaningful, while trends that come out as repeatable and remain distinct from expected uncertainty will provide stronger evidence of actual variations in the lunar environment. This approach helps prevent instrument noise calibration drift, or operational effects from being incorrectly interpreted as scientific changes and preserves confidence in comparisons made across different locations and operating conditions.

4. Data Summary Table

Instrument	Expected data product(s)	Expected scientific outcome	Associated science objective(s)
LISTER	Subsurface temperature profiles (T vs. depth); thermal pulse dissipation curves (T vs. time).	measurement of local geothermal heat flow and regolith thermal transport parameters at Nobile Rim 2.	Objective 2: Subsurface thermal profiling and heat flow quantification.
MECA Electrometer	Electrostatic charge decay rates; electric field strength; triboelectric voltage spikes.	characterization of regolith charging behavior and local electrostatic dust mobilization hazards.	Objective 3: High-latitude plasma characterization and electrostatic dust hazard analysis.
MSolo	Time and location referenced mass	Identification and comparison of	Objective 1: Characterize the

	spectra; mass-to-charge peaks and relative signal intensities for target volatile species.	volatile species and isotopic signatures in lunar regolith.	chemical composition and distribution of lunar volatiles near PSRs.
TENG	Time-series voltage and current measurements; calculate electrical power and cumulative harvested energy during interaction with charged particles.	Demonstrate and quantify triboelectric/electrostatic energy harvesting from rover-regolith interactions for potential supplemental power generation.	Identify feasibility of harvest using electrostatic charge at PSR boundary terrain as a supplemental power source for low power lunar operations.
Geophones	Seismic velocities throughout the lunar subsurface.	Differentiation between acoustic signals from the rover and seismic activity within the lunar surface. These measurements can be used to derive the Shear Modulus and Young Modulus to further help study slope stability within the craters rims.	Objective 1: passive seismic monitoring near the area of study and rover. Passive monitoring can help better constrain and understand how waves propagated through each layer can further detail the lunar subsurface.

Figure 72: Insert figure caption here

5. Example Data Figures

LISTER:

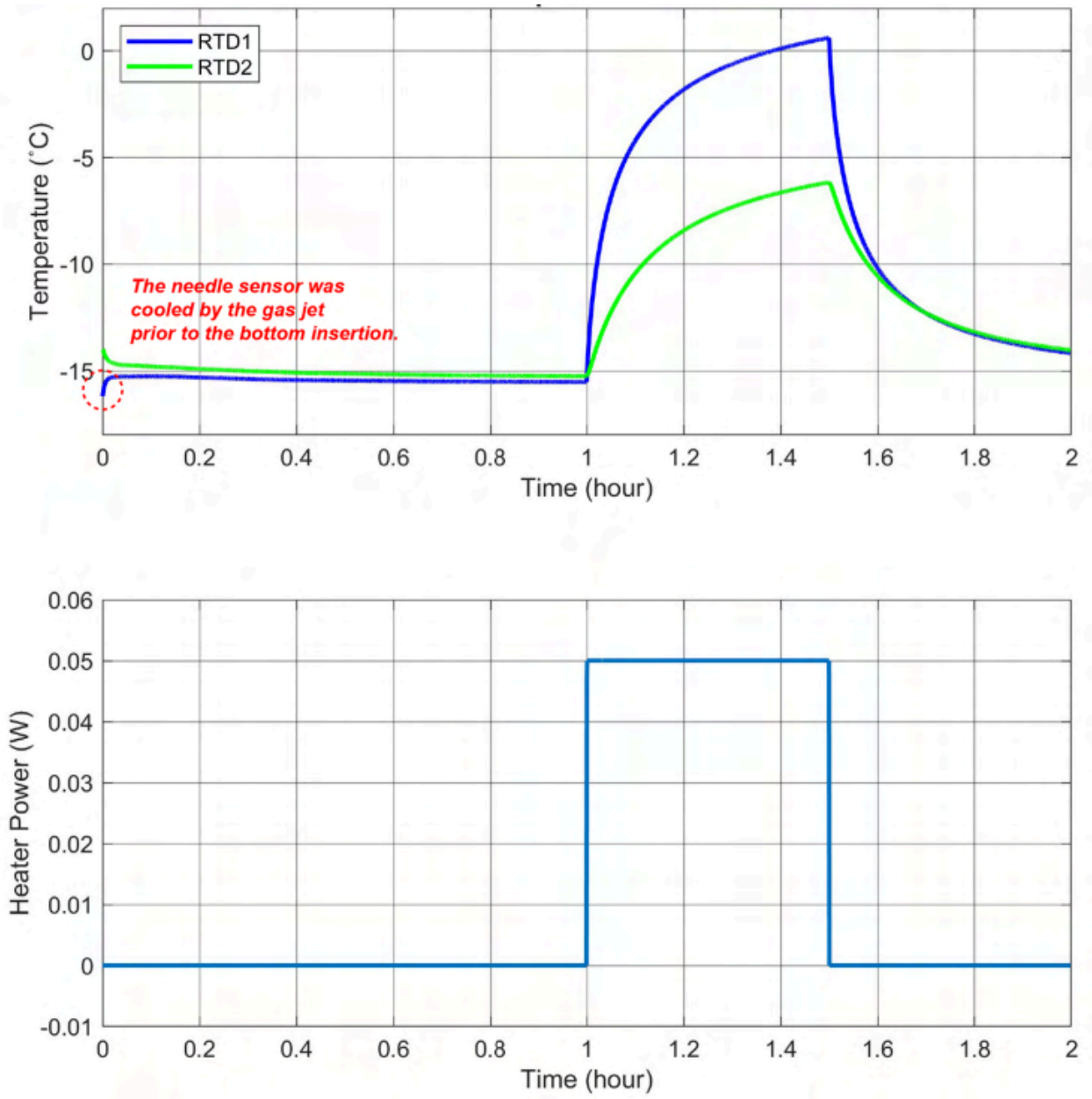


Figure 73: Illustrates typical active thermal measurements acquired at a discrete depth (0.329m). The top panel displays temperature readings from RTD1 (midpoint sensor) and RTD2 (gas nozzle sensor), capturing initial cooling from adiabatic gas jet expansion followed by thermal equilibration with the surrounding regolith. The bottom panel shows a controlled 0.05 W heating pulse applied between hours 1.0 and 1.5. In our mission, the slope and curvature of this temperature rise and subsequent dissipation curve will be fitted to heat-conduction equations to directly calculate bulk regolith thermal conductivity (k).

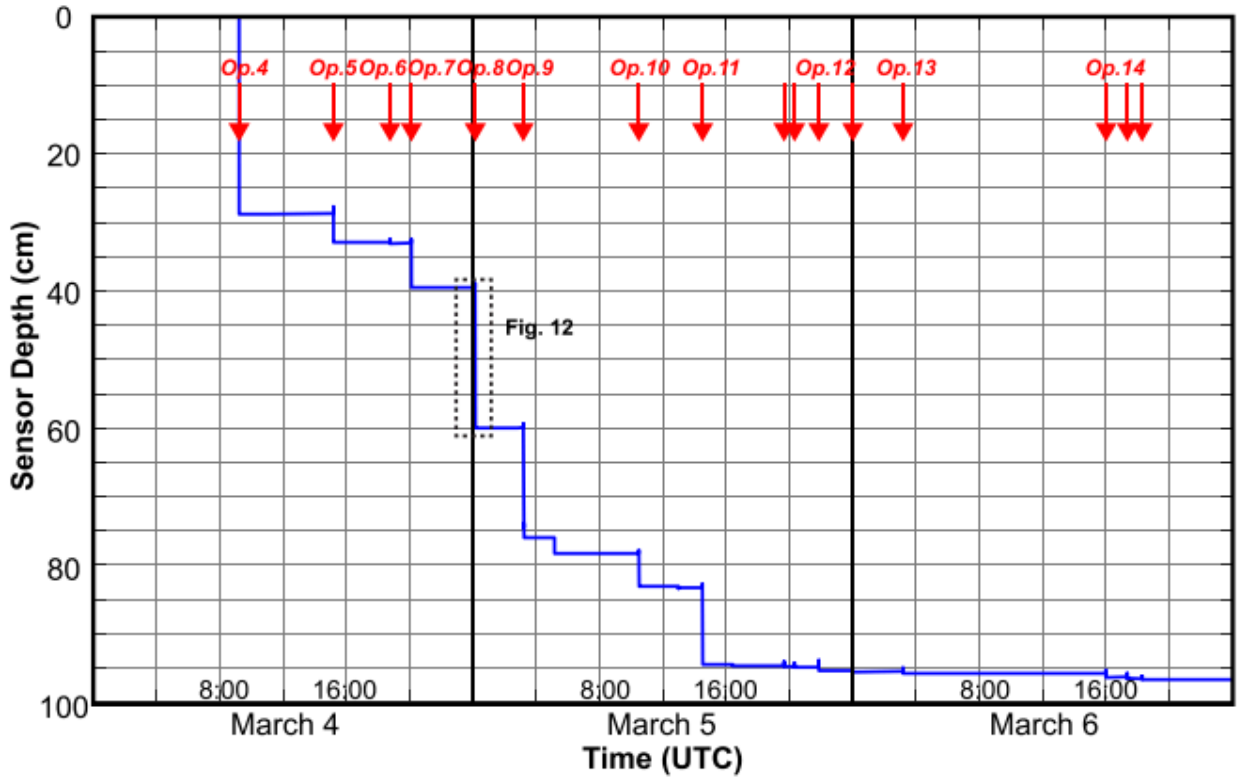

Op.4
Op.5
Op.6
Op.7
Op.8
Op.9
Op.10
Op.11
Op.12
Op.13
Op.14
Fig. 12
Sensor Depth (cm)
0
20
40
60
80
100
8:00
16:00
8:00
16:00
8:00
16:00
March 4
March 5
March 6
Time (UTC)

Figure 74: provides a graph summarizing the downward progress of LISTER during the first 3 days of surface operation (sensor depth in cm vs. UTC time). The step-wise curve shows the probe's incremental insertion down toward a target depth of 100 cm, with red arrows marking the precise timing of the pneumatic gas-jet excavation operations. Tracking penetration depth over time allows our science team to map the mechanical resistance and compaction state of the upper regolith column while establishing the exact depth locations for subsequent thermal conductivity and equilibrium temperature measurements.

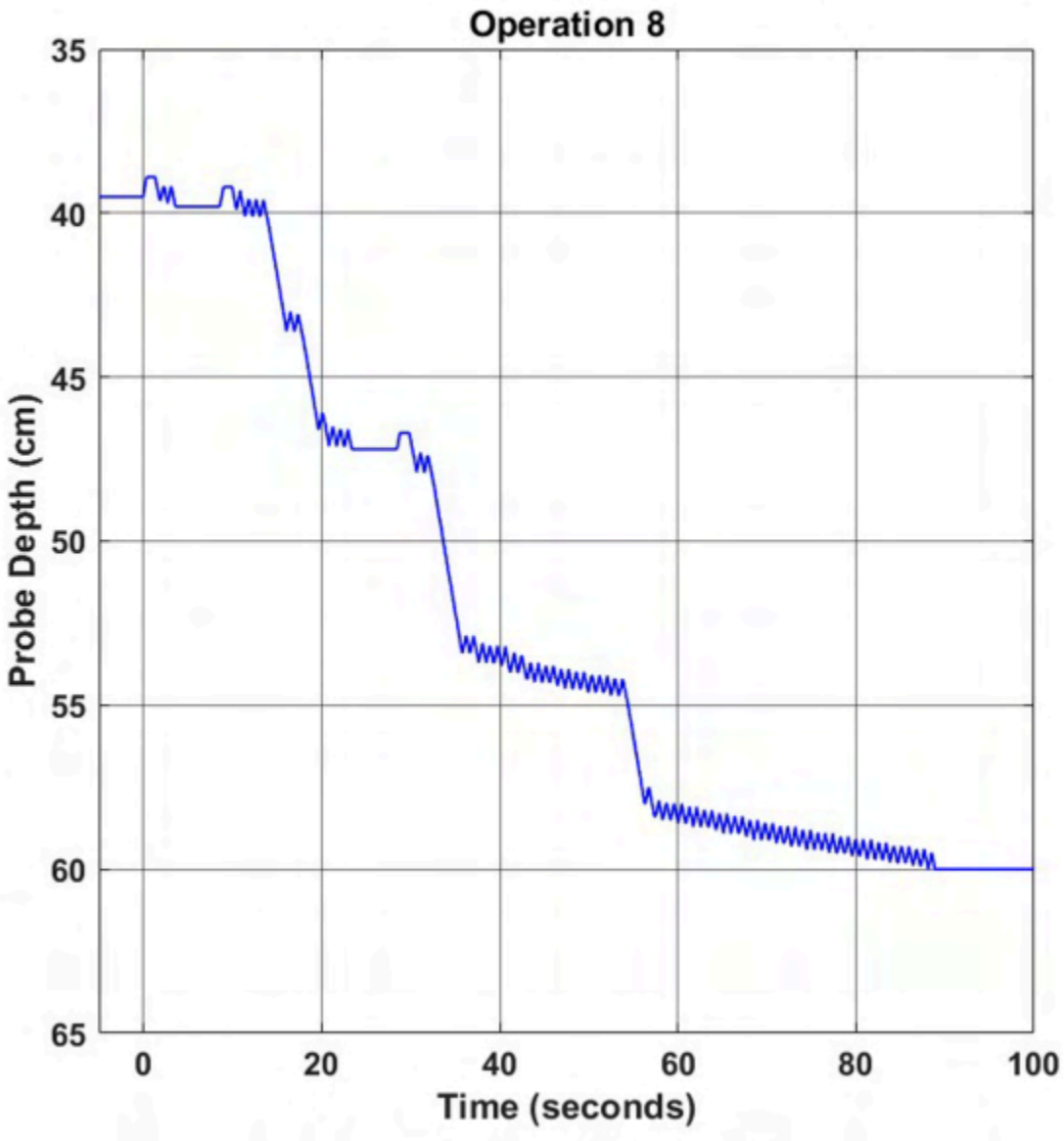


Figure 75: Provides a high-resolution time-series (100 seconds) of sensor depth during an active excavation interval (Operation 8, penetrating from 40 cm to 60 cm). The characteristic zigzag pattern highlights automated dithering maneuvers used to overcome physical regolith friction. Monitoring these high-frequency depth variations during our mission will help verify probe clearance and sensor placement stability prior to initiating active thermal pulse cycles.

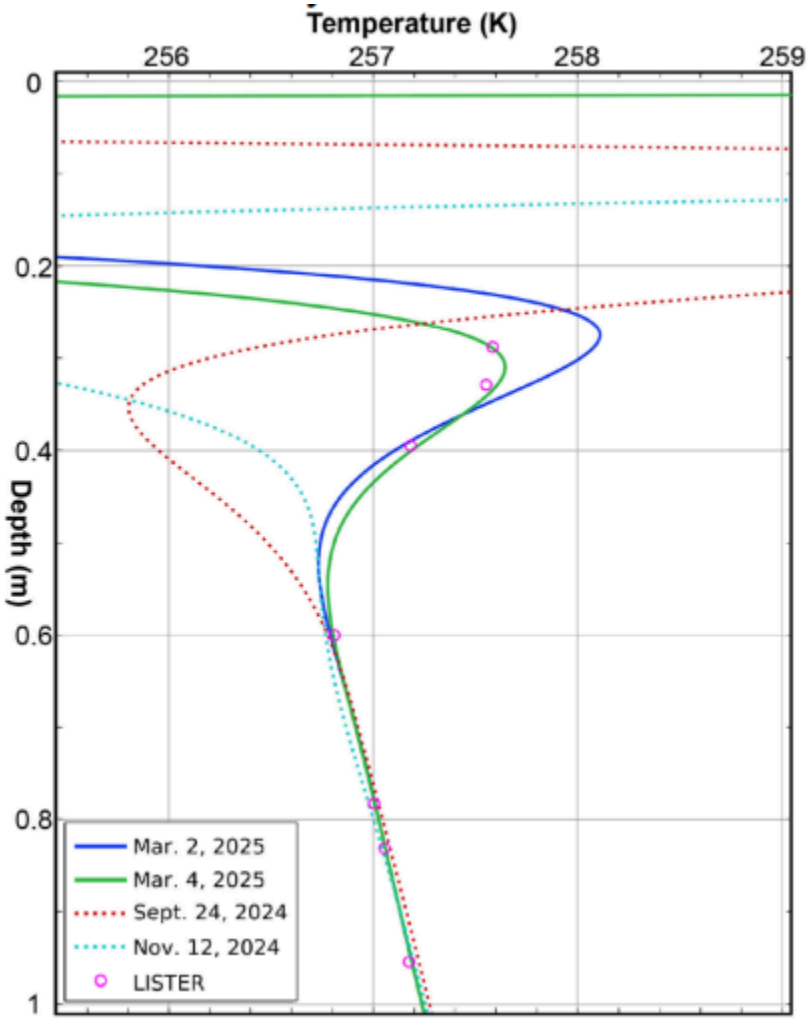


Figure 76: Displays the equilibrium subsurface temperatures measured by LISTER (magenta circles) plotted alongside model-predicted temperature versus depth curves for the day of operation (March 4, 2025) and preceding dates. Comparing in-situ discrete depth measurements with theoretical thermal conduction models allows our science team to constrain surface insulation effects and account for time-dependent thermal wave propagation, enabling an accurate inversion calculation to derive the local endogenic geothermal heat flow.

MECA Electrometer:

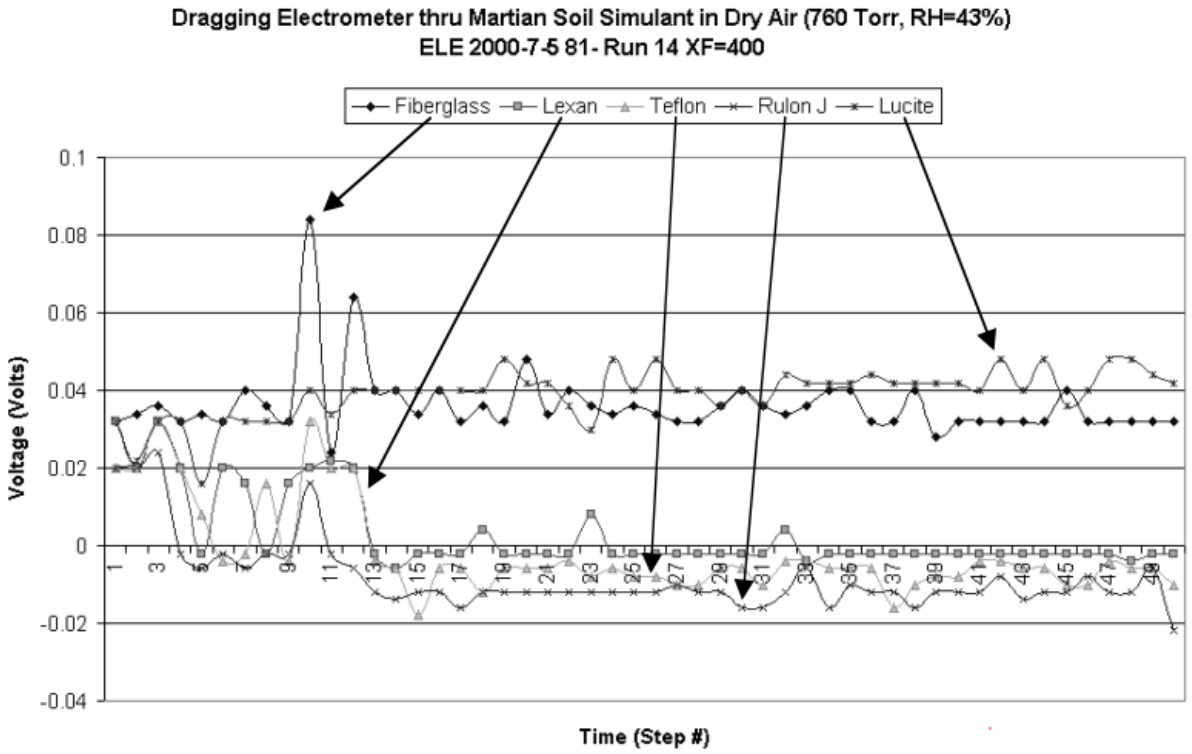


Figure 77: Presents output voltages versus time (1.4 -second sampling intervals) for five insulator materials dragged across soil simulant at 10 mm/s. The Rulon J insulator exhibited the largest voltage shift (-0.04 V), corresponding to a charge buildup of -10pC, which approaches the 0.5 pC/bit A/D converter resolution limit. In our mission, monitoring these discrete voltage shifts during mechanical contact will reveal which materials generate the highest electrostatic charges when interacting with polar lunar regolith.

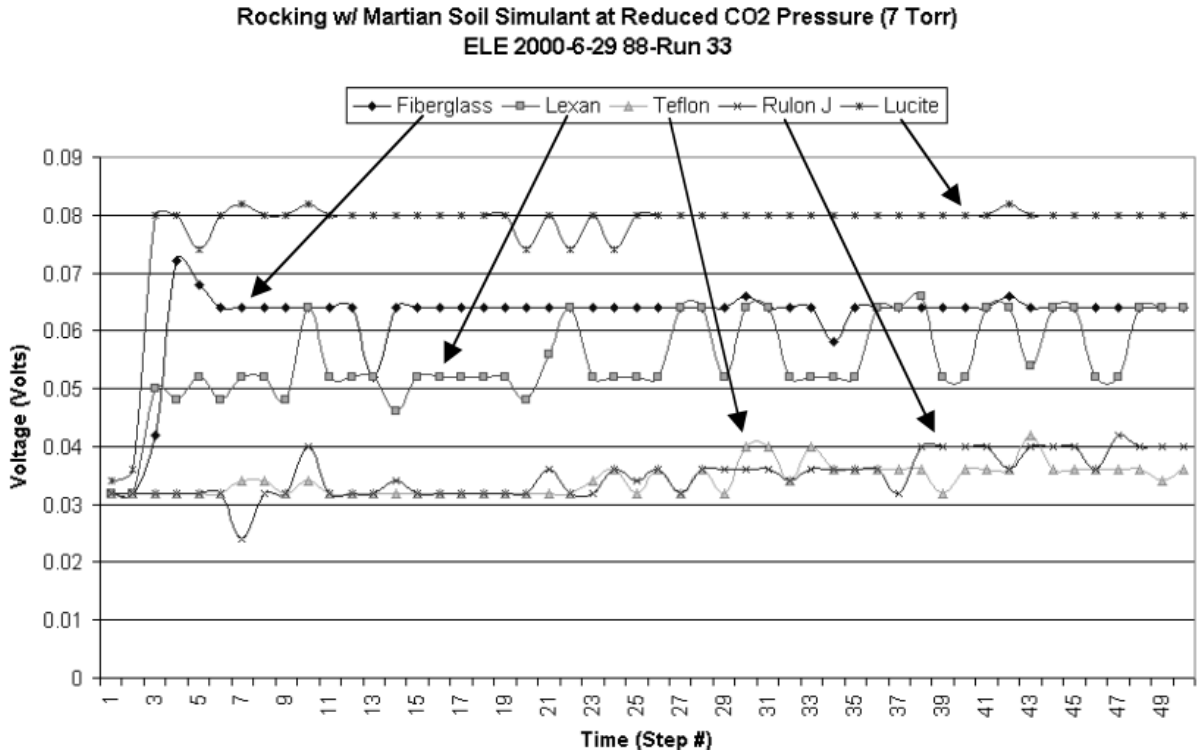


Figure 78: Displays triboelectric responses across the five insulator materials during a rocking agitation experiment in a reduced-pressure environment (7 Torr CO2). An air ionizer neutralized residual charges between runs to establish steady-state sensor behaviors. Measuring these response profiles during surface operations allows the team to model triboelectric charge saturation limits under near-vacuum conditions.

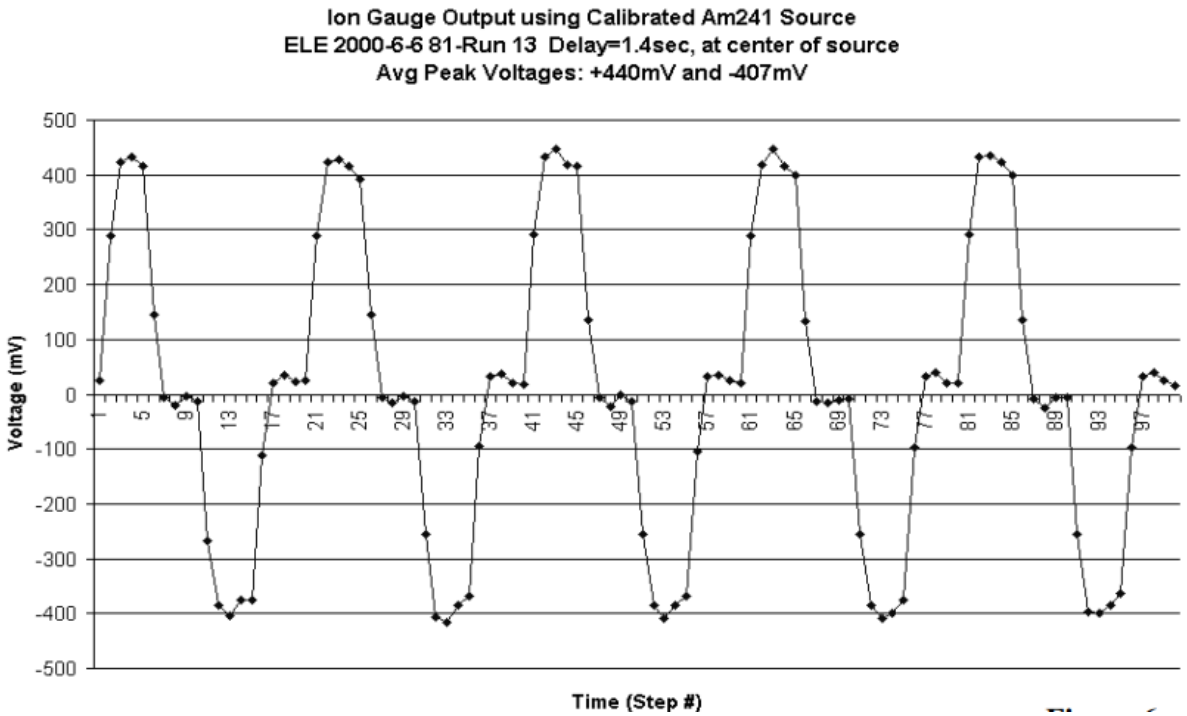


Figure 79: Illustrates a typical ion gauge response curve recorded at 5-second time intervals using a calibrated 241Am ionization source, yielding average peak voltages of +440 mV and -407 mV. Applying the electrometer circuit gain factor (42.2 pA/V), a 420 mV peak corresponds to an ion current of 18 pA. Analyzing these peak voltage waveforms enables the team to convert telemetry signals directly into precise atmospheric and surface ion current measurements.

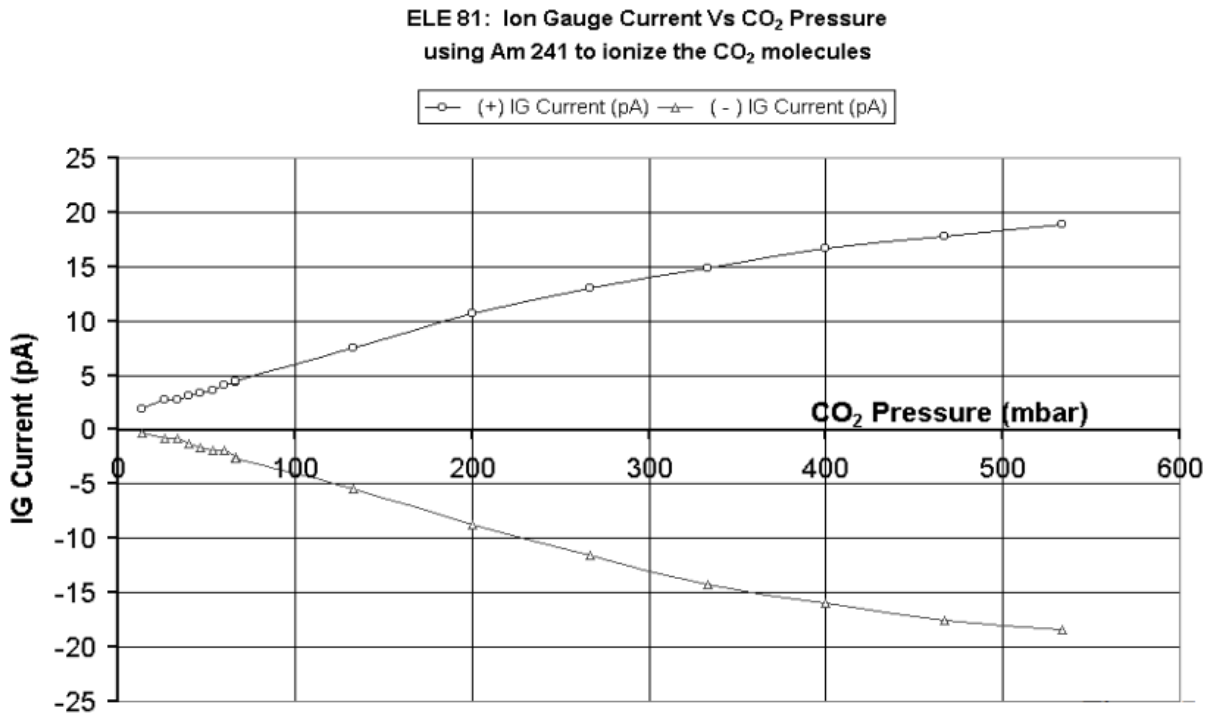


Figure 80: Plots the dependence of positive and negative deduced ion gauge currents (pA) across varying background pressures (mbar). This relationship serves as a baseline calibration curve, allowing the science team to differentiate between pressure-induced background current shifts and localized electrostatic charging events during surface operations.

4. Mission Risk Management

4.1. Safety and Hazard Overview

Team Fontus identifies mission risks and environmental hazards by connecting the lunar south pole operating environment to the rover's mission objectives, subsystem design, and planned surface operations. The mission must operate in and near permanently shadowed regions, where extreme temperatures, rough terrain, regolith, dust, radiation, electrostatic charging, communication limitations, and contamination hazards can affect rover survival and science return. These hazards are important because the mission is designed to characterize environmental and exploration hazards in lunar south pole permanently shadowed regions, while also collecting temperature, thermal conductivity, volatile, electrostatic, mobility, and seismic data.

Risk management is important throughout mission development and operations because the same environmental hazards that create scientific value also create engineering risk. Extreme temperature conditions can affect batteries, electronics, heaters,

thermal interfaces, and payload instruments. Regolith and dust can reduce traction, damage mechanisms, contaminate measurements, and reduce solar power generation. Radiation can affect avionics, onboard data storage, sensors, and payload electronics. These risks will be evaluated through the Risk Log, risk matrix, FMEA, personnel hazard analysis, and planetary protection assessment so that mitigation actions can be tracked and updated as the design matures.

The mission design already includes several hazard-driven controls. The MDR identifies nine Kapton polyimide heaters, multilayer insulation blankets, and operating and survival temperature limits for thermally vulnerable hardware. It also identifies a radiation-hardened GR712 flight computer, Aitech S-A1760 co-processor, 128 GB nonvolatile storage, dual-path CAN command bus, and SpaceWire payload data path for CDH functions. The power system uses GaAs solar panels, a lithium-ion battery system, and a regulated 28 V DC bus to support operations without RTG-derived hardware.

Risk Management Approach

Team Fontus identifies risks by reviewing mission requirements, subsystem designs, environmental hazards, landing-zone assumptions, interface needs, and planned science operations. Each risk is analyzed by likelihood and consequence, then prioritized in the risk matrix. The matrix shows the current assessed risk level, and planned mitigation does not lower a rating until analysis, inspection, or testing confirms improvement. Mitigate risks require active action, while Watch risks are monitored for changes.

Risks will be monitored throughout the mission lifecycle using the Advanced Risk Log, subsystem reviews, verification results, and design updates. The Advanced Risk Log records each risk’s owner, rating, mitigation actions, review dates, status, and closure criteria. Risks with high consequence, such as loss of thermal survival, insufficient power, mobility failure, flight computer upset, or loss of science data, will receive greater attention because they can directly affect mission success. The MDR requirements also establish major constraints that shape risk decisions, including the 175 kg mass limit, 1.5 m × 1.5 m × 1.0 m stowed envelope, $250 million cost cap, no RTG-derived hardware, integration readiness by December 1, 2028, and launch readiness by February 1, 2029.

Mitigation strategies will be used to maintain an acceptable mission risk posture. Thermal risks will be reduced through heaters, insulation, thermal-vacuum testing, temperature monitoring, and safe-mode heater responses. Power risks will be reduced through solar power margin tracking, battery thermal protection, sequential scheduling of high-power activities, cable management, and electrical testing. Mobility risks will be reduced through conservative traverse planning, wheel/regolith-simulant testing, and alternate route selection. CDH risks will be reduced through radiation-tolerant processing, EDAC, ECC, watchdog recovery, redundant data paths, and prioritized downlink. Science

risks will be reduced through calibration, contamination control, cross-checking between instruments, and useful nondetection planning.

Planetary Protection

Planetary protection is applicable because this is a lunar surface mission operating near the lunar south pole and permanently shadowed regions. NASA guidance treats lunar missions as planetary protection Category II, with additional lunar surface distinctions for Category IIa and IIb; current NASA lunar planetary protection materials identify PSR and lunar pole missions as Category IIb, requiring contamination-control attention such as volatile-release analysis and spacecraft organic inventory documentation. Specific mission requirements would be confirmed by the NASA Planetary Protection Officer or the selected mission provider, because NASA policy states that requirements are assigned for each planned mission.

For Team Fontus, the main planetary protection concern is not biological contamination of a life-detection target, but preservation of the scientific integrity of volatile and regolith measurements. The mission's science objectives include characterizing thermal properties, heat flow, volatile chemistry, water ice deposits, electrostatic conditions, and environmental hazards near PSRs. The MDR also lists contamination detection as an instrument performance need, including the ability to distinguish native lunar volatiles from rover or lander volatiles and the use of onboard calibration gas to separate native signals from lander signals.

To comply with the intent of lunar planetary protection and contamination-control expectations, Team Fontus should document materials and processes that could release contaminants near the science site. This includes an inventory of spacecraft materials with contamination risk, a review of lubricants, adhesives, coatings, cleaning agents, vented gases, battery materials, and payload-contact materials, and an assessment of any volatiles released by the lander or rover. The mission should also use clean handling procedures, contamination-control covers where appropriate, pre-launch cleanliness checks, material compatibility reviews, and operational keep-out practices to reduce contamination of PSR science targets.

4.1.1. Risk Analysis

The primary goal of Team Fontus's risk management process is to systematically identify, analyze, and control technical, programmatic, operational, and environmental risks across all mission lifecycle phases. Operating under NASA-standard systems engineering guidelines, the team continuously tracks potential failure modes to prevent adverse impacts on project cost, schedule, and safety parameters. Risks are identified

during early design reviews and trade studies, evaluated using standardized likelihood and consequence scales, and updated continuously through weekly technical reviews and formal change control board audits.

Team Fontus assesses subsystems, interfaces, hardware, and mission workflows to identify mission risks. Team Fontus starts risk identification during initial trade studies and design reviews. Mission assurance and lead engineers review and analyze the vulnerability of the thermal, power, and structural limits of baseline configurations. Team Fontus uses the Condition-Departure-Asset-Consequence (CDAC) framework to define baseline, departure, asset, and consequence for proposed risks of the mission.

Proposed risks (candidates) are collected in the project's risk log, and evaluated using Failure Mode and Effects Analysis (FMEA). The active and non-punitive risk reporting environment of Team Fontus allows all members to contribute. There are multiple ways to enumerate and capture risk, such as trade studies, lead times, and working sessions. Team Fontus uses these to assess and react to risk prior to the fabrication of mission and project hardware. The risk enumeration environment allows unexplored risks of mission and project margins to be rapidly identified and submitted by team members.

Risk Narrative & Log

The risk matrix ranks active risks by likelihood and consequence to show which items require the most attention. The matrix shows the current assessed risk level, and planned mitigation does not lower a rating until analysis or testing confirms improvement. Mitigate risks require active action, while Watch risks are monitored for changes. The Advanced Risk Log records each risk's owner, rating, mitigation actions, review dates, status, and closure criteria.

The highest-priority risks are tied to mission-enabling functions, especially power availability, thermal survival, mobility, and landing-zone access. POW-R1 Peak Power Demand Exceeds Available Power During PSR Operations is one of the most important risks because the rover must power thermal control, mobility, CDH, communications, and payload operations during surface activities. If peak demand exceeds available power, the rover may need to shut down subsystems, shorten science activities, reduce traverse time, or enter a power-conserving mode. This risk is managed by prioritizing essential subsystems and scheduling high-power activities sequentially.

TH-01 Inability to Maintain Survival Temperatures in Permanently Shadowed Regions is also high priority because the rover must protect batteries, avionics, electronics, and science instruments from extreme cold near PSRs. If the thermal control system cannot maintain survival or operating temperatures, hardware could degrade, shut down, or fail permanently. Mitigation includes MLI, Kapton heaters, heater redundancy,

distributed temperature sensors, autonomous thermal control, thermal-vacuum testing, and safe-mode heater responses.

MCH-R1 Wheel or Mobility Hardware Loses Traction is a major mission-level risk because mobility is required to reach the Exploration Zone and complete science objectives. The risk includes wheel slip, reduced traction, regolith obstruction, and degraded wheel performance. The six-wheel layout and rigid aluminum grouser wheels reduce some risk, but they do not remove uncertainty in wheel-regolith interaction. Mitigation includes low-speed traverse operations, regolith-simulant testing, slip/current monitoring, alternate route planning, and avoiding steep or hazardous terrain.

Landing-zone risks remain important because Mobile Rim 2 is favorable but still requires mission-specific confirmation. The 14-day operating duration is treated as a Team Fontus mission requirement, so sites with only 11–12 consecutive sunlit days and 10–11 days with simultaneous Sun and Earth visibility require additional verification. Insufficient sunlight could reduce power generation and battery recharge, unresolved slopes or craters could block traverse access, lack of an accessible PSR target within 250 m could reduce science access, and volatiles below detection limits could reduce expected science return. These risks are managed through updated illumination analysis, Direct-to-Earth communication checks, terrain mapping, alternate route planning, alternate Exploration Zones, multiple science targets, and useful nondetection planning.

Overall, the matrix shows that the strongest risks are connected across subsystems. A landing-zone illumination issue can reduce power margin, reduced power can limit heater use, poor thermal control can affect batteries and electronics, mobility limits can reduce science access, and CDH issues can reduce data return. The matrix will continue to be updated as terrain analysis, illumination monitoring, power budgets, thermal models, procurement status, and verification testing mature.

Risk Matrix

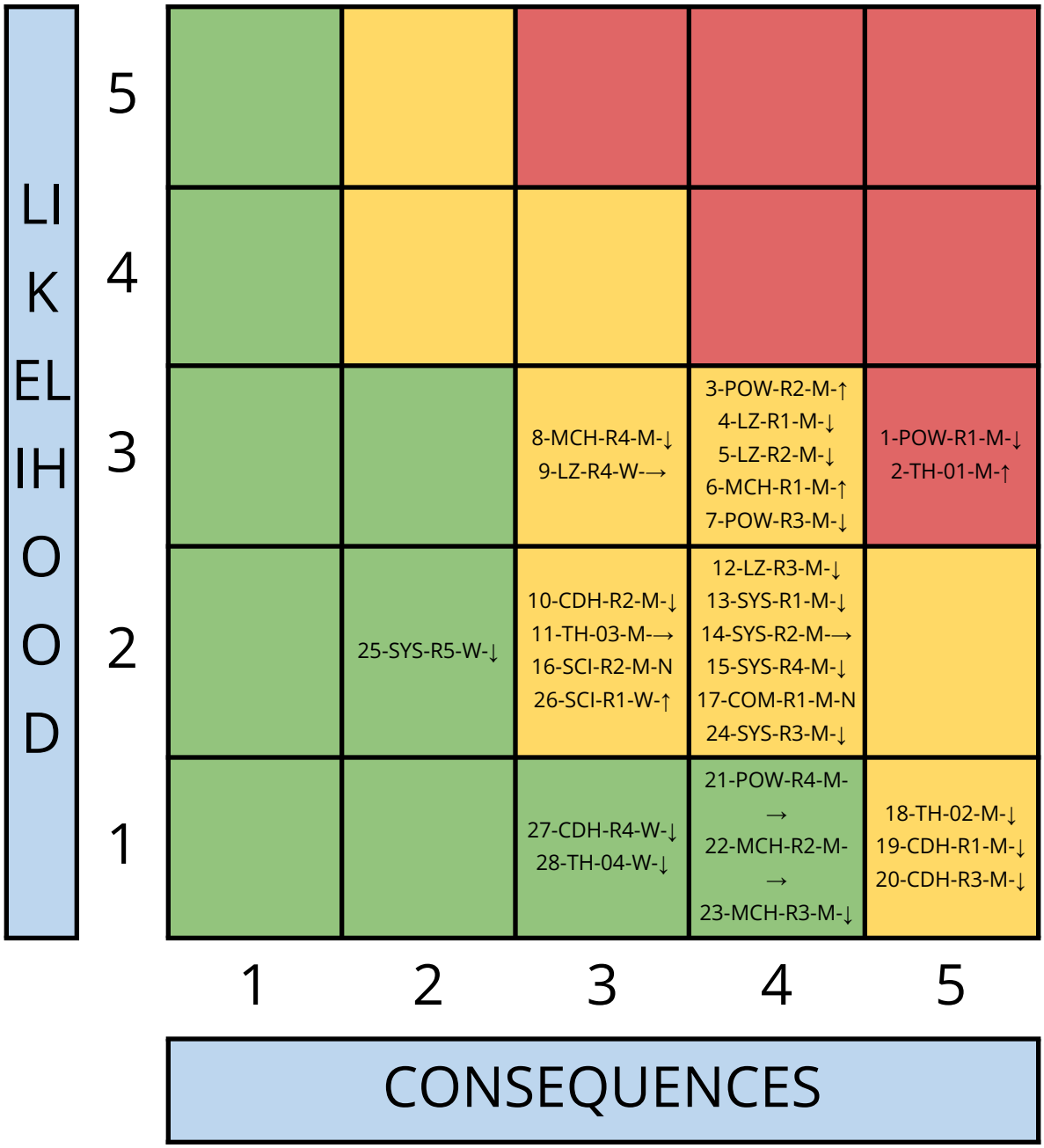


Figure 81: Risk Matrix for Team Fontus MDR, showing active mission risks ranked by likelihood and consequence to identify the highest-priority risks requiring mitigation or continued monitoring.

Rank	Risk ID #	WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement
Prioritization order of risks to addressed	Unique ID assigned to each risk, may have a precursor code to designate what level the risk is tracked at	What WBS element the risk affects the most	Which team member is assigned responsibility over seeing this risks closure	What effects will this risk cause? An increase in costs, a loss of schedule margin or launch dates, or loss of technical data to be collected?	What is the timeframe until this risk will be addressed? Short = 1-3 months Medium = 3-12 months Long = 12+ months	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]."
1	POW-R1	Power	Power Lead	Scope	Short Term	Peak Power Demand Exceeds Available Power	Given that the rover must operate power-intensive subsystems during PSR operations, there is a possibility that peak power demand exceeds available power, adversely impacting the power subsystem and surface operations, which can result in subsystem shutdown, reduced science time, or failure to complete planned operations.
2	TH-01	Thermal	Thermal Lead	Scope	Short Term	Inability to Maintain Survival Temperatures in PSRs	Given that the rover will operate near permanently shadowed regions, there is a possibility that the thermal subsystem cannot maintain survival temperatures, adversely impacting batteries, avionics, payload instruments, and electronics, which can result in shutdown, degraded performance, or permanent hardware damage.
3	POW-R2	Power	Power Lead	Scope	Medium Term	Solar Panel Regolith Accumulation	Given that lunar regolith can accumulate on solar panels, there is a possibility that solar power generation is reduced, adversely impacting the rover power subsystem, which can result in slower battery recharge, reduced operating time, or reduced science return.
4	LZ-R1	Landing Zone / Power	Mission Design and Power Leads	Scope	Medium Term	Insufficient Sunlight at Landing Location	Given that the Nobile Rim 2 landing area may provide limited date-specific illumination, there is a possibility that the landing location does not provide enough sunlight for expected rover power generation, adversely impacting the power subsystem and surface operations, which can result in reduced operating time, slower recharge, shortened traverses, or reduced science return.

Figure 82 : Advanced Risk Log for Risks 1–4, showing Rank through Risk Statement columns.

L	C	Rating	Approach	Trend	Mitigation Plans	Status of Mitigation Plans
Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle	Specific actions that will be implemented to mitigate the risk by either lowering the L and/or the C. Add or remove as many mitigation actions as needed and assign a new line and number (1,2,3...) to them. Carry this number across the next few rows where mitigation plans are monitored and controlled	Updates on the mitigation plans throughout their life cycle
3	5	High	Mitigate (M)	↓ Decrease	1. Prioritize power to essential subsystems. 2. Schedule high-power activities sequentially. 3. Update the power budget using worst-case PSR operations. 4. Maintain power and battery margins.	Open; power sequencing and margin analysis are active priority items because peak load directly affects surface operations.
3	5	High	Mitigate (M)	↑ Increase	1. Use multilayer insulation and Kapton heaters. 2. Validate thermal models through thermal-vacuum testing. 3. Monitor temperatures continuously. 4. Use safe-mode heater operation for abnormal temperatures.	Open; thermal survival remains a priority because PSR conditions affect multiple critical subsystems.
3	4	Moderate	Mitigate (M)	↑ Increase	1. Minimize regolith accumulation through panel placement. 2. Use protective covers where practical. 3. Include dust-related power loss in the power budget. 4. Analyze solar output under dust coverage.	Planned; solar dust-loss assumptions will be included in the power budget as the design matures.
3	4	Moderate	Mitigate (M)	↓ Decrease	1. Verify date-specific illumination conditions. 2. Update the power budget using expected sunlight windows. 3. Maintain battery and power margins. 4. Confirm solar-array geometry against Sun angles.	Planned; site-specific illumination will be checked against the mission operations timeline and power assumptions.

Figure 83: Advanced Risk Log for Risks 1–4, showing Likelihood through Status of Mitigation Plans columns.

Trigger Date	Trigger Event	Closure Date	Closure Event	Status Updates
The date that a mitigation plan should have an update by	A specific event that will be associated with the trigger date	The expected date the risk will be closed out	A specific event that will be associated with the closure date	The date followed by a short description of any changes or status updates to this risk Example: (10/22/2024): The risk's L was changed from a 2 to a 1 after completing mitigation plan #1.
Before PDR	Peak power budget review	Before CDR	Power budget supports required PSR operations with acceptable margin.	Current: This remains one of the highest-priority power risks because mission operations depend on maintaining enough power during PSR activity.
Before PDR	Thermal model and heater-zone review	Before thermal-vacuum qualification	Critical hardware remains within survival and operating temperature limits.	Current: This risk remains high because thermal survival is mission-enabling and must be verified through modeling and thermal-vacuum testing.
Before PDR	Solar dust-loss power budget update	Before environmental verification	Solar power margin remains acceptable under expected regolith accumulation.	Current: Risk is controlled as a moderate power risk because dust exposure is expected, but design margin and operational planning can reduce impact.
Before landing site review	Date-specific illumination analysis complete	Before CDR	Landing site supports required power generation and operating timeline.	Current: Risk is tracked to the Nobile Rim 2 site selection and will be reduced as illumination analysis and operations planning mature.

Figure 84: Advanced Risk Log for Risks 1–4, showing Trigger Date through Status Updates columns.

5	LZ-R2	Landing Zone / Mobility	Mission Design and Mechanical Leads	Schedule/Scope	Medium Term	Unresolved Slopes, Rocks, or Craters Obstruct Route	Given that available landing-zone data may not resolve rover-scale slopes, rocks, craters, or local obstructions, there is a possibility that the planned route is unsafe or blocked, adversely impacting rover mobility and science access, which can result in delayed operations, reduced traverse distance, or loss of access to science targets.
6	MCH-R1	Mechanical / Mobility	Mechanical Lead	Scope	Medium Term	Wheel or Mobility Hardware Loses Traction	Given that the rover must traverse loose lunar regolith near PSR boundary terrain, there is a possibility that wheel or mobility hardware loses traction, experiences excessive slip, or becomes partially obstructed by regolith, adversely impacting the mobility subsystem, which can result in reduced traverse capability or inability to reach science targets.
7	POW-R3	Power / Thermal	Power and Thermal Leads	Scope	Medium Term	Battery Temperature Drops During PSR Operations	Given that battery temperatures may drop during PSR operations, there is a possibility that battery performance degrades, adversely impacting the power subsystem, which can result in reduced capacity, instrument shutdown, or shortened surface operations.
8	MCH-R4	Mechanical / Procurement	Mechanical Lead	Schedule	Short Term	Mobility Component Procurement Delay	Given that motors and mobility components may have long procurement lead times, there is a possibility that required parts are delayed, adversely impacting mobility subsystem testing, which can result in schedule delay or reduced test margin.
9	LZ-R4	Landing Zone / Science	Science Payload Lead	Scope	Medium Term	Volatiles Below Detection Limits	Given that water ice or other volatiles at Nobile Rim 2 may be absent, patchy, deeply buried, or below instrument detection limits, there is a possibility that MSolo, Ground Penetrating Radar, and LISTER do not obtain the expected combined volatile and thermal characterization, adversely impacting science return, which can result in reduced confidence in volatile distribution and ISRU resource assessment.
10	CDH-R2	CDH	CDH Lead	Scope	Medium Term	Co-Processor Lower Radiation Tolerance	Given that the co-processor may have lower radiation tolerance than the main flight computer, there is a possibility that radiation exposure degrades or upsets co-processor performance, adversely impacting onboard processing and NeuPAN operations, which can result in reduced autonomy performance or increased safe-mode events.

Figure 85: Advanced Risk Log for Risks 5–10, showing Rank through Risk Statement columns.

3	4	Moderate	Mitigate (M)	↓ Decrease	1. Map primary and alternate traverse routes. 2. Verify slope, rock, crater, and shadow hazards. 3. Use conservative route planning. 4. Update route plans as higher-resolution terrain data becomes available.	Planned; route verification will mature as terrain analysis and traverse planning improve.
3	4	Moderate	Mitigate (M)	↑ Increase	1. Use low-speed traverse operations. 2. Conduct regolith-simulant wheel testing. 3. Monitor wheel slip, rover attitude, wheel speed/current, and commanded versus actual motion. 4. Use alternate paths if mobility performance degrades.	Planned; mobility testing and conservative operating rules will reduce this risk as the design matures.
3	4	Moderate	Mitigate (M)	↓ Decrease	1. Use thermally insulated battery enclosures. 2. Monitor battery temperature with sensors. 3. Include battery limits in thermal analysis. 4. Verify battery performance in cold-temperature testing.	Planned; battery thermal protection will be verified through thermal analysis and environmental testing.
3	3	Moderate	Mitigate (M)	↓ Decrease	1. Identify long-lead mobility parts early. 2. Order development hardware before final flight-like hardware when possible. 3. Maintain alternate suppliers. 4. Use representative surrogate hardware for early fit checks and mobility tests.	Planned; procurement tracking will begin early to protect mobility testing schedule margin.
3	3	Moderate	Watch (W)	→ Neutral	1. Survey multiple targets. 2. Define useful nondetection results. 3. Cross-check MSolo, GPR, and LISTER data. 4. Prioritize target diversity in operations planning.	Planned; target selection and nondetection criteria will be watched as science operations planning matures.
2	3	Moderate	Mitigate (M)	↓ Decrease	1. Evaluate shielding at PDR. 2. Use power isolation. 3. Limit co-processor exposure during high-risk events where possible. 4. Verify fallback operations through the rad-hard flight computer.	Planned; shielding and fallback operations will be evaluated as the CDH design matures.

Figure 86: Advanced Risk Log for Risks 5–10, showing Likelihood through Status of Mitigation Plans columns.

Before traverse planning review	Primary and alternate route assessment complete	Before mission operations readiness review	Route plan supports safe access to the Exploration Zone.	Current: Risk is tracked as a moderate landing-zone and mobility concern because rover-scale route hazards still require verification.
Before mobility testing	Regolith-simulant wheel test begins	Before integrated rover testing	Rover mobility meets traction, slip, and traverse requirements.	Current: Risk is tracked as a moderate mobility concern because mobility is mission-enabling, but conservative traversal and testing reduce the likelihood.
Before thermal model review	Battery thermal margin review complete	Before environmental verification	Battery remains within acceptable temperature and power margins.	Current: Risk is linked to both power and thermal performance and will be controlled through battery margin tracking and cold-temperature verification.
Before procurement review	Long-lead mobility parts identified	Before mobility subsystem test readiness	Mobility hardware is available without delaying planned testing.	Current: Risk is being managed as a schedule risk because early procurement and alternate suppliers can reduce the chance of test delay.
Before science operations planning review	Target diversity and nondetection criteria reviewed	Before science operations readiness review	Science plan defines how nondetections will be interpreted.	Current: Risk remains under watch because volatile presence is a science uncertainty that can be reduced through target diversity and data cross-checking.
Before PDR	Co-processor shielding and isolation review	Before CDR	Co-processor risk is reduced or fallback operation is verified.	Current: Risk is tracked as a moderate CDH concern because the co-processor supports autonomy but can be protected by shielding, isolation, and fallback operations.

Figure 87: Advanced Risk Log for Risks 5-10, showing Trigger Date through Status Updates columns.

11	TH-03	Thermal / Mechanical	Thermal and Mechanical Leads	Scope	Medium Term	Thermal Cycling Degrades Components	Given that the rover will experience repeated thermal cycling, there is a possibility that components degrade, adversely impacting structures, harnesses, mechanisms, electronics, and payload interfaces, which can result in degraded performance or hardware failure.
12	LZ-R3	Landing Zone / Science	Mission Design and Science Leads	Scope	Medium Term	No Accessible PSR Science Target Within 250 m	Given that the landing site must support access to an Exploration Zone within 250 meters, there is a possibility that no suitable PSR science target is accessible within the required distance, adversely impacting science operations, which can result in reduced thermal, volatile, and subsurface characterization.
13	SYS-R1	Programmatics / Cost	Project Management and Resources Leads	Cost	Long Term	Cost Cap Pressure	Given that the mission must remain within the $250 million life-cycle cost allocation for Phases C through F, there is a possibility that subsystem growth, testing needs, procurement delays, or integration rework increase cost, adversely impacting programmatics, which can result in descoping, reduced reserves, or cost-cap pressure.
14	SYS-R2	Programmatics / Integration	Systems Engineering Lead	Schedule	Long Term	System Integration Readiness Delay	Given that the Lunar Surface Vehicle system must be ready for system integration by December 1, 2028, there is a possibility that subsystem development, procurement, or verification delays prevent integration readiness, adversely impacting schedule, which can result in reduced test margin, late rework, or missed integration milestones.
15	SYS-R4	Systems / Mechanical	Systems and Mechanical Leads	Cost/Schedule /Scope	Medium Term	Mass Growth Above 175 kg	Given that the Lunar Surface Vehicle may not exceed a total mass of 175 kg, there is a possibility that subsystem mass growth exceeds available margin, adversely impacting system design and launch compatibility, which can result in redesign, descoping, added cost, or reduced payload capability.
16	SCI-R2	Science / Payload Verification	Science Payload Lead	Scope	Medium Term	Payload Calibration or Performance Shortfall	Given that payload instruments must meet required measurement ranges, sensitivities, and calibration needs, there is a possibility that one or more instruments do not meet expected performance during verification, adversely impacting science payload operations, which can result in reduced data quality, reduced confidence in measurements, or incomplete science objectives.

Figure 88: Advanced Risk Log for Risks 11–16, showing Rank through Risk Statement columns.

2	3	Moderate	Mitigate (M)	→ Neutral	1. Conduct qualification testing. 2. Use conservative design margins. 3. Inspect components after thermal cycling. 4. Update materials or interfaces if degradation is found.	Planned; thermal-cycle testing will be included in qualification planning.
2	4	Moderate	Mitigate (M)	↓ Decrease	1. Confirm primary and alternate Exploration Zones within the required distance. 2. Verify accessibility using slope, illumination, and communication constraints. 3. Identify backup science targets.	Planned; PSR target access will be confirmed before final operations planning.
2	4	Moderate	Mitigate (M)	↓ Decrease	1. Track cost by subsystem. 2. Maintain management reserve. 3. Review cost impacts during change control. 4. Identify descoping options if cost margin decreases.	Planned; cost tracking will continue as subsystem designs and procurement assumptions mature.
2	4	Moderate	Mitigate (M)	→ Neutral	1. Track subsystem readiness against the integration milestone. 2. Monitor procurement, software, payload, thermal, power, and mobility schedule margin. 3. Escalate late items through change control. 4. Protect time for integrated testing.	Planned; integration readiness tracking will mature as subsystem schedules are finalized.
2	4	Moderate	Mitigate (M)	↓ Decrease	1. Track mass growth by subsystem. 2. Maintain mass margin and reserve. 3. Review payload and structure mass at each design milestone. 4. Identify mass-reduction options if margin decreases.	Planned; subsystem mass tracking will continue as mechanical design and payload accommodations mature.
2	3	Moderate	Mitigate (M)	New	1. Verify instrument performance against measurement requirements. 2. Complete calibration planning before payload verification. 3. Track unresolved performance margins. 4. Develop alternate operations or data-quality flags if performance is limited.	Planned; payload calibration and performance checks will mature during instrument verification.

Figure 89: Advanced Risk Log for Risks 11–16, showing Likelihood through Status of Mitigation Plans columns.

Before thermal-cycle testing	Thermal-cycle test begins	Before environmental verification completion	Components show no unacceptable degradation after thermal cycling.	Current: Risk is moderate because thermal cycling can affect multiple parts, but qualification testing and design margins provide clear controls.
Before science target review	Primary and alternate PSR targets identified	Before mission operations readiness review	At least one primary and one alternate PSR target are confirmed accessible.	Current: Risk is tracked to the 250 m Exploration Zone requirement and will be reduced through target confirmation and alternate site planning.
Before cost review	Subsystem cost margin update	Before CDR	Mission cost remains within the $250 million cap with acceptable reserve.	Current: Risk is traceable to the MDR cost cap and will be managed through cost tracking, reserves, and change control.
Before integration planning review	Integration readiness schedule review	Before December 1, 2028	Subsystems are ready for planned system integration.	Current: Risk is traceable to the December 1, 2028 integration-readiness requirement and is controlled through schedule tracking and milestone reviews.
Before mass properties review	Subsystem mass margin update	Before CDR	Total rover mass remains within the 175 kg limit with acceptable margin.	Current: Risk is traceable to the 175 kg mass constraint and will be managed through mass margin tracking and design reviews.
Before payload verification planning	Payload calibration review	Before science operations readiness review	Payload instruments meet required performance or approved operational limits are documented.	Current: Risk is traceable to MDR science measurement requirements and will be reduced through calibration and payload verification.

Figure 90: Advanced Risk Log for Risks 11-16, showing Trigger Date through Status Updates columns.

17	COM-R1	Communications / Landing Zone	Communications Lead	Schedule/Scope	Medium Term	Limited Direct-to-Earth Visibility	Given that the Nobile Rim 2 operating area may have limited Direct-to-Earth visibility depending on location, terrain, and operations date, there is a possibility that communications windows are shorter or less reliable than expected, adversely impacting command, telemetry, and science data return, which can result in delayed operations, reduced downlink, or reduced science return.
18	TH-02	Thermal	Thermal Lead	Scope	Medium Term	Heater Failure	Given that heaters are required to protect thermally vulnerable hardware, there is a possibility that a heater fails, adversely impacting the thermal subsystem and protected hardware, which can result in localized overcooling, degraded performance, or shutdown of critical components.
19	CDH-R1	CDH	CDH Lead	Scope	Medium Term	Radiation-Caused Flight Computer Upset	Given that radiation may cause events in the lunar south pole environment, there is a possibility that the flight computer experiences an upset, adversely impacting CDH, which can result in corrupted processing, safe-mode entry, or temporary loss of command execution.
20	CDH-R3	CDH / Data Storage	CDH Lead	Scope	Medium Term	Radiation-Caused Storage Corruption	Given that radiation may affect onboard memory and storage, there is a possibility that stored mission data becomes corrupted, adversely impacting science and engineering data return, which can result in loss of science products or reduced confidence in mission data.
21	POW-R4	Power	Power Lead	Scope	Medium Term	Electrical Short Circuit or Wiring Fault	Given that rover power distribution depends on wiring and electrical connections, there is a possibility of short circuits or wiring faults, adversely impacting the power subsystem, which can result in loss of electrical power, damaged hardware, or reduced mission operations.
22	MCH-R2	Mechanical / Interface	Mechanical Lead	Schedule/Scope	Medium Term	Launch Vehicle Mechanical Interface Mismatch	Given that the rover must meet launch vehicle ICD dimensional and fastener requirements, there is a possibility that the mechanical interface does not meet ICD requirements, adversely impacting launch vehicle integration, which can result in redesign, delayed integration, or inability to integrate with the launch vehicle.

Figure 91: Advanced Risk Log for Risks 17–22, showing Rank through Risk Statement columns.

2	4	Moderate	Mitigate (M)	New	1. Perform Direct-to-Earth communication analysis for the exact landing site and traverse route. 2. Plan communication windows. 3. Use store-and-forward data handling. 4. Prioritize critical data for downlink.	Planned; communication coverage will be verified as landing-site and traverse planning mature.
1	5	Moderate	Mitigate (M)	↓ Decrease	1. Use heater redundancy. 2. Monitor heater performance continuously. 3. Add safe-mode heater logic. 4. Verify heater function before environmental testing.	Planned; heater redundancy and monitoring will be verified during thermal testing.
1	5	Moderate	Mitigate (M)	↓ Decrease	1. Use a radiation-hardened processor. 2. Use EDAC. 3. Use watchdog reset. 4. Verify safe-mode recovery.	Planned; CDH fault management will be verified before integrated testing.
1	5	Moderate	Mitigate (M)	↓ Decrease	1. Use ECC protection. 2. Prioritize downlink of critical data. 3. Verify data integrity checks. 4. Maintain storage margin for retransmission or duplicate files where practical.	Planned; storage protection and downlink priority will be verified in CDH testing.
1	4	Low	Mitigate (M)	→ Neutral	1. Use insulated wiring. 2. Apply proper cable management. 3. Conduct electrical continuity and fault testing. 4. Inspect harness routing before integration.	Planned; wiring checks will be included in subsystem and integrated testing.
1	4	Low	Mitigate (M)	→ Neutral	1. Verify through CAD inspection. 2. Perform drawing checks. 3. Complete dimensional inspection. 4. Include interface compliance in integration reviews.	Planned; interface compliance will be verified before launch vehicle integration.

Figure 92: Advanced Risk Log for Risks 17–22, showing Likelihood through Status of Mitigation Plans columns.

Before operations planning review	Direct-to-Earth communication coverage review	Before mission operations readiness review	Communication plan supports required command, telemetry, and science data return.	Current: Risk is traceable to landing-zone and operations assumptions requiring Direct-to-Earth communication support for command, telemetry, and science data return.
Before thermal subsystem testing	Heater functional test begins	Before thermal-vacuum qualification	Heater zones maintain required temperatures or recover through redundancy.	Current: Risk is controlled as a moderate thermal risk because redundancy and monitoring reduce the likelihood of mission-level impact.
Before CDH design review	Fault-management review complete	Before integrated system verification	Flight computer detects, corrects, or recovers from representative upset cases.	Current: Risk is controlled through the rad-hard flight computer architecture, EDAC, watchdog protection, and recovery testing.
Before CDH verification	Storage integrity test begins	Before mission operations readiness review	Critical data are protected and downlink priority is verified.	Current: Risk is controlled through ECC protection, data-integrity checks, and prioritized downlink of critical data.
Before electrical integration	Harness and wiring inspection begins	Before integrated system verification	Power wiring passes continuity, insulation, and fault-protection checks.	Current: Risk is managed through standard electrical integration controls, continuity testing, and harness inspections.
Before interface review	CAD and drawing interface check complete	Before launch vehicle integration readiness	Mechanical interface meets ICD dimensional and fastener requirements.	Current: Risk is traceable to launch-vehicle interface requirements and is controlled through CAD, drawing, dimensional, and integration reviews.

Figure 93: Advanced Risk Log for Risks 17-22, showing Trigger Date through Status Updates columns.

23	MCH-R3	Mechanical / Structure	Mechanical Lead	Scope	Medium Term	Fasteners Loosen During Launch or Rover Operations	Given that launch vibration and rover operations can load mechanical joints, there is a possibility that fasteners loosen or back out, adversely impacting rover structure and mechanisms, which can result in reduced structural reliability, damaged hardware, or failed mechanisms.
24	SYS-R3	Programmatics / Launch Readiness	Project Management Lead	Schedule	Long Term	Launch Readiness Delay	Given that the Lunar Surface Vehicle must be ready for launch by February 1, 2029, there is a possibility that late subsystem issues, verification delays, or integration rework prevent launch readiness, adversely impacting programmatics, which can result in delayed launch preparation, reduced test closure time, or failure to meet the launch-readiness milestone.
25	SYS-R5	Systems / Mechanical Packaging	Systems and Mechanical Leads	Schedule/Scope	Medium Term	Stowed Envelope Constraint	Given that the Lunar Surface Vehicle must remain within the 1.5 m by 1.5 m by 1.0 m stowed envelope, there is a possibility that payloads, mobility hardware, sensors, antennas, or solar hardware exceed the allowed packaging volume, adversely impacting mechanical integration, which can result in redesign, added testing, or delayed integration.
26	SCI-R1	Science / Payload	Science Payload Lead	Scope	Medium Term	Payload Contamination Affects Volatile Measurements	Given that volatile measurements are sensitive to spacecraft outgassing, dust, leaked materials, or disturbed regolith, there is a possibility that the sample environment becomes contaminated, adversely impacting MSolo and related science measurements, which can result in incorrect volatile composition results or reduced confidence in water ice assessment.
27	CDH-R4	CDH / Payload Interfaces	CDH Lead	Scope	Medium Term	Instrument Data Path Severed by Launch or Thermal Cycling	Given that launch conditions and thermal cycling can stress electrical and data interfaces, there is a possibility that an instrument data path is severed or degraded, adversely impacting payload data transfer, which can result in partial loss of science data or reduced payload functionality.
28	TH-04	Thermal / Sensors	Thermal Lead	Scope	Medium Term	Temperature Sensor Malfunction	Given that thermal control depends on temperature sensor feedback, there is a possibility that a temperature sensor malfunctions, adversely impacting thermal monitoring and control, which can result in incorrect heater response, reduced thermal margin, or delayed detection of unsafe temperatures.

Figure 94: Advanced Risk Log for Risks 23–28, showing Rank through Risk Statement columns.

1	4	Low	Mitigate (M)	↓ Decrease	1. Use required locking features. 2. Apply calibrated torque procedures. 3. Verify thread engagement. 4. Inspect critical joints after vibration testing.	Planned; fastening process controls will be included in assembly and verification.
2	4	Moderate	Mitigate (M)	↓ Decrease	1. Maintain launch-readiness milestone tracking. 2. Protect final verification and closeout time. 3. Monitor open risks and action items before launch readiness. 4. Escalate late changes through change control.	Planned; launch-readiness tracking will mature after integration and verification planning are finalized.
2	2	Low	Watch (W)	↓ Decrease	1. Perform CAD packaging checks. 2. Track payload, sensor, antenna, and solar hardware placement. 3. Review stowed configuration at each design update. 4. Identify alternate packaging layouts if needed.	Planned; packaging will be watched as CAD design and subsystem placement mature.
2	3	Moderate	Watch (W)	↑ Increase	1. Use contamination-control procedures. 2. Review materials and outgassing sources. 3. Perform background checks before volatile measurements. 4. Cross-check MSolo results with other payload data.	Planned; contamination controls will be refined as payload operations planning matures.
1	3	Low	Watch (W)	↓ Decrease	1. Use redundancy in bus architecture. 2. Conduct qualification testing. 3. Inspect instrument data paths after vibration and thermal-cycle testing. 4. Verify payload data transfer before integration closeout.	Planned; bus redundancy and interface testing will be verified during payload interface testing.
1	3	Low	Watch (W)	↓ Decrease	1. Use multiple sensors. 2. Include fault detection. 3. Verify sensor readings during thermal testing. 4. Compare readings across nearby sensors.	Planned; sensor redundancy and fault detection will be verified during thermal testing.

Figure 95: Advanced Risk Log for Risks 23–28, showing Likelihood through Status of Mitigation Plans columns.

Before structural assembly	Critical joint inspection procedure approved	Before vibration qualification	Critical fasteners remain secure after required testing.	Current: Risk is controlled through locking features, torque procedures, thread-engagement checks, and post-test inspection.
Before launch-readiness planning review	Launch-readiness milestone review	Before launch readiness review	All launch-readiness actions and verification closeouts are complete.	Current: Risk is traceable to the February 1, 2029 launch-readiness requirement and will be managed through milestone tracking and formal review.
Before packaging review	Stowed configuration review	Before CDR	Rover stowed configuration fits within the required envelope with acceptable clearance.	Current: Risk is traceable to the MDR stowed-envelope constraint and will remain under watch through packaging checks.
Before payload operations review	Contamination-control plan review	Before science operations readiness review	MSolo background and contamination limits are verified.	Current: Risk is traceable to MSolo volatile-measurement objectives and will be watched through contamination-control planning.
Before payload interface testing	Payload data-path test begins	Before integrated system verification	Instrument data paths remain functional after qualification testing.	Current: Risk is kept under watch because bus redundancy and qualification testing reduce the likelihood of mission-level data loss.
Before thermal subsystem testing	Temperature sensor check begins	Before thermal-vacuum qualification	Temperature sensors provide reliable data or trigger fault detection.	Current: Risk remains under watch because multiple sensors and fault detection reduce the likelihood of a single sensor causing mission-level impact.

Figure 96: Advanced Risk Log for Risks 23-28, showing Trigger Date through Status Updates columns.

1.1.1. Failure Mode and Effect Analysis (FMEA)

1. FMEA Overview

The Failure Mode and Effects Analysis identifies mission-level failures that could affect rover survival, power availability, thermal control, command and data handling, and science return. The highest-priority risks from the Advanced Risk Log were selected based on likelihood, consequence, risk rating, and impact to mission-enabling functions. The FMEA focuses on failures that could interrupt operations, reduce data return, damage hardware, or prevent completion of mission objectives

2. Failure Modes & Effects Analysis

Function	Failure Mode	Effects	Sev	Cause	Occ	Prevention	Det	RPN	Actions
Power Generation and Distribution	Solar panel dust accumulation	Reduced solar power generation, slower battery recharge, shorter traverses, and reduced science time	5	Fine lunar dust settles on exposed solar panels during surface operations	4	Solar panel orientation planning, dust-loss power margin, and dust exposure analysis	3	60	Complete dust coverage analysis, update the power budget, and verify acceptable solar output under dust-degraded conditions
	Dust on electrical and power interfaces	Reduced power efficiency, intermittent subsystem performance, increased resistance, and shortened mission operations	5	Abrasive lunar dust collects on exposed electrical contacts, power interfaces, or mechanical connection points	4	Covers, shielding, protected routing, interface inspection criteria, and dust exposure testing	3	60	Identify all exposed power interfaces and test interface resistance after dust exposure
	High compute power draw	Reduced science time, shorter traverses, increased power demand, and reduced power margin	3	NeuPAN and the perception pipeline draw more power than expected on the GR712RC + Aitech S-A1760 compute stack	4	Compute benchmarking, power budget updates, duty cycling, and reduced-rate processing if needed	3	36	Benchmark compute stack power draw and update the operations plan if power use exceeds allocation
Thermal Survival and Hardware Protection	Thermal control underperformance	Battery capacity loss, electronics degradation, instrument shutdown, and possible hardware failure	5	Thermal system cannot maintain safe temperatures across illuminated and PSR boundary conditions	3	Thermal modeling, insulation, heaters, radiator sizing, and thermal-vacuum testing	4	60	Update the thermal model, verify component temperature limits, and complete thermal-vacuum testing
	Thermal cycling damage	Cracking, loosening, material fatigue, harness damage, and mechanism degradation	4	Repeated hot-to-cold transitions between illuminated terrain and PSR boundary regions	3	Material selection, thermal-cycle testing, and inspection of structures, harnesses, joints, coatings, and mechanisms	4	48	Complete thermal-cycle testing and update material or joint design if unacceptable degradation is found
	Battery degradation in PSR	Reduced traverse time, instrument shutdown, and reduced ability to complete planned science operations	3	Battery loses capacity in very cold PSR terrain	2	Battery thermal margin tracking, heater/insulation analysis, and cold-temperature battery testing	3	18	Verify battery performance during cold-temperature testing and update power/thermal margins
Environmental Verification	Insufficient integrated environmental testing	Unexpected surface failures, reduced science return, and unverified subsystem interactions	5	Limited environmental testing does not reveal cold, dust, low-light, software, mobility, or payload interactions	4	Integrated test plan, thermal-vacuum testing, dust exposure testing, low-light testing, mobility testing, software testing, and payload verification	4	80	Protect environmental test time, link verification planning to schedule and facility access, and complete integrated system testing
	Out-of-state testing facility access delay	Reduced test time, rushed verification, and delayed environmental testing	3	Facility scheduling conflict, transit delay, or maintenance extension	3	Early facility reservation, backup test windows, alternate facility planning, and schedule tracking	3	27	Confirm facility access early and identify alternate testing options
	Schedule margin loss	Rushed testing, reduced mission readiness, and increased chance of missed verification issues	4	Delays in payload procurement, software development, environmental testing, or subsystem integration	3	Critical-path tracking, weekly schedule reviews, and early escalation of schedule slips	3	36	Track schedule margin regularly and escalate risks that threaten integration or launch-readiness milestones
Communications and CDH	Communication loss near PSR terrain	Delayed commands, delayed telemetry, reduced science data return, and slower operations	4	Crater rims or PSR boundary terrain block or weaken line-of-sight communication	3	Communication windows, antenna placement analysis, relay/store-and-forward buffering, and onboard autonomy	3	36	Complete communication coverage analysis and verify data handling during temporary communication gaps
	Radiation-induced electronics upset	Corrupted data, safe-mode events, or temporary loss of processor, sensor, or payload function	4	Solar particles or cosmic radiation cause single-event upsets in electronics	3	Radiation-tolerant parts, fault detection, reset procedures, error checking, and safe-mode recovery	3	36	Verify CDH fault detection, recovery logic, and data protection before integrated testing
	Low-light sensor performance loss	Unsafe path planning, reduced traverse speed, and lower navigation confidence	2	Cameras or vision sensors lose hazard-detection performance in deep shadows or high-contrast lighting	2	Low-light testing with stereo nav cameras, Hazcams, IMU, wheel encoders, and illumination system	3	12	Validate navigation sensor performance in low-light and high-contrast PSR boundary conditions
Science Payload Performance	MSolo contamination	Incorrect volatile composition results and reduced confidence in water ice and ISRU measurements	3	Spacecraft outgassing, dust, leaked materials, or disturbed regolith contaminates sample environment	2	Contamination-control procedures, material controls, background checks, and cross-checking with other payload data	3	18	Finalize contamination-control plan and verify MSolo background limits before science operations
	GPR signal interference	Uncertainty in ice depth, dielectric boundaries, and ISRU resource estimates	4	Rough subsurface layers, antenna placement issues, or rover electronics noise distort radar returns	3	Antenna placement review, calibration, electronics interference testing, and signal filtering	3	36	Test GPR signal quality with representative regolith layering and rover electronics noise
	LISTER probe deployment failure	Incomplete heat-flow and thermal conductivity measurements	2	Compacted, rocky, or uneven regolith prevents proper probe deployment	2	LISTER deployment testing in lunar regolith simulant and verification of the full 2-hour measurement sequence	3	12	Verify that LISTER can insert the thermal needle, equilibrate for 1 hour, heat for 30 minutes, and cool for 30 minutes at each target depth
Mobility and Surface Operations	Wheel wear from regolith	Reduced traction, higher motor load, shortened traverse capability, and reduced access to science targets	4	Sharp, abrasive lunar regolith wears wheels and mobility joints faster than expected	3	Wheel wear testing, traction testing, motor current tracking, and mobility design margin	3	36	Complete wheel and joint wear testing in lunar regolith simulant
	Rover immobilization	Loss of access to science targets and inability to complete planned traverses	2	Rover encounters loose regolith, slopes, crater rims, or shadowed terrain near PSR boundaries	1	Terrain analysis, slope testing, wheel slip testing, and conservative route planning	3	6	Verify mobility performance in representative terrain and update route planning rules
	Probe or rover disturbs science site	Less accurate interpretation of natural PSR thermal and volatile conditions	3	Drilling, probing, or rover movement disturbs regolith near volatile-rich targets before measurement	2	Rover approach planning, measurement sequencing, and documentation of disturbed areas	3	18	Finalize science operations sequence to protect undisturbed regolith before LISTER and MSolo measurements

Figure 97: Failure Mode and Effects Analysis for mission-level thermal, power, and command and data handling risks.

Thermal control underperformance:

This failure mode occurs if the thermal subsystem cannot maintain thermally vulnerable hardware within safe operating or survival limits. The likely causes are heater damage or degraded insulation performance. If this occurs, batteries, avionics, electronics, or payload instruments may degrade, shut down, or fail, reducing the rover's ability to operate near permanently shadowed region conditions.

Thermal cycling damage:

This failure mode occurs when the rover repeatedly transitions between illuminated regions and permanently shadowed region boundary conditions. The cause is repeated exposure to hot-to-cold and cold-to-hot thermal changes. These transitions can stress sensors, insulation, joints, mechanisms, and payload interfaces, which could result in

uncertain science measurements, increased power demand, reduced mobility performance, or reduced confidence in volatile measurements.

Inadequate power generation:

This failure mode occurs if the rover cannot generate enough power to recharge batteries or operate effectively in illuminated regions. Causes include lunar regolith accumulation on solar panels, poor solar panel orientation, solar cell degradation, or panel damage. The affected subsystem is the power subsystem, and the mission-level impact could include reduced operating time, limited mobility, reduced science activities, or repositioning the rover toward better sunlight.

Battery underperformance:

This failure mode occurs if the rover batteries do not provide the expected capacity or operating duration. Causes include battery degradation, low battery temperatures during PSR operations, or peak power demand exceeding available battery output. This could shorten PSR operating time, reduce battery capacity, or force the rover into safe mode or survival-power mode, limiting science return and surface operations.

Electrical short circuit:

This failure mode occurs if wiring damage or connector failure causes a short circuit in the rover electrical system. The affected subsystem is power distribution, but the impact could spread to other rover components if the fault is not isolated. Potential effects include loss of electrical power, hardware damage, and interruption of operations.

Flight computer failure:

This failure mode occurs if a radiation-induced single-event effect disrupts or halts the flight computer. The affected subsystem is CDH, and the effects could include loss of commanding, interruption of operations, safe-mode entry, or possible mission loss if recovery is unsuccessful. The radiation-hardened GR712RC, EDAC memory protection, and hardware watchdog reduce this risk.

Co-processor failure:

This failure mode occurs if the co-processor experiences radiation exposure beyond its rated tolerance. The main effect is loss or reduction of the NeuPAN demonstration and related autonomy-processing capability. Since the main flight computer can continue regular mission functions, this failure mode is not expected to cause full mission loss, but it could reduce technology-demonstration performance.

Onboard storage failure:

This failure mode occurs if radiation-induced bit errors corrupt stored science or engineering data. The affected subsystem is CDH data storage, and the mission-level impact is loss of important data before it can be downlinked. ECC protection and prioritized downlink reduce the chance of permanent data loss.

Data bus failure:

This failure mode occurs if transceiver damage or thermal cycling causes loss of a subsystem data path. The affected subsystem is CDH, and the effects include loss of internal data flow and increased interface complexity. Redundant bus architecture and qualification testing reduce this risk, and traffic can be moved to the redundant path if needed.

3. Mitigation Summary

The identified failure modes are reduced through design features, redundancy, recovery strategies, and operational procedures. Thermal risks are reduced through heaters, insulation, temperature sensors, passive thermal control, thermal-vacuum testing, and heater adjustment based on the thermal environment. Power risks are reduced through protective solar panel covers, solar panel orientation planning, battery thermal insulation, battery health monitoring, insulated wiring, cable management, circuit protection, and power sequencing. CDH risks are reduced through the radiation-hardened GR712RC flight computer, EDAC and ECC protection, watchdog recovery, redundant data paths, power isolation, and prioritized downlink.

The highest remaining failure modes are inadequate power generation, battery underperformance, thermal control underperformance, and flight computer failure because they directly affect rover survival, command execution, operating time, and science return. Future design phases should focus on verifying solar power margins, improving battery thermal protection, confirming heater and insulation performance, testing electrical fault protection, and validating CDH recovery functions. These actions will help reduce likelihood, improve detection, and lower mission risk before final integration and readiness.

1.1.2. Personnel Hazards and Mitigations

1. Safety Overview

Personnel safety during mission development will be managed through hazard identification, required training, safe work procedures, and facility-specific safety rules. The team will identify hazards before manufacturing, assembly, integration, testing, laboratory work, cleanroom operations, and office work begin. Hazards will be tracked by activity and location so that each risk has a clear mitigation strategy, responsible team member, and required safety control. Safety procedures will be reviewed before major test events, especially those involving electrical systems, batteries, thermal-vacuum testing, lifting, machining, or hazardous materials.

2. Personnel Hazards & Mitigations

During mission development, personnel may be exposed to hazards from fabrication, electronics integration, battery handling, thermal testing, laboratory work, cleanroom work, and general office activities. These hazards are separate from the lunar surface risks described in the mission risk matrix. The personnel hazards focus on protecting team members, lab users, technicians, and facility staff during the design, build, integration, and test phases.

Electrical hazards are a major concern during power subsystem integration, electronics testing, battery testing, and CDH hardware checkout. Personnel may be exposed to electrical shock, short circuits, overheating, arc flash, or damaged equipment if power systems are handled incorrectly. Mitigations include using lockout/tagout procedures when appropriate, insulated tools, current-limited power supplies, proper grounding, non-conductive work surfaces, buddy-system verification, and pre-test checklists before power is applied.

Battery handling also presents safety risks, especially during charging, storage, transportation, and testing. Lithium-based batteries can overheat, vent gas, rupture, or catch fire if they are damaged, overcharged, shorted, or exposed to unsafe temperatures. Mitigations include using approved battery storage containers, battery management systems, temperature monitoring, charge/discharge limits, lithium-rated fire suppression methods, and clear emergency procedures for battery swelling, leakage, smoke, or thermal runaway.

Manufacturing and mechanical assembly activities introduce hazards from machining, drilling, cutting, sanding, deburring, sharp edges, moving parts, and flying debris. Personnel may experience lacerations, puncture wounds, eye injuries, burns, or crush injuries if tools and machinery are not used properly. Mitigations include machine shop training, safety glasses, face shields when required, gloves when appropriate, guards, machine interlocks, proper clamping, debris shields, tool inspection, and deburring all sharp edges before handling hardware.

Thermal testing and environmental testing can expose personnel to hot surfaces, cold surfaces, vacuum equipment, cryogenic materials, pressurized lines, and oxygen-depletion hazards. Thermal-vacuum chambers, cold plates, heaters, and test fixtures may cause burns, frostbite, or pressure-related injuries if handled incorrectly. Mitigations include thermal gloves, warning labels, temperature verification before handling, oxygen sensors, pressure relief devices, vacuum-rated hardware, restricted access during tests, and trained facility operators.

Cleanroom and laboratory work can also create personnel hazards. Cleanroom garments may limit movement or visibility, and contamination-control procedures may involve solvents, wipes, adhesives, lubricants, or cleaning chemicals. Laboratory work may also include small tools, fragile hardware, electrical test equipment, and chemical exposure. Mitigations include chemical safety training, Safety Data Sheet review, proper ventilation, gloves, eye protection, cleanroom gowning procedures, labeled containers, spill kits, and proper waste disposal.

System integration creates lifting, material-handling, and ergonomic hazards. Rover structures, test equipment, batteries, payload hardware, and support fixtures may be heavy, awkward, or fragile. Improper lifting can cause musculoskeletal injuries, dropped hardware, crush injuries, or damage to flight-like equipment. Mitigations include two-person lift rules, mechanical hoists, carts, lift plans, hardware tie-downs, clearly marked pinch points, and keeping walkways clear during integration.

Office and computer-based work also creates safety concerns, even though it is lower risk. Long periods of computer work can cause eye strain, wrist pain, back pain, and fatigue. Poor cable management or cluttered work areas can also create trip hazards. Mitigations include ergonomic workstation setup, regular breaks, safe cable routing, clean workspaces, proper lighting, and reporting unsafe office conditions early.

3. *Hazard Summary Table*

Hazard	Activity/Location	Potential Consequences	Mitigation
High-voltage or High-current Electrical Exposure	Power Subsystem Integration / Electronics Lab	Electrical shock, arc flash, thermal burns, short circuits, or damaged test equipment	Use insulated tools, current-limited supplies, proper grounding, lockout/tagout when applicable, non-conductive mats, pre-test checklists, and buddy-system verification before applying power
Battery Handling and Charging Hazards	Battery Storage and Testing / Power Lab	Cell rupture, chemical burns, toxic gas exposure, overheating, fire, or thermal runaway	Use approved battery storage containers, battery management systems, temperature monitoring, charge/discharge limits, lithium-rated fire suppression methods, and emergency response procedures
Machining and Manufacturing Equipment	Machine Shop / Fabrication Area	Lacerations, puncture wounds, eye injuries, burns, or impact injuries from flying debris	Require machine shop training, safety glasses, face shields when needed, guards, interlocks, proper clamping, debris shields, and tool inspections
Sharp Tools and Hardware Edges	Mechanical Assembly / Integration Area	Cuts, punctures, torn gloves, or damage to cables and harnesses	Deburr hardware edges, use cut-resistant gloves when appropriate, store tools safely, cover sharp edges, and inspect parts before handling
Lifting and Material Handling	System Integration / Cleanroom Assembly Floor	Back strain, dropped hardware, crush injuries, fractures, or damaged rover components	Use two-person lifts, mechanical hoists, carts, lift plans, hardware restraints, and marked keep-out zones around suspended loads
Thermal Hazards from Hot Surfaces	Thermal Testing / Electronics Testing / Heater Testing	Skin burns, damaged PPE, or accidental contact	Use thermal gloves, warning labels, cooldown periods, temperature checks, restricted access, and test

		with hot test articles	conductor approval before handling hardware
Cold Surface or Cryogenic Hazards	Thermal-vacuum Facility / Environmental Test Area	Frostbite, cold burns, oxygen depletion, or pressure-related injury	Use cryogenic PPE, oxygen-depletion sensors, trained facility operators, controlled access, vacuum-rated equipment, and pressure relief devices
Pressurized Systems and Vacuum Equipment	Thermal-vacuum Chamber / Environmental Testing	Implosion, flying debris, pressure release injury, or hearing damage	Use pressure-rated hardware, pressure relief valves, chamber inspection, exclusion zones, trained operators, and controlled test procedures
Hazardous Materials and Chemicals	Laboratory / Cleanroom / Bonding and Cleaning Stations	Chemical burns, skin irritation, inhalation exposure, fire hazard, or contamination	Review Safety Data Sheets, use gloves and eye protection, provide ventilation, label chemicals, maintain spill kits, and follow hazardous-waste disposal procedures
Cleanroom and Contamination-control Hazards	Cleanroom Assembly / Payload Handling	Slips, restricted movement, visibility limits, chemical exposure, or contamination of hardware	Require cleanroom training, proper gowning, cleanroom-compatible tools, controlled chemical use, clean walkways, and contamination-control procedures
Electrical Test Equipment Misuse	CDH Testing / Payload Electronics Checkout	Equipment damage, electric shock, incorrect readings, or unsafe test configuration	Use approved test procedures, verify wiring before power-on, label cables, use fuses/current limits, and require peer review of test setups
Trip and Slip Hazards	Labs, Office Areas, Integration Floor	Falls, sprains, dropped hardware, or damage to test equipment	Keep walkways clear, manage cables, clean spills immediately, use floor markings, and store equipment in assigned locations
Ergonomic Hazards	Office Work / CAD Design /	Eye strain, wrist pain, back pain,	Use ergonomic workstations, proper monitor height,

	Documentation / Long Analysis Sessions	neck strain, fatigue, or reduced productivity	regular breaks, adjustable chairs, wrist support, and task rotation when possible
Noise Exposure	Machine Shop / Environmental test Facility	Hearing damage, distraction, or communication errors	Use hearing protection when required, post noise-warning signs, restrict access during loud operations, and follow facility noise limits
Fire Hazards	Battery Lab / Electronics Lab / Chemical Storage Area	Burns, smoke inhalation, equipment damage, or facility evacuation	Keep fire extinguishers accessible, use lithium-rated response procedures, store flammables properly, keep ignition sources controlled, and maintain emergency evacuation routes

Table 99: Personnel hazard summary for mission development, integration, laboratory work, cleanroom operations, and testing activities.

4. Safety Management

Personnel safety procedures will be implemented through activity-specific checklists, training requirements, facility rules, and pre-test reviews. Before manufacturing, integration, or testing begins, the team will identify the hazards associated with the activity and confirm the required controls. Activities involving electrical power, batteries, thermal-vacuum testing, machining, lifting, chemicals, or cleanroom operations will require appropriate supervision and facility approval. Team members will only operate equipment or enter controlled test areas after receiving the required training.

Required safety practices include PPE use, buddy-system checks for energized testing, clear labeling of hazardous materials, inspection of tools and test setups, safe lifting procedures, and documentation of test readiness before major operations. The team will use standard PPE such as safety glasses, gloves, lab coats, cleanroom garments, thermal gloves, hearing protection, and closed-toe shoes as required by the activity. Any incident, near miss, equipment damage, or unsafe condition will be reported and reviewed so procedures can be updated before work continues.

Remaining personnel hazards that require the most attention are battery testing, energized electronics work, thermal-vacuum testing, lifting during system integration, and L'SPACE Program 71 machining activities. These activities have higher consequences because they can cause serious injury or damage critical hardware. As the project moves into later design and test phases, the team will continue updating safety procedures and confirming that each high-risk activity has trained personnel, approved facilities, proper PPE, emergency response plans, and clear stop-work authority.

2. Activity Plan

2.1. Project Management Approach

1. Mission Team Overview

The mission team is organized using a matrixed management structure developed using and aligned with NASA/SP-2016-6105 Rev 2, NASA/SP-2010-576, and NPR 7120.5F governance standards. Executive oversight is led by the Project Manager (PM), who holds final authority over cost, schedule, overall project execution, and baseline requirements, directly managing central reserve allocations and vendor contract oversight. Technical and systems authority is led by the Lead Systems Engineer (LSE), who maintains centralized authority over technical baselines, master schedules, interface control, and contingency reserves. Under the LSE's direct technical jurisdiction fall the primary engineering and operational execution teams: the Rover Bus Subsystem Engineering Team, Flight Software

& CDH Team, Launch-Site Operations Team, and Mission Operations & Ground Data Systems Team. Technical guidance on science traceability, instrument requirements, and mission data collection goals is provided by the Chief Scientist, who directly oversees the Payload & Science Instruments Team in coordination with the LSE for interface compliance. Specialized functional leads govern individual subsystem domains, granting technical autonomy to subteams while enforcing strict interface compliance. This hybrid organization directly supports seamless development, integration, surface operations, and mission closeout by streamlining decision-making, ensuring continuous cross-subteam communication, and maintaining rigorous risk and configuration control through every mission phase.

2. Personnel & Functional Teams

To execute Phase C (Final Design and Fabrication) through Phase F (Closeout), the mission requires a full maximum workforce of approximately 88 Full-Time Equivalents across seven specialized functional teams. Executive governance is provided by the Project Management Office (5 FTE), led by the Project Manager alongside Program Analysts, Financial/EVMS Specialists, and Subcontract Managers who oversee project cost, master scheduling, vendor contracts, earned value management, and formal Change Control Board (CCB) reviews. Working directly alongside management, the Systems Engineering & Safety Team (10 FTE) operates under the Lead Systems Engineer and comprises Systems Integration Engineers, Requirements & ICD Managers, Reliability & Risk Mitigation Engineers, Safety & Mission Assurance (S&MA) Lead Engineers, and Quality Assurance (QA) Inspectors who manage overall system baselines, safety verification, and hardware quality.

Hardware and system execution are carried out by dedicated engineering teams operating under the direct technical jurisdiction of the Lead Systems Engineer. The Rover Bus Subsystem Engineering Team (22 FTE) includes Mechanical/Structural Engineers, Thermal Control Engineers, Power Systems Engineers (EPS), and Mobility/GNC Engineers responsible for physical chassis fabrication, thermal protection, power routing, and locomotion. Software and command structures are built by the Flight Software & CDH Team (12 FTE), which consists of Real-Time Operating System (RTOS) Developers, Embedded Systems Programmers, Cyber-Security Analysts, and Autonomy Specialists focused on NeuPAN algorithm integration and processing architecture. Scientific research and instruments are managed by the Payload & Science Instruments Team (18 FTE), reporting jointly to the Chief Scientist for science traceability and the Lead Systems Engineer for interface compliance. This team houses Instrument Scientists, Bioastronautics Specialists, Electrical Engineers, and Physical Integration Specialists organized into

dedicated payload subteams for LISTER, MSolo, TENG, the Geophone Array, and the MECA Electrometer.

As hardware matures toward deployment, the Launch-Site Operations Team (11 FTE) operates under the Lead Systems Engineer during Assembly, Integration, and Test (AI&T), utilizing Cleanroom Integration Technicians, Environmental Testing Specialists for thermal-vacuum (TVAC) and vibration testing, and Launch Vehicle Integration Interfaces to prepare the rover for launch. Finally, surface execution and mission completion are driven by the Mission Operations & Ground Data Systems Team (10 FTE), which reports to the Lead Systems Engineer and Project Manager during surface operations. Composed of Flight Controllers, DSN Telecommunications Engineers, Rover Command Planners, and Ground Data Systems (GDS) Engineers, this group manages daily surface command sequences during Phase E and executes data archiving, final reporting, and closeout during Phase F. Subteam leads hold localized budget authority for pre-approved Work Breakdown Structure (WBS) line items, but unallocated financial, mass, and power reserves are managed centrally by the PMO; any trade impacting reserves or cross-subteam ICD boundaries requires formal CCB approval.

3. Mission Organizational Chart

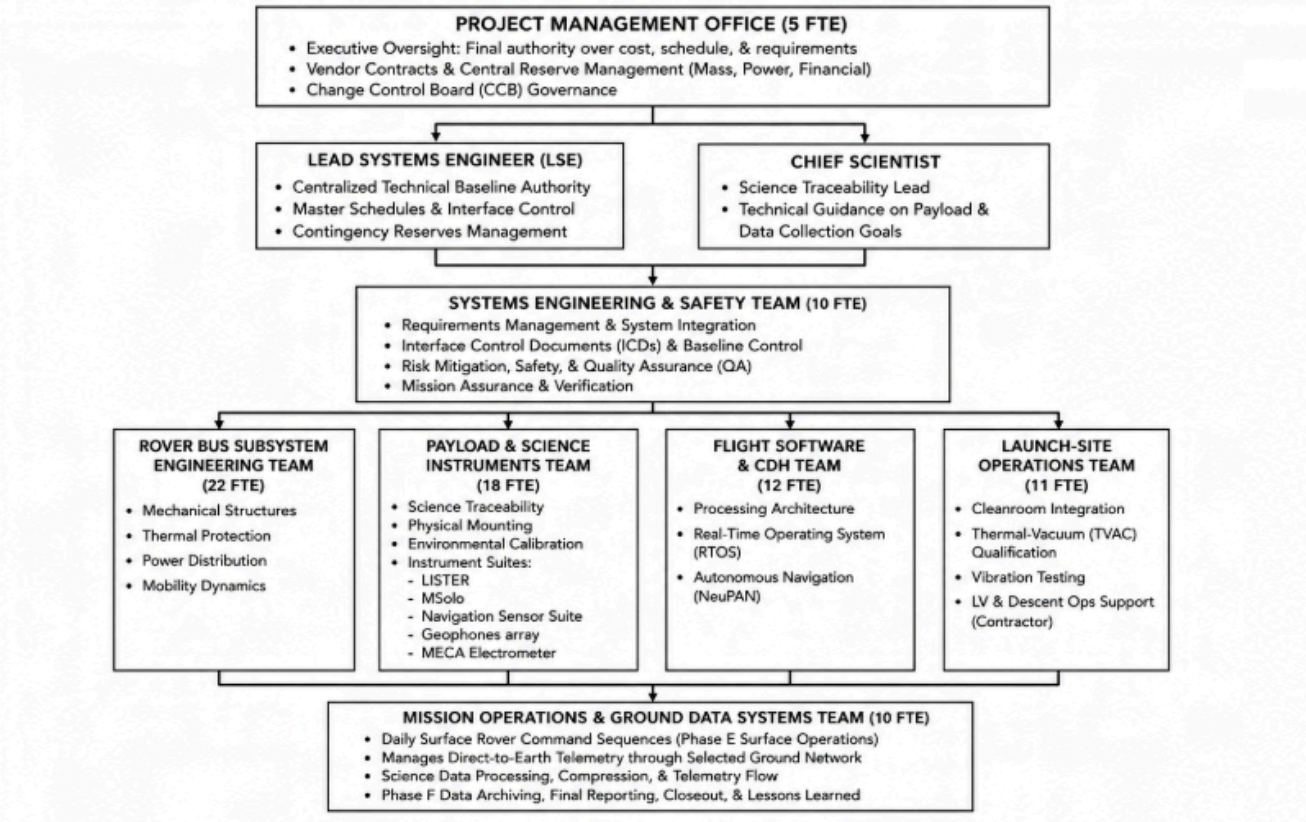


Figure 99: Project Organizational & Work Breakdown Structure (WBS) Organizational hierarchy detailing reporting structures, lead authorities, and full maximum workforce allocations (88 FTE) across Phase C through Phase F mission execution.

2.2. Mission Schedule

2.2.1. Schedule Basis of Estimate

Ground Rules: The Schedule Basis of Estimate (BOE) was developed for the post-Preliminary Design Review(PDR) part of the mission and applies to the NASA Phases C through F. Pre-Phase A, Phase A, and Phase B activities are considered completed and are treated as a sunk schedule in accordance with the customer mission constraints.. With the project schedule mainly focusing on NASA Phases C through F, while also including the final design and fabrication, system integration and testing, launch preparation, mission operations, and mission closeout (Customer-defined Integration and Launch-Readiness constraints).

The activity durations were estimated using a combination of analogous estimating, engineering judgment, and logical predecessor-sucessor relationships. Comparable NASA lunar missions, like the VIPER, Firefly Blue Ghost Mission 1, CLPS missions, and LCROSS, were reviewed to establish more realistic durations for design, fabrication, integration, environmental testing, and launch preparation activities. With the estimates being refined using engineering judgment based on the complexity of the proposed rover mission and organized within the project's Work Breakdown Structure (WBS).

This schedule was made using precedence relationships for important engineering activities to make sure that an executable project timeline is developed. So, when the subsystem fabrication begins, subsystem fabrication precedes system integration, and integrated system verification happens which allows the environmental qualification testing and launch readiness to happen so that the final design activities can be completed. To maintain acceptable technical risk programmatic activities and independent engineering are planned to run concurrently where practical to improve schedule efficiency (NASA Systems Engineering Handbook, Sec. 6.1.1.2.3, p. 121; NASA Schedule Management Handbook, Rev. 1, Sec. 5.5, pp. 77–80)

Schedule Constraints:

- The Post-PDR schedule includes only NASA Phases C through F.
- System ready for integration by December 1, 2028
- Launch readiness by February 1, 2029

● Personnel travel required for Key Decision Point reviews, environmental qualification testing, and launch operations is treated as a customer mission constraint and incorporated into the schedule.

● With Launch and Entry, Descent, and Landing (EDL) dates are externally constrained and therefore treated as fixed milestones.

● Delays beyond the required Launch Readiness Date (LRD) may require postponement until the next available launch opportunity, significantly impacting the overall project schedule (Customer Requirements).

● With the preliminary schedule being maintained using the calendar dates while also remaining consistent with anticipated fiscal-year funding availability and NASA life-cycle review milestones.

Key Schedule Assumptions: Several planning assumptions and the current mission concept are used for the project’s current level of maturity that are based on the preliminary schedule that is used. Resources are assumed to remain available throughout the project's execution for the engineering staffing, project funding, environmental qualification facilities, launch services, and supporting program. The objectives for the mission and the rover's architecture are assumed to remain consistent with the current baseline of the projects unless modified through future engineering analyses or approved design changes (NASA Space Flight Program and Project Management guidance, Sec. 4.3.4.3, p. 194). Payload selection, subsystem configuration, and supporting technologies remain pending on further development of engineering trade studies. Thus, future changes from payload down-selection, subsystem modifications, technology maturation, or updated engineering analyses will be added into future revisions of the Schedule Basis of Estimate as well as the project’s schedule when additional information becomes available. The continuous development and refining of the schedule estimates as the project’s maturity increases is consistent with the NASA guidance that recommends progressive refinement (NASA Schedule Management Handbook, Rev. 1, Sec. 5.4.1, pp. 74–75).

Major Schedule Drivers: The preliminary schedule is governed by several critical task dependencies that establish the logical flow of project execution. Before the slight hardware can be fabricated, the completion of the final design activities is required. Before the progression of full rover system integration, the following fabrication, payload integration, and subsystem verification must be completed. The successful system integration is required before starting the environmental qualification testing, which is needed to verify that the rover has been integrated and can withstand launch vibration, thermal-vacuum conditions, lunar dust exposure, and operational environments representative of the lunar south pole. Completion of environmental qualification testing

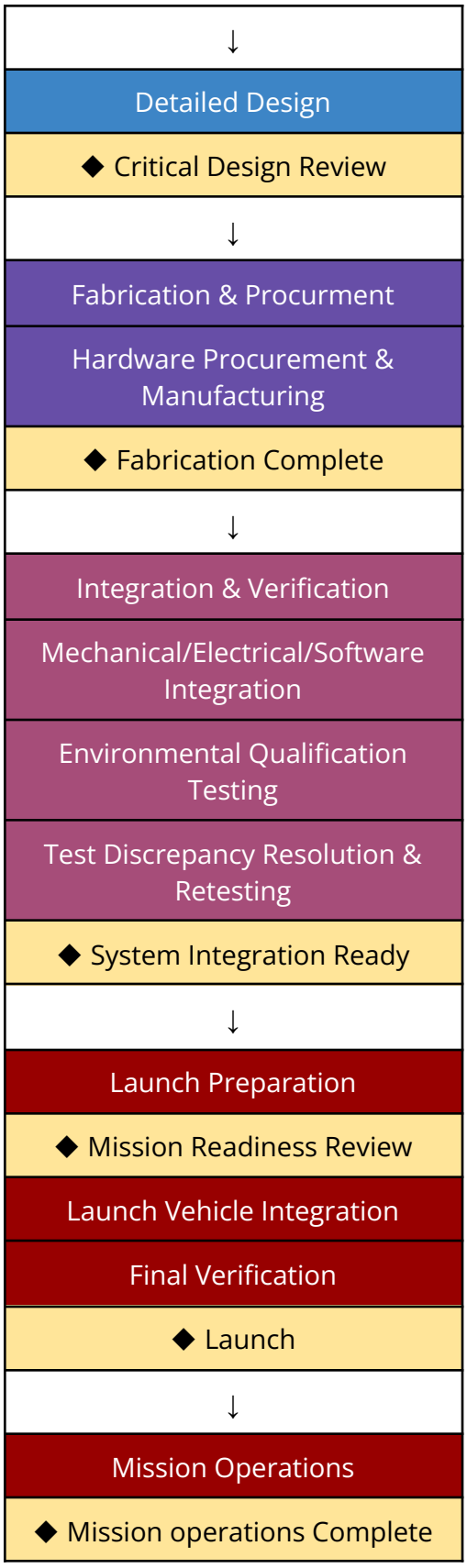
establishes the project's critical path by enabling the Flight Readiness Review (FRR), Launch Readiness Review (LRR), delivery to Kennedy Space Center, and achieving the February 1, 2029 Launch Readiness Date. Because these activities happen sequentially, any delays in the sequence of tasks will directly affect the schedule's performance downstream (NASA Systems Engineering Handbook, Sec. 6.1.1.2.3, p. 121; NASA Schedule Management Handbook, Rev. 1, Sec. 5.5, pp. 77–80). Additional scheduled drivers to consider are development and verification of the autonomous navigation system, Long-lead procurement of flight hardware ,integration of the scientific payloads, completion of mission operations planning, and preparation for launch activities. Coordination across the multiple engineering disciplines and the significant contributors is required for the overall project duration because they all directly affect the mission readiness and operational success of the project (NASA Schedule Management Handbook, Rev. 1, Sec. 5.5, pp. 77–80).

Estimate Maturity and Future Refinement: As the subsystem trade studies are completed, project risks are reduced, and additional technical information becomes available, activity durations, resource allocations, and predecessor-successor relationships will be refined within the Integrated Master Schedule (IMS). Future revisions of the Schedule Basis of Estimate will reflect updated engineering analysis while maintaining consistency with NASA schedule management guidance (NASA Schedule Management Handbook, Rev. 1, Sec. 5.4, pp. 72–77; NASA Space Flight Program and Project Management Handbook, Sec. 4.3.4.3, p. 194).

2.2.2. Mission Schedule

Critical Path: Project's preliminary critical path begins with the completion of the Preliminary Design Review (PDR) and continues through the detailed design and the Critical Design Review (CDR). Following the CDR, the critical path progresses through the hardware procurement and manufacturing, hardware completion, mechanical, electrical, and software integration, environmental qualification testing, test discrepancy resolution and retesting, and the System Integration Ready milestone. The path then continues through the launch preparation activities, including the Launch Readiness Review, launch vehicle integration, and final verification, leading to the planned February 1, 2029 launch. Following the launch, mission operations and mission closeout will complete the preliminary critical path. With delays in these schedules driving activities could directly affect subsequent milestones and ultimately delay the planned launch or completion of the mission.

◆ Preliminary Design Review


↓
Detailed Design
◆ Critical Design Review
↓
Fabrication & Procurment
Hardware Procurement & Manufacturing
◆ Fabrication Complete
↓
Integration & Verification
Mechanical/Electrical/Software Integration
Environmental Qualification Testing
Test Discrepancy Resolution & Retesting
◆ System Integration Ready
↓
Launch Preparation
◆ Mission Readiness Review
Launch Vehicle Integration
Final Verification
◆ Launch
↓
Mission Operations
◆ Mission operations Complete

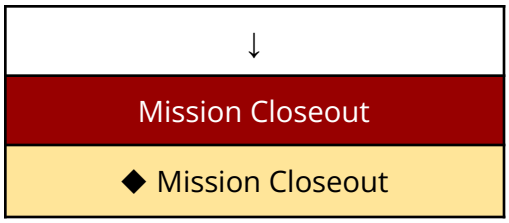


Figure 100:

The preliminary mission schedule establishes the sequence of activities required to successfully develop, integrate, launch, and operate the proposed lunar south pole rover mission. The schedule begins with the completion of the Preliminary Design Review (PDR), with all Pre-Phase A, Phase A, and Phase B activities assumed completed in accordance with the customer mission constraints. The schedule will then focus mainly on project execution during the NASA Phases C through F while also providing logical progression from final design and fabrication through mission closeout. All the activities are developed with the customer’s mission requirements and NASA schedule management guidance to ensure that appropriate sequential relationships, realistic activity durations, and with continuous management of technical and programmatic risks through the project's completion (NASA Schedule Management Handbook, Rev. 1, Sec. 5.5, pp. 77–80).

1. **Schedule Assumptions & Constraints**

The duration of the schedule was estimated using a combination of analogous NASA missions, engineering judgment, NASA Systems Engineering guidance, and project-specific technical assessments. The durations of the activities are refined through an iterative process as the subsystem designs matured, and more additional mission constraints are identified. Logical precedence relationships were established between engineering procurement, manufacturing, integration, verification, and operational activities to develop the preliminary Integrated Master Schedule (IMS).

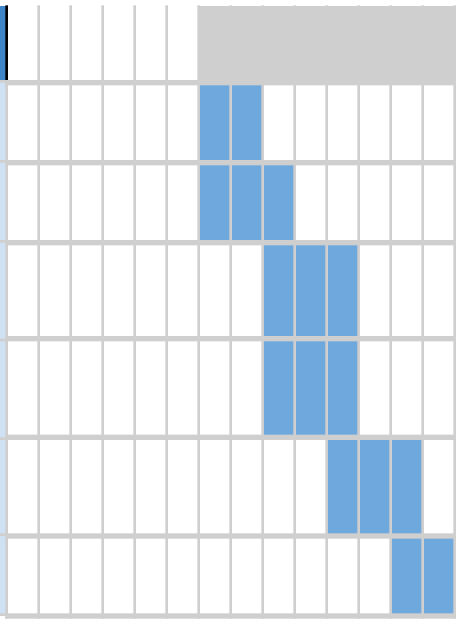
2.2.2.1. Mission Phase C Overview

Phase C begins upon the completion of the Preliminary Design Review (PDR) and focuses on completing the detailed design, fabrication, and procurement activities required to prepare the rover for integration and verification in Phase D. During the detailed design portion of Phase C, subsystem designs are matured, engineering analyses are completed, and design documentation is finalized in preparation for the Critical Design REview (CDR). This work includes CAD completion, electrical design, flight software development, navigation software, structural analysis, documentation, and long-lead procurement planning. Following the successful completion of CDR, Phase C transitions into fabrication and procurement activities, including hardware ordering, parts manufacturing, electronics

assembly, and payload fabrication. These activities lead to the Hardware Complete milestone, which establishes readiness to transition into integration and verification activities in Phase D. The Phase C schedule begins immediately following the June 30, 2027 PDR, with detailed design activities beginning in July 2027. This sequence ensures that the preliminary design is reviewed and approved before the project proceeds into detailed design and subsequent fabrication activities.

With activity durations being estimated using analogous NASA missions, engineering judgement, subsystem complexity, and predecessor-successor relationships identified in the Work Breakdown Structure. The development of the Phase C timeline was estimated using analogous NASA lunar missions (VIPER, Firefly Blue Ghost Mission 1, CLPS missions, and LCROSS), engineering judgment, subsystem complexity, and Work BreakDown Structure (WBS) predecessor-successor relationships. CAD completion, flight software development, structural analysis, and documentation durations were individually estimated based on the expected engineering effort required to achieve Critical Design Review (CDR) readiness. With the long-lead procurement planning and supplier identification are assumed to begin during the final detailed design activities so that hardware fabrication can begin immediately following the successful completion of the CDR. With schedule margins being maintained prior to major milestones to accommodate design refinements, engineering trade studies, and technical risks identified during subsystem development. As shown in the Phase C Gannt chart, with the detailed design beginning in July 2027, and continues through CDR on February 29, 2028, followed by fabrication and procurement activities leading to the Hardware Complete milestone on June 28, 2028.

4	**Detailed Design**	7/1/27	2/29/28
4.1	CAD completion	7/1/27	8/13/27
4.2	Electrical design	7/20/27	9/3/27
4.3	Flight software development	9/6/27	11/17/27
4.4	Navigation software	9/20/27	11/11/27
4.5	Structural analysis	11/18/27	1/5/28
4.6	Documentation	1/8/28	2/18/28



4.7	Long-Lead Procurement Planning	11/18/27	2/18/28
4.8	Schedule Margin	2/18/28	2/29/28
4.9	◆ Critical Design Review	2/29/28	2/29/28

5	**Fabrication & Procurement**	3/1/28	6/28/28
5.1	Order hardware	3/1/28	3/28/28
5.2	Manufacture parts	3/31/28	5/23/28
5.3	Assemble electronics	5/1/28	6/20/28
5.4	Payload fabrication	5/10/28	6/16/28
5.5	Schedule Margin	6/20/28	6/28/28
5.6	◆ Hardware Complete	6/28/28	6/28/28

Figure 101:

2.2.2.2. Mission Phase D Overview

The focus of Phase D will be on the completion of Integration & Verification and Launch Preparation activities that will prepare the lunar rover for launch and transition into Phase E. Phase D begins following the Hardware Complete milestone and includes mechanical, electrical, and software integration, followed by functional , thermal, mobility, and environmental testing. Test discrepancy resolution and retesting are included following environmental testing to provide time for corrective actions and verification of identified issues before reaching the System Integration Ready milestone. Following the System Integration Ready milestone on December 1, 2028, the schedule will transition into Launch Preparation. This will include the Mission Readiness Review, flight software freeze, launch vehicle integration, final verification, and schedule margin leading to the planned February 1, 2029 launch.

The timeline for Phase D was developed using NASA schedule management guidance, engineering judgment, and analogous lunar missions such as NASA's VIPER, Firefly Blue Ghost Mission 1, CLPS missions, and LCROSS to estimate the activities durations and task sequencing. These missions were used as comparative references for

the sequencing and relative duration of integration, environmental testing, verification, and launch preparation activities. Engineering judgment was then used to adjust the schedule for the rover's specific subsystem complexity, testing requirements, and planned launch date. With the schedule margin and dedicated test discrepancy resolution and retesting time were included to account for integration issues, testing anomalies, corrective actions, and final verification activities. As shown in the Phase D Gannt Chart, Integration & Verification begins on June 30, 2028 and progresses through the System Integration Ready milestone on December 1, 2028. Launch Preparation will then continue from December 4, 2028 through the planned February 1, 2029 launch.

6	**Integration & Verification**	6/30/28	12/1/28
6.1	Mechanical integration	6/30/28	8/18/28
6.2	Electrical integration	8/18/28	9/29/28
6.3	Software integration	8/18/28	10/5/28
6.4	Functional testing	10/6/28	10/20/28
6.5	Thermal testing	10/20/28	10/31/28
6.6	Mobility testing	11/1/28	11/10/28
6.7	Environmental testing	11/13/28	11/30/28
6.8	Test Discrepancy Resolution & Retesting	11/24/28	11/28/28
6.9	Schedule Margin	11/20/28	11/30/28
6.10	◆ System Integration Ready	12/1/28	12/1/28
7	**Launch Preparation**	12/4/28	2/1/29
7.1	Flight software freeze	12/4/28	1/8/29
7.2	Launch vehicle integration	12/7/28	1/8/29

7.3	Final verification	1/9/29	1/22/29	
7.4	Launch Ready	1/23/29	1/31/29	
7.5	◆ Launch Readiness Review	1/23/29	1/23/29	
7.6	Schedule Margin	1/24/29	2/1/29	
7.7	◆ Launch	2/1/25	2/1/29	

Figure 102:

2.2.2.3. Mission Phase E Overview

The Focus of Phase E is the mission operations, during which the lunar south pole rover performs commissioning, engineering checkout, nominal science operations, and routine mission sustainment activities. Following the successful February 1, 2029 launch, the mission will transition into rover activation and initial checkout from February 2 through February 16, followed by surface commissioning from February 17 through March 2 and initial traverse and instrument checkout from March 3 through March 30. These activities establish that the rover and its instruments are functioning as expected before primary science operations. As shown in the Phase E Gannt chart, primary science operations are planned from March 31 through August 31, with the science data downlink and preliminary analysis through September 14. With the mission operation concluding following schedule margin with the Mission operations Complete milestone on September 28, 2029.

The Phase E schedule reflects the mission's Concept of Operations by progressing from initial rover checkout and commissioning into surface traverses, science data collection, and transmission of mission data to Earth. With the schedule being developed using NASA schedule management guidance, engineering judgement, and analogous lunar missions including NASA's VIPER, Firefly Blue Ghost Mission 1, CLPS missions, and LCROSS. These analogous missions were used to inform the sequencing and relative durations of commissioning, checkout, surface operations, science activities, and data return rather than to directly duplicate their missions timelines. Engineering judgement was then applied to adapt these estimates to the proposed rover's operational concept and mission objectives. Additionally the schedule margin from September 15 through September 27 provides time to accommodate operational uncertainties, communication delays, and potential replanning of science activities before the Mission Operations Complete milestone.

8	Mission Operations	2/2/29	9/28/29
8.1	Rover Activation and initial checkout	2/2/29	2/16/29
8.2	Surface commissioning	2/17/29	3/2/29
8.3	Initial traverse and instrument checkout	3/3/29	3/30/29
8.4	Primary science operations	3/31/29	8/31/29
8.5	Science Data downlink & preliminary analysis	3/31/29	9/14/29
8.6	Schedule Margin	9/15/29	9/27/29
8.7	◆ Mission operations Complete	9/28/29	9/28/29

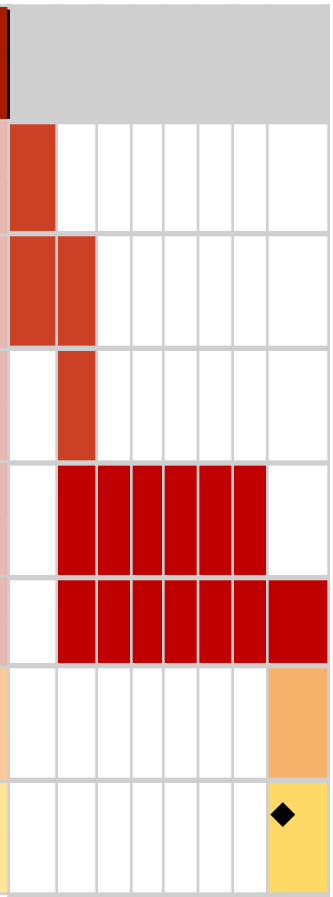


Figure 103:

2.2.2.4. Mission Phase F Overview

The focus of Phase F is the Mission Closeout activities that will formally conclude the lunar south pole rover mission following the completion of surface operations. As shown in the Phase F Gannt chart, Phase F begins on October 1, 2029, following the completion of mission operations, and continues through the Mission Closeout milestone on March 29, 2030. During the Mission Closeout the team will confirm mission completion, validate engineering and scientific data, archives mission products, performs post-mission analysis, and documents lessons learned activities leading to the Mission Closeout milestone. These activities ensure that all technical documentation, scientific data products, and programmatic deliverables are reviewed, finalized, and archived for future mission reference and to support future lunar exploration.

The Phase F schedule was developed using NASA schedule management guidance, NASA program/project life-cycle guidance, engineering judgement and historical mission closeout practices to estimate activity durations and task sequencing. The duration of the data validation and archival activities was based on the expected time required to verify

mission products, organize engineering and scientific data, complete final reporting and satisfy NASA documentation requirements. The sequencing of these activities reflects their predecessor-successor relationships, with data validation supporting final archiving and subsequent post-mission analysis and reporting. With the schedule margin being maintained from March 24 through March 28, 2030, to accommodate documentation revisions, final data validation, administrative closeout actions, and remaining programmatic requirements before the March 29, 2030 Mission Closeout milestone.

9	**Mission Closeout**	10/1/29	3/29/30
9.1	Mission completion confirmation	10/1/29	10/11/29
9.2	Science and engineering data validation	10/14/29	12/14/29
9.3	Final data archiving	12/16/29	1/31/30
9.4	Post-mission analysis and final reporting	2/3/30	3/7/30
9.5	Lessons Learned Report	3/10/30	3/21/30
9.6	Schedule Margin	3/24/30	3/28/30
9.7	◆ Mission Closeout	3/29/30	3/29/30

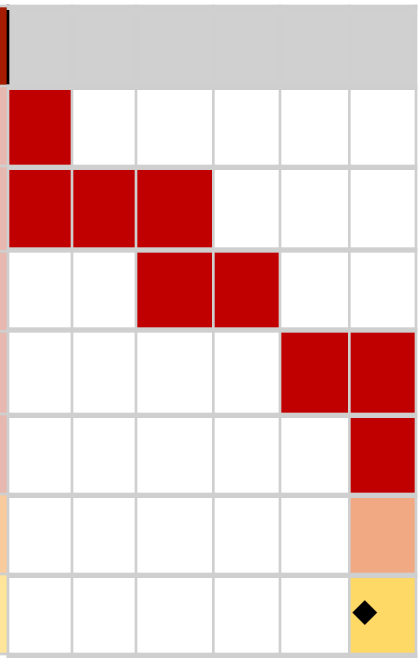


Figure 104:

2.3. Mission Budget

2.3.1. Budget Basis of Estimate

The Budget Basis of Estimate encompasses all post-Preliminary Design Review lifecycle costs across NASA Phases C through F (Fiscal Year 2026 through Fiscal Year 2030), treating Pre-Phase A through Phase B formulation as sunk costs in accordance with customer guidelines. The mission is strictly bounded by a customer-mandated lifecycle cost cap of $250,000,000.00 (Requirement MR-10 / SYS-12), fixed delivery dates requiring system integration readiness by December 1, 2028, and launch readiness by February 1, 2029, as well as explicit prohibitions against radioactive power sources exceeding 5 g. To ensure programmatic resilience against cost growth, the total allocated margin for the mission includes an explicit $17,000,000.00 spacecraft technical cost reserve embedded directly within direct subsystem budgets, operating alongside $38,987,327.00 (15.59%) in

unallocated headroom below the allowable cap to absorb technical redesigns, inflation, or expanded Phase D qualification testing.

Estimates are baselined in Fiscal Year 2026 (FY26) dollars and apply compound annual inflation rates of 0.0% in FY1 (FY26–27), 2.6% in FY2 (FY27–28), 5.2% in FY3 (FY28–29), and 7.8% in FY4 (FY29–30). Standardized base salary bands are applied across five functional labor categories (Project Management at $120k/yr; Engineering and Science at $80k/yr; Technicians and Administration at $60k/yr), incorporating an Employee Related Expenses (ERE) fringe rate of 27.91% and a 10.0% Facilities & Administrative (F&A) rate applied to Modified Total Direct Costs. Flight instrumentation with high Technology Readiness Levels (TRL 6–7), such as LISTER and MSolo, are assumed to leverage mature heritage hardware and commercial modifications to minimize non-recurring engineering costs, while the low-TRL TENG demonstration (TRL 4) is assumed to undergo in-house prototype fabrication and environmental characterization. The cost baseline funds one primary flight-ready rover system along with dedicated engineering development units (EDUs), breadboards, and subsystem test fixtures to support Phase D integration and qualification before launch vehicle delivery. A funded spacecraft design reserve of $17.00M (8.06% of total estimated cost) and a 10.0% test facility margin ($5.10M) are carried within direct costs to absorb scope growth and test anomalies, operating beneath an additional $33.55M in unallocated ceiling headroom.

The primary driver of mission cost is the development, modification, and environmental qualification of the high-precision scientific payload suite, which accounts for $51.00M (~24.17% of total expenditures), led by the LISTER thermal probe ($12.00M), seismic geophone array ($8.50M), and MSolo mass spectrometer ($4.50M). Extensive Phase D environmental qualification testing constitutes the second largest cost driver at $53.06M, driven by necessary specialized cleanroom usage and extensive thermal-vacuum (TVAC), vibration, and abrasive regolith dust chamber testing to verify vehicle survivability down to cryogenic temperatures in Permanently Shadowed Regions. Additional secondary cost drivers include prime contractor development for high-efficiency GaAs triple-junction solar panels and radiation-hardened avionic components required to meet strict spaceflight environmental standards.

As subsystem trade studies conclude, component selections are baselined, and hardware prototypes undergo Phase D environmental testing, all cost models, activity allocations, and estimating assumptions will be continuously updated within the project's Integrated Master Budget. Future iterations leading to the Critical Design Review (CDR) will progressively refine parametric calculations using updated vendor quotes and verified

technical parameters in accordance with the NASA Cost Estimating Handbook and NASA Space Flight Program and Project Management Handbook guidance.

2.3.2. Total Mission Cost

The lifecycle mission budget for the lunar surface vehicle system encompasses all post-Preliminary Design Review activities across NASA Phases C through F (Fiscal Years 2026 through 2030), treating Pre-Phase A through Phase B formulation as sunk costs. The total estimated mission cost is $211,012,673.00, which includes direct spacecraft bus engineering, payload instrument procurement, manufacturing and environmental testing facilities, labor, travel, public outreach, and indirect Facilities & Administrative costs. Compared to the customer-defined lifecycle cost cap of $250,000,000.00 mandated by requirement SYS-12 (MR-10), the baseline estimate is fully compliant and operates $38,987,327.00 below the allowable limit (utilizing 84.41% of the allocated ceiling). This remaining 15.59% unallocated capacity provides executive project management with programmatic headroom to absorb supply chain fluctuations or extended integration testing, complementing the explicit $17.00M spacecraft technical cost reserve embedded directly within subsystem budgets.

Budget

Line Item / Budget Category	FY 1 (Phase C)	FY 2 (Phase C)	FY 3 (Phase D–E)	FY 4 (Phase F)	Cumulative Total (USD)	% of Total Cost
PERSONNEL HEADCOUNT (FTE)						
Science Personnel	8	6	7	9	—	—
Engineering Personnel	18	20	21	12	—	—

Technicians	10	12	11	4	—	—
Administration Personnel	3	3	3	3	—	—
Project Management	4	4	4	4	—	—
Total Funded Headcount (FTEs)	43	45	47	32	—	—
PERSONNEL COSTS						
Science Personnel	$640,000.00	$492,480.00	$637,280.00	$776,160.00	$2,581,920.00	1.19%
Engineering Personnel	$1,440,000.00	$1,641,600.00	$1,767,360.00	$1,034,880.00	$5,883,840.00	2.72%
Technicians	$600,000.00	$738,720.00	$694,320.00	$258,720.00	$2,291,760.00	1.06%
Administration Personnel	$180,000.00	$184,680.00	$189,360.00	$194,040.00	$748,080.00	0.35%
Project Management	$480,000.00	$492,480.00	$504,960.00	$517,440.00	$1,994,880.00	0.92%

Total Base Salaries	$3,340,000.00	$3,549,960.00	$3,829,280.00	$2,781,240.00	$13,500,480.00	6.24%
Total ERE (27.91%)	$932,194.00	$990,794.00	$1,068,752.00	$776,244.00	$3,767,984.00	1.74%
Personnel Margin (10.0%)	$427,219.00	$454,075.00	$489,803.00	$355,748.00	$1,726,845.00	0.80%
TOTAL PERSONNEL	$4,699,413.00	$4,994,829.00	$5,387,835.00	$3,913,232.00	$18,995,309.00	8.78%
TRAVEL COSTS						
Total Flights Cost	$94,500.00	$94,500.00	$283,500.00	$157,500.00	$630,000.00	0.29%
Total Hotel Cost	$108,000.00	$108,000.00	$324,000.00	$180,000.00	$720,000.00	0.33%
Total Transportation Cost	$21,600.00	$21,600.00	$64,800.00	$36,000.00	$144,000.00	0.07%
Total Per Diem Cost	$45,900.00	$45,900.00	$137,700.00	$76,500.00	$306,000.00	0.14%
Inflation Adjustment (0.0%, 2.6%, 5.2%, 7.8%)	$0.00	$7,020.00	$42,120.00	$35,100.00	$84,240.00	0.04%

Travel Margin (10.0%)	$27,000.00	$27,702.00	$85,212.00	$48,510.00	$188,424.00	0.09%
TOTAL TRAVEL COSTS	$297,000.00	$304,722.00	$937,332.00	$533,610.00	$2,072,664.00	0.96%
OUTREACH COSTS						
Total Outreach Materials	$90,000.00	$112,500.00	$157,500.00	$90,000.00	$450,000.00	0.21%
Total Outreach Venue Costs	$72,000.00	$90,000.00	$126,000.00	$72,000.00	$360,000.00	0.17%
Total Outreach Travel Costs	$54,000.00	$67,500.00	$94,500.00	$54,000.00	$270,000.00	0.12%
Total Outreach Services Costs	$72,000.00	$90,000.00	$126,000.00	$72,000.00	$360,000.00	0.17%
Total Outreach Personnel Costs	$72,000.00	$90,000.00	$126,000.00	$72,000.00	$360,000.00	0.17%
Inflation Adjustment (0.0%, 2.6%, 5.2%, 7.8%)	$0.00	$11,700.00	$32,760.00	$28,080.00	$72,540.00	0.03%

Outreach Margin (10.0%)	$36,000.00	$46,170.00	$66,276.00	$38,808.00	$187,254.00	0.09%
TOTAL OUTREACH COSTS	$396,000.00	$507,870.00	$729,036.00	$426,888.00	$2,059,794.00	0.95%
DIRECT COSTS: SPACECRAFT SUBSYSTEMS						
Mechanical Subsystem	$1,275,000.00	$2,550,000.00	$3,400,000.00	$1,275,000.00	$8,500,000.00	3.93%
Power Subsystem (EPS)	$2,550,000.00	$5,100,000.00	$6,800,000.00	$2,550,000.00	$17,000,000.00	7.85%
Thermal Control Subsystem (TCS)	$1,275,000.00	$2,550,000.00	$3,400,000.00	$1,275,000.00	$8,500,000.00	3.93%
Command & Data Handling (CDH)	$2,040,000.00	$4,080,000.00	$5,440,000.00	$2,040,000.00	$13,600,000.00	6.28%
Guidance, Navigation, & Control (GNC)	$750,000.00	$1,500,000.00	$2,000,000.00	$750,000.00	$5,000,000.00	2.31%

Science Instrumentation Suite	$7,650,000.00	$15,300,000.00	$20,400,000.00	$7,650,000.00	$51,000,000.00	23.56%
Spacecraft Cost Margin / Technical Reserve	$4,250,000.00	$4,250,000.00	$4,250,000.00	$4,250,000.00	$17,000,000.00	7.85%
Total Spacecraft Direct Costs	$19,790,000.00	$36,248,580.00	$48,065,880.00	$21,333,620.00	$125,438,080.00	57.95%
DIRECT COSTS: FACILITIES & TESTING						
Manufacturing Facility Cost	$2,295,000.00	$4,590,000.00	$6,120,000.00	$2,295,000.00	$15,300,000.00	7.07%
Test Facility Cost (TVAC / Vibe / Dust)	$5,355,000.00	$10,710,000.00	$14,280,000.00	$5,355,000.00	$35,700,000.00	16.49%
Facility Cost Margin (10.0%)	$765,000.00	$1,530,000.00	$2,040,000.00	$765,000.00	$5,100,000.00	2.36%
Total Facilities Direct Costs	$8,415,000.00	$17,227,800.00	$23,500,800.00	$9,011,700.00	$58,155,300.00	26.87%

TOTAL DIRECT COSTS	$28,205,000.00	$53,476,380.00	$71,566,680.00	$30,345,320.00	$183,593,380.00	84.82%
INDIRECT & FINAL CALCULATIONS						
Modified Total Direct Costs (MTDC Base)	$24,273,390.00	$35,816,430.00	$45,917,390.00	$25,802,870.00	$131,810,080.00	—
Facilities & Administrative (10.0% MTDC)	$2,427,339.00	$3,581,643.00	$4,591,739.00	$2,580,287.00	$13,181,008.00	6.09%
FINAL ESTIMATED MISSION COST	$36,024,752.00	$62,865,444.00	$83,212,622.00	$37,802,337.00	$219,905,155.00	100.00 %
Customer Budget Cap (SYS-12 / MR-10)	—	—	—	—	$250,000,000.00	—
Available Unallocated Headroom Under Cap	—	—	—	—	$30,094,845.00	12.04%

Total Programmatic Reserves (Funded Margins + Headroom)	—	—	—	—	$54,297,368.00	21.72%

Figure 105: Comprehensive Master Lifecycle Budget across Phases C through F (FY26–FY30).

The primary driver of overall spacecraft cost is Science Instrumentation, which accounts for $51,000,000.00 (~24.17% of total mission expenditure). This cost is concentrated in high-precision, flight-proven lunar instruments, including the LISTER Thermal Probe ($12.0M lifecycle estimate), the Mass Spectrometer Observing Lunar Operations (MSolo) ($4.5M lifecycle estimate), the MECA Electrometer suite ($5.0M), and the seismic geophone array ($8.5M NICM CER scaling). Ground Penetrating Radar was formally descoped from the payload baseline per approved Change Request CR-10-005 to respect vehicle mass limits.

Spacecraft bus direct subsystems comprising Electrical Power Subsystem ($17.00M), Command & Data Handling and Communications ($13.60M), Mechanical & Mobility ($8.50M), Thermal Control ($8.50M), Guidance, Navigation & Control ($5.00M), and Spacecraft Cost Margin ($17.00M) collectively sum to $125,438,080.00. Specialized testing and manufacturing facilities represent another major financial investment at $53,055,300.00 ($15.30M for manufacturing cleanrooms/machining and $35.70M for thermal-vacuum and vibration test chambers). Personnel labor totals $17,918,705.00, reflecting engineering, scientific, technician, administrative, and project management personnel scaled across Phases C through F. The largest financial uncertainties and potential cost drivers center on Phase D environmental verification facility usage and payload thermal/power qualification for operations in cryogenic Permanently Shadowed Regions (PSRs). Secondary cost drivers include lead-time volatility for radiation-hardened electronic components and potential COTS modifications required to sustain operating temperature swings between -45°C and +50°C.

2.3.3. Personnel Budget

Execution of the mission requires a multidisciplinary workforce distributed across five core disciplines: Engineering, Science, Technicians, Project Management, and Administration. Funded staffing begins at 43 Full-Time Equivalents (FTEs) in Phase C / FY1 (18 Engineers, 8 Scientists, 10 Technicians, 4 Project Managers, and 3 Administration

personnel) and expands to 45 FTEs in FY2 as detailed design and procurement accelerate. Concurrently funded personnel reach a lifecycle peak of 47 FTEs during Phase D–E / FY3 (21 Engineers, 7 Scientists, 11 Technicians, 4 Project Managers, and 3 Administration personnel) to support intensive assembly, integration, environmental qualification, and launch operations. Following surface operations, workforce allocations scale down to 32 FTEs in Phase F / FY4 closeout (12 Engineers, 9 Scientists, 4 Technicians, 4 Project Managers, and 3 Administration personnel), maintaining critical science and engineering capacity for data validation and archiving while phasing down technician support.

Budget Template

Mission Phase	**Phase C**	**Phase C**	**Phase D-E**	Phase F	
Year	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Cumulative Total**
PERSONNEL					
Science Personnel	$ 640,000	$ 492,480	$ 637,280	$776,160	$2,581,920
Engineering Personnel	$1,440,000	$1,641,600	$1,767,360	$1,034,880	$5,883,840
Technicians	$600,000	$738,720	$694,320	$258,720	$2,291,760
Administration Personnel	$ 180,000	$ 184,680	$189,360	$194,040	$748,080
Project Management	$480,000	$492,480	$504,960	$517,440	$1,994,880
Total Salaries	$3,340,000	$3,549,960	$3,829,280	$2,781,240	**$13,500,480**
Total ERE	$932,194	$990,794	$1,068,752	$776,244	**$3,767,984**
Personnel Margin (10%)	$427,219.00	$454,075.00	$489,803.00	$355,748.00	**$1,726,845.00**

TOTAL PERSONNEL	$4,699,413.00	$4,994,829.00	$5,387,835.00	$3,913,232.00	$18,995,309.00

Figure 106:

Personnel cost estimates were developed using standard NASA labor rate baselines and a phased Work Breakdown Structure (WBS) staffing model aligned with lifecycle requirements across Phases C through F, as detailed in Figure 2.2.3. Standardized annual base salaries were applied to each functional category: Project Management ($120,000/year), Engineering ($80,000/year), Science ($80,000/year), Technicians ($60,000/year), and Administration ($60,000/year). All labor expenditures incorporate an Employee Related Expenses (ERE) fringe benefit rate of 27.91% and compound annual inflation adjustments (0.0% in FY1, 2.6% in FY2, 5.2% in FY3, and 7.8% in FY4). A dedicated 10% personnel cost reserve is applied to accommodate unexpected labor surge requirements or testing delays during development.

Funded headcount evolves dynamically over the four-year fiscal timeline to reflect changing project demands across each lifecycle phase. Active staffing begins at 43 Full-Time Equivalents (FTEs) in FY1 (Phase C detailed design) and expands to 45 FTEs in FY2 as fabrication and component procurement commence. Concurrently funded staffing reaches its peak in FY3 (Phase D–E) at 47 FTEs. This funded peak is fully consistent with the organizational structure defined in Section 1.6: the 88-person total workforce identified in the mission organizational plan represents the cumulative roster of individual specialists, domain consultants, and contractor personnel engaged across distinct, non-overlapping lifecycle phases, whereas 47 FTEs represents the maximum concurrent full-time equivalent labor funded simultaneously during any single fiscal year. This FY3 staffing surge is justified by the intensive labor requirements of Assembly, Integration, and Testing (AI&T), multi-facility environmental qualification testing (thermal-vacuum, vibration, and regolith dust exposure), and launch operations at Kennedy Space Center. In FY4 (Phase F), active headcount scales down to 32 FTEs following the completion of surface operations, retaining essential Engineering (12 FTEs), Science (9 FTEs), Project Management (4 FTEs), Administration (3 FTEs), and reduced Technician support (4 FTEs) to process telemetry, perform planetary data validation and PDS archiving, and finalize closeout reporting.

Total personnel expenditures across lifecycle Phases C through F equal $19,710,576.00, which encompasses $13,500,480.00 in base salaries, $3,767,984.00 in Employee Related Expenses (ERE), compound annual inflation scaling, and an explicit 10% labor contingency reserve ($1,791,871.00) to cover potential staffing surges. Key assumptions include standardized annual salary bands across five labor categories, full

availability of required FTEs, and continuous federal fringe and inflation adjustments applied across FY26 through FY30. Primary cost uncertainties center on potential schedule anomalies during Phase D environmental qualification testing that could require additional technician or engineering shift hours prior to launch vehicle integration.

2.3.4. Travel Budget

Travel is required throughout lifecycle Phases C through F to support major project review milestones, hardware manufacturing oversight, environmental testing, launch site integration, surface operations, and mission closeout. Mandatory travel includes attendance at Key Decision Point reviews, the Critical Design Review (CDR), System Integration Review (SIR), and the Flight Readiness Review with Standing Review Boards and NASA program leadership. Mission-specific travel activities encompass engineer and technician transit to specialized supplier cleanrooms, out-of-state thermal-vacuum (TVAC) and vibration testing facilities, extended deployment to Kennedy Space Center in Cape Canaveral, Florida for final launch vehicle integration, and post-mission closeout meetings. Travel demands scale from $270,000.00 in FY1 to a peak of $852,120.00 in FY3 to support intensive pre-launch testing and surface operational synchronization.

Total Flights Cost	$94,500	$94,500	$283,500	$157,500	**$630,000**
Total Hotel Cost	$108,0000	$108,000	$324,000	$180,000	**$720,000**
Total Transportation Cost	$21,600	$21,600	$64,800	$36,000	**$144,000**
Total Per Diem Cost	$45,900	$45,900	$137,700	$76,500	**$306,000**
Travel Margin	$27,000.00	$27,702.00	$85,212.00	$48,510.00	$188,424.00
Total Travel Costs	$297,000.00	$304,722.00	$937,332.00	$533,610.00	**$2,072,664.00**

Figure 107. Travel Cost Budget Summary Table

Travel cost estimates across lifecycle Phases C through F were developed using federal travel parameters, standard U.S. General Services Administration (GSA) per diem benchmarks, and City Pair Program (CPP) high-fare commercial airline rates. Estimates account for round-trip commercial coach airfare, lodging, ground transportation (rental vehicles), per diem allowances (Meals and Incidental Expenses [M&IE] calculated at 100% for full mission days and 75% for first and last travel days), and compound annual inflation adjustments (0.0% in FY1, 2.6% in FY2, 5.2% in FY3, and 7.8% in FY4). Over the four-year

post-PDR lifecycle, total baseline travel expenditures equal $1,884,240.00 comprising $630,000.00 for commercial airfare, $720,000.00 for lodging, $144,000.00 for rental transportation, and $306,000.00 for M&IE with an explicit 10% travel contingency margin ($188,424.00) allocated to absorb dynamic flight adjustments and itinerary extensions, yielding a total funded travel budget of $2,072,664.00.

- Phase C Detailed Design & Formulation (FY1 / Year 1 & FY2 / Year 2): Travel in FY1 ($270,000.00 base) and FY2 ($277,020.00 inflation-adjusted base) is dedicated to formal NASA gateway reviews and supplier synchronization. This includes sending delegation teams to Washington, D.C. (NASA Headquarters) for 5-day Key Decision Point (KDP-C) and Critical Design Review (CDR) briefings, with flights budgeted at ~$1,500.00/person round-trip, lodging at GSA standard rates ($180.00–$250.00/night), ground rental vehicles ($60.00/day), and per diem ($79.00/day). In FY2, supplemental trips support early supplier milestone checks at component fabrication facilities.
- Phase D Integration, Environmental Testing & Phase E Launch Operations (FY3 / Year 3): Travel expenditures peak in FY3 at $852,120.00 base ($937,332.00 with 10% margin) to support extensive Assembly, Integration, and Test (AI&T) deployments. Funding covers extended travel for engineering and technician teams to specialized vendor cleanrooms (Honeybee Robotics for LISTER, NASA JPL/KSC for MECA, and Frontgrade Gaisler for avionics) and out-of-state thermal-vacuum (TVAC), vibration, and abrasive dust test chambers. FY3 also funds the primary launch campaign deployment to Kennedy Space Center (flying into Orlando International Airport, MCO). In accordance with customer mission requirements, the launch team is supported for a 5-day / 4-night stay arriving 2 days prior to the February 1, 2029 launch date and departing 2 days post-launch, utilizing GSA Brevard County benchmark rates ($166.00–$208.00/night lodging and $74.00/day M&IE).
- Phase F Mission Closeout & Science Archiving (FY4 / Year 4): Travel totals $485,100.00 base ($533,610.00 with 10% margin) to support post-mission scientific workshops, final Planetary Data System (PDS) data validation panels, and closeout reporting with NASA program leadership.

Total funded travel expenditures across lifecycle Phases C through F equal $2,072,664.00, which includes a baseline cost of $1,884,240.00 comprising $630,000.00 for commercial airfare, $720,000.00 for lodging, $144,000.00 for ground transportation, and $306,000.00 for per diem stipends plus an explicit 10% travel contingency margin of $188,424.00. Key assumptions include commercial coach-class airfare priced using

high-fare City Pair Program benchmarks, federal GSA lodging and per diem rates for Washington, D.C. reviews and Cape Canaveral launch deployments ($166.00/night lodging and $74.00/day M&IE), shared rental vehicles to maximize efficiency, and compound annual inflation scaling applied across FY26 through FY30.

2.3.5. Outreach Budget

The public outreach campaign for Team Fontus's lunar South Pole rover mission translates the technical milestones of volatile exploration and robotic engineering into accessible educational programs for diverse public audiences. Planned activities include interactive K–12 STEM workshops ("Lunar South-Pole Rover Challenge"), university and community symposia ("Engineering the Mission"), public science center scale-model exhibits, and a bi-weekly digital video series ("Lunar Science Digest"). These activities directly support the mission's outreach goals by promoting public stewardship of space exploration, simplifying NASA systems engineering methodology, and expanding educational access for historically underrepresented and Title I student communities.

Total Outreach Materials	$90,000	$112,500	$157,500	$90,000	**$450,000**
Total Outreach Venue Costs	$72,000	$90,000	$126,000	$72,000	**$360,000**
Total Outreach Travel Costs	$54,000	$67,500	$94,500	$54,000	**$270,000**
Total Outreach Services Costs	$72,000	$90,000	$126,000	$72,000	**$360,000**
Total Outreach Personnel Costs	$72,000	$90,000	$126,000	$72,000	**$360,000**
Outreach Margin	$36,000.00	$46,170.00	$66,276.00	$38,808.00	**$187,254.00**
Total Outreach Costs	$396,000.00	$507,870.00	$729,036.00	$426,888.00	**$2,059,794.00**

Figure 108: Outreach Cost Budget Summary across Phases C through F

Outreach cost estimates were developed using commercial market pricing, vendor catalog benchmarks for printing and fabrication, freelance service standard rates, and public venue rental fees, scaled across the 4-year project lifecycle with annual inflation

adjustments (0.0% in FY1, 2.6% in FY2, 5.2% in FY3, and 7.8% in FY4). Total cumulative outreach expenditures equal $1,872,540.00 across lifecycle Phases C through F, directly reflecting the resource allocations defined in the Science Communication and Public Outreach Plan (Section 1.12 of MDR).

The budgeted line items directly fund the four primary outreach initiatives. The outreach materials with a $450,000.00 cumulative represent the largest financial investment, funding physical supplies for Title I elementary/middle school hands-on workshops (interactive budget tokens, trade-off activity kits, pre/post survey forms), 3D-printed scale models of the rover chassis and payload instruments for museum exhibits, and high-contrast, screen-reader-friendly educational print materials in English and Spanish. The venue costs with a $360,000.00 cumulative cover rental fees for public science center exhibit booths, university auditorium hosting fees for public systems engineering symposia, and community center demonstration spaces across local school districts. The outreach travel costs with a $270,000.00 cumulative support team member transportation, lodging, and per diem to conduct in-person workshops at Title I schools, present technical poster sessions at university symposia, and manage interactive exhibit booths. Outreach services costs with a $360,000.00 cumulative and personnel costs with a $360,000.00 cumulative cover external website hosting, domain licensing, graphic design services, closed-captioning accessibility software for the "Lunar Science Digest" video logs, and dedicated student stipend support for digital media coordination.

Total funded public outreach expenditures across lifecycle Phases C through F equal $2,059,794.00, which includes a baseline cost of $1,872,540.00 (approximately 0.89% of the mission baseline) plus an explicit 10% outreach contingency margin of $187,254.00 to accommodate potential material price fluctuations and venue expansions. Major assumptions include the full utilization of open-access digital platforms (such as YouTube, LinkedIn, and public web portals) to minimize media distribution overhead, strategic partnerships with regional Title I school districts and local science centers to subsidize facility rental costs, and compound annual inflation scaling applied through FY30.

2.3.6. Direct Costs

Direct costs encompass all non-labor, non-travel expenditures required to design, manufacture, procure, integrate, qualify, and operate the spacecraft bus, payload instruments, and associated Ground Support Equipment (GSE) across NASA Phases C through F. These expenses are grouped into two primary domains: spacecraft direct hardware subsystems (comprising Mechanical, Electrical Power, Command and Data

Handling, Thermal Control, Guidance, Navigation and Control, Science Instrumentation, and Spacecraft Technical Reserve) and specialized manufacturing and testing facility rentals. Overall estimates were derived through a hybrid parametric and analogous framework using the NASA Instrument Cost Model (NICM Version 9c), vendor catalog pricing for space-qualified Commercial Off-The-Shelf (COTS) components, and historical hardware development analogies.

As detailed in Figure 2.2.6 below, total direct costs equal $183,593,380.00, which includes $125,438,080.00 for spacecraft bus and payload systems, $53,055,300.00 for manufacturing and environmental test facility leases, and an explicit 10% test facility margin of $5,100,000.00 to account for potential re-testing anomalies. The primary assumptions anchoring these estimates are an FY26 economic base with compound annual inflation scaling (0.0% in FY1, 2.6% in FY2, 5.2% in FY3, and 7.8% in FY4), single-flight unit production with dedicated engineering development units (EDUs), and full compliance with the 175 kg mass and 1.5 m × 1.5 m × 1.0 m stowed configuration envelopes. Primary uncertainties involve cryogenic thermal-vacuum testing iterations and supply chain volatility for radiation-hardened electronic components.

Mechanical Subsystem	$1,275,000	$2,550,000	$3,400,000	$1,275,000	$8,500,000
Power Subsystem	$2,550,000	$5,100,000	$6,800,000	$2,550,000	$17,000,000
Thermal Control Subsystem	$1,275,000	$2,550,000	$3,400,000	$1,275,000	$8,500,000
Comms & Data Handling Subsystem	$2,040,000	$4,080,000	$5,440,000	$2,040,000	$13,600,000
Guidance, Nav, & Control Subsystem	$750,000	$1,500,000	$2,000,000	$750,000	$5,000,000
Science Instrumentation	$7,650,000	$15,300,000	$20,400,000	$7,650,000	$51,000,000
Spacecraft Cost Margin	$4,250,000	$4,250,000	$4,250,000	$4,250,000	$17,000,000

Total Spacecraft Direct Costs	$19,790,000	$36,248,580	$48,065,880	$21,333,620	$125,438,080
Manufacturing Facility Cost	$2,295,000	$4,590,000	$6,120,000	$2,295,000	$15,300,000
Test Facility Cost	$5,355,000	$10,710,000	$14,280,000	$5,355,000	$35,700,000
Facility Cost Margin	$765,000.00	$1,530,000.00	$2,040,000.00	$765,000.00	$5,100,000.00
Total Facilities Costs	$8,415,000.00	$17,227,800.00	$23,500,800.00	$9,011,700.00	$58,155,300.00
Total Direct Costs	$28,205,000.00	$53,476,380.00	$71,566,680.00	$30,345,320.00	**$183,593,380.00**

Figure 109: Direct Costs Budget Summary across Phases C through F, detailing spacecraft subsystems, testing facilities, and funded margins.

Subsystem Cost Estimates

Mechanical Subsystem ($8,500,000.00): Estimated using vendor build-to-print quotes, historical analogies from small lunar rovers, and engineering judgment. The cost basis covers precision Computer Numerical Control (CNC) machining of the 6061-T6 aluminum chassis frame, aluminum honeycomb body panels, rocker-bogie suspension linkages, six rigid aluminum grouser wheels, mounting brackets, and brushless DC drive motors with planetary reduction gearboxes. Key assumptions include high-heritage aerospace alloy procurement and standard build-to-print contracting via Land Aero Group and MIC Group. The primary cost drivers and uncertainties are machining lead times and tight dimensional tolerances for the suspension pivot assemblies and deployment lock mechanisms.

Power Subsystem ($17,000,000.00): Estimated via prime contractor pricing and space-qualified Commercial Off-The-Shelf (SQCOTS) component catalogs. The estimate funds custom Gallium Arsenide (GaAs) triple-junction solar panels (Boeing/Redwire), an integrated space-qualified lithium-ion battery system delivering 1,296 Wh of usable capacity (EaglePicher), and distributed Maximum Power Point Tracking (MPPT) power distribution units (AAC Clyde Space). Assumptions include full qualification of the 28 V regulated DC bus and battery thermal containment for cold-trap operations. Major cost drivers include custom solar panel substrate manufacturing and cryogenic cell qualification.

Command & Data Handling Subsystem ($13,600,000.00): Estimated using component catalog pricing and mission-level analogies for radiation-hardened avionic suites. Direct flight hardware procurement accounts for $1.20M—anchored by dual Frontgrade Gaisler GR712RC radiation-hardened flight computers ($400k unit cost), the Aitech S-A1760 co-processor ($150k), stacked 3D Plus radiation-tolerant NAND flash storage modules ($200k), CAN transceivers, and SpaceWire harnesses—with the remaining funds covering Direct-to-Earth (DTE) X-band radio transceivers, diplexers, patch antennas, and engineering test units ($350k). The primary cost driver and uncertainty is the radiation tolerance verification and flight packaging of the commercial Jetson-based S-A1760 co-processor hosting the NeuPAN autonomy software.

Thermal Control Subsystem ($8,500,000.00): Estimated through SQCOTS vendor pricing and aerospace material databases. The cost basis covers tailored Multilayer Insulation (MLI) blankets ($49.8k; Sheldahl/Dunmore), 99.99% high-purity indium foil thermal interface materials ($1.10M to $1.30M; Indium Corp), nine distributed Kapton polyimide survival and operational heaters ($846.63 to $1,063.71; Minco Products), LTC2986 digital temperature sensor ICs, and cryogenic platinum Resistance Temperature Detectors (Minco S240880). The estimate assumes standard aerospace flight heritage for all sensors and heaters. The primary cost driver is custom blanket fabrication and vacuum outgassing qualification (ASTM E595) to eliminate contamination of volatile science targets.

Guidance, Navigation, & Control Subsystem ($5,000,000.00): Estimated through analogous flight hardware scaling and supplier quotes. Costs cover mast-mounted stereo navigation cameras, body-mounted hazard avoidance cameras (Hazcams), a tactical-grade Inertial Measurement Unit (IMU), optical wheel encoders, and front-facing LED surface illumination arrays required for vision-based navigation in permanently shadowed craters. The baseline assumes sensor synchronization with the onboard flight computer. Major cost drivers include sensor optical calibration and vibration-tolerant lens housing fabrication.

Science Instrumentation Suite ($51,000,000.00): Estimated using the NASA Instrument Cost Model (NICM Version 9c) parametric Cost Estimating Relationships (CERs) combined with analogous mission builds. Expenditures fund the LISTER subsurface thermal probe ($12.00M; Honeybee Robotics build-to-spec), the MSolo quadrupole mass spectrometer ($4.50M; adapted from INFICON/VIPER commercial hardware), the planetary seismic geophone array and GSB digitizer ($8.50M; Geospace/Paulsson), the MECA Electrometer package ($5.00M; JPL/KSC Mars Phoenix heritage), and the university-led Triboelectric Nanogenerator demonstration ($800k; Georgia Tech cleanroom prototype). Ground Penetrating Radar was formally descoped per Change Request CR-10-005 to respect vehicle

mass and budget limits. Primary cost drivers include the LISTER pneumatic gas drill qualification and MSolo mass spectrometer calibration gas systems.

Additional non-flight hardware direct expenses total $58,155,300.00 (including funded margin) across Phases C through F. Manufacturing facility support ($15,300,000.00) funds ISO Class 7 cleanroom rentals ($25,000 to $50,000/month), precision Computer Numerical Control (CNC) machining centers, surface treatment baths, and specialized Printed Circuit Board (PCB) assembly infrastructure. Environmental qualification testing facilities ($35,700,000.00) fund critical Phase D test campaigns, encompassing thermal-vacuum (TVAC) chamber runs simulating extreme lunar temperature swings (-230°C in shadow to +50°C in sunlight), launch vehicle acoustic/vibration shake-table profiles (VIBE), and abrasive lunar regolith dust exposure chambers. Ground Support Equipment (GSE) covers mechanical assembly holding fixtures ($5,000 to $30,000 per unit), breakout boxes, electrical harness continuity test sets, current-limited power supplies, and specialized handling carts. A dedicated 10% facility margin ($5,100,000.00) is included to provide reserves for potential test discrepancy resolutions, chamber re-pumping cycles, and extended facility access during Phase D integration.

Direct cost estimates were generated through a combination of the NASA Instrument Cost Model (NICM Version 9c), vendor catalog pricing for space-qualified Commercial Off-The-Shelf (SQCOTS) components, build-to-print/build-to-spec manufacturing contracts, and analogous mission cost comparisons. NICM Version 9c parametric Cost Estimating Relationships (CERs) were applied across the scientific instruments (MSolo, LISTER, Geophones, MECA) utilizing physical parameters such as instrument mass, maximum power draw, and operational design life. Analogous mission comparisons were utilized for the MSolo volatile spectrometer (derived from NASA VIPER mission modifications of commercial INFICON hardware) and the LISTER thermal probe (scaled from NASA CLPS Blue Ghost Mission 1 contract baselines). Vendor catalog data and engineering quotes were used for commercial electronics, structural fasteners, solar arrays, and batteries. These methods are appropriate for a Phase B/PDR lifecycle review as they balance empirical historical mission data with verified vendor quotes, providing high technical credibility while establishing defensible reserves.

Total direct costs across spacecraft hardware and facility infrastructure equal $183,593,380.00, driven primarily by Science Instrumentation ($51.00M), Environmental Test Facilities ($35.70M), Spacecraft Technical Reserves ($17.00M), and the Electrical Power Subsystem ($17.00M). The largest single cost drivers are the complex pneumatic drilling mechanics of the LISTER probe, mass spectrometer vacuum flight qualification, and

multi-week cryogenic thermal-vacuum chamber operations. The highest-uncertainty areas involve battery cell packaging iterations for cryogenic PSR sorties, radiation shielding verification for commercial co-processors, and prototype iterations for the low-TRL TENG demonstration. As subsystem detailed designs, Computer-Aided Design (CAD) models, and electrical schematics mature through Phase C, these direct cost estimates will be progressively refined using finalized vendor purchase orders and test facility contracts.

2.4. Scope Management

2.4.1. Change Control Management

Change Control Overview

Configuration and change management maintain system integrity, programmatic discipline, and technical traceability across every phase of the mission lifecycle. Team Fontus utilizes a formal, closed-loop change control process governed by the to systematically evaluate, document, and baseline all proposed engineering and programmatic modifications prior to implementation. Formal CCB review and approval are mandatory for any modifications impacting system or subsystem requirement baselines, Interface Control Documents (ICDs), mass and power resource allocations, baseline budget allocations, or critical path schedule milestones. This rigorous governance ensures complete cross-functional alignment, prevents uncoordinated subsystem drift, and protects project resources against uncontrolled scope growth.

Change Request Process

The formal modification process begins when a subsystem engineer first identifies the need for a design change. The engineer documents this by formalizing the technical baseline change using the official Team Fontus Change Request (CR) Form. The CR Form includes the results of a comprehensive engineering impact analysis regarding the proposed changes and the immediate effects (if any) on mass, power balance, funding, and critical-path impact. The completed CR Form is subject to an initial quality check by the Mission Assurance team. Simran Kaur and Quinterrel Battle conduct verification of baseline compliance and elimination of immediate risk concerns. After the initial quality check, the CR Form is presented to the Change Control Board (CCB). The final disposition of the CR Form rests with the Project Control Board. The formal authorization signed by Simm Vallaban, Project Manager, and Yassin Vila, Deputy Project Manager of Resources, is required to effect any formal change on systems within the scope of the change request. No change may be physically or electronically implemented or executed on the project

systems without first removing or authorizing the restriction from the Project Control Board.

Change Tracking & Communication

Design baselines, requirements revisions, and technical decisions are managed through a sequential, closed-loop tracking workflow across designated configuration tools. Once a Change Request (CR) is authorized by the Project Control Board, the change workflow progresses through three integrated levels of configuration management:

1. Central Change Log Tracking: The Mission Assurance team logs the approved CR into the Master Configuration Change Log, assigning a unique Tracking ID, recording the approval timestamp, and summarizing the affected parameters (mass, power, budget, or schedule margin).
2. Document & Database Synchronization: The designated subsystem engineer updates the active technical baseline—modifying the relevant Interface Control Documents (ICDs), subsystem models, or requirements allocation matrices. All modified files receive an incremented version identifier and are uploaded to the version-controlled project repository, archiving prior revisions to ensure full auditing traceability.
3. Cross-Team Stakeholder Notification: Automated release notifications and updated baseline briefs are distributed to all functional subsystem leads, ensuring immediate synchronization across all disciplines and guaranteeing that subsequent engineering work proceeds strictly against the newly verified baseline.

RFAs & ADVs

External stakeholder, mentor, and review board feedback generated during lifecycle milestone reviews (e.g., SRR, MDR, PDR) is formally tracked, evaluated, and resolved through the mission review action database. To ensure effective resource allocation and technical compliance, incoming feedback items are classified and prioritized into two distinct categories:

- Requests for Action (RFAs) — Priority 1 (Mandatory / Binding): RFAs designate critical technical non-compliances, missing deliverables, requirement violations, or significant design risks that impede mission feasibility. RFAs carry the highest priority and represent blocking constraints. Each RFA requires a formal Root Cause Analysis (RCA), an explicit Corrective Action Plan (CAP) authored by the responsible subsystem lead, review by Mission Assurance, and final verification sign-off from the Change Control Board (CCB) prior to baselining or advancing to the subsequent mission phase.

- Advisories (ADVs) — Priority 2 (Non-Binding / Enhancement): ADVs provide recommendations, technical suggestions, formatting refinements, or design optimizations aimed at enhancing overall deliverable quality and operational maturity. ADVs are assessed by the relevant functional engineering leads for technical merit and implementation feasibility. They are incorporated based on available engineering capacity, provided they do not adversely impact critical path schedule slack, cost margins, or mass and power allocations.

All resolutions, whether addressing mandatory RFAs or incorporated ADVs, are documented with explicit closure rationales in the Master Review Disposition Matrix, ensuring stakeholder feedback directly drives design maturity and verification planning for future development phases.

Documentation

Team Fontus: Change Request Form

CR NUMBER:	CR-2026-XXXX	DATE SUBMITTED:	YYYY-MM-DD
ORIGINATOR:	[Name / Lead Title]	SUBSYSTEM:	[e.g., CD&H / Power]

1. CHANGE DESCRIPTION & REASON FOR CHANGE

Title	[Short title describing proposed modification]
Description	[Detailed description of proposed physical, operational, or software change]
Justification	[Why is this change necessary? (e.g., thermal mitigation, mass reduction)]

2. TECHNICAL & SYSTEM IMPACT ANALYSIS

Mass Margin	[+/- kg / % change]	Power Allocation	[+/- Watts / % change]
Data / Telemetry	[Data rate / storage requirements]	Interfaces	[Pinouts, mounting, CAD bounds]

3. PROGRAMMATIC IMPACT ANALYSIS

Cost Impact	[+$0 / Reallocation]	Schedule Impact	[Days delay / Milestones]	Risk Profile	[New / Mitigated risks]

4. BOARD APPROVAL & CONCURRENCE SIGN-OFFS

Role	Signature	Date
[ ] Project Manager (PM)	_____	_____
[ ] Deputy PM of Resources (DPMR)	_____	_____

Role	Signature	Date
[ ] Lead Systems Engineer (LSE)	_____	_____
[ ] Chief Scientist	_____	_____
[ ] Mission Assurance (MA) Lead	_____	_____

DISPOSITION: [ ] APPROVED [ ] REJECTED [ ] DEFERRED FOR REVISION

Figure 110: Team Fontus: Change Request Form

Team Fontus: Change Log

The central historical database managed exclusively by Mission Assurance to provide full trace and audit capability for all engineering deviations.

CR Number	Date Submitted	Originator	Description of Change	Primary Subsystem	CCB Status	Date Resolved
CR-2026-001	2026-06-12	Avionics Lead	Swapped MCU for higher low-temp efficiency	Avionics	Approved	2026-06-14
CR-2026-002	2026-07-02	Struct Lead	Aluminum to carbon fiber brackets (Mass savings)	Structures	Approved	2026-07-05
CR-2026-003	2026-07-18	Power Eng	Secondary battery pack addition (Over mass limit)	Power	Rejected	2026-07-19

Figure 111:

Team Fontus: Configuration Tracking Document

This tracking matrix monitors active hardware baselines, Interface Control Document (ICD) statuses, and resource margins against the team's allocation ceilings.

Subsystem	Component	Baseline Part/Version	ICD Status	Current Mass	Mass Margin Remaining	Current Power	Power Margin Remaining
Mechanical/ Structure	Chassis	v1.2	Approved	450g	12%	0.0W	*N/A*
Power (EPS)	LiPo_3S	v1.0	Pending CR	180g	8%	36.0Wh (Cap)	15%
Avionics/ CDH	MicroCtrl	v2.1	Approved	45g	20%	1.8W	22%
Payload/ Sensors	OpticalCam	v1.1	Approved	85g	—	—	—

Figure 112:

Team Fontus: Change Approval Workflow

Step	Phase	Actions & Responsibilities
1	Configuration Identification	Subsystem Engineer identifies an optimization/deviation.
2	CR Entry & Impact Analysis	Originator drafts CR Form; assesses Mass, Power, & Schedule.
3	Quality Gate Review	Submitted to Mission Assurance (S. Kaur / Q. Battle) to check tracking logs and verify ICD compliance.
4	Board Review & Disposition	The Change Control Board (CCB) convenes to review system impact. Board: S. Vallaban (PM) & Y. Vila (Deputy PM of Resources).

Path A: CR Rejected (System Disapproval)	Path B: CR Approved (System Sign-Off)
CCB issues a rejection notice with feedback.	PM & Deputy PM sign authorization; MA verifies risk compliance logs.

5	Baseline Update (Path B)	MA updates Config Tracking Document and locks updated Interface Drawings.

Final Stage: Master Change Ledger Update
CR Status set to final state ("Approved" / "Rejected"). Closed file archived in the Master Configuration Repository.

Figure 113:

Change Control Flowchart

The complete lifecycle of an engineering or programmatic change; from initial request through review, approval, implementation, and verification, is illustrated in the change control workflow diagram below. The process ensures that no modification is introduced to the active technical baseline without rigorous multi-disciplinary impact analysis, Mission Assurance quality verification, and executive authorization by the Project Control Board. Once approved, changes flow through a closed-loop sequence of baseline document synchronization, physical or software implementation, and independent verification before the change record is officially closed and archived.

Team Fontus: Configuration Control & Change Request Workflow

Standard Operating Procedure | Engineering Change Management

Step	Role / Personnel	Process Description & Action Items
1. Identification	Subsystem Engineer	Subsystem Engineer identifies need for design / baseline modification.
2. Initiation & Impact	Originating Lead	Author official Team Fontus CR Form: assess mass, power, cost, schedule, risk.
3. Quality Assurance	Mission Assurance	Mission Assurance quality check: verify compliance & eliminate risks.
4. CCB Review	Change Control Board (CCB)	CCB evaluates technical merit & cross-system impact.
5. PCB Disposition	Project Control Board	Project Control Board review & vote (Approved / Disapproved).

Branch A: Disapproved / Rejection Path		
Step	Role / Personnel	Action Items
6A. Rejection	CCB / Secretariat	Log disposition rationale; archive in ECP database.
7A. Closure	Project Lead	Notify originating lead (no baseline change).

Branch B: Approved / Implementation Path		
Step	Role / Personnel	Action Items
6B. Baseline Update	Configuration Management	Update Master Change Log, ICDs, allocations & workbooks.
7B. Implementation	Subsystem Leads & Engineers	Broadcast updates to leads; subsystems execute changes.
8. Verification	Mission Assurance	Mission Assurance verification testing & inspection of change.
9. Formal Closure	Project Leadership & QA	Sign-off & archive final record in central project repository.

Figure 114:

2.4.2. Scope Control Management

Team Fontus will manage changes in mission scope through the established change control process to protect mission objectives, cost, schedule, technical performance, safety, and verification requirements. Any proposed scope change must be reviewed for its effect on mission success, subsystem interfaces, risk, cost growth, schedule margin, and technical performance. Scope changes that affect requirements, subsystem design, procurement, testing, or mission operations must be documented and approved before implementation. This process helps prevent uncontrolled design growth while preserving the mission's required functions.

If the mission must be downscoped due to budget, schedule, or technical constraints, Team Fontus will use a clear priority order to reduce scope while protecting mission success. Required verification, qualification, and safety testing will remain protected. Only duplicated, optional, or nonrequired testing may be considered for reduction after confirming that the change does not increase unacceptable technical or safety risk. The downscope priority order is:

1. Remove optional outreach or demonstration features that do not affect mission success.
2. Reduce nonessential operational activities that are not required to meet core mission or science objectives.
3. Simplify noncritical subsystem features while preserving required interfaces, safety controls, and mission functionality.
4. Reduce optional data products, extended operations, or secondary analysis activities that are not required for mission success.
5. Adjust manufacturing or procurement methods if lower-cost or faster options still meet technical and safety requirements.
6. Defer lower-priority capabilities that are not required for the primary mission.
7. Consider simplifying or removing lower-priority science instrument functions only if the core science objectives remain achievable.

Scope reduction decisions will be evaluated against mission requirements before approval. For example, a subsystem design may be simplified if the change reduces cost or schedule pressure without reducing required performance. Manufacturing or procurement methods may also be adjusted if the selected alternative still satisfies interface, reliability, and safety needs. Mission operations may be modified by reducing nonessential activities or

shortening lower-priority operational sequences, but changes must preserve the ability to complete the primary mission objectives.

If additional budget, schedule margin, or technical resources become available, Team Fontus will consider scope growth only when it improves mission performance, reduces risk, or strengthens mission assurance. Scope growth will not be used to add complexity just because resources are available. Any proposed growth must still go through the change control process so the team can evaluate impacts to cost, schedule, mass, power, subsystem interfaces, verification, and risk before approval.

The scope-growth priority order is:

1. **Increase margin in mission-critical subsystems.** Additional resources should first improve power, thermal, mobility, CDH, and communications margins because these subsystems directly affect rover survival, commanding, surface operations, and science return.
2. **Add redundancy or fault tolerance where it reduces major risk.** Redundancy should be considered for high-consequence risks such as heater performance, power distribution, data paths, or command recovery.
3. **Strengthen verification and qualification.** Extra resources should support thermal-vacuum testing, environmental testing, integrated subsystem testing, and fault-recovery testing because these activities provide evidence that planned mitigation actually reduces risk.
4. **Improve operational flexibility.** Additional resources may support alternate traverse routes, contingency procedures, additional communication planning, or larger data-management margins.
5. **Expand science operations only after required mission functions are protected.** Additional science activities may be added if they do not reduce power, schedule, or operational margin needed for the core mission.
6. **Add optional demonstrations last.** Technology demonstrations or outreach features should only be added if required mission objectives, safety, verification, and schedule margin remain protected.

This priority order ensures that scope growth is used first to reduce mission risk and improve reliability before adding optional capability. A proposed scope increase will not be approved if the added cost, mass, power demand, schedule pressure, or integration complexity outweighs the expected mission benefit.

Scope Management Lifecycle

Scope management becomes more restrictive as the mission moves through each lifecycle phase because changes become more expensive and harder to verify later in

development. During formulation and concept development, scope changes are easiest to implement because requirements, subsystem concepts, procurement plans, and verification plans are still flexible. At this stage, changes can be used to improve mission feasibility, clarify requirements, or remove unnecessary complexity before the design becomes locked.

During preliminary design and detailed design, scope changes require stronger justification because subsystem interfaces, mass budgets, power budgets, thermal design, software architecture, and supplier decisions are becoming more mature. A change in one subsystem can affect multiple other subsystems, so proposed changes must be reviewed for interface impacts and verification impacts. During integration and test, scope changes are much more difficult because hardware, software, test procedures, and qualification plans are already established. Late changes can delay testing, reduce schedule margin, require rework, or introduce new technical risk.

During launch readiness and operations, scope changes should be limited to mission-critical corrections, risk reductions, or approved operational updates. At this stage, the main goal is to protect verified hardware, approved procedures, safety controls, and mission readiness. Any late scope change must show clear mission benefit and must not compromise required verification, safety, launch readiness, or mission-critical performance.

Team Fontus’s scope control strategy is to protect required mission objectives while allowing controlled changes when cost, schedule, technical, or risk conditions change. Downscoping will remove optional or nonessential items first, while required verification, qualification, safety testing, and mission-critical functions remain protected. Scope growth will prioritize risk reduction, added margin, redundancy, and stronger verification before expanding optional science activities or demonstration features.

This approach keeps scope decisions tied to mission success instead of convenience or preference. By requiring all scope changes to be documented, reviewed, and approved, Team Fontus can control cost, schedule, mass, power, technical performance, and safety while still allowing practical changes that improve mission reliability.

2.5. Outreach Summary

The Team Fontus public outreach plan cultivates widespread public interest in lunar science, broadens participation across diverse student populations, and articulates the enduring value of robotic planetary exploration. All public engagement modules directly trace to the Science Traceability Matrix (STM), translating the mission's core investigations—specifically characterizing the abundance, composition, and distribution of volatiles in Permanently Shadowed Regions (PSRs) and profiling near-surface regolith thermal behavior—into accessible educational content. Mechanical and mobility systems, including rover chassis resilience and wheel tread design, are presented not as isolated objectives, but as supporting engineering enablers that demonstrate how robotic architectures adapt to extreme cryogenic and abrasive terrain. By maintaining visibility across the project lifecycle, the program strengthens public scientific literacy and communicates the vital role of robotic precursors in long-term solar system exploration.

To maximize educational impact and promote equitable access, the outreach strategy engages five key target audiences:

- K–12 Students and Educators: Selected to spark early curiosity in aerospace disciplines through hands-on, curriculum-aligned engineering design challenges.
- Historically Underrepresented and Underserved Communities in STEM: Prioritized to reduce socioeconomic barriers to aerospace careers through direct partnerships with Title I schools, public library networks, and diversity-focused professional student chapters.
- Colleges, Universities, and Early-Career Researchers: Targeted to build academic workforce pipelines through technical symposia, open-source dataset workshops, and instrument science briefings.
- Regional Communities and Broad Public Demographics (County & State Fairs, Festivals): Selected to reach non-traditional, multi-generational audiences beyond standard academic environments through high-foot-traffic community venues.
- Informal Learning Centers, Planetariums, and Science Museums: Engaged to deploy immersive, high-volume public exhibits and interactive displays.

Outreach Activities

Outreach initiatives dynamically evolve across the project lifecycle, ensuring educational content mirrors real-time development milestones while reinforcing the mission's science goals:

1. **Formulation & Concept Development (Phases A–B) — Planetary Science Fundamentals & Physical Analogies:**
    - *Elementary School ("Cold-Trap Thermos" Sublimation Lab):* To explain how Permanently Shadowed Regions (PSRs) preserve volatile compounds, students place ice samples inside containers with varying insulating materials exposed to external heat lamps. Measuring melt and sublimation rates introduces young learners to phase changes, thermal gradients, and the physical principles of how polar craters at Nobile Rim 2 trap ancient water ice.
    - *Middle & High School (Orbital Mapping Workshops):* Students use public planetary GIS tools (such as JMARS) to evaluate slope, illumination, and temperature data, learning how mission constraints dictate landing site selection and safe approach corridors.

2. **Final Design, Fabrication, & Integration (Phases C–D) — Engineering Enablers, Prototyping, & Public Fairs:**
    - *Interactive County and State Fair Discovery Pavilion:* Team Fontus deploys interactive public booths at high-traffic county and state fairs to reach diverse, multi-generational demographics. The booth features 5- to 10-minute "Micro-Lectures" explaining volatile prospecting at Nobile Rim 2, tactile simulant regolith touch-tables, and a scaled rover prototype demonstration. Attendees receive educational take-home souvenirs (such as mission patch stickers, 3D-printed miniature wheel models, or infographic field cards) to reinforce long-term mission recall and scientific interest.
    - *K–12 Regolith Mobility & Tread Design Challenge:* Student teams design and test scaled wheel treads in simulated abrasive lunar regolith testbeds. The activity demonstrates how mechanical engineering solutions mitigate wheel sinkage, slippage, and high drive-motor power consumption on steep crater slopes.
    - *University Technical Seminars & Cleanroom Virtual Tours:* Undergraduate engineering and planetary science students attend technical webinars covering flight hardware integration, environmental testing protocols (TVAC and vibration), and the operational physics of instruments such as the volatile mass spectrometer (MSolo) and subsurface thermal probe (LISTER).
3. **Surface Operations & Science Execution (Phase E) — Real-Time Discovery & Public Mission Operations:**

- *Live Milestone Landing & Deployment Broadcasts:* Public livestreams feature landing telemetry, initial mobility drive-off, and instrument deployment milestones with commentary from project scientists and engineers.

- *Public Surface Traverse & Science Dashboard:* An open-access web dashboard visualizes real-time rover telemetry, including cumulative distance traversed, local surface illumination status, and non-proprietary volatile detection indicators as the rover navigates across daylight-to-shadow boundaries.
- *Classroom "Ask a Mission Specialist" Virtual Panels:* Bi-weekly live Q&A sessions allow K–12 classrooms to engage directly with subsystem engineers and scientists, exploring daily operational problem-solving and incoming environmental data.

MISSION LIFECYCLE OUTREACH EVOLUTION		
PHASES A & B Formulation & Concept Design	PHASES C & D Design, Build, Test & Launch	PHASES E & F Surface Ops & Data Legacy
• Cold-Trap Thermos Labs • JMARS Landing Site Workshops • Title I Partner Onboarding • Bilingual Lesson Kit Rollouts	• County & State Fair Booths • Regolith Mobility Challenges • Virtual Cleanroom Tours • Educational Souvenir Outreach	• Live Landing & Traverse Feed • Real-Time Volatile Telemetry • Public Q&A Science Panels • Open Data Legacy Repositories

Figure 115: Team Fontus Outreach Evolution Across NASA Mission Lifecycle Phases.

Community Engagement

To broaden participation in aerospace, Team Fontus actively partners with Title I school districts, public library networks, agricultural/county fair organizing committees, and student chapters of professional diversity organizations, including NSBE, SHPE, and SWE. All activity guides, visual graphics, and booth informational materials are published bilingually (English and Spanish) and formatted for accessibility. Fair demonstrations and classroom labs are intentionally engineered around low-cost, portable items, ensuring equitable participation across both urban centers and rural agricultural communities.

Outreach Implementation

The outreach plan is executed collaboratively through academic departments, regional science centers, Title I educational coordinators, fairground event organizers, and public libraries. These established venues provide exhibition booths, auditorium facilities, virtual streaming networks, and community classrooms without placing logistical or financial burdens on primary spacecraft development. All distributed materials, physical displays, and souvenir media undergo review by the Mission Assurance team to guarantee strict compliance with ITAR/EAR export regulations and intellectual property safeguards prior to public release.

Summary

The Team Fontus outreach plan extends the impact of lunar exploration beyond technical milestones, turning high-stakes robotic exploration into an open catalyst for educational equity and scientific curiosity. By translating extreme-environment engineering challenges and lunar volatile science into inclusive, hands-on learning pathways across community fairs, classrooms, and digital platforms, the mission equips the next generation of researchers with the skills, tools, and inspiration to lead humanity's ongoing expansion into the solar system.

3. Conclusion

The Preliminary Design Review has advanced Team Fontus's lunar surface mission from a defined mission concept into a preliminary engineering design for investigating lunar volatiles, subsurface thermal properties, seismic activity, and environmental conditions near the Nobile Rim 2 exploration region. During the PDR phase, the mission architecture was further matured through the development of integrated mechanical, power, command and data handling, thermal, and payload subsystem designs. The rover architecture was refined around a six-wheel rocker-bogie mobility system, non-deployable solar arrays, and integrated science instrumentation, while subsystem requirements, verification approaches, procurement strategies, interfaces, and recovery considerations were developed to establish a preliminary technical baseline. The current design has a total mass of 158.9 kg within the 175 kg system limit and maintains the required 1.5 m × 1.5 m × 1.0 m customer-defined vehicle envelope. The mission also remains within the $250 million lifecycle cost constraint and maintains the planned December 1, 2028 system integration readiness and February 1, 2029 launch readiness dates. Together, these efforts demonstrate substantial progression from the MDR baseline toward a technically integrated preliminary design.

The next phase of development will focus on closing the remaining technical and programmatic gaps identified during PDR and preparing the mission for the final MCA

deliverable and subsequent Critical Design Review (CDR). Work that will take priority will include finalizing subsystem interfaces and integrated CAD, validating the mass and power budgets against the physical design, completing thermal and structural analyses, refining the launch vehicle interface, and developing more detailed verification and validation plans. Special importance will be given to the remaining mission risks and requirements that are currently partially met or unresolved, including the two-week operational requirement, thermal performance within permanently shadowed regions, launch vehicle compatibility, subsurface measurements, and electrostatic characterization. These risks and their associated mitigation strategies will be baselined in the final MCA deliverable to provide a clear record of the remaining technical uncertainties and the work required to close them.

With additional development time, the mission would benefit from higher-fidelity thermal, structural, power, and mobility analyses; additional testing and simulation of rover operations within permanently shadowed regions; and further refinement of science operations and instrument performance. Additional analysis of LISTER deployment and drilling operations, geophone placement and seismic measurements, MSolo sampling, and the TENG/electrostatic investigation would also improve confidence in the mission's ability to achieve its science objectives. Further refinement of subsystem interfaces, component procurement and manufacturing schedules, cost estimates, and verification plans would also reduce remaining technical and programmatic uncertainty. Overall, the work completed during PDR establishes a substantially more integrated and traceable mission design and provides a strong foundation for closing the remaining risks and advancing Team Fontus toward the final MCA deliverable and subsequent phases of mission development.

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Appendix: Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the team used ChatGPT/Claude to identify relevant search terms and potential scholarly or government sources, identify missing information, and suggest revisions for clarity and completeness. ChatGPT/Claude was also used to compare selected draft sections against the Mission Concept Review instructions and recommend areas requiring additional research or explanation. After using this tool, the team independently reviewed and edited the content as needed and takes full responsibility for the content of the deliverable.

ChatGPT/Claude was used as a research and writing-support tool rather than as a primary technical source. The goal was to help the team begin research more efficiently, identify possible NASA documents, peer-reviewed publications, government resources, and relevant technical terminology, and organize the collected information according to the required MCR sections. The tool was also used to review draft material for completeness, consistency, organization, and alignment with the Mission Task Document and MCR template.

AI-generated responses were not automatically included in the deliverable. The team selected only information that was relevant to the mission concept, edited the selected material as necessary, and supported factual and technical statements with the original sources cited in the bibliography.

The team verified AI-assisted information through the following process:

1. Every factual or technical statement used in the deliverable was compared with the original L'SPACE Mission Task Document, NASA publications, peer-reviewed research, government websites, university resources, or other credible professional sources.
2. Each suggested source was opened and reviewed by a team member to confirm that it existed, was credible, and supported the associated statement. ChatGPT was not cited as the source of scientific or technical information.
3. Source titles, authors, publication dates, URLs, numerical values, and technical conclusions were checked against the original publications.
4. Unsupported, inaccurate, outdated, or unverifiable information was removed or revised.
5. Relevant science, engineering, and programmatic content was reviewed by the appropriate subteam members before inclusion in the final deliverable.
6. Citations and bibliography entries were checked to ensure that the original sources were properly credited using the required Chicago citation style.

The complete prompts submitted to ChatGPT/Claude and the corresponding ChatGPT/Claude responses are included in this appendix. The specific sentences,

recommendations, source suggestions, or other portions that contributed to the final deliverable are highlighted. Material included in the transcript but not highlighted was reviewed, but was not directly used in the submitted deliverable.